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Maria Stella Mongiovì
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Non-equilibrium Thermodynamics of Superfluid Helium and Quantum Turbulence

The Richness of Non-Fourier Heat Transport

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Chapter 1
Introduction



In this book, we provide a general vision of four main aspects of the low-temperature liquid phase of helium-4, often named He II or Helium II or superfluid helium, and quantum turbulence from the perspective of non-equilibrium thermodynamics. The first one is a comparison of the standard two-fluid model and a one-fluid extended model of Helium II and their respective merits in providing complementary descriptions of this fascinating physical system. The second aspect is a thorough analysis of the rich phenomenology of heat transport in superfluid helium, including an analysis of quantum turbulence, its generation, its physical features, and its consequences on heat transport. The third aspect is devoted to a comparison of quantum and classical turbulence, from the points of view of the closure of the hierarchy of moments of fluctuations and of the main features of energy cascades in turbulence. The fourth aspect is a discussion of some analogies between the thermodynamics of turbulent vortex tangles and the thermodynamics of cosmic strings, its cosmological consequences and some dualities between the thermodynamics of photons and of cosmic string loops. Emphasis is provided on non-equilibrium thermodynamic methods, superfluid heat transport phenomenology, macroscopic dynamics of quantum turbulence, and formal explorations on temperature and entropy beyond local equilibrium states.

1.1 From Classical Fluids to Quantum Fluids

Helium-4 was first liquified by Kammerling Onnes in 1908 [1–4]. This was a remarkable achievement, as it was the last pure gas to be liquefied (hydrogen had been liquefied by Dewar in 1898), but the perspectives opened by it were much wider and surprising than expected. Indeed, the liquefaction temperature of helium at standard pressure is 4.15 K, and it behaves as a normal, viscous, heat-conducting fluid, but when submitted to lower temperatures it turns out that below 2.17 K (the so-called λ temperature, where the specific heat of liquid helium as a function of temperature has

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a very sharp peak evocating the form of λ as shown in Fig. 1.1) it shows completely new aspects as compared to those of simple Newtonian heat-conducting fluids.

As it was found in 1937 by Kapitza, it flows along narrow channels almost without viscosity, whereas in rotational oscillating viscometer it shows a usual value for the viscosity, of the order of tenth of that of air. In 1938, Allen and Jones found the so-called fountain effect, namely, they observed that when two vessels containing Helium II are connected by a very narrow capillary, heating of one vessel leads not only to a difference of temperature between both vessels, but also to a flow of liquid into the hotter vessel, as a consequence of a difference of pressure between both vessels, in a direction opposing that of the heat flow (see Fig. 1.2). Liquid helium also shows a tendency to flow up to the walls of the containers keeping it and to partially escape from them.

On the other hand, Helium II transports heat in a very efficient way, in different forms than usual heat conduction described by Fourier's law. In 1946, in the so-called counterflow experiment (an external heat is supplied to Helium II in a channel closed from one side as shown in Fig. 1.3) it was experimentally found temperature waves propagating in superfluid helium almost without attenuation called second sound to differentiate them from the usual propagating pressure waves of sound, also dubbed

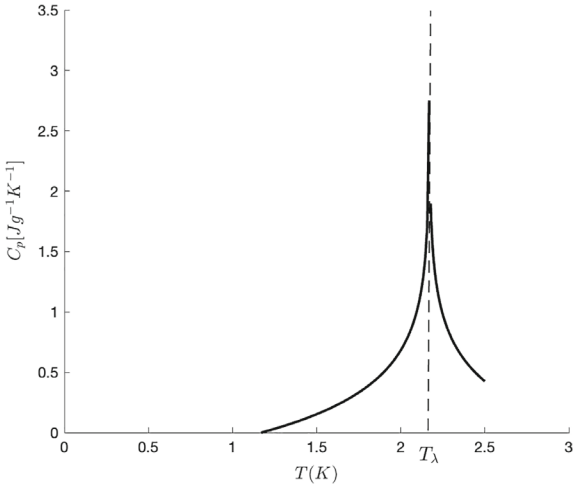


Fig. 1.1 The specific heat C of liquid helium as a function of temperature T. Note the cusp behaviour near 2.17 K (the so-called λ temperature). Below such temperature, liquid helium shows superfluid behaviour

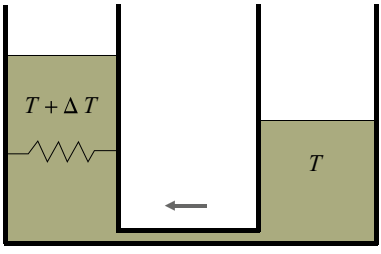


Fig. 1.2 The fountain effect: helium flows from the lower (T) to the higher ($T + \Delta T$) temperature, until the pressure differences, shown by the different height of the surface of the fluid, compensate the temperature difference. The zigzag line on the left container represents a heat source, and the arrow the helium flow between the containers

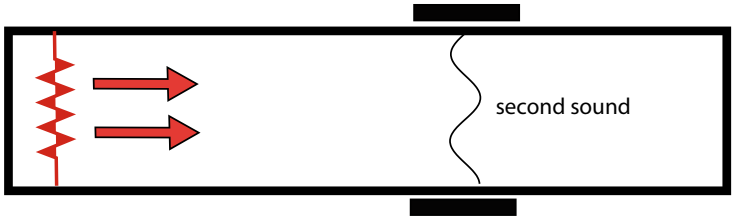


Fig. 1.3 Counterflow experiment and second sound wave: the red arrows represent steady heat flux, flowing from the zigzag heat source; horizontal black rectangles are emitter and receptor of probe heat wave (second sound) represented by the wavy line

as first sound. Two other characteristic effects are the mechanocaloric effect in which superfluid motion produces a temperature gradient [5], and the thermo-mechanical effect in which a temperature gradient produces motion [6]. Also, there is a nonlinear relation between heat flux and mechanical stress.

With those discoveries, a new field of physics began: that of quantum fluids. Indeed, such features were explained in terms of coherent collective effects due to long-range quantum correlations between particles at low temperatures, when the typical quantum wavelength related to the thermal motion of particles becomes longer than the separation among particles. Later, such effects were also found, at temperatures of the order of milliKelvin, in Helium 3, and still at lower temperature, of the order of nanoKelvin, in so-called Bose–Einstein condensates of alkali atoms. In fact, such peculiar effects are found in particles of Bosonic character (namely, with integral values of the spin, following the Bose–Einstein statistics) but, in some circumstances, fermionic particles (with half-integral spin, following the Fermi-Dirac statistics) may form Bosonic pairs collectively exhibiting these features.

Some particularly well-recognized and influential discoveries in the field of quantum fluids, having been awarded with Nobel Prizes in Physics, are the following ones: in 1913 the prize was awarded to Heike Kamerlingh Onnes "for his investigations on the properties of matter at low temperatures which led to the production of liquid helium". In 1962, Lev D. Landau received the award "for his pioneering theories for condensed matter, especially liquid helium". In 1978 Piotr L. Kapitza shared the prize "for his basic inventions and discoveries in the area of low-temperature physics" and in particular for his discovery (1938) that at very low temperatures liquid helium flows without resistance, which is the basis of the name "superfluidity". David M. Lee, Douglas D. Osheroff, and Robert C. Richardson were awarded the 1996 Prize for their discovery of superfluidity in ^{3}He. In 2003, Alexei A. Abrikosov, Vitaly L. Ginzburg, and Anthony J. Leggett were awarded "for pioneering contributions to the theory of superconductors and superfluids". In 2001, Eric A. Cornell, Wolfgang Ketterle, and Carl E. Wieman received the award "for the achievement of Bose–Einstein condensation in dilute gases of alkali atoms, and for early fundamental studies of the properties of the condensates". These particular discoveries and landmarks illustrate the relevance and intensity of the research in this field [1–4, 7].

Two kinds of models have been proposed to describe the observations on Helium II and make new predictions: the two-fluid model, initially proposed by Tisza and Landau in 1940, and widely used since then, and an extended one-fluid model with a vectorial internal degree of freedom, closely related to the heat flux proposed in 1990. Rather than radically different alternatives we see them as complementary ones, each of them with its own motivations and advantages, discussed below.

In the two-fluid model, Helium II is considered as a mixture of a normal component and a superfluid component, each of them having their own velocity, $\mathbf{v}_n$ and $\mathbf{v}_s$. In the one-fluid model [8–11], the velocity $\mathbf{v}$ and the heat flux $\mathbf{q}$ are chosen as fundamental variables instead of $\mathbf{v}_n$ and $\mathbf{v}_s$. Both models may be related to each other because $\mathbf{v}$ and $\mathbf{q}$ are closely related to $\mathbf{v}_n$ and $\mathbf{v}_s$, but they may be directly and easily observed, in contrast to $\mathbf{v}_n$ and $\mathbf{v}_s$.

Using such additional independent variable (either $\mathbf{v}_n - \mathbf{v}_s$, or $\mathbf{q}$) requires two things: obtaining an evolution equation for it, and specifying its contribution to the fundamental thermodynamic equation of the system, as, for instance, to the internal energy or to the entropy. Both topics (evolution equations, entropy contributions) are mutually related, because the evolution equations for the variables should be compatible with the second law of thermodynamics. A framework where this kind of questions has been discussed at length is, for instance, extended thermodynamics, where the fluxes are taken as additional independent variables [12–15]. From a macroscopic point of view, an extended approach to thermodynamics is required in Helium II because the relaxation time of heat flux is comparable to the evolution times of the other variables. In these circumstances, a constitutive description of the heat flux field in terms of the traditional field variables (namely, as a function of the temperature gradient) is impossible, so that it must be treated as an independent field of the thermodynamic process.

The superfluidity of ^{3}He has several outstanding differences with that of ^{4}He [4, 7, 16]. In contrast to the Bosonic character of ^{4}He (whose atoms have 0 spin), the atoms of ^{3}He have half-integral spin $1/2$ and therefore they are fermionic particles. Below 3.3 K, ^{3}He becomes a normal liquid, down to 0.8 K. From 0.8 K to 2.6 mK it behaves as a Fermi quantum liquid. Below 2.6 mK the atoms pair up and form Bosonic pairs, especially with spin triplet pairing (spin $S = 1$, rather than spin $S = 0$) and has a transition to superfluid states, which are found in two different phases, A and B, which have a small (but not extremely small) viscosity, higher than that of superfluid ^{4}He. Phase A is anisotropic, and it has the states $+1$ and -1 of the spin triplet, and a magnetic susceptibility independent of the temperature; phase B is isotropic, the three states of the spin triplet are found $(+1, 0, -1)$, and its magnetic susceptibility decreases with temperature. The superfluid states of ^{3}He are too expensive and too cold to be used for cryogenic applications (in contrast, non-superfluid mixtures of ^{4}He and ^{3}He are considerably used in cryogenics), but they are of basic interest because their phenomenology is even richer than that of ^{4}He and because it allows for some analogies in cosmology or in particle physics.

The Bose–Einstein condensates of alkali atoms share some aspects with Helium II (superfluidity, quantized vortices) but they differ from it in several substantial aspects [4, 7]. For instance, the Bose–Einstein condensates are found at much lower

temperature than superfluid helium of the order of nanoKelvin, they consist of a relatively small number of particles (of the order of 10^5 or 10^6, much lower, up to now, than the usual liquid helium samples), they are much more dilute and compressible (as they are gases rather than liquids), and they are not confined in material containers but in optical or magnetic traps, whose size is some three or four orders of magnitude smaller than the containers used for experiments with superfluid helium. Furthermore, since a number of different species of atoms may be used, in contrast to the case of ^{4}He or ^{3}He, this allows for a wider variety of physical parameters (mass, spin, spin-orbit couplings) to be explored.

1.2 From Classical Heat Conduction to Inertial, Hydrodynamic, and Turbulent Heat Transport

Heat transport deserves a particular attention in superfluid helium, not only from a practical perspective but also from the conceptual one. Indeed, the high values of thermal conductivity and specific heat at constant pressure, as well as its very low resistance to flow in narrow channels, make it very useful for cryogenic applications [17]. But besides these practical aspects there is a wide wealth of conceptual new aspects in heat transport in superfluid helium which make it probably the richest system in this perspective. These new aspects may be conventionally dubbed as inertial, hydrodynamic, and turbulent heat transport.

By inertial transport it is meant that the relaxation time of the heat flux is very long, namely, that its inertia with respect to changes in the temperature gradient is high. This has as a particularly relevant consequence the wavelike propagation of heat by means of second sound. To incorporate this phenomenon, the equation relating $\mathbf{q}$ and ∇T must incorporate an additional term, proportional to the time derivative of the heat flux.

By hydrodynamic transport it is meant that heat transport exhibits remarkable nonlocal effects, as a consequence of long quantum correlations. This means, for instance, that the equation relating $\mathbf{q}$ to ∇T must contain terms in the Laplacian of $\mathbf{q}$, analogous to the term in the Laplacian of velocity in the Navier–Stokes equation describing the evolution of the velocity in viscous fluids. This has as a consequence, for instance, that the effective thermal conductivity of fluid helium along channels depends on the size and the form of the cross section of the channel and that heat flow may have vortices. This has served as inspiration for the description of the so-called phonon hydrodynamics found in solids since the decades of 1960 [18–20].

By turbulent transport we refer to two relevant features: that for values of q higher than some critical value, the heat flux acts as a source of disordered tangle of quantized vortices in the superfluid and, second, that such quantized vortices produce an additional resistance to the flow of heat, of the order of q^3 (q being the modulus of $\mathbf{q}$), thus reducing the advantages of superfluid helium in cryogenic situations. These two features are not found in other physical systems, in contrast to heat inertia

and heat nonlocal features, which may be found in solids and nanosystems. The description of such turbulent effects requires that the equation relating $\mathbf{q}$ to ∇T must contain a term describing the heat resistance due to the vortices and that there must be a coupling of the heat flux with an additional equation for the dynamics of quantized vortices produced by the heat flux.

Superfluid helium is much used in cryogenic techniques, because it flows with much ease, and because it has very high thermal conductivity and specific heat. It is used, for instance, to refrigerate the superconducting coils carrying high electric current and producing high magnetic fields used in particle accelerators, in nuclear fusion experiments, and in magnetic resonance devices in chemical analysis and medical images. It is also used to refrigerate quantum computers and sensitive measuring devices in observation satellites. Helium II has been the paradigmatic model of superfluidity, which has been later discovered in helium 3, and in Bose–Einstein condensates. Superfluidity is thought to be present also in neutron stars, and it may provide an heuristic model for some properties of the quantum vacuum in cosmology [21].

1.3 From Classical Turbulence to Quantum Turbulence

Another peculiar behaviour of Helium II is the presence of quantized vortices [3, 22]. The first observations of some peculiar aspects of rotation in superfluids arose in the late 1950s, when it was realized that vorticity may appear inside superfluids and that it is quantized, its quantum of circulation κ being $\kappa = h/m_4$, with h the Planck constant and m_4 the atomic mass of ^{4}He.

Some experiments about propagation of second sound in rotating systems showed that when second sound propagates at right angles with respect to the rotation axis, it suffers an extra attenuation compared to the propagation in non-rotating vessels, in an amount proportional to the angular velocity. In contrast, a negligible attenuation of the second sound is found when the propagation is parallel to the axis of rotation. The large increase of attenuation observed when the liquid is rotated can be explained by the mutual friction between the flow of excitations (phonons and rotons) and the quantized vortex filaments. Such vortices have been directly observed and quantitatively studied.

Since vortices are always characterized by the same quantum of circulation, it follows that for higher rotation rates the total length of the vortices per unit volume, L, increases. In rotating systems the vortices are seen to form a regular array of almost parallel lines. This has strong similarities with electrical current vortex lines appearing in superconductors submitted to a high enough external magnetic field. This analogy has fostered the interest in vortices in superfluids, which allow one to get a better understanding of the practically relevant vortices in superconductors [23, 24].

Quantized vortices also play a central role in superfluid turbulence. If the average separation between quantized vortices, namely, $L^{-1/2}$, is denoted as ℓ, it may be said that the energy spectrum of quantum turbulence has a hydrodynamic region, for wavevectors k such that $k\ell \ll 1$, where quantum vortices are irrelevant or they

form bundles of quantized vortices which mimic classical eddies leading to strong analogies to the usual hydrodynamic turbulence, a quantum region for $k\ell \gg 1$, where quantized vortices are dominant and which strongly differs from usual turbulence, and an intermediate region, for $k\ell \approx 1$. Thus, the energy cascade among vortices of different sizes will exhibit different features in these different regions of values of $k\ell$.

Another difference between classical turbulence and quantum turbulence is that in Helium II two main kinds of turbulence arise, namely, coflow turbulence, when the barycentric velocity $\bar{\mathbf{v}}$ exceeds a critical value of a classical Reynolds number $\rho v d/\eta$, with η the viscosity of Helium II and counterflow turbulence, where $\bar{\mathbf{v}} = 0$ but the heat flux $\mathbf{q}$ exceeds a critical value of a quantum Reynolds number $qd/sT\kappa$, with s the entropy per unit volume and κ the quantum of circulation. The macroscopic typology is basically the same but they differ in some details of their respective energy cascades.

In counterflow experiments, when the heat flux exceeds a critical value q_c, a random array of vortex filaments appears [25–28]. The presence of vortices considerably reduces the efficiency of heat transport in the superfluid, as compared to the value one could expect in the absence of vortices. Vortex tangles also appear when the average barycentric velocity of Helium II with respect to the walls of the channel exceeds some critical value. In these circumstances, the mechanical resistance to the flow considerably increases. The formulation of transport equations introducing all these complexities is a topic in transport theory.

1.4 From Thermodynamics of Vortex Tangles to Thermodynamics of Cosmic Strings

Still a fourth aspect dealt with in the present book and not yet presented in other books is an exploration of formal analogies between the thermodynamics of vortex loops in turbulent superfluid helium and the thermodynamics of cosmic string loops. These up to now hypothetical physical entities are assumed to be linear topological defects of the cosmic spacetime, acting as sources of a repulsive interaction which accelerates the rate of cosmic expansion and are candidates to models of dark energy.

In this analysis, we do not only explore some aspects of this analogy, related to a negative thermodynamic pressure for the gas of cosmic strings, and to an adiabatic heating in expansions instead of the adiabatic cooling of radiation, but we also deal with a formally appealing thermodynamic duality between a gas of photons and a gas of cosmic string loops, by using a common thermodynamic formalism for both kinds of physical entities, as well as for some possible hypothetical intermediate entities.

We also explore some consequences of such thermodynamic duality on the thermodynamics of hypothetical mini black holes respectively dual to the thermodynamics of macro-black holes. Whereas in Hawking's theory small black holes are

expected to evaporate in a very fast, almost explosive way, in the duality version they turn out to evaporate very slowly, thus leading to a very different behaviour than the classical one.

The relation between superfluids and some particular ingredients of cosmological models should not come as a surprise, but it is well known and explored in theoretical physics, from other perspectives than the one examined in this book [29, 30]. For instance, it has been proposed to consider the quantum vacuum from the perspectives of low-temperature condensed matter, where symmetry and topology acquire a fundamental role. A special attention is devoted to superfluid ^{3}He in A phase, because it belongs to the same universality class of the Fermi liquids, where an effective gravity and gauge fields appear in the low-temperature limit, leading to some correspondences between phenomena in superfluid ^{3}He A and those in relativistic particle physics. Also superfluid ^{4}He, a Bose liquid, exhibits interesting analogies with a sort of effective gravitational fields appearing at very low energies, thus providing the simplest model for an effective field theory with a low number of effective fields. For instance, it has been worked out a view of the two-fluid model as an effective field theory incorporating the collective motion of the superfluid background (the ground state, playing an analogy with a gravitational field) and the quasi-particle excitations (phonons, rotons) composing the so-called normal component (playing an analogy with matter). Whereas in Helium II all the excitations are Bosonic, in ^{3}He they may be fermionic and Bosonic, thus providing a view closer to the known physical observations. Though we devote only a small part of the book to this topic, its potential interest should not be dismissed.

1.5 Structure of the Present Book

In Chaps. 2 and 3, we present, discuss, and compare the two-fluid model and a one-fluid extended model for superfluid helium, and show how they are able to describe the phenomenology observed in laminar superfluid flows (i.e. in the absence of vortices). In a linear and non-dissipative approximation, both models are practically equivalent, but in nonlinear settings some differences arise. The two models are not inconsistent with each other, but take two different sets of independent physical quantities as their respective bases. Depending on the physical circumstances and kinds of experiments, one choice may be more direct and convenient than the other.

Chapters 4–7 deal with quantized vortices arising in rotation and in turbulent flows of superfluid helium, from a macroscopic perspective, as well as their coupling with heat transport. We examine several forms of the evolution equation for the vortex line density L, describing the production and destruction of the vortices, as well as the effects of the walls and of rotation. We also work out inhomogeneous, polarized, and anisotropic situations in which the vortex tangle must be described by a second-order tensor instead of the single scalar L and which arise, for instance, in radial heat flow with axial or spherical symmetry, in heat flow along channels with inhomogeneous cross section and in two-dimensional superfluid flows with polarized vortices, in

which there is a mutual interaction between heat flow **q** and polarization **p**. We also describe the resistance presented by the vortices to the heat flux across the superfluid, and discuss the couplings between the equations for the velocity **v**, the heat flux **q**, and the vortex line density L.

Chapter 8 provides a glimpse to microscopic aspects of superfluid helium and of lambda transition from the basic quantum perspective. We clarify the basic foundations of the topic, we discuss the dynamics of the quantized vortices from a microscopic perspective, consider second sound near the lambda transition and comment on the influence of the vortices on the lambda transition from normal behaviour to superfluid behaviour. In Chap. 9, a microscopic view of the dynamics of vortex lines is presented to provide a basic understanding of the main aspects of the evolution equations for L provided in Chaps. 4–7.

Chapter 10 provides a comparison between classical and quantum turbulence. A generalization of classical models of turbulence to quantum turbulence is proposed; in particular, we go from the $K-\epsilon$ model for classical turbulence to a generalized $K-\epsilon-L$ model for quantum turbulence, taking into account the couplings between the fluctuations of **v** and the fluctuations of **q** and of L. Furthermore, we discuss the energy cascades in the classical and quantum case, commenting the differences between the hydrodynamic region and the quantum region of the energy cascades. Both kinds of analyses are phenomenological and try to describe some essential physical features without exploring their microscopic details, which are deserving attention in current specialized literature.

Chapter 11 provides a discussion of some thermodynamic features of the vortex tangle, dealing with the entropy and temperature which may be assigned to it as a peculiar physical entity. Such entropy and temperature are not directly related to the entropy and temperature of the underlying superfluid helium, but exhibit some peculiar features, as, for instance, an effective temperature for tangles which is several orders of magnitude higher than the temperature of the superfluid (this is not surprising as we are dealing with a system far from equilibrium instead of a system at equilibrium), and a negative thermodynamic pressure. These features are of relatively small practical interest for applications of superfluid helium, but from a conceptual perspective they provide a bridge towards analogies between the thermodynamics of vortex tangles and that of cosmic strings. In particular, this leads to the basis for a dual thermodynamic formalism for photons and cosmic strings, and an analogous thermodynamic duality between macroscopic and microscopic black holes. On the basis of such dualities, we suggest a duality-invariant version of the de Broglie relation of quantum mechanics, making it invariant under the change of λ/l_p and l_p/λ, with λ the wavelength and l_p the Planck length, which could alleviate some inconsistency problems between quantum physics and general relativity.

Chapter 12 is devoted to the final discussions regarding the values of the physical parameters and the boundary and initial conditions of the variables in the one-fluid extended model of superfluid helium, in order to support the researchers interested to improve the model or implement the model by numerical simulations. For these reasons, we analyse the parameters appearing in the model and hope that new

experiments will help a better understanding of the parameters as well as the results. On the other hand, the chapter wants to assist the researcher on the boundary conditions for the fields involved.

We are aware of the considerable number of textbooks and monographs devoted to superfluid helium 4 and to superfluidity in general (incorporating that in helium 3 and in Bose–Einstein condensates) [7, 16, 17, 29, 31–40]. However, the number of books paying a detailed attention to non-equilibrium thermodynamic methods applied to superfluidity and heat transport in superfluids is very small [15]. This is also the case of books incorporating the application of non-equilibrium thermodynamic methods as applied to quantum turbulence, which is a relatively new aspect, or comparing thermodynamics of photons, of quantum strings, and of vortex loops.

The book emphasizes didactical views over specialistic details, and may be used as an introduction to non-equilibrium thermodynamics of superfluid helium and its astonishing heat transport properties (second sound, nonlocal transport, nonlinear transport). It will also be of interest to researchers in superfluid helium and in heat transport, as it provides a wide overview of the rich connections between heat transport and quantum turbulence. The last two chapters will be appealing to researchers interested in the thermodynamics of cosmic strings and of black holes and their consequences in cosmological models. The diversity and complexity of physically founded equations will be inspiring for researchers in mathematical physics.

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Chapter 2
The Two-Fluid Model of Superfluid Hydrodynamics: Concepts, Equations, and Applications



To explain the complex phenomenology of flowing helium described in the introduction, Landau (1941) and Tisza (1940) independently proposed the so-called two-fluid model, according to which Helium II is considered as a mixture of a normal component (viscous, with non-vanishing entropy) and a superfluid component (with zero viscosity and vanishing entropy) [1–5]. This idea had a double motivation: from the microscopic view, it was based on the Bose–Einstein condensation and from a macroscopic view, it provided a possibility of describing apparently contradictory results, as zero viscosity in narrow channels and normal viscosity in torsional viscometers.

The conceptual basis for this splitting is the Bose–Einstein condensation (proposed by Einstein in 1925 and applied by London to superfluid helium in 1938). This phenomenon consists of a macroscopic occupation of the fundamental state of the system, namely, the lowest energy single-particle state with zero momentum below a critical temperature. In such a state, the particles would behave in a macroscopically coherent way, thus exhibiting a null (or a very small) entropy and a very small resistance to the flow. In contrast, the uncondensed atoms would constitute a normal fluid with a non-vanishing viscosity (instead of the uncondensed atoms one now considers that the normal component is constituted by two kinds of quasi-particles, phonons, and rotons, corresponding to excitations of the system).

In his work London argued that liquid helium is not an ideal Bose gas, owing to the strong interactions between its particles. Avoiding the difficult problem of a rigorous treatment of an interacting Bose–Einstein fluid, Tisza, in the same year, using London's results, formulated a macroscopic description of helium as a condensing Bose–Einstein gas, which is now known as the two-fluid model [6]. Landau considered the basic idea of an ideal Bose gas not applicable to liquid helium, and pointed out that this theory does not explain superfluidity [7]. His treatment is inspired by the quantum theory of low-temperature solids. Following Landau, there are in Helium II modes of vibration corresponding to sound waves, and modes of vibration corresponding to thermal waves. Thus Landau constructed the energy spectrum of Helium II, introducing two types of elementary excitations (phonons

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and rotons) (see Sect. 8.1 and Fig. 10.3.39). The lower position of the phonon states with respect to the roton states in the spectrum of excitations, and the existence of an energy gap between them, leads to superfluidity.

As Tisza theory, Landau theory leads to a two-fluid description, but in the view of Landau the two-fluid model must be regarded only as a convenient way of saying that the liquid as a whole is capable of two types of motion simultaneously. Thus, the terms "normal" and "superfluid" were associated with the effective masses of the two types of motion. In fact, the fluid cannot be physically separated in both parts, but this model provides a convenient description and a conceptual framework for describing in a systematic way the observed phenomenology. The point of view of Landau is the one considered today, and is the starting point for the construction of the one-fluid extended model of Helium II derived from Extended Thermodynamics, that will be the object of Chap. 3.

2.1 Short Introduction to the Two-Fluid Model

The normal and superfluid components of the two-fluid model have mass densities ρ_n and ρ_s, respectively, such that $\rho_n + \rho_s = \rho$, with ρ being the total observable mass density of the fluid. Both ρ_n and ρ_s depend on the temperature T in such a way that $\rho_s = 0$ at $T = T_\lambda$ (with T_λ the lambda temperature) and $\rho_s = \rho$ in the limit when T tends to zero. The behaviour of ρ_n is the opposite one, namely, $\rho_n = \rho$ at $T = T_\lambda$ and $\rho_n = 0$ in the limit when T tends to zero. The behaviour of ρ_s in terms of T is roughly described as $\rho_s = \rho[(T_\lambda - T)/T_\lambda]^{1/3}$. The values of $\rho_s(T)$ and $\rho_n(T)$ are usually obtained from measurements of the viscosity in torsional viscometers which, in contrast to the flow in narrow channels (with almost vanishing resistance), show a non-vanishing viscosity, related to ρ_n. The ratio ρ_s/ρ_n may also be obtained from the speed of second sound, as it will be seen in Sect. 2.8 (Fig. 2.1).

The normal and the superfluid components are assumed to move with respective speeds $\mathbf{v}_n$ and $\mathbf{v}_s$, related to the barycentric speed $\mathbf{v}$ as

$$\rho_n \mathbf{v}_n + \rho_s \mathbf{v}_s = \rho \mathbf{v}. \tag{2.1.1}$$

These velocities, in general, are not directly observable, but they may be obtained from $\mathbf{v}$ and $\mathbf{q}$. The heat flux is given by

$$\mathbf{q} = \rho s T (\mathbf{v}_n - \mathbf{v}), \tag{2.1.2}$$

with s the specific entropy, that in this model is ascribed only to the normal component ($\rho s = \rho_n s_n$). Indeed, since the specific entropy of the superfluid component is assumed to be zero, the only flow of heat in helium at rest is related to the motion of the normal component. The barycentric contribution in $\mathbf{v}$ is subtracted in order

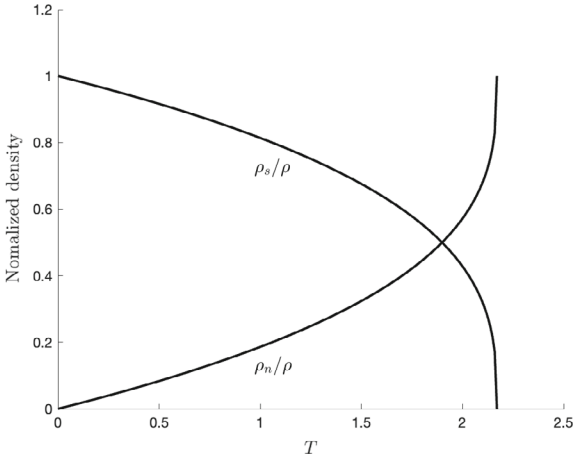


Fig. 2.1 Density ratios ρ_s/ρ and ρ_n/ρ as a function of temperature. At lambda temperature and above, $\rho_s = 0$ and $\rho_n = \rho$, because the superfluid component disappears

Table 2.1 Values of the densities ρ, ρ_n, ρ_s, the ratios ρ_n/ρ, ρ_s/ρ, the entropy s per unit mass, the specific heat c_V per unit mass, and the shear viscosity η of the normal component; Data from [8, 9]

	2.0 K	1.7 K	1.5 K	1.2 K	1.0 K	0.7 K	0.5 K
ρ(g/cm^3)	0.1456	0.1453	0.1452	0.1451	0.1451	0.1451	0.1451
ρ_n(g/cm^3)	0.0805	0.0332	0.0162	0.0037	0.0010	0.0000	0.0000
ρ_s(g/cm^3)	0.0651	0.1121	0.1290	0.1414	0.1441	0.1451	0.1451
ρ_n/ρ	0.553	0.229	0.111	0.026	0.007	0.000	0.000
ρ_s/ρ	0.447	0.771	0.889	0.974	0.993	1.000	1.000
s(J/g K)	0.9621	0.3950	0.1965	0.0052	0.0016	0.0002	0.0000
c_V(J/g K)	5.190	2.187	1.133	0.335	0.105	0.009	0.002
η(Pa · s $\times 10^6$)	1.468	1.290	1.346	1.736	3.873		

that $\mathbf{q}$ does not contain the purely convective contribution of the global motion of the system. If ρ and $\mathbf{v}$ are written in terms of ρ_n and ρ_s and of $\mathbf{v}_n$ and $\mathbf{v}_s$, (2.1.2) becomes

$$\mathbf{q} = \rho_s T s(\mathbf{v}_n - \mathbf{v}_s) = \rho_s T s \mathbf{V}_{ns}, \tag{2.1.3}$$

where the so-called counterflow velocity $\mathbf{V}_{ns} = \mathbf{v}_n - \mathbf{v}_s$ has been introduced.

Though the system cannot be physically decomposed into these two separate components, ρ_n and ρ_s may be measured (see Sect. 2.8 and Table 2.1), and $\mathbf{v}_n$ and $\mathbf{v}_s$ may be mathematically obtained from the values of $\mathbf{v}$ and $\mathbf{q}$. In the last years, there has been much progress in visualization techniques trying to visualize the flow of the normal component through its effects on very small particles, which are drifted by normal flow (Sect. 6.6).

2.2 Evolution Equations of the Model

To describe the evolution of the system, dynamic equations for the variables are needed and their consistency with the second law of thermodynamics must be examined. The results of such proposed equations in several different physical situations must be later compared to the experimental results in order to check their validity.

In this section, a brief introduction directly going to the essential physical features of the basic variables of the two-fluid model is presented without dissipative effects, which are introduced in the next section.

2.2.1 Landau Evolution Equations

We shall present here the Landau evolution equations that describe the behaviour of liquid Helium II in the zero temperature limits, where dissipative phenomena are absent [2, 10].

Dynamical Equations for ρ, $\mathbf{v}$, and $\mathbf{v}_s$
The density ρ and the mass flux $\rho\mathbf{v}$ must satisfy the equation of continuity

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \tag{2.2.4}$$

and the balance of the total momentum

$$\frac{\partial (\rho \mathbf{v})}{\partial t} + \nabla \cdot (\rho \mathbf{v}\mathbf{v} + p\mathbf{U}) = \mathbf{0} \tag{2.2.5}$$

with p the thermodynamic pressure and $\mathbf{U}$ the identity tensor.

Besides the general balance Eqs. (2.2.4) and (2.2.5), a particular equation describing the evolution of $\mathbf{v}_s$ is needed. In order that $\nabla \times \mathbf{v}_s = 0$ is always zero, the time derivative of $\mathbf{v}_s$ must be the gradient of some scalar function. In Landau's model it is taken [2, 10]

$$\frac{\partial \mathbf{v}_s}{\partial t} + \nabla \cdot \left(\frac{1}{2} \mathbf{v}_s^2 + \mu \right) = \mathbf{0} \tag{2.2.6}$$

with μ some scalar, identified a posteriori with the chemical potential.

Since on general thermodynamic grounds one has for the chemical potential μ the relation $\nabla \mu = \frac{1}{\rho} \nabla p - s \nabla T$, Eq. (2.2.6) may be written as

$$\rho_s \left(\frac{\partial \mathbf{v}_s}{\partial t} + \mathbf{v}_s \cdot \nabla \mathbf{v}_s \right) = -\frac{\rho_s}{\rho} \nabla p + \rho_s s \nabla T. \tag{2.2.7}$$

This has the form of the Euler equation for perfect fluids but with the additional term in ∇T. In the presence of nonlinear terms, Eq. (2.2.7) includes an additional term, as we will see in the following paragraph.

Dynamical Equations for Entropy and Energy

Besides evolution equations for mechanical quantities ρ, $\mathbf{v}$, and $\mathbf{v}_s$, evolution equations for thermodynamical quantities, namely, specific internal energy e per unit mass and specific entropy s per unit mass, are also needed.

Landau complemented Eqs. (2.2.4)–(2.2.6) imposing that in Helium II the entropy is conserved in the absence of dissipative effects, and wrote for the entropy density s the following equation :

$$\frac{\partial(\rho s)}{\partial t} + \nabla \cdot (\rho s \mathbf{v}_n) = 0. \tag{2.2.8}$$

Equations (2.2.4), (2.2.5), (2.2.6), and (2.2.8) describe the whole evolution of the system, in the absence of dissipation; in them ρ, μ, and s are functions not only of the usual thermodynamic variables p and T, but also of V_{ns}^2, the squared relative velocity of normal and superfluid components.

Energy and entropy are not independent of each other, but related by a fundamental equation of state. For the specific energy e_0 in a coordinate system in which the velocity of the superfluid component is zero, Landau writes [2, 10]

$$d(\rho\epsilon_0) = \mu d\rho + T d(\rho s) + \mathbf{V}_{ns} \cdot d\mathbf{j}_0, \tag{2.2.9}$$

with the momentum $\mathbf{j}_0$ in such system being $\mathbf{j}_0 = \rho_n \mathbf{V}_{ns}$. Incidentally, (2.2.9) implies for the pressure p the relation

$$p = -\rho\epsilon_0 + \rho s T + \rho\mu + \rho_n V_{ns}^2. \tag{2.2.10}$$

Differentiating (2.2.10) and using (2.2.9) one obtains

$$d\mu = -s dT + \frac{1}{\rho} dp - \frac{\rho_n}{\rho} \mathbf{V}_{ns} \cdot d\mathbf{V}_{ns}, \tag{2.2.11}$$

and also $\nabla\mu = \frac{1}{\rho}\nabla p - s\nabla T + \frac{\rho_n}{2\rho}\nabla V_{ns}^2$; substituting in (2.2.6), one gets

$$\rho_s \left(\frac{\partial \mathbf{v}_s}{\partial t} + \mathbf{v}_s \cdot \nabla \mathbf{v}_s \right) = -\frac{\rho_s}{\rho}\nabla p + \rho_s s \nabla T + \frac{\rho_s \rho_n}{2\rho}\nabla V_{ns}^2. \tag{2.2.12}$$

The additional nonlinear term is analogous to the Bernoulli contribution to the pressure, but with the relative velocity instead of the barycentric velocity.

The specific internal energy ϵ in a coordinate system in which $\mathbf{v}_s$ is not zero must incorporate the corresponding kinetic contributions related to $\mathbf{v}_s$, namely:

$$\rho\epsilon = \frac{1}{2}\rho\mathbf{v}_s^2 + \mathbf{j}_0 \cdot \mathbf{v}_s + \rho\epsilon_0. \tag{2.2.13}$$

In the absence of viscous dissipation, the balance equation for ϵ reads as

$$\frac{\partial\rho\epsilon}{\partial t} + \nabla \cdot \mathbf{Q} = p\nabla \cdot \mathbf{v}, \tag{2.2.14}$$

with $\mathbf{Q}$ the energy flux density. In the Landau–Lifshitz's book [10] $\mathbf{Q}$ is taken as

$$\mathbf{Q} = (\rho\epsilon + p)\,\mathbf{v} + \rho sT\mathbf{v}_n + \rho_n\left[(\mathbf{V}_{ns} \cdot \mathbf{v}_n)\mathbf{v}_n - V_{ns}^2\mathbf{v}\right], \tag{2.2.15}$$

where $\rho sT\mathbf{v}_n = \mathbf{q}$ is the purely conduction term, and the other two terms are of convective nature.

2.3 The Two-Fluid Model in the Framework of the Rational Theory of Fluid Mixtures: Dissipative Processes and Superfluid Entropy

The model presented in this section generalizes the standard (Landau and Tisza) two-fluid model allowing that a small amount of entropy is carried by the superfluid component. Indeed, as observed by Puttermann [11], the statement that the entropy of the superfluid fraction vanishes has not been demonstrated theoretically, so that it must be regarded as an additional simplifying postulate. The experimental results [12, 13] simply tell us that entropy carried by superfluid component is smaller than 2% with respect to entropy carried by normal component. Here a more general two-fluid model, where also the superfluid component carries a small amount of entropy, is presented (see also [14, 15]], where theoretical microscopic motivations in favour of a small superfluid entropy are advanced). Dissipative effects are also taken into account.

The phenomenological model of helium as a two-fluid mixture can be viewed in the general frame of the rational theory of fluid mixtures [16]. This theory allows the determination of very strong constraints on the functional form of constitutive equations through a systematic exploitation of second law of thermodynamics and material objectivity principle. The following equations

$$\frac{\partial \rho_\alpha}{\partial t} + \nabla \cdot (\rho_\alpha \mathbf{v}_\alpha) = \tau_\alpha, \tag{2.3.16}$$

$$\frac{\partial \rho_\alpha \mathbf{v}_\alpha}{\partial t} + \nabla \cdot (\rho_\alpha \mathbf{v}_\alpha \mathbf{v}_\alpha + \mathbf{P}_\alpha) = \mathbf{m}_\alpha, \tag{2.3.17}$$

$$\frac{\partial \rho \epsilon}{\partial t} + \nabla \cdot (\rho \epsilon \mathbf{v} + \mathbf{q}) = \mathbf{P} : \nabla \mathbf{v} \tag{2.3.18}$$

(where $\alpha = s, n$) are, respectively, the equations of balance of mass and momentum of the two components and the equation of balance of energy of the whole mixture; ρ_α and $\mathbf{v}_\alpha$ are the partial densities and velocities; $\rho = \rho_s + \rho_n$ is the total density; $\mathbf{v}$ the barycentric velocity; ϵ is the specific internal energy; and $\mathbf{q}$ the internal energy flux of the mixture. In these equations, τ_α and $\mathbf{m}_\alpha$ are the mass and the momentum production of two components (note that in situations with temperature changes, ρ_n and ρ_s also change, in such a way that raising temperature means $\tau_n > 0$ and $\tau_s < 0$). Further $\mathbf{P}_n$, $\mathbf{P}_s$, and $\mathbf{P}$ denote, respectively, the stresses on the normal component, on the superfluid component, and on the mixture. The partial stresses are decomposed as $\mathbf{P}_n = p_n \mathbf{U} + \mathbf{P}_n^v$ and $\mathbf{P}_s = p_s \mathbf{U} + \mathbf{P}_s^v$, p_n and p_s being the partial pressures, and $\mathbf{P}_n^v$ and $\mathbf{P}_s^v$ the partial viscous stresses. The stress $\mathbf{P}$ on the whole mixture is decomposed in an intrinsic term, a part due to the relative motion between the two components, and in a dissipative contribution. This latter is simply $\mathbf{P}^v = \mathbf{P}_s^v + \mathbf{P}_n^v$. The non-viscous part $\mathbf{P}_0$ of stress on the mixture is [16]

$$\mathbf{P}_0 = \sum_{\alpha=s,n} \left(p_\alpha \mathbf{U} - \rho_\alpha \mathbf{u}_\alpha \mathbf{u}_\alpha \right), \tag{2.3.19}$$

where $\mathbf{u}_\alpha = \mathbf{v}_\alpha - \mathbf{v}$ are the velocities relative to the barycentric velocity. As $\rho_s \mathbf{u}_s + \rho_n \mathbf{u}_n = 0$, here we consider only the superfluid relative velocity $\mathbf{u}_s = \mathbf{v}_s - \mathbf{v}$ that in the following will be denoted simply as $\mathbf{u}$. This relative velocity $\mathbf{u}$ is linked to the counterflow velocity $\mathbf{V}_{ns} = \mathbf{v}_n - \mathbf{v}_s$ by the relation: $\mathbf{u} = -\frac{\rho_n}{\rho} \mathbf{V}_{ns}$. With these positions, the stress $\mathbf{P}$ on the whole mixture is

$$\mathbf{P} = p\mathbf{U} + \frac{\rho \rho_s}{\rho_n} \mathbf{u}\mathbf{u} + \mathbf{P}^v = p\mathbf{U} + \frac{\rho_s \rho_n}{\rho} \mathbf{V}_{ns} \mathbf{V}_{ns} + \mathbf{P}^v, \tag{2.3.20}$$

where $p = p_n + p_s$ is the total pressure.

The laws of conservation of mass and momentum imply $\tau_s + \tau_n = 0$ and $\mathbf{m}_s + \mathbf{m}_n = 0$. Thus, from (2.3.16)–(2.3.17) we immediately obtain the balance equations of mass and momentum of the mixture.

In (2.3.18), the specific internal energy ϵ is sum of the specific energies of the two components, which are decomposed into an intrinsic part and a term depending on the relative motion [16]:

$$\epsilon = \sum_{\alpha=s,n} \frac{\rho_\alpha}{\rho} \left(\epsilon_\alpha + \frac{1}{2} u_\alpha^2 \right) = \epsilon^I + \frac{1}{2} \frac{\rho_s}{\rho_n} u^2 = \epsilon^I + \frac{1}{2} \frac{\rho_s \rho_n}{\rho^2} V_{ns}^2, \tag{2.3.21}$$

where $\rho\epsilon^I = \rho_s\epsilon_s + \rho_n\epsilon_n$. The internal energy flux $\mathbf{q}$ of the mixture is decomposed into an intrinsic part and a term depending on the relative motion [16]:

$$\mathbf{q} = \sum_{\alpha=s,n} \mathbf{q}_\alpha + \rho_\alpha \left(\epsilon_\alpha + \frac{1}{2}u_\alpha^2\right)\mathbf{u}_\alpha - \mathbf{u}_\alpha \cdot \mathbf{P}_\alpha \tag{2.3.22}$$

with $\mathbf{q}_\alpha$ the partial energy fluxes. It is useful to split this expression in the following way:

$$\mathbf{q} = \mathbf{q}^I - \frac{\rho_n\rho_s(\rho_n - \rho_s)}{2\rho^2}V_{ns}^2\mathbf{V}_{ns} + \frac{\rho_s\rho_n}{\rho}\mathbf{V}_{ns}\cdot\left(\frac{\mathbf{P}_s^v}{\rho_s} - \frac{\mathbf{P}_n^v}{\rho_n}\right), \tag{2.3.23}$$

with

$$\mathbf{q}^I = -\lambda\nabla T - \frac{\rho_s\rho_n}{\rho}\left(\epsilon_s - \epsilon_n + \frac{p_s}{\rho_s} - \frac{p_n}{\rho_n}\right)\mathbf{V}_{ns}, \tag{2.3.24}$$

where we have put $\mathbf{q}_s + \mathbf{q}_n = -\lambda\nabla T$, with λ the thermal conductivity.

Using the principle of material objectivity and the second law of thermodynamics, assuming that the constitutive quantities are linear in the vectorial and tensorial quantities, the following expression can be derived for the momentum productions of the two components of the mixture [16]:

$$\mathbf{m}_\alpha - \tau_\alpha\mathbf{v}_\alpha = \sum_{\beta=s,n}\left(\frac{\partial p_\alpha}{\partial\rho_\beta} - \rho_\alpha\frac{\partial\mu_\alpha^I}{\partial\rho_\beta}\right)\nabla\rho_\beta + M_u^\alpha\mathbf{u} + M_T^\alpha\nabla T \qquad (\alpha = s, n), \tag{2.3.25}$$

where μ_n^I and μ_s^I represent the intrinsic chemical potentials of the normal and superfluid components, connected with the chemical potentials by the relation [16]:

$$\mu_n - \mu_s = \mu_n^I - \mu_s^I - \frac{1}{2}V_{ns}^2. \tag{2.3.26}$$

2.3.1 Mathematical Characterization of Liquid Helium II

In what follows, we call s_n, s_s, and s the entropies of the normal component, of the superfluid component, and of the whole mixture, respectively. These quantities are related by

$$\rho s = \rho_s s_s + \rho_n s_n. \tag{2.3.27}$$

According to the Nernst–Planck formulation of third law of thermodynamics, the entropy of the whole mixture is zero for $T = 0$. Because the normal component is absent at this temperature, we conclude that $s_s = 0$ for $T = 0$. Landau assumed that this condition is valid also for $T \neq 0$, so that he associated a non-zero entropy only to the normal component. Here, for a more complete comparison with the results of

the following chapter, we will consider the non-conventional two-fluid model, where the entropy is expressed by Eq. (2.3.27).

The experimental observations show a very free and easy transition from the superfluid component to the normal component and vice versa, thus showing the absence of dynamical interaction between the two components. This leads to an important mathematical consequence in superfluid helium. The source term $\mathbf{m}_s$ in (2.3.17) is caused only by the mass exchange between the two components, i.e.:

$$\mathbf{m}_s - \tau_s \mathbf{v}_s = 0. \tag{2.3.28}$$

Then, from mass and momentum conservation laws ($\tau_s + \tau_n = 0$, $\mathbf{m}_s + \mathbf{m}_n = 0$), we obtain

$$\mathbf{m}_n - \tau_n \mathbf{v}_n = \tau_n(\mathbf{v}_s - \mathbf{v}_n) = -\tau_n \mathbf{V}_{ns} = \frac{\rho}{\rho_n}\tau_n \mathbf{u}. \tag{2.3.29}$$

From this, taking into account (2.3.25), we get $M_T^s = M_T^n = 0$, $M_u^s = 0$, $M_u^n = \frac{\rho}{\rho_n}\tau_n$, and the important relationship:

$$\frac{\partial p^\alpha}{\partial \rho_\beta} = \rho_\alpha \frac{\partial \mu_\alpha^I}{\partial \rho_\beta}, \qquad (\alpha = s, n). \tag{2.3.30}$$

From (2.3.30) several important consequences can be deduced; mixtures obeying the previous condition were called by Müller simple mixtures (see [16]).

The second law of thermodynamics, when applied to simple mixtures, leads to the following functional forms for the constitutive quantities:

$$p_\alpha = p_\alpha(\rho_\alpha, T), \qquad \mu_\alpha^I = \mu_\alpha^I(\rho_\alpha, T), \qquad \epsilon_\alpha = \epsilon_\alpha(\rho_\alpha, T), \qquad s_\alpha = s_\alpha(\rho_\alpha, T). \tag{2.3.31}$$

Moreover, in a simple mixture, the intrinsic chemical potentials coincide with the partial specific free enthalpies, and there is a Gibbs equation for each component of the mixture:

$$\mu_\alpha^I = \epsilon_\alpha - T s_\alpha + \frac{p_\alpha}{\rho_\alpha}, \qquad T ds_\alpha = d\epsilon_\alpha - \frac{p_\alpha}{\rho_\alpha^2} d\rho_\alpha. \tag{2.3.32}$$

From this, one gets

$$d\mu_\alpha^I = \frac{1}{\rho_\alpha} dp_\alpha - s_\alpha dT, \tag{2.3.33}$$

and the Gibbs–Duhem equation and the Gibbs equation for the whole mixture are, respectively,

$$\rho(\epsilon^I - Ts) = \rho_s \mu_s^I + \rho_n \mu_n^I - p, \tag{2.3.34}$$

$$T d(\rho s) = d(\rho \epsilon^I) - \mu_s^I d\rho_s - \mu_n^I d\rho_n; \tag{2.3.35}$$

we have also

$$Tds = d\epsilon^I - \frac{p}{\rho^2}d\rho + (\mu_n^I - \mu_s^I)d\left(\frac{\rho_s}{\rho}\right). \tag{2.3.36}$$

Introducing the total internal energy ϵ, Eqs. (2.3.34) and (2.3.36) can be written as

$$\rho(\epsilon - Ts) = \rho_s\mu_s^I + \rho_n\mu_n^I - p + \frac{1}{2}\frac{\rho_s\rho_n}{\rho^2}V_{ns}^2, \tag{2.3.37}$$

$$Tds = d\epsilon - \frac{p}{\rho^2}d\rho + \left[\mu_n^I - \mu_s^I - \frac{1}{2}V_{ns}^2\right]d\left(\frac{\rho_s}{\rho}\right) - \frac{1}{2}\frac{\rho_s}{\rho_n}du^2. \tag{2.3.38}$$

Another very important consequence of the complete absence of dynamical interaction between normal and superfluid components is that the chemical equilibrium between the two phases is reached instantaneously, therefore this process is completely reversible. As a consequence, the production of entropy during this process must be exactly zero. This implies that, using (2.3.26)

$$\mu_n = \mu_s \qquad \Rightarrow \qquad \mu_n^I - \mu_s^I - \frac{1}{2}V_{ns}^2 = 0. \tag{2.3.39}$$

Equation (2.3.39) has several important consequences. Indeed the left-hand side of this equation is a function of ρ, ρ_s, T, and V_{ns}. Its vanishing provides an algebraic dependence between these fields.

In this hypothesis, the Gibbs-Duhem equation and Gibbs equation for helium can be written as

$$\mu_s^I = \epsilon^I - Ts + \frac{p}{\rho} - \frac{1}{2}\frac{\rho_n}{\rho}V_{ns}^2, \tag{2.3.40}$$

$$Td(\rho s) = d(\rho\epsilon^I) - \mu_s^I d\rho - \frac{1}{2}V_{ns}^2 d\rho_n, \tag{2.3.41}$$

from which we get

$$d\mu_s^I = -sdT + \frac{1}{\rho}dp - \frac{1}{2}\frac{\rho_n}{\rho}dV_{ns}^2. \tag{2.3.42}$$

As seen from (2.3.40)–(2.3.42), the chemical potential μ_s^I, the energy ϵ, and the entropy s of helium depend not only on the equilibrium quantities, but also on the counterflow velocity $\mathbf{V_{ns}}$. Hence they are non-equilibrium quantities. If we assume as field variable the pressure p, instead of the total density ρ, assuming $\mu_s^I = \mu_s^I(p, T, V_{ns})$ we get from (2.3.42):

$$s = -\frac{\partial\mu_s^I}{\partial T}, \qquad \frac{1}{\rho} = \frac{\partial\mu_s^I}{\partial p}, \qquad \frac{\rho_n}{\rho} = -2\frac{\partial\mu_s^I}{\partial V_{ns}^2}. \tag{2.3.43}$$

Integration of (2.3.42) shows that up to the second order in $\mathbf{V}_{ns}$ one has

$$\mu_s^I(p, T, \mathbf{V}_{ns}) \simeq \mu_s^I(p, T, 0) - \frac{1}{2}\frac{\rho_n}{\rho}{V_{ns}}^2. \tag{2.3.44}$$

Differentiating (2.3.44) with respect to T and to p, and using results (2.3.43), it is found

$$s(p, T, \mathbf{V_{ns}}) = -\frac{\partial \mu_s^I}{\partial T} \simeq s(p, T, 0) + \frac{1}{2}V_{ns}^2\frac{\partial}{\partial T}\left(\frac{\rho_n}{\rho}\right), \tag{2.3.45}$$

$$\rho(p, T, \mathbf{V_{ns}}) = \frac{1}{\partial \mu_s^I/\partial p} \simeq \rho(p, T, 0) - \frac{1}{2}V_{ns}^2\frac{\partial}{\partial p}\left(\frac{\rho_n}{\rho}\right). \tag{2.3.46}$$

Equation (2.3.45) is of interest for the comparison with the next chapter, as it shows that the entropy depends on the relative velocity, which may be considered as an internal variable, and that the contribution of $\mathbf{V}_{ns}$ to the entropy is quadratic in the internal variable $\mathbf{V}_{ns}$.

Introducing the total internal energy ϵ, from (2.3.40) and (2.3.41), we get also

$$\mu_s^I = \epsilon - Ts + \frac{p}{\rho} - \frac{1}{2}\left(\frac{\rho_n}{\rho} + \frac{\rho_s\rho_n}{\rho^2}\right)V_{ns}^2, \tag{2.3.47}$$

$$Tds = d\epsilon - \frac{p}{\rho^2}d\rho - \frac{1}{2}\frac{\rho_s}{\rho_n}du^2. \tag{2.3.48}$$

Using Eqs. (2.3.32) and (2.3.39) one can write the internal energy flux (2.3.23) in the following way:

$$\mathbf{q} = -\lambda\nabla T + \frac{\rho_n\rho_s}{\rho}\left[T(s_n - s_s)\mathbf{V}_{ns} + \frac{\rho_s}{\rho}V_{ns}^2\mathbf{V}_{ns} + \mathbf{V}_{ns}\cdot\left(\frac{\mathbf{P}_s^v}{\rho_s} + \frac{\mathbf{P}_n^v}{\rho_n}\right)\right]; \tag{2.3.49}$$

note that the convective term linear in the counterflow velocity in this equation can be written as $\rho T(s - s_s)(\mathbf{v}_n - \mathbf{v})$, and coincides with the expression (2.1.2) introduced by Landau, if we assume $s_s = 0$.

2.3.2 *Constitutive Relations for the Viscous Pressure Tensor*

In accord with Khalatnikov (see [17]), we assume that the shear viscosity is absent in the superfluid component, and choose for the dissipative contributions for the partial viscous stress introduced over (2.3.19)):

$$\mathbf{P}^v = \mathbf{P}_n^v + \mathbf{P}_s^v = -\rho_s\zeta_1\nabla\cdot\mathbf{V}_{ns} - \zeta_2\nabla\cdot\mathbf{v}_n - 2\eta\langle\nabla\mathbf{v}_n\rangle, \tag{2.3.50}$$

$$\mathbf{P}_s^v = -\rho_s\zeta_3\nabla\cdot\mathbf{V}_{ns} - \zeta_4\nabla\cdot\mathbf{v}_n, \tag{2.3.51}$$

with η the shear viscosity of the normal component and ζ_1, ζ_2, ζ_3, and ζ_4 expressing bulk viscous effects.

From Onsager reciprocity principle, it follows $\zeta_1 = \zeta_4$ and $\zeta_2\zeta_3 \geq \zeta_1^2$. Thus this dissipative two-fluid model contains five independent kinetic coefficients: η, ζ_3, ζ_1, ζ_2, and λ. They cover a wide range of experimental situations. In [18], it was shown that, taking into account the explicit expressions of the kinetic coefficients obtained by Khalatnikov (see [17], Chaps. 9 and 21), one realizes that these coefficients satisfy the equality $\zeta_2\zeta_3 = \zeta_1^2$. Thus this dissipative two-fluid model contains only four independent kinetic coefficients: η, ζ_3, ζ_4, and λ.

2.3.3 *Field Equations*

To determine the system of field equations, recall that the left-hand side of Eq. (2.3.39) is a function of ρ, ρ_s, T, and V_{ns}. Its vanishing provides an algebraic dependence between these fields, so that one of them can be expressed in terms of the remaining ones. As a consequence of the reduction in the number of independent fields, to establish the field equations, we shall use only the equations of balance of mass and momentum, obtained immediately from (2.3.16)–(2.3.17):

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0, \tag{2.3.52}$$

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \cdot (\rho \mathbf{v}\mathbf{v} + \mathbf{P}) = 0, \tag{2.3.53}$$

the balance equation for the energy of the whole system (2.3.18), and the momentum balance equation for the superfluid component

$$\frac{\partial \rho \epsilon}{\partial t} + \nabla \cdot (\rho \epsilon \mathbf{v} + \mathbf{q}) = \mathbf{P} : \nabla \mathbf{v}, \tag{2.3.54}$$

$$\frac{\partial \rho_s \mathbf{v}_s}{\partial t} + \nabla \cdot (\rho_s \mathbf{v}_s \mathbf{v}_s + \mathbf{P}_s) = \mathbf{m}_s. \tag{2.3.55}$$

These equations must be complemented by the constitutive quantities (2.3.20), (2.3.23), (2.3.28), (2.3.29), (2.3.49), (2.3.50), (2.3.51). The equation of balance of mass for the superfluid component can be used, once the fundamental fields are known, to determine the mass production τ_s of the superfluid appearing in (2.3.19).

The evolution equations of velocities of the two components are

$$\frac{\partial \mathbf{v}_s}{\partial t} + (\mathbf{v}_s \cdot \nabla)\mathbf{v}_s + \frac{1}{\rho_s}\nabla p_s + \frac{1}{\rho_s}\nabla \cdot \mathbf{P}_s^v = \frac{1}{\rho_s}(\mathbf{m}_s - \tau_s \mathbf{v}_s) = 0, \tag{2.3.56}$$

$$\frac{\partial \mathbf{v}_n}{\partial t} + (\mathbf{v}_n \cdot \nabla)\mathbf{v}_n + \frac{1}{\rho_n}\nabla p_n + \frac{1}{\rho_n}\nabla \cdot \mathbf{P}_n^v = \frac{1}{\rho_n}(\mathbf{m}_n - \tau_n \mathbf{v}_n) = -\frac{1}{\rho_n}\tau_n \mathbf{V}_{ns}. \tag{2.3.57}$$

Note that the right-hand side of Eq. (2.3.57) depends on the mass production τ_n. This term is usually neglected in the literature. However, it is known that normal and superfluid components can easily transform one in the other when the temperature changes, and the mass supply $\tau_n (= -\tau_s)$ of the normal component, in general, cannot be neglected. Since $-\tau_n \mathbf{V}_{ns} = \left(\frac{\partial \rho_s}{\partial t} + \nabla \cdot (\rho_s \mathbf{v}_s)\right) \mathbf{V}_{ns}$, Eq. (2.3.57) may be rewritten as

$$\frac{\partial \mathbf{v}_n}{\partial t} + (\mathbf{v}_n \cdot \nabla)\mathbf{v}_n + \frac{1}{\rho_n}\nabla p_n + \frac{1}{\rho_n}\nabla \cdot \mathbf{P}_n^v = -\frac{1}{\rho_n}\left(\frac{\partial \rho_s}{\partial t} + \nabla \cdot (\rho_s \mathbf{v}_s)\right)\mathbf{V}_{ns}. \tag{2.3.58}$$

Now, by using (2.3.33), we obtain

$$\frac{1}{\rho_s}\nabla p_s = \nabla \mu_s^I + s_s \nabla T, \qquad \frac{1}{\rho_n}\nabla p_n = \nabla \mu_n^I + s_n \nabla T, \tag{2.3.59}$$

from which

$$\frac{\partial \mathbf{v}_s}{\partial t} + (\mathbf{v}_s \cdot \nabla)\mathbf{v}_s + \nabla \mu_s^I + s_s \nabla T + \frac{1}{\rho_s}\nabla \cdot \mathbf{P}_s^v = 0; \tag{2.3.60}$$

then, taking into account Eq. (2.3.41), we obtain

$$\nabla \mu_s^I = -s\nabla T + \frac{1}{\rho}\nabla p - \frac{1}{2}\frac{\rho_n}{\rho}\nabla V_{ns}^2. \tag{2.3.61}$$

Substituting in (2.3.56), we get the following evolution equation for the superfluid velocity:

$$\rho_s \left(\frac{\partial \mathbf{v}_s}{\partial t} + (\mathbf{v}_s \cdot \nabla)\mathbf{v}_s\right) - \rho_s (s - s_s)\nabla T + \frac{\rho_s}{\rho}\nabla p - \frac{1}{2}\frac{\rho_s \rho_n}{\rho}\nabla V_{ns}^2 + \nabla \cdot \mathbf{P}_s^v = 0. \tag{2.3.62}$$

We can also obtain an evolution equation of the velocity of the normal component. One arrives to the following equation:

$$\rho_n \left(\frac{\partial \mathbf{v}_n}{\partial t} + (\mathbf{v}_n \cdot \nabla)\mathbf{v}_n\right) + \frac{\rho_n}{\rho}\nabla p - \rho_n (s - s_n)\nabla T + \frac{\rho_s \rho_n}{2\rho}\nabla V_{ns}^2 + \nabla \cdot \mathbf{P}_n^v = -\tau_n \mathbf{V}_{ns} \tag{2.3.63}$$

that however always contain the unknown quantity $-\tau_n \mathbf{V}_{ns} = \tau_s \mathbf{V}_{ns}$.

Note that both Eqs. (2.3.62) and (2.3.63) contain a term proportional to the temperature gradient. Observing that $\rho_n (s - s_n) = -\rho_s (s - s_s)$, we recognize that the term linear in the temperature gradient appears in both equations with an opposite sign, meaning that it transfers motion between both components. The nonlinear terms are analogous to the Bernoulli contribution to the pressure, but with the relative velocity instead of the barycentric velocity.

A special attention deserves Eq. (2.3.60). In fact, also in the absence of viscosity, from this equation follows that the superfluid component is curl-free only if the superfluid entropy s_s is constant. Note that this in contrast with the paper [15]. These authors, in fact, assume ab initio that the superfluid component is curl-free.

2.4 Consequences of Entropy Conservation and Comparison with Landau Model

We recall that Landau, in his derivation of Helium II evolution equations, associated a non-zero entropy only to the normal component; therefore to compare with it we must assume zero the entropy of the superfluid component ($s_s = 0$), i.e.:

$$s_{Landau} = \frac{\rho_n}{\rho} s_n. \tag{2.4.64}$$

Landau assumed also the conservation of the entropy, so imposing zero all the kinetic dissipative coefficients introduced in Sect. 2.3.2 ($\eta = 0$, $\zeta_2 = 0$, $\zeta_3 = 0$, $\zeta_4 = 0$, and $\lambda = 0$).

Under these hypotheses the expression (2.3.49) of heat flux $\mathbf{q}$ becomes

$$\mathbf{q} = \left(\rho_s T s + \frac{\rho_s^2 \rho_n}{\rho^2} V_{ns}^2\right) \mathbf{V}_{ns} = \left(\rho T s - \rho \frac{\rho_n}{\rho_s} u_n^2\right) \mathbf{u}_n = \rho T s \mathbf{u}_n + \rho_n \mathbf{u}_n (\mathbf{u}_n^2 - \mathbf{u}_n \cdot \mathbf{u}_s), \tag{2.4.65}$$

where $\mathbf{u}_\alpha$ are the relative velocities.

In order to compare the model deduced in Sect. 2.3 with Landau model, we observe, first, that mass and momentum balance equations are identical with the first two Landau equations.

The evolution equation for the superfluid component (2.3.60) (assuming $s_s = 0$ and zero all the dissipative coefficients) becomes

$$\frac{\partial \mathbf{v}_s}{\partial t} + (\mathbf{v}_s \cdot \nabla)\mathbf{v}_s + \nabla \mu_s^I = 0 \tag{2.4.66}$$

recalling that it is $(\mathbf{v}_s \cdot \nabla)\mathbf{v}_s = \mathbf{v}_s \times (\nabla \times \mathbf{v}_s) + \nabla \left(\frac{1}{2} v_s^2\right)$, one deduces, by taking the curl of this quantity, that $\frac{\partial}{\partial t}(\nabla \times \mathbf{v}_s) = 0$; therefore, if $\nabla \times \mathbf{v}_s$ is initially zero, it remains zero at any instant of time. Putting therefore $\nabla \times \mathbf{v}_s = 0$ in Eq. (2.4.66), the latter equation can be written in the form (2.2.6) postulated by Landau:

$$\frac{\partial \mathbf{v}_s}{\partial t} + \nabla \cdot \left(\frac{v_s^2}{2} + \mu_s^I\right) = 0. \tag{2.4.67}$$

Comparing this equation with Landau equation for the velocity of the superfluid component (2.2.6), we are led to identify the chemical potential μ introduced by Landau with the intrinsic part μ_s^I of the chemical potential of the superfluid component. If $s_s \neq 0$, Eq. (2.4.67) must be generalized as

$$\frac{\partial \mathbf{v}_s}{\partial t} + \nabla \cdot \left(\frac{v_s^2}{2} + \mu_s^I + T s_s \right) = 0. \tag{2.4.68}$$

Consider now Eq. (2.2.10), where ϵ_0 is the specific energy of the mixture in the frame where the superfluid centre of mass is at rest, and compare it with (2.3.40). We deduce that the energy density $\rho\epsilon_0$ is connected to the intrinsic internal energy density $\rho\epsilon^I$ appearing in Eq. (2.3.40), through the relation:

$$\rho\epsilon_0 = \rho\epsilon^I + \frac{1}{2}\rho_n V_{ns}^2. \tag{2.4.69}$$

The balance equation of total energy is given by Eq. (2.3.54); in it the total energy flux can be written as

$$\mathbf{Q} = \rho\epsilon\mathbf{v} + \mathbf{q} = \rho\epsilon\mathbf{v} + \rho T s\mathbf{u}_n + \rho_n\mathbf{u}_n(\mathbf{u}_n^2 - \mathbf{u}_n \cdot \mathbf{u}_s), \tag{2.4.70}$$

where Eq. (2.4.65) has been used. Making use of Eq. (2.3.40), one gets also

$$\mathbf{Q} = \rho\epsilon\mathbf{v} + \mathbf{q} = \mu_s^I\mathbf{J} + \left(\rho T s - p + \frac{\rho_n\rho_s}{\rho} V_{ns}^2 \right)\mathbf{v} + \left(\rho_s T s + \frac{\rho_s^2\rho_n}{\rho^2} V_{ns}^2 \right)\mathbf{V}_{ns}, \tag{2.4.71}$$

where we have put $\mathbf{J} = \rho\mathbf{v}$. Finally, noting that $\mathbf{v} + \frac{\rho_s}{\rho}\mathbf{V}_{ns} = \mathbf{v}_s$, one get too

$$\mathbf{Q} = \mu_s^I\mathbf{J} - p\mathbf{v} + \rho T s\mathbf{v}_n + \frac{\rho_n\rho_s}{\rho} V_{ns}^2\mathbf{v}_n = \mu_s^I\mathbf{J} - p\mathbf{v} + \rho T s\mathbf{v}_n + \rho_n(\mathbf{u}_n^2 - \mathbf{u}_n \cdot \mathbf{u}_s)\mathbf{v}_n. \tag{2.4.72}$$

Finally, we show that the balance law for entropy, postulated by Landau, can be deduced from the two-fluid model discussed before, when the entropy of the superfluid component and all dissipative coefficients are set equal to zero. Using these latter equations and the relation:

$$T\dot{s} = \dot{\epsilon} - \frac{p}{\rho^2}\dot{\rho} - \frac{1}{2T}\frac{\rho_s}{\rho_n}\dot{u^2} \tag{2.4.73}$$

which can be deduced from (2.3.48), we obtain after some algebraic manipulations:

$$\rho\dot{s} + \nabla \cdot (\rho s\mathbf{v}_n) = 0 \tag{2.4.74}$$

Equation (2.4.74) is just the entropy conservation law, postulated by Landau as a balance equation in his model for superfluid helium.

2.5 Mass Flow Along Narrow Channels

One of the most striking properties of liquid Helium II is its ability to flow through very narrow constrictions (the so-called "superleaks", as thin capillaries or porous media) without any resistance [1–3, 19]. This is the origin of the denomination "superfluid". Indeed, it has long been known that, even in the absence of applied forces or pressure gradient, Helium II can flow slowly through narrow channels "for as long as the experimenter's patience lasts" [20]. A typical example is the liquid gyroscope [21]: an axisymmetric container of Helium II filled with powder (in order to provide very narrow channels for the liquid) in which Helium II keeps its initial rotating flow for weeks.

This shows that some aspects of Helium II may be associated to a vanishing viscosity. However, the situation is more complicated because of two reasons: (a) when the channels are slightly larger than superleaks, some amount of viscous resistance is observed; (b) when the motion of rotational pendula inside liquid Helium II is studied, an attenuation of the amplitude of the oscillations is observed, analogous to that in viscous fluids. For instance, experiments with oscillating metal disc show a decrease of the period of oscillation below T_λ, indicating that not all the fluid is being dragged by the discs. In this way, one may measure ρ_n/ρ as function of temperature. The dependence of ρ_s/ρ is easily obtained from the relation $\rho_s/\rho = 1 - \rho_n/\rho$.

The presence of aspects apparently contradictory, some ones with seemingly vanishing viscosity and other ones with non-zero viscosity (oscillating discs, vibrating wires, rotation viscometers) given in Table 2.1, was the first motivation for the two-fluid model, which found almost immediately a microscopic support on the idea of an imperfect Bose–Einstein transition.

2.6 Heat Transfer in Counterflow Experiments

Another example of non-classical behaviour is heat transfer in counterflow experiments. Consider a channel with a heater at a closed end and open to a helium bath at the other end, with a heat flux $\mathbf{q}$ present in the channel. Using an ordinary classical fluid (such as He I), a temperature gradient can be measured along the channel, which indicates a finite thermal conductivity. If Helium II is used, and the heat flux inside the channel is lower than some critical value, the temperature gradient is so small that it cannot be measured, so indicating an extremely high thermal conductivity. This effect explains the remarkable ability of Helium II to remove heat and makes it important in cryogenic applications [22].

This shows the independent character of the heat flux, which is not determined by Fourier's law. This effect was the first motivation for the use of the one-fluid extended model that uses the heat flux $\mathbf{q}$ as an independent variable (Fig. 2.2).

The two-fluid model interprets this behaviour in the following way: in the absence of net mass flux ($\rho_n \overline{\mathbf{v}_n} + \rho_s \overline{\mathbf{v}_s} = 0$, where the overline denotes average over the chan-

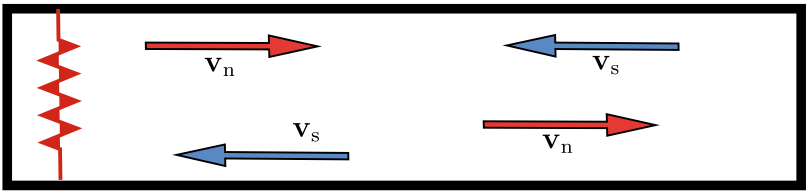


Fig. 2.2 Counterflow experiment in the two-fluid view: there is a vanishing mass flow but a non-vanishing heat flow. $\mathbf{v}_n$ (red arrow) and $\mathbf{v}_s$ (blue arrow), respectively, stand for the average velocities of normal and superfluid components, which compensate the respective mass flows in opposite direction. The zigzag line represents a periodic heat source

nel transversal section), in Helium II the heat is carried towards the bath by the normal fluid only, and $\overline{\mathbf{q}} = \rho s T \overline{\mathbf{v}_n}$. Being the net mass flux zero, there is superfluid motion towards the heater ($\overline{\mathbf{v}_s} = -(\rho_n/\rho_s)\overline{\mathbf{v}_n}$), hence there is a net internal counterflow $\overline{\mathbf{v}_{ns}} = \overline{\mathbf{v}_n} - \overline{\mathbf{v}_s} = \overline{\mathbf{q}}/(\rho_s s T)$ which is proportional to the applied heating power.

Considering that $\mathbf{v}_n$ moves in the usual viscous way, with vanishing velocity on the walls, one may find the Poiseuille profile for the flow of $\mathbf{v}_n$ and later multiply such flow times $\rho s T$, to have the heat flow. Using Eq. (2.3.58) for v_n, and assuming $\tau_n = 0$, it turns out that the total heat flow along a cylindrical channel is

$$\dot{Q} = -\frac{\pi R^4 s T^2}{8\eta} \nabla T. \tag{2.6.75}$$

This implies that the effective thermal conductivity, relating $\mathbf{q} = \dfrac{\dot{Q}}{\pi R^2}$ to ∇T, is $R^2 s T^2/8\eta$. It is seen that it does depend not only on material properties of the system, but also on the size through the factor R^2. This is a typical effect of hydrodynamic heat transport, because this factor comes from the factor R^4 in (2.6.75), which is typical of hydrodynamic Poiseuille flow. Some characteristic values of η are reported in Table 2.1.

2.7 Thermomechanical Effects

Thermomechanical effects relate a difference of temperature to mechanical forces, which may produce a flow of fluid, or deflect a sensor in a quiescent superfluid [1–3, 19]. Consider first the "fountain effect" first observed in 1938 by Allen and Jones. Two open vessels (I and II) containing helium at a temperature lower than the λ point are connected by a capillary and in thermal contact with two heat reservoirs at temperatures T^I and T^{II} ($T^I < T^{II}$) (see Fig. 1.2). It is experimentally seen that the level of the liquid contained in the vessel at the higher temperature raises. Measurements show that the pressure difference at the same level in the two vessels

is proportional to the temperature difference. The observed phenomenon is known as the fountain effect: by terminating the vessel II into a thin channel, the pressure difference is able to produce a jet of helium similar to a fountain.

A related effect is the so-called thermomechanical effect: when helium flows out of a vessel through a very narrow capillary and goes to another vessel, temperature in the first vessel rises and that of the second vessel decreases. The explanation is that the flow along the capillary is mainly of superfluid component which has zero entropy, and temperature rises in the first vessel, since in it the relative proportion of the normal component increases, and it becomes smaller in the second vessel, where the relative fraction of the normal component decreases. Mechanical equilibrium is established faster than thermal equilibrium. The final equilibrium condition is established by the equality of the chemical potential in systems 1 and 2, namely:

$$\mu(p_1, T_1) = \mu(p_2, T_2). \tag{2.7.76}$$

From Eq. (2.3.41), we see that, if the differences $p_1 - p_2$ and $T_1 - T_2$ are sufficiently small, they will be related at equilibrium through

$$p_1 - p_2 = \rho s(T_1 - T_2). \tag{2.7.77}$$

Thermomechanical pumps, based on the fountain effect and the thermomechanical effects in superfluid helium, are used in space technology, to keep at very low temperature some delicate instruments demanding a very low level of fluctuations and a high sensitivity. Thermomechanical pumps are advantaged over traditional mechanical pumps because they do not have moving parts, which may have mechanical wear and breakdown.

Another anomalous phenomenon is the Kapitza experiment. Kapitza [23–25] observed that suspending a movable vane in motionless helium opposite to the open end of a Dewar flask, when the heating element inside the flask was switched on, the vane was deflected. Hall [26] found that the force experienced by such a hanging plate is proportional to the square of the heat flux.

Such a relationship emerges spontaneously in the nonlinear extension of the theory [27]. In fact, when nonlinear terms in the heat flux are taken into account, the (non-equilibrium) pressure p depends on the square of the heat flux $\mathbf{q}$, namely, $p = p_0 + p_1 q^2$ [28].

In the two-fluid model, this phenomenon is explained by the presence of the quadratic term in the counterflow velocity in Eq. (2.3.20), and by the link between the internal energy flux and counterflow velocity appearing in Eq. (2.3.49).

2.8 Second Sound (Temperature Wave)

The usual temperature waves in physical systems are in general diffusive and externally driven, rather than purely propagative. This is a consequence of the second

law of thermodynamics, according to which heat may only go from higher to lower temperatures. It came thus as a surprise the observation that in superfluid helium there were propagative temperature waves, with very small attenuation.

In the two-fluid model, this is interpreted as a relative and out-of-phase motion of superfluid component (without entropy) and normal component (with entropy), keeping constant the mass density but yielding entropy oscillations, manifested as temperature waves. This was named "second sound" in contrast to the well-known density and pressure waves, which constitute the "first sound".

To consider wave propagation of ρ and s in the superfluid, we start from the balance equation of total mass and total momentum, omitting nonlinear terms and viscosity:

$$\frac{\partial \rho}{\partial t} = -\rho \nabla \cdot \mathbf{v}, \tag{2.8.78}$$

$$\rho \frac{\partial \mathbf{v}}{\partial t} = -\nabla p; \tag{2.8.79}$$

combining them and neglecting second-order terms, one gets

$$\frac{\partial^2 \rho}{\partial t^2} = \nabla^2 p. \tag{2.8.80}$$

To obtain the second equation, one writes an equation for the relative velocity $\mathbf{u} = \mathbf{v}_s - \mathbf{v}$, by using (2.8.79) and (2.3.62). Neglecting nonlinear and dissipative terms, one gets

$$\frac{\partial \mathbf{u}}{\partial t} = -(s - s_s)\nabla T. \tag{2.8.81}$$

This equation must be combined with the evolution equation of entropy (obtained from Eq. (2.2.8)) neglecting nonlinear and dissipative terms, one gets

$$\frac{\partial s}{\partial t} = -\frac{\rho_s}{\rho}(s - s_s)\nabla \cdot \mathbf{u}. \tag{2.8.82}$$

Combining (2.8.81) and (2.8.82) one arrives to

$$\frac{\partial^2 s}{\partial t^2} = \frac{\rho_s}{\rho_n}(s - s_s)^2 \nabla^2 T. \tag{2.8.83}$$

Expressing p and T in (2.8.80) and (2.8.83) in terms of ρ and s, it is found

$$\frac{\partial^2 \rho}{\partial t^2} = \left(\frac{\partial p}{\partial \rho}\right)_s \nabla^2 \rho + \left(\frac{\partial p}{\partial s}\right)_\rho \nabla^2 s, \tag{2.8.84}$$

$$\frac{\partial^2 s}{\partial t^2} = \frac{\rho_s}{\rho_n}(s - s_s)^2 \left[\left(\frac{\partial T}{\partial \rho}\right)_s \nabla^2 \rho + \left(\frac{\partial T}{\partial s}\right)_\rho \nabla^2 s\right]. \tag{2.8.85}$$

Studying perturbations of the form

$$\rho = \rho_0 + \rho' \exp[\mathrm{i}(\omega t - kz)] \quad \text{and} \quad s = s_0 + s' \exp[\mathrm{i}(\omega t - kz)] \tag{2.8.86}$$

one obtains by making zero the determinant of these perturbations in (2.8.84) and (2.8.85) that there are two waves with speeds V_1^2 and V_2^2 given by

$$V_1^2 = \left(\frac{\partial p}{\partial \rho}\right)_s \quad \text{and} \quad V_2^2 = \frac{\rho_s}{\rho_n}(s - s_s)^2 \left(\frac{\partial T}{\partial s}\right)_\rho \approx \frac{\rho_s}{\rho_n}(s - s_s)^2 \frac{T}{C_p}. \tag{2.8.87}$$

The first one is the usual first sound, density, or pressure wave, and the second is the second sound, namely, temperature or entropy wave. From its wave speed and from $s - s_s$, T, and c_p one may determine ρ_s/ρ_n. Since ρ_s tends to zero when T tends to the lambda temperature, V_2 tends also to zero, whereas V_1 does not change very much in the transition. Near 0 K, V_1 is of the order of 240 m/s and the extrapolated value of V_2 is of the order of 140 m/s [8, 9]. Note that when T tends to zero, ρ_n and c_p in the denominator of V_2 tend to zero, and that s and T in the numerator also tend to zero, whereas ρ_s tends to ρ. Note that this analysis of second sound is valid for situations in which the wavevector is much longer than the mean free path. But when T approaches zero the mean free path becomes very long, in such a way that the present analysis of second sound is no longer valid, so that one must be careful in the extrapolation.

The proportion of first sound and second sound produced depends very much on the method for production. Usual mechanical oscillators produce only a very small proportion of second sound as compared with that of first sound. To produce second sound, it is better to use some device with an oscillating temperature. In this case, the proportion of second sound is considerably higher than that of the first sound. As commented in Chaps. 4 and 5, second sound has turned out a very valuable tool in the study of quantized vortices and quantum turbulence.

In this section, it has been assumed that both normal and superfluid components may move. If one goes to sufficiently narrow capillaries, the normal component cannot move, because it is stopped by friction against the walls. In this case, one must consider the mass balance equation, the equation of motion for $\mathbf{v}_s$ and the balance of entropy density and one obtains for the speed V_4 of the corresponding thermal waves (also known as fourth sound) the expression

$$V_4^2 = \frac{\rho_s}{\rho} V_1^2 + \frac{\rho_n}{\rho} V_2^2. \tag{2.8.88}$$

Another aspect of interest concerning the propagation of thermal signals is the possibility of shock waves for temperature, analogous to shock waves for pressure. Thus, the two kinds of waves (first sound and second sound) also extend to two kinds of shock wave (Fig. 2.3 and Table 2.2).

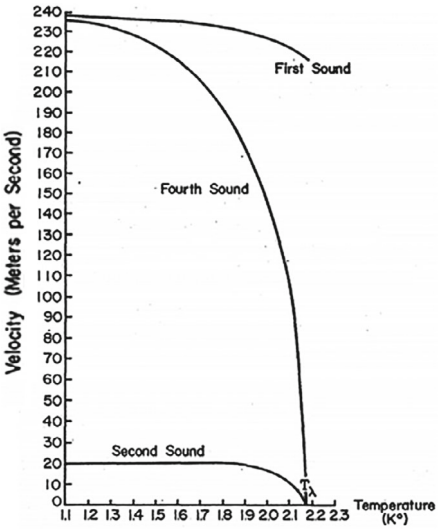


Fig. 2.3 Sketch of the first, second, and fourth sound velocities as a function of temperature. Note that the velocities of second and fourth sound vanish at the lambda point, where the corresponding waves do no longer propagate [8, 29]

Table 2.2 Values of the velocities of the first, second, and fourth sounds V_1, V_2, V_4 from [8, 9]

	2.0 K	1.7 K	1.5 K	1.2 K	1.0 K	0.7 K	0.5 K
V_1 (cm/s)	226.2	233.1	235.5	237.3	237.8	238.2	238.2
V_2 (cm/s)	16.78	20.33	20.00	18.51	18.93	47.07	130.93
V_4 (cm/s)	152.3	205.1	222.2	234.5	236.9	238.1	238.2

2.9 Third Sound, Fourth Sound, and Other Waves

Besides the first and the second sounds studied in the previous section, there are other kinds of characteristic perturbations in superfluid helium, dubbed as third sound, fourth sound, fifth sound, and sixth sound in the order of their respective discovery. Here we give a few details on them. *Third sound*: Third sound is a thickness long-wavelength fluctuation on a superfluid helium film in which the normal component remains stationary and the superfluid component oscillates parallel to the wall. The corresponding temperature fluctuations imply a periodic evaporation and condensation of the liquid film, which must be taken into account for a precise description. Helium II can "creep" along surfaces forming a 30-nm-thick film, called Rollin film [30]. Usually, surface waves cannot occur on thin films because viscosity inhibits their motion. However, in thin films of Helium II [31] and of superfluid ^{3}He [32], and in superfluid ^{3}He – ^{4}He mixtures [33], surface waves do exist [34], and are

known as third sound. Their propagation is governed by the same equation as gravity waves in shallow water, and is associated with van der Waals interaction between the film and the substrate.

Fourth sound: When Helium II is forced to flow in thin capillaries ($10 - 20$ nm) or in porous media, a wave in which all thermodynamic fields vibrate is propagated in it; this wave, anticipated by Pellam [35] and Atkins [36] and later detected experimentally [37, 38], is known as *fourth sound*. As in the third sound, the normal component remains stationary and the superfluid oscillates parallel to the wall, but in contrast to third sound, the thickness of the system is constant because the lateral walls are fixed, instead of having a free surface. The velocity of the fourth sound is given by (2.8.88). Thus, since ρ_s tends to zero when temperature tends to the lambda temperature, and V_2 also tends to zero, V_4 also drops to zero when it approaches to T_λ.

Fifth sound: Fifth sound is a thermal wave that propagates in a clamped superfluid under pressure-released boundary conditions [39–41]. It can also propagate in Helium II films [33]. Its velocity V_5 is related to the speed of second sound as $V_5^2 = (\rho_n/\rho)V_2^2$. Thus, V_5 tends to zero at $T = T_\lambda$ and at $T = 0$ (because $\rho_n = 0$ at $T = 0$).

Sixth sound: In [15, 42], it was shown that the hypothesis of a small but non-vanishing entropy content of the superfluid fraction of liquid helium would give rise to a new sound mode in ring-shaped superleaks. In [43], discontinuity evolution and sonic propagation for the two-fluid model with small superfluid entropy were studied. The fourth sound and the sixth sound were identified as the propagation of discontinuity. Discriminations between the fourth and sixth sounds may be meaningful for future experiments to identify the small superfluid entropy, and compare the results with those in [44, 45] from the attenuation of fourth sound.

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Chapter 3
The One-Fluid Extended Model of Superfluid Hydrodynamics: Concepts, Equations, and Applications



After presenting the essential physical phenomenology of superfluid helium and its description in terms of the two-fluid model we introduce a one-fluid extended model with $\mathbf{v}$ (barycentric velocity) and $\mathbf{q}$ (heat flux) instead of $\mathbf{v}_n$ and $\mathbf{v}_s$ (the velocities of the normal and of the superfluid components of the two-fluid model, respectively), as vectorial independent variables. A motivation of the one-fluid extended model is that, from the purely mesoscopic point of view, one sees only a single fluid rather than two physically separable fluids.

Another motivation is that $\mathbf{v}_n$ and $\mathbf{v}_s$ are only indirectly measured, usually from the measurements of $\mathbf{q}$ and $\mathbf{v}$. The internal degree of freedom corresponding to the relative motion of the two components of the two-fluid model, namely, $\mathbf{v}_n - \mathbf{v}_s$, is here described by the dynamics of the heat flux. An alternative way of a one-fluid extended model would be considering a vectorial internal variable $\mathbf{w}$, not identified a priori. When identified a posteriori, such variable is found to be related to $\mathbf{q}$. Such an approach has the advantage of generality—since $\mathbf{w}$ is in principle general—but, on the other hand, it lacks the motivation of a physical interpretation in terms of directly measurable quantities [1–4].

Since the proportionality coefficient relating $\mathbf{w}$ to $\mathbf{q}$ may depend on temperature and pressure rather than being a constant, it follows that though in a linear approximation the choice of $\mathbf{w}$ or of $\mathbf{q}$ may lead to the same dynamics, it will be not so in nonlinear approaches, introducing additional couplings in the dynamical equations.

The evolution equations describing the behaviour of $\mathbf{v}$ and $\mathbf{q}$ are obtained by means of a mesoscopic theory of non-equilibrium processes [5–8] which uses fluxes, besides the traditional variables, as independent fields, known as Extended Thermodynamics (E.T.). Indeed, since the relaxation time of the heat flux is very long, of macroscopic order, it must be treated as an independent field [1, 4], in contrast to the classical theory of irreversible processes [7, 9, 10], where $\mathbf{q}$ is expressed in terms of the gradients of intensive thermodynamic variables. A relevant part of the chapter is devoted to the analysis of hydrodynamical and inertial effects of heat transport in

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superfluid helium. Inertial effects lead to the second sound and hydrodynamical effects to a size and form dependence of the effective thermal conductivity.

3.1 Introduction to the One-Fluid Extended Model

The fundamental fields in extended thermodynamics of simple fluids are the mass density ρ, the velocity $\mathbf{v}$, the specific internal energy ϵ (or, alternatively, the temperature T), the non-equilibrium stress tensor $\mathbf{P_v}$, and the heat flux $\mathbf{q}$. Note that all these fields are mesoscopic quantities, i.e. quantities averaged over a mesoscopic volume Λ. Whether such small volume is in local thermodynamic equilibrium or not is a relevant question in non-equilibrium thermodynamics beyond local equilibrium.

3.1.1 Dynamical Equations

The corresponding evolution equations for these variables are, on the one side, the balance equations of mass, momentum, and internal energy, which have the general form [6, 7, 9, 10]:

$$\dot{\rho} + \rho \nabla \cdot \mathbf{v} = 0, \tag{3.1.1}$$

$$\rho \dot{\mathbf{v}} + \nabla p + \nabla \cdot \mathbf{P_v} = \mathbf{0}, \tag{3.1.2}$$

$$\rho \dot{\epsilon} + \nabla \cdot \mathbf{q} + [p\mathbf{U} + \mathbf{P_v}] : \nabla \mathbf{v} = 0. \tag{3.1.3}$$

Here, p is the thermostatic pressure, and the non-equilibrium stress $\mathbf{P_v}$ may be split into its trace p_V (non-equilibrium pressure) and its deviator, that is,

$$\mathbf{P_v} = p_V \mathbf{U} + \langle \mathbf{P_v} \rangle. \tag{3.1.4}$$

Angular brackets denote the deviatoric part of the corresponding tensors, (i.e. $(\langle \nabla \mathbf{q} \rangle)_{ij} = \frac{1}{2}\frac{\partial q_i}{\partial x_j} + \frac{1}{2}\frac{\partial q_j}{\partial x_i} - \frac{1}{3}\frac{\partial q_k}{\partial x_k}\delta_{ij}$) and the two dots between two tensors stand for their double contraction.

To close the system, the quantities $\mathbf{q}$, p_V, and $\langle \mathbf{P_v} \rangle$ (heat flux, bulk, and shear viscous stress, respectively) must be specified by three additional equations. In classical irreversible thermodynamics, constitutive equations relate $\mathbf{q}$, p_V and $\langle \mathbf{P_v} \rangle$ to temperature and velocity gradients. Instead, in extended thermodynamics, the fluxes are considered independent fields, and evolution equations are written for them. Neglecting nonlinear terms and up to the first order in the gradients of the independent variables, these equations are [6]

$$\tau_0 \dot{p}_V + \lambda_0 \nabla \cdot \mathbf{v} - \beta' T \lambda_0 \nabla \cdot \mathbf{q} = -p_V, \tag{3.1.5}$$

$$\tau_2\langle\dot{\mathbf{P}}_\mathbf{v}\rangle + 2\eta\langle\nabla\mathbf{v}\rangle - 2\beta T\eta\langle\nabla\mathbf{q}\rangle = -\langle\mathbf{P}_\mathbf{v}\rangle, \tag{3.1.6}$$

$$\tau_1\dot{\mathbf{q}} + \lambda_1\nabla T - \beta' T^2\lambda_1\nabla p_V - \beta T^2\lambda_1\nabla\cdot\langle\mathbf{P}_\mathbf{v}\rangle = -\mathbf{q}. \tag{3.1.7}$$

The coefficients τ_0, τ_2, and τ_1 are the relaxation times of p_V, $\mathbf{P}_\mathbf{v}$ and $\mathbf{q}$, respectively; λ_0, η, and λ_1 are the coefficients which (in a normal fluid) can be identified, respectively, with the bulk viscosity, shear viscosity, and heat conductivity, and β and β' are coefficients appearing in the general expression of the entropy flux in extended thermodynamics (see Eq. (3.1.9)) and describing nonlocal effects in (3.1.5)–(3.1.7). Note that when τ_0, τ_1, τ_2, β, and β' are zero, these equations, respectively, reduce to Stokes equations for p_V and $\langle\mathbf{P}_\mathbf{v}\rangle$, and Fourier equation for $\mathbf{q}$.

3.1.2 Thermodynamic Equations

To discuss thermodynamic behaviour, the contribution of the fluxes to the entropy and the entropy flux must also be stated. In extended thermodynamics the specific entropy s per unit mass and the entropy flux $\mathbf{J}^S$ are the classical ones plus non-equilibrium contributions of second order in the fluxes. The expressions for s, $\mathbf{J}^S$, and the entropy production σ^S are [6, 8]:

$$s = s_0(\rho, T) - \frac{1}{2\rho T^2}\frac{\tau_1}{\lambda_1}\mathbf{q}\cdot\mathbf{q} - \frac{1}{2\rho T}\frac{\tau_0}{\lambda_0}p_V^2 - \frac{1}{4\rho T}\frac{\tau_2}{\eta}\langle\mathbf{P}_\mathbf{v}\rangle : \langle\mathbf{P}_\mathbf{v}\rangle, \tag{3.1.8}$$

$$\mathbf{J}^S = \rho s_0\mathbf{v} + \frac{1}{T}\mathbf{q} + \beta' p_V\mathbf{q} + \beta\langle\mathbf{P}_\mathbf{v}\rangle\cdot\mathbf{q}, \tag{3.1.9}$$

$$\sigma^S = \frac{1}{\lambda_1 T^2}\mathbf{q}\cdot\mathbf{q} + \frac{1}{\lambda_0 T}p_V p_V + \frac{1}{2\eta T}\langle\mathbf{P}_\mathbf{v}\rangle : \langle\mathbf{P}_\mathbf{v}\rangle. \tag{3.1.10}$$

The three last terms in (3.1.8) and the two last terms in (3.1.9) have not an equivalent in local equilibrium thermodynamics and they describe deviations with respect to local equilibrium which are confirmed by several microscopic theories, as the kinetic theory of gases or information theoretic approaches [5, 6, 8]. Those in (3.1.8) formally vanish when the relaxation times τ_i tend to zero, and those in (3.1.9) when β and β' tend to zero, in which case they reduce to the classical local equilibrium entropy and entropy flux.

The specific entropy s (3.1.8) satisfies the generalized Gibbs equation (approximated to the second order in the fluxes) [6]:

$$Tds = d\epsilon - \frac{p}{\rho^2}d\rho - \frac{1}{\rho T}\frac{\tau_1}{\lambda_1}\mathbf{q}\cdot d\mathbf{q} - \frac{1}{\rho}\frac{\tau_0}{\lambda_0}p_V dp_V - \frac{1}{2\rho}\frac{\tau_2}{\eta}\langle\mathbf{P}_\mathbf{v}\rangle : d\langle\mathbf{P}_\mathbf{v}\rangle. \tag{3.1.11}$$

The entropy principle, namely, the positive character of the entropy production σ^S, and the thermodynamical stability requirement $s - s_0 \leq 0$, respectively, lead to

$$\lambda_0 > 0, \qquad \lambda_1 > 0, \qquad \eta > 0, \tag{3.1.12}$$

$$\tau_0 > 0, \qquad \tau_1 > 0, \qquad \tau_2 > 0. \tag{3.1.13}$$

The signs of β and β' are not specified by the second law, but a further consideration shows them to be negative [see (3.4.74)], as they may be identified, in some simple situations, as $\beta = \beta' = -1/(ST^2)$, with S the entropy density per unit volume, namely, $S = \rho s$ (see discussion in Sect. 3.4).

3.1.3 Evolution Equations for Helium II

Experiments [11] show that in Helium II the relaxation time τ_1 of heat flux is comparable to the evolution times of the classical variables, while the relaxation times τ_0 and τ_2 of p_V and $\langle \mathbf{P_v} \rangle$ are very small. Also in Helium II the heat conductivity is extremely high. Thus, the fundamental fields needed to describe the dynamical behaviour of Helium II are the scalar fields ρ and T and the vector fields $\mathbf{v}$ and $\mathbf{q}$. Constitutive relations for p_V and $\langle \mathbf{P_v} \rangle$ are determined by taking in the field Eqs. (3.1.5) and (3.1.6) τ_0 and τ_2 equal to zero. One gets

$$p_V = -\lambda_0 \nabla \cdot \mathbf{v} + \beta' T \lambda_0 \nabla \cdot \mathbf{q}, \tag{3.1.14}$$

$$\langle \mathbf{P_v} \rangle = -2\eta \langle \nabla \mathbf{v} \rangle + 2\beta T \eta \langle \nabla \mathbf{q} \rangle. \tag{3.1.15}$$

These equations contain, in addition to classical terms proportional to the velocity gradient, contributing to the dissipation of mechanical origin and to the attenuation of the first sound, terms depending on the gradient of the heat flux, contributing to the dissipation of thermal origin and to the attenuation of second sound.

Neglecting body forces, the field equations of Helium II are then the balance Eqs. (3.1.1)–(3.1.3) and the evolution equation for $\mathbf{q}$ (3.1.7), which may be rewritten for further convenience as [12, 13]

$$\dot{\rho} + \rho \nabla \cdot \mathbf{v} = 0, \tag{3.1.16}$$

$$\rho \dot{\mathbf{v}} + \nabla p + \nabla \cdot \mathbf{P_v} = 0, \tag{3.1.17}$$

$$\rho \dot{\epsilon} + \nabla \cdot \mathbf{q} + p \nabla \cdot \mathbf{v} + \mathbf{P_v} : \nabla \mathbf{v} = 0, \tag{3.1.18}$$

$$\dot{\mathbf{q}} + \zeta \nabla T + \zeta \nabla \cdot \mathbf{P_q} = \sigma^{\mathbf{q}}, \tag{3.1.19}$$

with $\zeta = \lambda_1/\tau_1$ and with the flux term $\mathbf{P_q}$ and the production term $\sigma^{\mathbf{q}}$. Analogously to the pressure tensor, $\mathbf{P_q}$ may be split into the trace p_q and the deviator $\langle \mathbf{P_q} \rangle$, respectively, given by

$$p_q = \lambda_0 \beta' T^2 \left[\nabla \cdot \mathbf{v} - \beta' T \nabla \cdot \mathbf{q} \right] \qquad \text{and} \qquad \langle \mathbf{P_q} \rangle = 2\eta \beta T^2 \left[\langle \nabla \mathbf{v} \rangle - \beta T \langle \nabla \mathbf{q} \rangle \right]. \tag{3.1.20}$$

For $\beta = \beta' = -1/(ST^2)$, these may be rewritten as

$$p_q = -\frac{\lambda_0}{S}\left[\nabla\cdot\mathbf{v} + \frac{1}{ST}\nabla\cdot\mathbf{q}\right] \quad \text{and} \quad \langle\mathbf{P_q}\rangle = -\frac{2\eta}{S}\left[\langle\nabla\mathbf{v}\rangle + \frac{1}{ST}\langle\nabla\mathbf{q}\rangle\right]. \tag{3.1.21}$$

Note that $q/(ST)$ has dimensions of velocity.

The production term $\sigma^{\mathbf{q}}$ on the right-hand side of Eq. (3.1.19d) is taken as

$$\sigma^{\mathbf{q}} = -\frac{1}{\tau_1}\mathbf{q}. \tag{3.1.22}$$

It will be generalized in the presence of rotation (see Eq. (4.2.6)) and in the turbulent regime (see Eq. (5.3.27) for isotropic turbulence and (7.3.33) for anisotropic one), incorporating the couplings of $\mathbf{q}$ with quantized vortices, a variable not yet included in (3.1.19). The equation for the entropy will be (3.1.8) with $\tau_0 = \tau_2 = 0$. Such expression is the analogous of (2.3.45), taking into account that $\mathbf{V}_{ns} = \mathbf{v}_n - \mathbf{v}_s$ in (2.3.45) is proportional to $\mathbf{q}$, as noted in (2.1.3), where $\mathbf{q} = \rho_s T s \mathbf{V}_{ns}$. In (2.3.45) the entropy is expressed as a function of p, T and $\mathbf{V}_{ns}$, whereas in (3.1.8) it is in terms of ρ, T and $\mathbf{q}$.

3.2 Laminar Heat Transport in Narrow Channels

As a first illustration of Eq. (3.1.19) for the heat flux we consider heat transport in narrow channels in quiescent Helium II, a topic of fundamental interest since 1950s, due to the peculiar ability of superfluid helium to flow along them [11, 14–21], and of practical interest for the cooling of nanosystems.

When the mean free path of the elementary excitations of helium (carrying the heat flow) is short as compared with the radius of the pipe, and the heat flux value is low enough in such a way that the flow is laminar, the total heat flow $\dot{Q}$ along a cylindrical channel of radius R in the presence of a temperature gradient ∇T is described, as said in Sect. 2.6 (expression (2.6.75)) and shown below [11, 17, 22, 23]:

$$\dot{Q} = -\frac{\pi R^4 S^2 T}{8\eta}\nabla T, \tag{3.2.23}$$

with η the viscosity of the normal component. Expression (3.2.23) is often referred as Landau expression.

Furthermore, $\dot{Q}$ depends on the transversal form of the pipe when applied to other flow geometries, as, for instance, channels of elliptical, square, or rectangular cross section. In contrast, for radial heat flow from a hot cylinder [see Sect. 6.3], the equation relating local heat flux to local temperature gradient is completely different from Fourier's law.

3.2.1 Mathematical Derivation

To derive (3.2.23) in steady situations, and neglecting the nonlinear convective terms in the derivatives of the field variables, the mass and energy balance Eqs. (3.1.1) and (3.1.3) reduce to $\nabla \cdot \mathbf{v} = 0$, $\nabla \cdot \mathbf{q} = 0$ which imply $p_V = 0$ in (3.1.14) and $p_q = 0$ in (3.1.20). Thus, equations for $\dot{\mathbf{v}} = 0$ and $\dot{\mathbf{q}} = 0$ become, after using (3.1.22)

$$\nabla p - \eta \nabla^2 \mathbf{v} - \frac{\eta}{ST} \nabla^2 \mathbf{q} = 0, \tag{3.2.24}$$

$$\lambda_1 \nabla T - \frac{\eta \lambda_1}{S} \nabla^2 \mathbf{v} - \frac{\eta \lambda_1}{TS^2} \nabla^2 \mathbf{q} = -\mathbf{q}, \tag{3.2.25}$$

where $\lambda_1 = \tau_1 \zeta$. For constant ST, (3.2.24) and (3.2.25) can be written as

$$\nabla p - \eta \nabla^2 \left(\mathbf{v} + \frac{1}{ST} \mathbf{q} \right) = 0, \tag{3.2.26}$$

$$S \nabla T - \nabla p = -\frac{S}{\lambda_1} \mathbf{q}. \tag{3.2.27}$$

When the right-hand side of Eq. (3.2.27) is negligible, namely, when λ_1 is high enough, as it is experimentally observed, and T changes along the pipe but is homogeneous in any transversal section of the pipe, $\mathbf{v}$ and $\mathbf{q}$ depend only on the radial coordinate r.

The same procedure adopted in classical hydrodynamics for Poiseuille flow leads for the solution of (3.2.26) to

$$v(r) + \frac{1}{ST} q\,(r) = \frac{R^2 S \nabla T}{4\eta} \left(\frac{r^2}{R^2} - 1 \right) + v_w + \frac{1}{ST} q_w, \tag{3.2.28}$$

where v_w and q_w are the values of v and q along the walls ($r = R$), which here we take as zero. We have also used $S \nabla T = \nabla p$ from (3.2.27) to express ∇p in terms of ∇T in (3.2.26). The effect of a non-vanishing slip heat flow along the walls in cylindrical channels will be considered in Sect. 3.5.

Now, let's compute the average value of the left-hand side of (3.2.28) on a transversal section of the channel:

$$\frac{1}{\pi R^2} \int_0^{2\pi} \int_0^R (ST v(r) + q(r))\, r\, dr d\theta = ST \bar{v} + \frac{1}{\pi R^2} \dot{Q} = \frac{1}{\pi R^2} \dot{Q}. \tag{3.2.29}$$

In the last member on the right-hand side, the term in $\bar{\mathbf{v}}$ vanishes because of the counterflow condition $\bar{\mathbf{v}} = 0$. Then, for $v_w = 0$ and $q_w = 0$, the total heat flow $\dot{Q}$ across any transversal section of the tube turns out to be

$$\dot{Q} = 2\pi \int_0^R q\,(r)\, r\, dr = -\frac{\pi R^4 S^2 T}{8\eta} \nabla T, \tag{3.2.30}$$

which is expression (3.2.23). Comparing to the Fourier law, from Eq. (3.2.30) we obtain the following effective thermal conductivity K_{eff}:

$$K_{\text{eff}} \equiv \frac{\dot{Q}}{\pi R^2} \frac{l}{\Delta T} = \frac{R^2 T S^2}{8\eta}. \tag{3.2.31}$$

This effective thermal conductivity depends not only on the material and the temperature, but also on the size of the system, given by R. This hydrodynamic analogy, emphasizing the role of the $\nabla^2 \mathbf{q}$ terms in Eq. (3.2.25), explains why such nonlocal terms often deserve the name of hydrodynamic effects and lead to size and form-dependent expressions for the effective thermal conductivity.

By following the same procedure we may obtain K_{eff} in several other geometries, by suitably using the parallelism with well-known results of classical hydrodynamics. In particular, for channels of elliptical cross section with semiaxes a and b, one has

$$K_{\text{eff}} = \frac{a^3 b^3\, S^2\, T}{4\eta (a^2 + b^2)}, \tag{3.2.32}$$

which for $a = b = R$ reduces to (3.2.31).

For rectangular channels of thickness a and width b for null heat flux on the walls, thermal conductivity K_{eff} is [24]

$$K_{\text{eff}} = \left(\frac{2 \int_0^{a/2} \int_0^b q\,(y,z)\,, y, z}{ab} \right) \frac{l}{\Delta T} = \frac{b^2}{12} \left(\frac{S^2 T}{\eta} \right) \left[1 - \sum_{n,\text{odd}}^{\infty} \frac{1}{n^5} \left(\frac{192}{\varphi \pi^5} \right) \tanh \left(n\pi \frac{\varphi}{2} \right) \right] \tag{3.2.33}$$

wherein $\varphi = a/b$ is the aspect ratio of the cross section.

For $b/a \to 0$, namely, for large ϕ (i.e. for thin and wide channels) one has [25]

$$K_{\text{eff}} = \frac{b^2 S^2\, T}{12\eta} \left[1 - 0.63 \frac{b}{a} \right]. \tag{3.2.34}$$

For channels of square cross section of size a

$$K_{\text{eff}} \approx \frac{1}{20.21} \frac{a^4\, S^2\, T}{\eta}. \tag{3.2.35}$$

More complicated expressions are obtained in other situations, as, for instance, K_{eff} for channels of annular cross section, or triangular cross section of size a. In the triangular case, we have

$$K_{\text{eff}} = \frac{\sqrt{3}}{320} \frac{a^4\, S^2\, T}{\eta}. \tag{3.2.36}$$

Also, expressions of effective thermal conductivity in curved channels could be derived.

3.2.2 *Refrigeration of an Array of Heat-Dissipating Cylinders*

From the practical point of view, for a simple illustration of the use of the effective thermal conductivity we briefly consider the refrigeration of arrays of heat-dissipating cylinders [24, 26–28]. Such cylinders (aiming to represent nanochips in quantum computers working at very low temperature) are assumed, for simplicity, to be distributed in a regular way between two parallel insulating plates of width a separated by a distance b. In this section, we consider the dissipation of heat produced by these cylinders, assuming an absence of mass flow, which will be the aim of Sect. 7.1.

The heat per unit time $\dot{Q}_j$ that each cylinder produces in its operations (as, for instance, the computations) induces a practical limit to efficient cooling set by the transition to quantum turbulence. For instance, if one has an array of $M \times (N+1)$ cylindrical devices, each one dissipating heat at some rate $\dot{Q}$, which will be the maximum total heat we will be able to extract from the system per unit time avoiding the appearance of quantum turbulence? Given $\dot{Q}$, for instance, which will be the maximum value of N we will be able to keep at constant temperature by extracting the dissipated heat? Or given a value of N, which will be the maximum value of $\dot{Q}$ we may manage to extract? And how do these limiting values depend on the radius of the cylinders, the separation of the cylinders, and the separation between the parallel plates?

Let's consider heat flowing through a lattice of transversal cylinders symmetrically distributed all over the channel (see Figs. 3.1 and 3.2 for a qualitative sketch). The

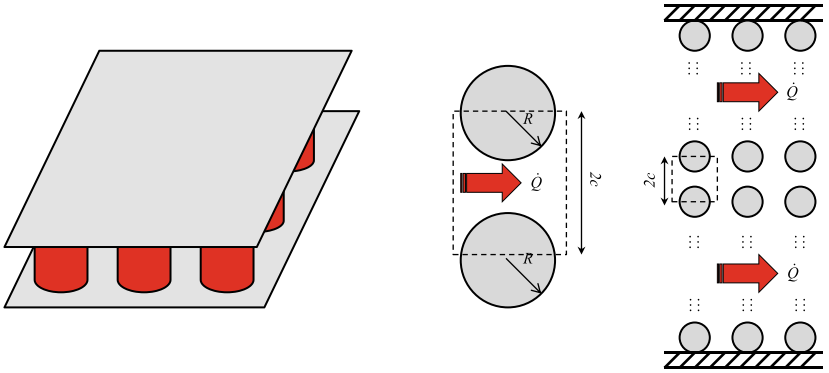


Fig. 3.1 Hot array of transversal heat-dissipating cylinders symmetrically distributed over a rectangular channel filled by superfluid helium, carrying out the dissipated heat. Three-dimensional view (left), horizontal view (right). Arrows in the right side show the total heat flux $\dot{Q}$ along the channel. Figures from [24]

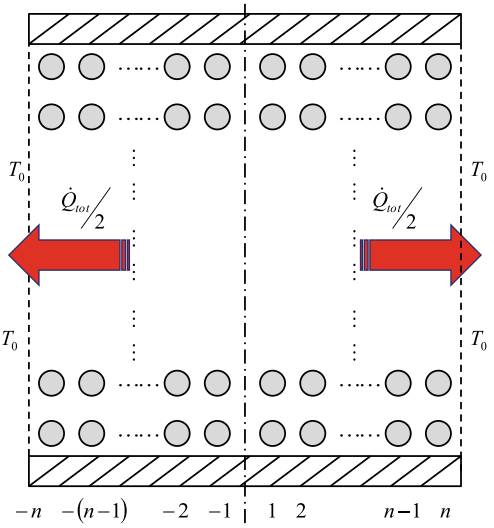


Fig. 3.2 Sketch of an array of $M = 2n$ columns of cylinders (n on the left and n on the right of the indicated vertical axis). Each cylinder radiates heat symmetrically from every side ($\dot{Q}/2$ towards the left and $\dot{Q}/2$ towards the right) in such a way that the total heat flux (represented by the red arrows) will flow towards both sides of the lattice (see $\dot{Q}_{\rm tot}/2$ in the figure). Above and below the lattice, two insulating walls prevent heat transport in the vertical direction. Figure from [24]

axis of each cylinder is perpendicular to the direction of the flow. Therein R and $2c$, respectively, stand for the radius of each cylinder and the distance between the axes of two consecutive cylinders, and we call $\phi = R/c$.

The total effective thermal conductivity is

$$\frac{1}{K_{\rm eff}^{\rm tot}} = \frac{1}{K_{\rm eff}^{\rm (plates)}} + \frac{1}{K_{\rm eff}^{\rm (cylinders)}}. \tag{3.2.37}$$

In the laminar counterflow regime, $K_{\rm eff}^{\rm (plates)}$ and $K_{\rm eff}^{\rm (cylinders)}$ may be obtained by using hydrodynamic analysis of heat transport [26].

Let us consider two main situations:

1. $1 - \phi \ll 1 \Leftrightarrow \phi \to 1$, i.e. the spacing $2(c - R)$ between the cylinders is much smaller than their radius R. In this case, the effective thermal conductivity for Helium II is [24]

$$K_{\rm eff} = \frac{1}{b\,(1-\phi)} \left\{ \frac{12}{b^3\,(1-\phi)\left(1 - \frac{0.63}{\varphi}\right)} + \frac{9\pi}{2bc^2} \left[\frac{\phi^2}{\left(1-\phi^2\right)^{5/2}} \right] \right\}^{-1} \left(\frac{S^2 T}{\eta} \right). \tag{3.2.38}$$

2. $\phi \ll 1 \Leftrightarrow \phi \to 0$, i.e. the spacing between the cylinders is much bigger than their radius R. In this other case, the effective thermal conductivity for Helium II is [24]

$$K_{\text{eff}} = \frac{1}{b\,(1-\phi)} \left\{ \frac{12}{b^3\,(1-\phi)\left(1-\frac{0.63}{\varphi}\right)} + \frac{3}{bc^2}\left[\frac{1}{\left(1-\phi^2\right)^2}\right] \right\}^{-1} \left(\frac{S^2 T}{\eta}\right). \tag{3.2.39}$$

More in detail, we consider M columns of $N+1$ cylinders symmetrically arranged in a channel with a high aspect ratio rectangular cross section (see Fig. 3.2 for a qualitative sketch of the geometry), with the subindices i and j denoting the position (i.e. the row and the column) of the cylinder in the array. We assume the cylinder at (i, j) position to dissipate an energy $\dot{Q}_{ij}$ per unit time as the consequence of its internal functioning. To keep the system at a constant temperature, the total power $\dot{Q}_{\text{tot}} = \sum_{i,j} \dot{Q}_{ij}$ produced by all the cylinders must be removed. For the sake of simplicity, we assume that each cylinder delivers the same amount of heat per unit time, i.e. $\dot{Q}_{ij} = \dot{Q}$.

For the sake of illustration, we assume that $M = 2n$ and that the origin of the reference frame is in the middle of the columns, in such a way that n columns are to the left and n columns are to the right (we consider $j = -n \ldots, -1, 1, \ldots, n$).

These results can be rearranged using a variable x spacing from the first column $(-n)$ to the last column (n), imagining the range of numbers to be continuous instead of discrete. In this case, the energy balance equation ($\rho \dot{u} = -\nabla \cdot \mathbf{q} + \dot{Q}$) giving T as function of x leads to

$$\frac{\mathrm{d}T}{\mathrm{d}x} = -x K_{\text{eff}}^{-1} \left(\frac{\dot{Q}'}{2}\right) \tag{3.2.40}$$

wherein $K_{\text{eff}}^{-1}\dot{Q}'/2$ is constant. By integrating Eq. (3.2.40) from 0 to a fixed distance x' we find

$$T\left(x'\right) = T\left(0\right) - K_{\text{eff}}^{-1}\left(\frac{x'^2}{2}\right)\left(\frac{\dot{Q}'}{2}\right) \tag{3.2.41}$$

where $T\,(0)$ is the temperature at the central axis, at $x = 0$, and $T\left(x'\right)$ is the temperature at $x = x'$. If x' is the last column of cylinders, and setting $x' = n$ and assuming that T is there the temperature of the external bath $T^{(\text{bath})}$, we find for the central temperature $T(0)$

$$T\left(0\right) = T^{(\text{bath})} + K_{\text{eff}}^{-1}\left(\frac{n^2}{2}\right)\left(\frac{\dot{Q}'}{2}\right). \tag{3.2.42}$$

This result gives the highest temperature of the lattice $T\,(0)$ in terms of the temperature of helium bath, the effective thermal conductivity K_{eff}, the number of cylinder columns from the centre to the side, and the total heat $\dot{Q}'$ released by unit time by a column of cylinders. It is crucial that $T(0)$ remains smaller than the lambda temperature T_λ over which helium is no longer superfluid (see Sect. 1.1). Otherwise, helium will not be able to carry out so much heat and the temperature of the array will increase very much, with danger for the integrity of the system.

Expression (3.2.41) can be also written by using the value (3.2.42) of $T\,(0)$ as

$$T\left(x'\right) = T^{(\text{bath})} + \frac{1}{2} K_{\text{eff}}^{-1} \left(n^2 - x'^2\right) \frac{\dot{Q}'}{2}. \tag{3.2.43}$$

In dimensionless form, by setting $\Theta = T\left(x'\right)/T^{(\text{bath})}$ and $\bar{x} = x'/n$, with

$$C = \left(\frac{K_{\text{eff}}^{-1}}{T^{(\text{bath})}}\right) \frac{\dot{Q}'}{2}, \tag{3.2.44}$$

expression (3.2.43) becomes

$$\Theta\left(\bar{x}\right) = 1 + \frac{nC}{2}\left(1 - \bar{x}^2\right) \tag{3.2.45}$$

which is plotted in Fig. 3.3 (Fig. 3.4).

When heat may be exchanged with the environment in the x- and y-directions and not only in the x-direction as above, we find instead of (3.2.43)

$$T\left(x, y\right) = T^{(\text{bath})} + \left[\left(n^2 + m^2\right) - \left(x^2 + y^2\right)\right] K_{\text{eff}}^{-1} \frac{\dot{Q}}{8}. \tag{3.2.46}$$

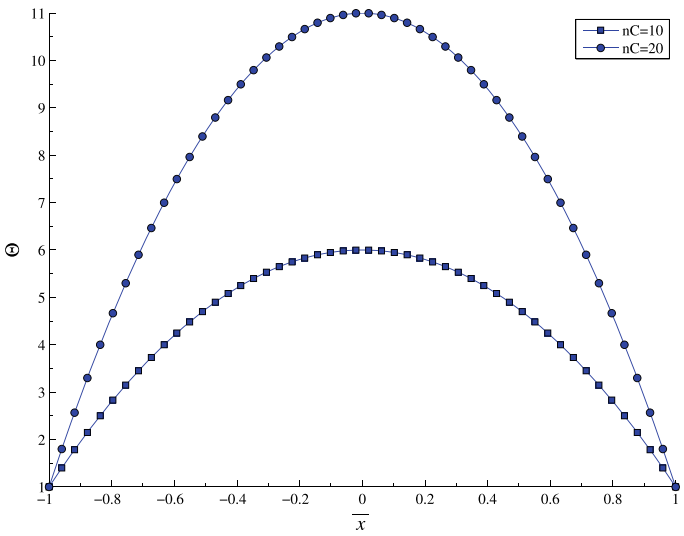


Fig. 3.3 Dimensionless temperature $\Theta = T/T^{(\text{bath})}$ (3.2.45) versus the dimensionless position $\bar{x}$ for two different values of nC. Figure from [24]

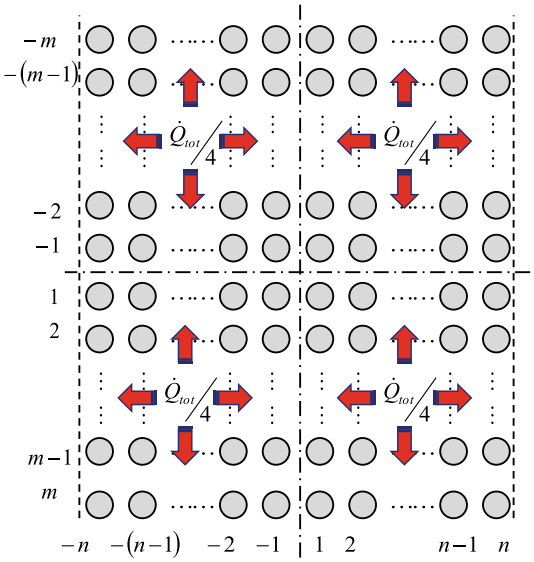


Fig. 3.4 Upper view of an array of hot cylinders symmetrically arranged in the channel ($2n$ columns of $2m$ cylinders). Each cylinder radiates heat symmetrically from every side ($\dot{Q}/4$ towards the left, $\dot{Q}/4$ towards the right, $\dot{Q}/4$ towards the top, and $\dot{Q}/4$ towards the bottom). Figure from [29]

The dimensionless version of Eq. (3.2.46) is

$$\Theta(\bar{x}, \bar{y}) = 1 + \frac{C(n^2 + m^2)}{8}\left[1 - \left(\frac{n^2}{n^2 + m^2}\bar{x}^2 + \frac{m^2}{n^2 + m^2}\bar{y}^2\right)\right], \tag{3.2.47}$$

where we have set $\Theta(x, y) = \dfrac{T(x, y)}{T^{(\text{bath})}}$, $\bar{x} = \dfrac{x}{n}$ and $\bar{y} = \dfrac{y}{m}$, and $C = \dfrac{\dot{Q}}{K_{\text{eff}} T^{(\text{bath})}}$. The maximum value of the temperature Θ_0 is in the middle of the system of arrays, namely, at $\bar{x} = \bar{y} = 0$. From Eq. (3.2.47) the expression of Θ_0 is $\Theta_0 = 1 + \dfrac{C(n^2 + m^2)}{8}$, and the expression (3.2.47) can be written as

$$\Theta(\bar{x}, \bar{y}) = 1 + (\Theta_0 - 1)\left[1 - \left(\frac{n^2}{n^2 + m^2}\bar{x}^2 + \frac{m^2}{n^2 + m^2}\bar{y}^2\right)\right] \tag{3.2.48}$$

which has the form of the parabolic profile (Fig. 3.5).

As in the previous situation, the most interesting result is the temperature at the centre $T(0, 0)$, the highest one in the system in terms of $T^{(\text{bath})}$, K_{eff}, $\dot{Q}$, n, and m. $T(0, 0)$ must be kept lower than the lambda temperature T_λ. This condition restricts the admissible size of the array or of the heat-dissipated power in it.

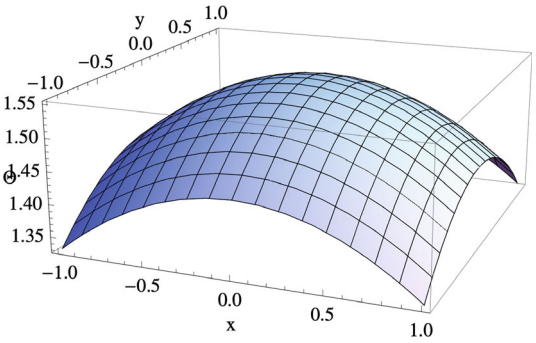


Fig. 3.5 Dimensionless temperature profile Θ (expression (3.2.48)) versus $\bar{x}$ and $\bar{y}$ for $\Theta_0 = 1.55$ (which means $T(0, 0) = 2.17$ K and $T^{(\text{bath})} = 1.4$ K) and $n = 10$ and $m = 20$

3.2.3 *Dependence on the Length of the Channel: Entrance Effects*

Expressions (3.2.31) and (3.2.34) correspond to very long channels, where the Poiseuille parabolic profile (3.2.28) for the heat flux is fully developed. For actual finite channels, the entrance region along which the profile of the heat flow changes from a flat initial profile to the developed parabolic profile should be taken into account. In this region, the thermal resistance is higher than in the regions with parabolic profile and it may be tentatively described by approximations analogous to those used in classical hydrodynamics. For instance, in [30, 31], the following generalization of (3.2.31) has been proposed:

$$K_{eff} = \frac{R^2\, T S^2}{8\eta}\left[0.214 X^{-\frac{1}{2}} + \frac{0.019 X^{-1} + 1 - 0.214 X^{-\frac{1}{2}}}{1 + 0.00021 X^{-2}}\right]^{-1}, \qquad (3.2.49)$$

with $X = \dfrac{\eta\pi S T l}{4\dot{Q}}$, being l the total length of the channel. For l long or $\dot{Q}$ small, X is high and (3.2.49) reduces to (3.2.31). In contrast, for short channels or for high $\dot{Q}$, one has $X \ll 1$, and (3.2.49) reduces to $K_{eff} = \dfrac{R^2 T S^2}{1.68\eta} X^{1/2}$, so that the effective thermal conductivity becomes very small. In this case, $\dot{Q}$ becomes proportional to $l^{-1/3}(\Delta T)^{2/3}$, in contrast to $l^{-1}\Delta T$ behaviour corresponding to (3.2.30).

In principle, the length-dependent correction (the X dependent factor in (3.2.49)) could depend on the form of the transversal section but, lacking such information, it may be tentatively used for other geometries.

3.3 Wave Propagation: Inertial Heat Effects

Beyond steady-state situations as the ones examined in Sect. 3.2, dynamical situations must be considered, because they provide much useful information as already said in the analysis of first, second, and fourth sounds in Sect. 2.8. A typical situation is the propagation of harmonic plane waves. Defining $\mathcal{U} = (p, T, \mathbf{v}, \mathbf{q})$, we look for solutions of the linearized Eqs. (3.1.16)–(3.1.19) having the form:

$$\mathcal{U} = \mathcal{U}_0 + \bar{\mathcal{U}} e^{i(K\mathbf{n}\cdot\mathbf{x}-\omega t)}, \tag{3.3.50}$$

where $\mathcal{U}_0 = (p_0, T_0, 0, 0)$ is the steady-state solution, $\bar{\mathcal{U}} = (\bar{p}, \bar{T}, \bar{v}_i, \bar{q}_i)$ is the small-amplitude perturbation to the steady solution (whose products may be neglected), ω is the real angular frequency, $K = k_r + ik_s$ is the complex wave number, and $\mathbf{n}$ is the unit vector orthogonal to the wavefront.

3.3.1 *Longitudinal Modes: First Sound and Second Sound*

Inserting (3.3.50) in (3.1.16)–(3.1.19), putting $\bar{v}_n = \bar{\mathbf{v}} \cdot \mathbf{n}$, $\bar{q}_n = \mathbf{q} \cdot \mathbf{n}$, and multiplying Eqs. (3.1.18) and (3.1.19) by the unit vector $\mathbf{n}$, the algebraic system for small amplitudes is obtained [13]:

$$-\omega\bar{p} - \omega[p_T]_0\bar{T} + K\left[\rho p_\rho\right]_0 \bar{v}_n = 0, \tag{3.3.51}$$

$$\left(-\omega + iK^2\left[\frac{1}{\rho}\left(\lambda_0 + \frac{4}{3}\eta\right)\right]_0\right)\bar{v}_n + K\left[\frac{1}{\rho}\right]_0 \bar{p} + iK^2\left[\frac{T}{\rho}\left(\lambda_0\beta' + \frac{4}{3}\eta\beta\right)\right]_0 \bar{q}_n = 0, \tag{3.3.52}$$

$$-\omega\bar{T} + K\left[\frac{Tp_T}{\rho c_p}\right]_0 \bar{v}_n + K\left[\frac{1}{\rho c_V}\right]_0 \bar{q}_n = 0, \tag{3.3.53}$$

$$\left(-\omega - i\left[\frac{1}{\tau_1}\right]_0 - iK^2\left[T^3\zeta\left(\lambda_0\beta'^2 + \frac{4}{3}\eta\beta^2\right)\right]_0\right)\bar{q}_n + \\ + K\left[\zeta\right]_0\bar{T} + iK^2\left[T^2\zeta\left(\lambda_0\beta' + \frac{4}{3}\eta\beta\right)\right]_0 \bar{v}_n = 0, \tag{3.3.54}$$

where $p_T = \partial p/\partial T$ and $p_\rho = \partial p/\partial\rho$.

If the thermal expansion $\partial\rho/\partial T = p_T/p_\rho$ is zero (namely, if p_T is negligible, which is a good approximation at vapour pressure for $T = 1.2$ K), assuming $k_r \gg k_s$ (small perturbation) and the frequency ω not too small, one finds two longitudinal waves (first sound: density and pressure wave; second sound: temperature and entropy wave) whose respective phase speeds u_i and attenuation coefficients $k_s^{(i)}$ are [13]

$$u_1^2 = \frac{\partial p}{\partial\rho} \equiv V_1^2, \qquad k_s^{(1)} = \frac{\omega^2}{2\rho u_1^3}\left(\lambda_0 + \frac{4}{3}\eta\right), \qquad \text{(first sound)} \tag{3.3.55}$$

Table 3.1 Values of the parameter $\zeta = \lambda_1/\tau_1 = \rho c_V V_2^2$ from experimental values for V_2, ρ and c_V listed in Tables 2.1 and 2.2 at different temperatures

	2.0 K	1.7 K	1.5 K	1.2 K	1.0 K	0.7 K	0.5 K
$\zeta\ 10^{-9}(\mathrm{Jm^{-1}s^{-2}\,k^{-1}})$	17.5278	13.3582	8.1224	2.6730	0.8550	0.0740	0.0165

$$u_2^2 = \frac{\zeta}{\rho c_p} \equiv V_2^2, \qquad k_s^{(2)} = \frac{1}{2u_2}\frac{1}{\tau_1} + \frac{\omega^2 T^3 \zeta}{2u_2^3}\left(\lambda_0 \beta'^2 + \frac{4}{3}\eta\beta^2\right). \qquad \text{(second sound)} \tag{3.3.56}$$

The attenuation coefficient $k_s^{(1)}$ is identical to the one deduced by Landau and Khalatnikov, using the two-fluid model [32]. The attenuation coefficient of the second sound contains a term proportional to the square of the frequency ω which is dominant for very long τ_1, but it is slightly different from the one obtained in [32]. In the two-fluid model ζ may be identified as $\zeta = \rho\rho_s/\rho_n T s^2$. In (2.8.87), the expressions for the speeds have been derived, but not those of the attenuation coefficients, because there we have used the non-dissipative version of the theory.

When thermal expansion is taken into account (p_T non negligible), first and second sounds become coupled: that is, in first sound, temperature and heat flux oscillations are also present and, in second sound, pressure and velocity oscillations also occur. Speeds and attenuation coefficients of first and second sounds in such case may be found in [33], and agree with experimental data (Table 3.1).

3.3.2 *Transversal Modes and a Macroscopic Two-Fluid Model*

The analysis of transverse modes connects in a natural way the one-fluid model with the two-fluid model. Thus, it provides the basis of a major conceptual conclusion relating both models. Projecting the vector equation for the small-amplitude perturbations of velocity and heat flux (3.3.50) on the plane of the wavefront, and denoting such projections with the π subscript, we obtain from Eqs. (3.1.18) and (3.1.19):

$$\left(-\omega - iK^2\frac{\eta}{\rho}\right)\bar{v}_\pi + iK^2\frac{\eta}{\rho}\beta T\bar{q}_\pi = 0, \tag{3.3.57}$$

$$iK^2\eta\beta T^2\zeta\bar{v}_\pi + \left(-\omega - i\frac{1}{\tau_1} - iK^2\eta\beta^2 T^3\zeta\right)\bar{q}_\pi = 0. \tag{3.3.58}$$

Note that here, in contrast to (3.2.26) and (3.2.27), we have not equated $\beta = \beta' = -(ST^2)^{-1}$, which is not necessarily valid in full generality, as it will be seen in Sect. 3.4, and which will allow a deeper understanding of the physics of the problem.

Imposing the vanishing of the determinant of (3.3.57) and (3.3.58), the corresponding dispersion equation is

$$\omega^2 + i\omega\left(\frac{1}{\tau_1} + K^2\eta\frac{P}{\rho}\right) - K^2\frac{1}{\tau_1}\frac{\eta}{\rho} = 0 \qquad \text{where} \qquad P \equiv 1 + \rho\beta^2\, T^3\zeta. \tag{3.3.59}$$

Since $1/\tau_1$ is extremely small, and neglecting it, Equation (3.3.59a) has the two solutions:

$$\omega_1 \simeq 0, \qquad \omega_2 \simeq -i\,K^2 2\eta\frac{P}{\rho}. \tag{3.3.60}$$

The modes associated to these two solutions will play a central role in Sect. 3.4. Substituting (3.3.60a) in (3.3.57) and (3.3.58), we get

$$-\,i\,K^2\frac{\eta}{\rho}\,(\bar{v}_\pi - \beta T\bar{q}_\pi) = 0. \tag{3.3.61}$$

We conclude that in this mode the vector field

$$\mathbf{u}^{(n)} \equiv \mathbf{v} - \beta T\,\mathbf{q} \tag{3.3.62}$$

is almost zero. Analogously, substitution of (3.3.60b) in (3.3.57) and (3.3.58) leads to:

$$i\,K^2\eta\beta T^2\zeta\left(\bar{v}_\pi + \frac{1}{\rho\beta T^2\zeta}\bar{q}_\pi\right) = 0, \tag{3.3.63}$$

whence we conclude that in this other mode the vector field

$$\mathbf{u}^{(s)} \equiv \mathbf{v} + \frac{\beta T}{P-1}\mathbf{q} \tag{3.3.64}$$

is almost zero. The first transverse mode corresponds to a very slow relaxation involving only $\mathbf{u}^{(s)}$ (since $\mathbf{u}^{(n)}$ is practically zero). Concerning the second mode, the dispersion relation (3.3.59) yields

$$k_r^2 - k_s^2 = -\frac{1}{\tau_1}\frac{\rho}{\eta}\omega^2\frac{P-1}{\frac{1}{\tau_1^2} + P^2\omega^2} \tag{3.3.65}$$

$$k_s = \frac{\rho}{2\eta}\frac{\omega}{k_r}\frac{\frac{1}{\tau_1^2} + P\omega^2}{\frac{1}{\tau_1^2} + P^2\omega^2} = \frac{\omega}{k_r}\frac{\rho}{2\eta P}. \tag{3.3.66}$$

Thus, we obtain

$$k_r = k_s = \sqrt{\frac{\omega\rho}{2\eta P}}. \tag{3.3.67}$$

Therefore the second mode corresponds to perturbations of $\mathbf{u}^{(n)}$ (since $\mathbf{u}^{(s)}$ is practically zero), which are attenuated within a small number of wavelengths, owing to the small value of η. The result of $\mathbf{u}^{(s)}$ mode with almost no attenuation and $\mathbf{u}^{(n)}$ mode with strong attenuation is conceptually relevant and will be worked out in the next two sections.

In the analysis above, the time evolution of the non-equilibrium stress tensor was neglected, putting the relaxation times τ_0 and τ_2 equal to zero. In [34], the fast dynamics of these fields is not neglected and the complete model with 14 fields (the system of Eqs. (3.1.1)–(3.1.7)) is studied, in the linear approximation. Also in this more complete model, the two vector fields $\mathbf{u}^{(s)}$ and $\mathbf{u}^{(n)}$ defined in (3.3.62) and (3.3.64) arise in a natural way.

The two fields $\mathbf{u}^{(s)}$ and $\mathbf{u}^{(n)}$ in (3.3.62) and (3.3.64) (almost no attenuation for $\mathbf{u}^{(s)}$, strong attenuation for $\mathbf{u}^{(n)}$) are related to $\mathbf{v}$ and $\mathbf{q}$ as follows:

$$\mathbf{v} = \frac{P-1}{P}\mathbf{u}^{(s)} + \frac{1}{P}\mathbf{u}^{(n)}, \qquad \mathbf{q} = \frac{1}{\beta T}\frac{P-1}{P}[\mathbf{u}^{(s)} - \mathbf{u}^{(n)}], \tag{3.3.68}$$

as obtained by inversion of (3.3.62) and (3.3.64).

Concerning the tangential boundary conditions on a wall, since the shear viscosity is absent from $\mathbf{u}^{(s)}$, no condition must be imposed on it. In contrast, we postulate the following condition for $\mathbf{u}^{(n)}$ [35]: "On the walls of a container, the tangential component of the vector field $\mathbf{u}^{(n)}$ is zero". In terms of $\mathbf{v}$ and $\mathbf{q}$, we can write this as

$$u_t^{(n)} = v_t - \beta T q_t = 0, \tag{3.3.69}$$

where the underscript t denotes the tangential component of the vectors. Boundary Eq. (3.3.69) is valid when the mean free path of the heat carriers is sufficiently smaller than the radius of the channel. When the mean free path is comparable or longer than radius, a different condition is taken into account (see Sect. 3.5).

3.4 From the One-Fluid Extended Model to a Generalized Two-Fluid Model

In the previous section, it has been shown, using only macroscopic considerations, that two vector fields $\mathbf{u}^{(s)}$ and $\mathbf{u}^{(n)}$ having dimensions of velocity arise, in a natural way, in the one-fluid extended model of liquid Helium II. It may be shown [see [36]] that, if we suppose $\beta = \beta'$ and $1/\tau_1$ negligible, dissipative phenomena are due only to $\mathbf{u}^{(n)}$. It is therefore natural interpreting $\mathbf{u}^{(s)}$ and $\mathbf{u}^{(n)}$ as the velocities of a superfluid and of a normal component, respectively. Starting from this identification, a comparison of the one-fluid extended model and the two-fluid model may be performed in more depth.

3.4.1 Velocities, Densities, and Entropies of the Two Physical Modes

First, observe that (3.3.68a) suggests the introduction of two scalar fields, $\rho^{(s)}$ and $\rho^{(n)}$, associated with $\mathbf{u}^{(s)}$ and $\mathbf{u}^{(n)}$, which can be interpreted as the "mass densities" of the superfluid and normal components in analogy to (2.1.1). In this way that, one sets

$$\frac{\rho^{(s)}}{\rho} \equiv \frac{P-1}{P}, \qquad \frac{\rho^{(n)}}{\rho} \equiv \frac{1}{P}. \tag{3.4.70}$$

These ratios are always between 0 and 1, because of (3.3.59b). From (3.4.70), it follows that $\frac{\rho^{(s)}}{\rho^{(n)}} = P - 1$.

Recall now that in its usual formulation the two-fluid model assumes not only that the superfluid component is not involved in dissipative phenomena but also that its entropy is zero. In the extended model, the "convective" terms of the entropy flux $\mathbf{J}^S$ may be written as [36]

$$[\mathbf{J}^S]_u = \rho s \mathbf{u}^{(n)} + \left(\rho s + \frac{1}{\beta T^2}\right) \frac{\rho^{(s)}}{\rho} (\mathbf{u}^{(s)} - \mathbf{u}^{(n)}). \tag{3.4.71}$$

From this expression, we see that both $\mathbf{u}^{(s)}$ and $\mathbf{u}^{(n)}$ are associated with convective entropy fluxes. In terms of $\rho^{(s)}$ and $\rho^{(n)}$, defined in (3.4.70), and recalling that $P - 1 = \rho\beta^2 T^3 \zeta$ (see (3.3.59b)) (3.4.71) can be rewritten as

$$[\mathbf{J}^S]_u = \rho^{(s)} \left(s + \frac{1}{\rho\beta T^2}\right) \mathbf{u}^{(s)} + \rho^{(n)} (s - \beta T \zeta)\, \mathbf{u}^{(n)}. \tag{3.4.72}$$

Consequently, from (3.4.72) the following quantities

$$s^{(s)} = s + \frac{1}{\rho\beta T^2}, \quad s^{(n)} = s - \beta T \zeta, \tag{3.4.73}$$

can be interpreted as the entropy per unit mass of the superfluid component and of the normal component, respectively, as they satisfy $\rho s = \rho^{(s)} s^{(s)} + \rho^{(n)} s^{(n)}$. Note that from (3.4.73a) if β takes the particular value:

$$\beta^* = -\frac{1}{\rho s T^2} \tag{3.4.74}$$

$s^{(s)}$ becomes zero and the "convective" part of the entropy flux reduces to the first term: $[\mathbf{J}^S]_u = \rho s \mathbf{u}^{(n)}$. This is just the expression of the entropy flux postulated by Landau. Indeed, under this hypothesis for β and $P - 1 = \rho\beta^2 T^3 \zeta$, one has from (3.3.62) and (3.3.64) that

$$\mathbf{u}^{(s)} = \mathbf{v} - \frac{s}{\zeta}\mathbf{q} = \mathbf{v}_s, \quad \mathbf{u}^{(n)} = \mathbf{v} + \frac{1}{\rho T s}\mathbf{q} = \mathbf{v}_n, \tag{3.4.75}$$

with

$$\zeta = \frac{P-1}{\rho\beta^2 T^3} = \rho\frac{\rho^{(s)}}{\rho^{(n)}}Ts^2. \tag{3.4.76}$$

In this linear approximation, Eqs. (3.1.16)–(3.1.19) of the one-fluid extended model, in the absence of dissipative phenomena, as well as the form of ζ are truly identical with those of the original Landau two-fluid model without any microscopic hypothesis [12, 37].

In contrast, if β is different from β^* defined in (3.4.74), the superfluid component carries some amount of entropy $s^{(s)}$, as discussed in Chap. 8 from a microscopic perspective. As observed by Puttermann [38], the statement that the superfluid component doesn't carry entropy must be regarded as an additional postulate since it has not been demonstrated theoretically. As evidenced by Schäfer and Fliessbach [39], the experimental results simply tell us that the entropy carried by the superfluid component is smaller than 2% with respect to the entropy carried by the normal component. These authors have advanced theoretical (microscopic) motivations in favour of a small non-vanishing superfluid entropy, and have proposed an extension of the two-fluid model, where a small entropy associated with the superfluid component is allowed. This is in agreement with the derivations carried above in (3.4.73) from the extended thermodynamic model of liquid Helium II.

Previous results allow us to give a microscopic meaning to the field $\mathbf{u}^{(s)}$ as

$$\mathbf{u}^{(s)} \equiv \mathbf{v} + \frac{1}{\rho\beta T^2\zeta}\mathbf{q}. \tag{3.4.77}$$

The small entropy carried by superfluid component influences several dynamical and thermodynamical properties of liquid Helium II, as, for instance, speeds and attenuations of the first and second sounds. Observe that from (3.4.73), keeping in mind (3.4.74), one may write

$$\frac{s^{(s)}}{s} = 1 - \frac{\beta^*}{\beta}. \tag{3.4.78}$$

This implies that the superfluid entropy $s^{(s)}$ influences only quantities depending explicitly on the parameter β. From (3.4.73a), we set

$$\beta = -\frac{1}{\rho(s - s^{(s)})T^2}. \tag{3.4.79}$$

To determine the influence of the superfluid entropy $s^{(s)}$ on normal and superfluid velocities, note that using (3.4.74), expressions (3.3.62) and (3.3.64) may be written as

$$\mathbf{u}^{(n)} = \mathbf{v} + \frac{\beta}{\beta^*}\frac{1}{\rho s T}\mathbf{q}, \qquad \mathbf{u}^{(s)} = \mathbf{v} - \frac{\beta^*}{\beta}\frac{s}{\zeta}\mathbf{q}, \tag{3.4.80}$$

and using relation (3.4.79), they can be also written as

$$\mathbf{u}^{(n)} = \mathbf{v} + \frac{1}{\rho(s - s^{(s)})T}\mathbf{q}, \qquad \mathbf{u}^{(s)} = \mathbf{v} - \frac{s - s^{(s)}}{\zeta}\mathbf{q}. \tag{3.4.81}$$

As one sees, the superfluid entropy $s^{(s)}$ modifies the speeds of normal and superfluid components for given values of $\mathbf{v}$ and $\mathbf{q}$. When $\beta = \beta^*$, these quantities coincide with the normal and superfluid velocities of the Landau two-fluid model.

As shown in [35], neither the velocity nor the attenuation of first sound depend perceptibly on $s^{(s)}$. In contrast, $s^{(s)}$ influences second sound velocity and attenuation. In fact, from (3.3.56) and (3.4.79) one can write V_2 in terms of $s - s^{(s)}$ instead of s:

$$V_2^2 = \frac{\rho^{(s)}}{\rho^{(n)}}\frac{T}{c_V}\left(s - s^{(s)}\right)^2. \tag{3.4.82}$$

This expression allows us to determine the small entropy $s^{(s)}$ associated to superfluid component only if $\rho^{(s)}/\rho^{(n)}$ can be calculated independently from V_2. Combining this with (3.4.70) one has

$$\frac{\rho^{(s)}}{\rho^{(n)}} = \frac{\zeta}{\rho T(s - s^{(s)})^2}. \tag{3.4.83}$$

By inversion of (3.4.81), we get for $\mathbf{v}$ and $\mathbf{q}$:

$$\mathbf{v} = \frac{\rho^{(n)}}{\rho}\mathbf{u}^{(n)} + \frac{\rho^{(s)}}{\rho}\mathbf{u}^{(s)}, \tag{3.4.84}$$

$$\mathbf{q} = \rho^{(s)}T(s - s^{(s)})(\mathbf{u}^{(n)} - \mathbf{u}^{(s)}). \tag{3.4.85}$$

Furthermore, from (3.4.79) it follows that

$$\frac{\rho^{(s)}}{\rho} = \frac{\zeta}{\zeta + \rho T(s - s^{(s)})^2}, \qquad \frac{\rho^{(n)}}{\rho} = \frac{\rho T(s - s^{(s)})^2}{\zeta + \rho T(s - s^{(s)})^2}, \tag{3.4.86}$$

which show the role of $s^{(s)}$ on $\rho^{(n)}$ and $\rho^{(s)}$. Recalling that at microscopic level, heat is carried by the elementary excitations, from (2.75) we deduce that, on the average, the energy carried by these quasi-particles is $\rho^{(s)}T(s - s^{(s)})$, in accord with the hypothesis that no energy transfer is associated to the condensate.

3.4.2 Evolution Equations in Terms of the Two Physical Modes

Neglecting in (3.1.16)–(3.1.19) the nonlinear terms $p\nabla\cdot\mathbf{v}+\mathbf{P}_{\mathbf{v}}:\nabla\mathbf{v}$, and assuming that some terms do not depend on spatial and temporal variables as βT, λ_0/ρ, and β'/P, we get the evolution equations for ρ, T, $\mathbf{u}^{(n)}$, and $\mathbf{u}^{(s)}$, which are of interest for a general dynamical comparison of the one-fluid extended model and the two-fluid model, where the evolution equations for $\mathbf{v}^{(s)}$ and $\mathbf{v}^{(n)}$ were given in (2.3.62) and (2.3.63).

In the particular case that $\beta=\beta^*$ (the results for the more general situation $\beta\neq\beta^*$ may be found in Mongioví et al. [36]), the equations are

$$\frac{\partial\rho}{\partial t}+\mathbf{v}\cdot\nabla\rho+\nabla\cdot\left[\rho^{(s)}\mathbf{u}^{(s)}+\rho^{(n)}\mathbf{u}^{(n)}\right]=0 \tag{3.4.87}$$

$$\frac{\partial\epsilon}{\partial t}+\mathbf{v}\cdot\nabla\epsilon+\nabla\cdot\left[\frac{1}{\beta T}\frac{\rho^{(s)}}{\rho}\left(\mathbf{u}^{(s)}-\mathbf{u}^{(n)}\right)\right]=0 \tag{3.4.88}$$

$$\begin{aligned}\frac{\partial}{\partial t}\mathbf{u}^{(n)}+(\mathbf{v}\cdot\nabla)\mathbf{u}^{(n)}=&-\frac{1}{\rho}\nabla p-\frac{\rho^{(s)}}{\rho^{(n)}}(s-s^{(s)})\nabla T+\frac{\lambda_0}{\rho^{(n)}}\nabla(\nabla\cdot\mathbf{u}^{(n)})+\\&+\frac{\eta}{\rho^{(n)}}\nabla^2\mathbf{u}^{(n)}+\frac{1}{\tau_1}\frac{\rho^{(s)}}{\rho}(\mathbf{u}^{(s)}-\mathbf{u}^{(n)}),\end{aligned} \tag{3.4.89}$$

$$\frac{\partial}{\partial t}\mathbf{u}^{(s)}+(\mathbf{v}\cdot\nabla)\mathbf{u}^{(s)}=-\frac{1}{\rho}\nabla p+(s-s^{(s)})\nabla T-\frac{1}{\tau_1}\frac{\rho^{(n)}}{\rho}(\mathbf{u}^{(s)}-\mathbf{u}^{(n)}) \tag{3.4.90}$$

where we have considered the relation (3.4.76) with $(s-s^{(s)})$ instead of s. The production terms will be generalized in the presence of vortices in (7.3.33) and (7.3.34).

Note that, whereas in the evolution equation for $\mathbf{u}^{(n)}$ the bulk (λ_0) and shear (η) viscosity terms appear, in the evolution equation for $\mathbf{u}^{(s)}$ the terms depending on η are lacking. This is the main reason that $\mathbf{u}^{(n)}$ shows a high attenuation (see (3.3.67)), in contrast to $\mathbf{u}^{(s)}$. In (7.3.35) and (7.3.36), the latter two equations (3.4.87) will be generalized to incorporate the effects of quantum turbulence.

Furthermore, if we write $\mathbf{v}=\mathbf{u}^{(n)}-\dfrac{\rho^{(s)}}{\rho}(\mathbf{u}^{(n)}-\mathbf{u}^{(s)})$ or $\mathbf{v}=\mathbf{u}^{(s)}+\dfrac{\rho^{(n)}}{\rho}(\mathbf{u}^{(n)}-\mathbf{u}^{(s)})$ Eqs. (3.4.89) and (3.4.90) become

$$\begin{aligned}\frac{\partial}{\partial t}\mathbf{u}^{(n)}+(\mathbf{u}^{(n)}\cdot\nabla)\mathbf{u}^{(n)}=&-\frac{1}{\rho}\nabla p^{(n)}-\frac{\rho^{(s)}}{\rho^{(n)}}(s-s^{(s)})\nabla T+\frac{\lambda_0}{\rho^{(n)}}\nabla(\nabla\cdot\mathbf{u}^{(n)})+\\&+\frac{\eta}{\rho^{(n)}}\nabla^2\mathbf{u}^{(n)}+\frac{1}{\tau_1}\frac{\rho^{(s)}}{\rho}(\mathbf{u}^{(s)}-\mathbf{u}^{(n)}),\end{aligned} \tag{3.4.91}$$

$$\frac{\partial}{\partial t}\mathbf{u}^{(s)}+(\mathbf{u}^{(s)}\cdot\nabla)\mathbf{u}^{(s)}=-\frac{1}{\rho}\nabla p^{(s)}+(s-s^{(s)})\nabla T-\frac{1}{\tau_1}\frac{\rho^{(n)}}{\rho}(\mathbf{u}^{(s)}-\mathbf{u}^{(n)}), \tag{3.4.92}$$

with the pressure terms $p^{(n)}$ and $p^{(s)}$ being

$$\nabla p^{(n)} = \nabla p - \rho^{(s)} \left((\mathbf{u}^{(n)} - \mathbf{u}^{(s)}) \cdot \nabla \right) \mathbf{u}^{(n)},$$
$$\nabla p^{(s)} = \nabla p + \rho^{(n)} \left((\mathbf{u}^{(n)} - \mathbf{u}^{(s)}) \cdot \nabla \right) \mathbf{u}^{(s)},$$

instead of the ones in the two-fluid model, namely: $\nabla p_n = \nabla p + (\rho_s/2)\nabla V_{ns}^2$, $\nabla p_s = \nabla p - (\rho_n/2)\nabla V_{ns}^2$ (see Eqs. (7.3.39)–(7.3.40)).

3.5 From Hydrodynamic to Ballistic Heat Transport: Slip Heat Flow

In Sects. 3.2 and 3.3, we have emphasized the inertial effects of heat transport (second sound) and the hydrodynamic effects (nonlocal terms leading to size and form dependence of the effective thermal conductivity along channels). Still another situation, the so-called ballistic heat transport, must be considered when the mean free path of phonons and rotons constituting the normal fluid becomes comparable to the radius of the channel (this happens at low temperature and in narrow channels). In this case, the two-fluid model breaks down, because the thermal resistance is no longer related to bulk properties (viscosity) but mainly to the collisions with the walls (ballistic effects).

Below 1 K (for channels with diameter of the order of 10^{-3}m) the excitations (rotons and phonons) cannot be considered as a normal viscous fluid, but as a rarefied gas because their mean free path is of the same order of the channel diameter. Indeed, the phonon mean free path ℓ increases when temperature is lowered (for low temperatures it behaves as $T^{-4.3}$ [18]), in such a way that for sufficiently low temperatures, it becomes comparable to (or higher than) the radius of the pipe (this happens below some 0.7 K for R of the order of 0.5 mm). In this case, the predominant collisions are not the phonon-phonon ones, as in viscous effects or in usual thermal conduction, but the phonon-wall collisions. In this case, it is experimentally found that the effective thermal conductivity K_{eff} relating $\dot{Q}$ and ∇T is not proportional to R^2, as in (3.2.23), but to R [18, 20]. This may be relevant for the design of microscale and nanoscale cryogenic circuits.

To describe the transition from the Landau regime to the ballistic regime, we consider the slip heat flow q_w along the walls appearing in Eq. (3.2.28). In (3.2.29)–(3.2.34), we have set $q_w = 0$, because of the viscous character of the normal component. In the ballistic regime and in the transition between the two regimes, instead q_w has to be taken into account. In rarefied phonon gas theory, q_w is given as

$$q_w = -C\ell \left(\frac{\partial q}{\partial r} \right)_{wall}, \tag{3.5.93}$$

with ℓ being the mean free path of the heat carriers, and C a positive numerical parameter which in kinetic theory of rarefied gases may be identified as $C = (2 - f)/f$ where f is the fraction of reflective phonon-wall collision [18, 20]. Since (3.5.93) is of the order of $(\ell/R)\bar{q}$, with $\bar{q}$ the average heat flux across the channel, q_w will be negligible for $\ell/R \ll 1$, as in Sects. 2.8 and 3.2.

When q_w is not negligible, but given by Eq. (3.5.93), and $q(r)$ is given from Eq. (3.2.28), and keeping in mind the counterflow condition, one has [25, 27]

$$\dot{Q}l = \int_{T_1}^{T_2} \pi R^2 \frac{R^2 S^2 T}{8\eta} \left(1 + 4C\frac{\ell}{R}\right) dT = \frac{\pi R^4 S^2 T}{8\eta} \left(1 + 4C\frac{\ell}{R}\right) \Delta T, \tag{3.5.94}$$

which yields the following effective thermal conductivity:

$$K_{\text{eff}} = \frac{R^2 S^2 T}{8\eta} \left(1 + 4C\frac{\ell}{R}\right). \tag{3.5.95}$$

The term in C was lacking in (3.2.30), where the normal component is considered as a viscous fluid with $\ell \ll R$. Comparison of (3.5.95) with (3.2.31) shows the transition from $K_{\text{eff}} \sim R^2$ to $K_{\text{eff}} \sim R$ for decreasing R or for decreasing T (which implies an increase of ℓ).

3.6 Nonlinear Relations Between Mechanical Stress and Heat Flux

In the linear non-dissipative approximation, the two-fluid model and the one-fluid extended model for superfluid helium yield very similar results to each other. In contrast, they differ in some nonlinear aspects, and therefore it is at this level that the models should be compared to decide which is better to describe some particular nonlinear phenomena.

As an illustration, we comment on two kinds of nonlinear aspects. One of them refers to the choice of an independent vectorial internal variable $\mathbf{m}$ instead of selecting directly $\mathbf{q}$, as in the model proposed in this chapter. Such internal variable will be proportional to $\mathbf{q}$, but the coefficient of proportionality will be in general a function of T and ρ, namely, $\mathbf{m} = \alpha(T, \rho)\mathbf{q}$, so that when one differentiates $\mathbf{m}$ the results will be different than those obtained by differentiating directly $\mathbf{q}$. Therefore one may ask what kinds of differences would arise $\mathbf{q}$.

The second illustration is a nonlinear relation between the stress $\mathbf{P}$ and the heat flux $\mathbf{q}$. This link was discovered by Kapitza in 1941 [40] by observing that a heat flux was able to produce a force on a surface, and to deviate and orient it and it was put into a quantitative form by Hall [41]. The concrete form was seen to be

$$\mathbf{P} = p\mathbf{U} + \frac{\rho_n}{\rho_s} \frac{1}{\rho T^2 s^2} \langle \mathbf{q}\mathbf{q} \rangle. \tag{3.6.96}$$

In the two-fluid model, such a term, referred to the relative velocity $\mathbf{v}_n - \mathbf{v}_s$, arises in a natural way, as a contribution of the relative motions of the constituents to the stress tensor, which is a rather common feature in the theory of mixtures.

Thus, in principle, one could conjecture that adding a term in $\langle \mathbf{qq} \rangle$ into expression (3.1.15) would be sufficient to modellize this effect, i.e. one could write

$$\langle \mathbf{P}_v \rangle = -2\eta \langle \nabla \mathbf{v} \rangle + 2\beta T \eta \langle \nabla \mathbf{q} \rangle + a \langle \mathbf{qq} \rangle \tag{3.6.97}$$

with a a temperature-dependent coefficient.

However, from a basic thermodynamic perspective, namely, from the requirement of consistency of the constitutive equations with the second principle of thermodynamics, this is not so easy, because when the technique of Lagrange multipliers used in rational thermodynamics is used, it follows that the coefficient a must be zero in order to avoid possible violations of the positive character of the entropy production [42].

This raises a subtle question about how to reach rigorous compatibility between (3.6.96) and the second law. To achieve this, one may use as variables some quantities $\mathbf{M}^i$, with i an integer label 0, 1, 2, 3, etc., such that $\mathbf{M}^i$ is the ith moment of the phase function (or of the probability distribution function) in terms of some variables (for instance, for an ideal gas one could take the moments of the distribution function with respect to the particle velocity). The quantities $\mathbf{M}^i$ are completely symmetric tensors of order i. In particular, for ideal gases, $\mathbf{M}^0 = \rho$, $\mathbf{M}^1 = \rho \mathbf{v}$.

The equations for $\mathbf{M}^i$ are assumed to have the form of a hierarchy of balance equations:

$$\frac{\partial \mathbf{M}^i}{\partial t} + \nabla \cdot \mathbf{F}^i = \mathbf{P}^i. \tag{3.6.98}$$

Here $\mathbf{F}^i = \mathbf{M}^i \mathbf{v} + \mathbf{G}^i$ is the flux of $\mathbf{M}^i$ (and has tensorial order $i + 1$), $\mathbf{G}^i$ is the reduced flux, and $\mathbf{P}^i$ is the production of $\mathbf{M}^i$. The quantities $\mathbf{M}^i$, $\mathbf{G}^i$, and $\mathbf{P}^i$ may be decomposed into intrinsic and convective parts. Denoting $\hat{\mathbf{M}}^i$, $\hat{\mathbf{G}}^i$, and $\hat{\mathbf{P}}^i$ the respective intrinsic parts of these quantities, it follows that in the kinetic theory of gases $\hat{\mathbf{M}}^0 = \rho$ is the density; $\mathbf{M}^1 = \rho \mathbf{v}$, with $\mathbf{v}$ the velocity; $\hat{\mathbf{F}}^1$ is the momentum flux; $\hat{\mathbf{G}^{(2)}}$ is the stress tensor; $Tr\left(\mathbf{M}^{(2)}\right) = 2E$, with E the internal energy; and $Tr'\left(\mathbf{F}^{(2)}\right)$ (where we denote by Tr' the trace of the corresponding tensor with respect to the two first components) is the energy density flux, whereas $Tr' \hat{\mathbf{G}^{(2)}} = 2\mathbf{q}$; and so on.

In view of their corresponding meaning, the set of Eq. (3.6.98) for $\mathbf{M}^{(0)}$, $\mathbf{M}^{(1)}$, $tr\left(\mathbf{M}^{(2)}\right)$, and $tr'\left(\mathbf{M}^{(3)}\right)$, i.e. for ρ, $\rho \mathbf{v}$, E, and $\mathbf{q}$, may be written as

$$\dot{\rho} + \rho \nabla \cdot \mathbf{v} = 0, \tag{3.6.99}$$

$$\rho \dot{\mathbf{v}} + \nabla \cdot \mathbf{P} = 0, \tag{3.6.100}$$

$$\dot{E} + E \nabla \cdot \mathbf{v} + \nabla \cdot \mathbf{q} + \mathbf{P} : \nabla \mathbf{v} = 0, \tag{3.6.101}$$

$$\dot{\mathbf{m}} + \mathbf{m} \nabla \cdot \mathbf{v} + \mathbf{q} \nabla \cdot \mathbf{v} + \hat{\mathbf{G}}^{(3)} : \nabla \mathbf{v} + \nabla \cdot \mathbf{P}_q - \frac{3}{2\rho} \langle \mathbf{M}^{(2)} \cdot \nabla \mathbf{P} \rangle = \mathbf{Q}^{(3)}. \tag{3.6.102}$$

The term $\langle \mathbf{M}^{(2)} \cdot \nabla \mathbf{P} \rangle$ in Eq. (3.6.102) may be rewritten as $\omega \nabla \cdot \mathbf{P}$, with ω a parameter related to $\mathbf{M}^{(2)}$ which will be specified later.

The several quantities $\mathbf{m}$, $\mathbf{P}$, $\mathbf{P_q}$, $\mathbf{M}^{(2)}$ may be expressed as

$$\mathbf{m} = \alpha \mathbf{q}, \tag{3.6.103}$$

$$\mathbf{P} = p\mathbf{U} + a\langle \mathbf{qq} \rangle, \tag{3.6.104}$$

$$\mathbf{P_q} = \beta \mathbf{U} + \gamma \langle \mathbf{qq} \rangle, \tag{3.6.105}$$

$$\mathbf{M}^{(2)} = \frac{2}{3} E\mathbf{U} + \delta \langle \mathbf{qq} \rangle. \tag{3.6.106}$$

with a, γ, β, and δ numerical coefficients and the components of $\hat{\mathbf{G}}^{(3)}$ are given by

$$\hat{\mathbf{G}}^{(3)}_{ijk} = \frac{2}{3} q_k \delta_{ij} + \nu \delta_{k<i} q_{j>} + \chi q_k q_{<i} q_{j>}. \tag{3.6.107}$$

Here, all the coefficients, α, β, ν, δ, ν, and χ, may be functions of T and ρ.

One of the main consequences of the application of the Lagrange multipliers method is, after long calculations, the following relation between coefficient a in (3.6.97) and coefficients of (3.6.107) [42]:

$$a = -\frac{1}{6} \left(10 + \nu + 4\chi q^2\right) \theta. \tag{3.6.108}$$

All the terms in this expression for a are related to $\hat{\mathbf{G}}^{(3)}$: the coefficient 10 comes from the first term in $\hat{\mathbf{G}}^{(3)}$, and the coefficients ν and χ come from the second and third terms in $\hat{\mathbf{G}}^{(3)}$ as given by (3.6.107). Thus, in a thermodynamic analysis based directly on $\mathbf{P}$, but not including $\mathbf{G}^{(3)}$, the coefficient a in (3.6.104) turns to be zero, but it may be different from zero provided $\mathbf{G}^{(3)}$ is taken into account.

To carry out a rigorous thermodynamic analysis, the consistent choice of variables is relevant. Indeed, since the effect we want to describe (namely, relation (3.6.96)) is related to the heat flux, which is a third-order moment of the distribution function, one needs incorporating variables up to the third order (as G^3) in the formalism, as done in (3.6.99)–(3.6.102). If one stops at second order (recall that $\mathbb{P}$ is of second order in kinetic theory) some relevant information will be lost, as, for instance, the non-zero value of the coefficient a.

The entropy and the entropy flux are found to be [42]

$$ds = \theta^{-1} du - \mu \theta^{-1} dr + \lambda \mathbf{q} \cdot d(\alpha \mathbf{q}), \tag{3.6.109}$$

$$\mathbf{J}^s = \left(\theta^{-1} + \lambda(\gamma - a\omega)\mathbf{q}^2\right) \mathbf{q}, \tag{3.6.110}$$

with $\omega = \frac{1}{3\rho} \left(5E + 2\delta q^2\right)$ which are terms related to $\mathbf{M}^{(2)}$ and which also appear in the coupling $\mathbf{M}^{(2)} \cdot \nabla \mathbf{P}$ which may be rewritten as $\nabla \mathbf{P}$.

Thus, in this formalism the coefficient a of the nonlinear relation between stress and heat flux is compatible with thermodynamics, but this requires that not only $\mathbf{P}$

contains a term in $\langle \mathbf{qq} \rangle$ but also $\mathbf{P}_q$ as in (3.6.105), and that the terms containing the nonlinear coupling of $\mathbf{G}^{(3)}$ with $\nabla \mathbf{v}$ in (3.6.99)–(3.6.102) and of $\mathbf{M}^{(2)}$ and $\nabla \mathbf{P}$ are considered. Furthermore, the coefficient a in (3.6.104) has a contribution to the entropy flux (3.6.110).

On the other side, the choice of a variable $\mathbf{m} = \alpha \mathbf{q}$ instead of $\mathbf{q}$ implies that instead of $\alpha d\mathbf{q}/dt$ a term in $(d(\alpha)/dt)\mathbf{q}$ should also be taken into account which is not present in the equation obtained starting from the heat flux. In the future, one should carefully study the predictions of this term and find some physical situation in which the consequences of such a term could be tested.

The analysis in this section shows that to have a non-zero value of the coefficient a in the pressure tensor (3.6.97) consistent with the second law, nonlinear effects must be considered, as it is the case with the term $\hat{\mathbf{G}}^{(3)} : \nabla \mathbf{v}$, to which the term in a is related through (3.6.108). This is only an illustration of the relevance of nonlinear terms, not only in their dynamical aspect, which establishes relations between diverse different effects, but also in thermodynamic aspects.

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Chapter 4
Superfluid Hydrodynamics in Rotating Systems: Quantized Vortices



Experimental studies on rotating Helium II led to the discovery of quantized vortices, namely, vortex lines whose circulation is $\kappa = \frac{h}{m_4}$, where h is Planck's constant, and m_4 the mass of one helium atom ($\kappa \simeq 9.97\ 10^{-4}\,\mathrm{cm}^2/\mathrm{s}$) [1–5]. In this chapter, we consider rotating helium, and in Chap. 6 we will consider combined rotation and heat transport, two sources of quantum vorticity interacting in a non-additive way. Quantized vortices also appear in turbulent superfluid helium, which will be explored in more detail in Chap. 5 [4, 5], and in superconductors. They were predicted for the first time by Onsager in 1949, in connection with rotating superfluid helium, interpreting them as topological defects. Rotating superfluids have not only a fundamental interest, but are also found in rotating neutron stars, where a fraction of the system is in superfluid form.

According to the two-fluid model [6, 7], the superfluid component cannot participate to a rigid rotation, owing to its irrotationality, namely, the "superfluid velocity field" is given by the gradient of the phase of the wave function (see (8.1.9)) and therefore it is curl-free (i.e. $\nabla \times \mathbf{v}_s = 0$). Consequently, owing to the temperature dependence of the normal component fraction, different forms of the free surface of the rotating fluid should be observed at different temperatures, but no such dependence is observed. Feynman [8] gave an explanation of the rigid rotation of Helium II in which following the suggestion of the quantization of circulation by Onsager [9], he supposed that the superfluid component, although irrotational at the microscopic level, creates quantized vortices at an intermediate level, being mathematically singular lines of quantum wave function of the system, as it will be discussed in Chap. 9, yielding non-zero value for the curl of the macroscopic velocity of the superfluid component around such singular lines. The condition that the integral of the phase around the singular line should be 2π or an integer multiple of 2π leads to the quantized character of circulation with the quantum $\kappa = h/m_n$.

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4.1 Vortices in Rotating Helium II

If superfluid helium rotates with angular velocity Ω around the z-axis, in accord with classical hydrodynamics, it should be expected that its free surface assumes a concave parabolic profile dependent only on the fraction of the fluid able to rotate, namely, the normal fraction. Thus, its surface is expected to be of the form:

$$z(r) = \frac{\Omega^2 r^2}{2g}\frac{\rho_n}{\rho}, \tag{4.1.1}$$

where r denotes the distance from the vertical z-axis, $z(r)$ is the vertical coordinate of the surface of the rotating helium, and g is the gravitational acceleration. However, experiments (carried out for the first time by Osborne in 1950) show that

$$z(r) = \frac{\Omega^2 r^2}{2g}, \tag{4.1.2}$$

i.e. both the normal and the superfluid components are rotating at the same rate.

This contradiction with the expectation that the superfluid fraction should not rotate can be explained [8, 9] assuming that in the liquid a regular array of vortex lines is present, with the quantum of circulation κ and a number of vortex lines per unit area given by $2\Omega/\kappa$ [10, 11]. There are approximately 1000 vortex lines in a container of radius $R = 1\,\text{cm}$, rotating at angular velocity Ω of about 1 rounds per minute.

Note that κ has dimension of length^2/time, which are the dimensions of diffusion (the diffusion coefficient D and the kinematic viscosity ν, for instance). As will be seen below, this makes that in some classical hydrodynamic expressions in which ν appears, it is replaced by κ in superfluid dynamics. This indicates that κ has a dynamical relevance and not only a steady-state meaning.

In 1950, it was discovered that turbulence in Helium II (discussed in Chaps. 5–7) consists of a disordered tangle of quantized vortex lines. Such lines increase the resistance of the system to heat flow and to barycentric motion because of the mutual friction of vortices and the normal fluid and it is very relevant in practical situations. In fact, if the rotation angular velocity becomes too high, the regular array of parallel vortex lines may become a polarized tangle of disordered vortices, as commented in Sect. 5.4 (Fig. 4.1).

From a microscopical point of view, the mutual friction results from the collision of the quasi-particles (phonons and rotons) with vortex lines [11–13]. The collision cross section is a function of the direction of the quasi-particle velocity relative to the vortex line: it is a maximum when the quasi-particle is travelling perpendicular to this line and a minimum (in fact zero) when it moves parallel to the line. The microscopic mechanism is the same in rotating Helium II as in counterflow turbulence. In both cases, the set of vortex lines is described by introducing a scalar quantity L, the average vortex line length per unit volume (briefly called *vortex line density* and

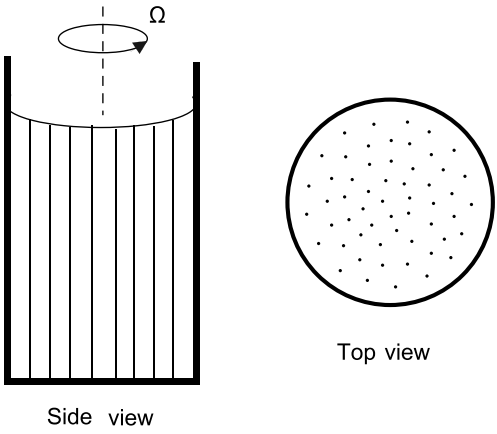


Fig. 4.1 Osborne experiment on rotating superfluid helium; the lines (left figure) and the dots (right figure) correspond to straight, vertical quantized vortices in a cylinder rotating with angular velocity Ω

whose dimensions are $(length)^{-2}$). In rotation, the structure of the vortex lines is an ordered array of lines parallel to the rotation axis, of areal density L_R proportional to the angular velocity Ω of the sample; in this case, the line density L equals the areal density L_R, and is given by

$$L = L_R = \frac{2\Omega}{\kappa}. \tag{4.1.3}$$

This result comes from equating the global circulation $\Omega R\ 2\pi R = 2\pi\Omega R^2$, to the sum of the circulation of the several vortices, $L\kappa\pi R^2$. Equation (4.1.3) is known as Feynman rule and it is valid provided that Ω exceeds a small critical value, of the order of

$$\Omega_{critic} = \frac{\kappa}{2\pi R^2} \ln\left(\frac{R}{a_0}\right), \tag{4.1.4}$$

where a_0 is the vortex core radius, of the order of 10^{-10} m in Helium II. Relation (4.1.4) is obtained by equating the circulation due to Ω (i.e. $2\pi\Omega R^2$) to κ, which would be the total vortex circulation if there were only a single vortex. The critical value of Ω is small, for $R = 1$, Ω_{critic} is of the order of 10^{-4} s^{-1}, so that it may be safely neglected in most cases. The vortex energy per unit length is given by the Feynmann formula:

$$\frac{E}{l} = \pi\rho_s \left(\frac{h^2}{m^2}\right) \ln\left(\frac{b}{a_0}\right), \tag{4.1.5}$$

with b the effective vortex radius, and it is of the order of 10^{-12} Joules/m.

4.2 Second Sound and Vortices in Rotating Helium

Another crucial experiment was performed by Hall and Vinen [14] about propagation of second sound in rotating systems. A resonant cavity is placed inside a vessel containing Helium II, and the whole setting rotates at constant angular velocity Ω. When the second sound propagates at right angles with respect to the rotation axis, it suffers an extra attenuation proportional to the angular velocity, compared to a non-rotating vessel. In contrast, no extra attenuation is found when the propagation direction is parallel to the axis of rotation. This large increase of attenuation when the liquid is rotated can be explained by the *mutual friction* between phonons and rotons and the quantized vortex filaments.

The most used method to detect the presence of vortices in liquid Helium II is the measurement of the attenuation of the second sound. The field equations for the analysis of the wave propagation in rotating helium are the system (3.1.16)–(3.1.19) used to study laminar flows in an inertial frame, plus the non-inertial forces, due to rotation, plus a production term $\sigma^{\mathbf{q}}$ in the heat flux Eq. (3.1.19) of the form:

$$\sigma^{\mathbf{q}} = -\frac{1}{\tau_1}\mathbf{q} + B_{HV}\frac{\Omega}{|\Omega|} \times \Omega \times \mathbf{q} + B'_{HV}\,\Omega \times \mathbf{q}, \tag{4.2.6}$$

where B_{HV} and B'_{HV} are the so-called Hall–Vinen coefficients that depend on temperature and that have been directly measured [14]. The first one determines the dissipation in the absence of rotation, the second one is coupled to a frequency shift of circularly polarized modes and to a friction resistance to the propagation of a perpendicular to the rotation axis, and the third one describes a lateral deviation of the heat flux perpendicular to the rotation axis [15]. When $\Omega = 0$ (4.2.6) reduces to the production term (3.1.22). Here we assume that q/τ_1 may be neglected. A further extension of (4.2.6) with a small attenuation B''_{HV} parallel to the rotating axis was studied in [16]. Some indicative values of B_{HV} and B'_{HV} are usually given through the dimensionless coefficients α and α', respectively, related to B_{HV} and B'_{HV} as $B_{HV} = (\rho/2\rho_n)\alpha$, $B'_{HV} = (\rho/2\rho_n)\alpha'$, given in Table 4.1.

Table 4.1 Values of the dimensionless coefficients α and α' in Eq. (4.2.6) versus temperature. Data are reported from [18]

T (K)	α	α'	B_{HV}	B'_{HV}
1.30	0.034	0.0138	1.526	0.616
1.50	0.072	0.0177	1.296	0.317
1.60	0.097	0.0161	1.193	0.198
1.70	0.126	0.0123	1.1	0.107
1.90	0.206	0.0083	0.980	0.040
2.17	1.577	−0.6358	3.154	−1.272

From Eqs. (3.1.16)–(3.1.19) with (4.2.6) for $\sigma^{\mathbf{q}}$, the wave propagation parallel and orthogonal to the rotation axis was analysed [17]. In the first case, since $\Omega \times \mathbf{q} = 0$, the second and third terms of (4.2.6) vanish and, neglecting the thermal dilatation of Helium II, the following dispersion relation is obtained:

$$(\omega^2 - k^2 V_1^2)(\omega^2 - k^2 V_2^2) = 0, \tag{4.2.7}$$

with ω the real frequency and $k = k_r + ik_s$ the complex wave number, and V_1 and V_2 the propagation speed of first and second sound, as calculated in (3.3.55) and (3.3.56). It is seen that the propagation of the first and second waves parallel to the rotation axis is not influenced by the rotation.

In contrast, when the direction of propagation of the waves is orthogonal to the rotation axis, the second term in (4.2.6) yields an additional resistance of the form $-B_{HV}|\Omega|\mathbf{q}$, and the following dispersion relation is obtained:

$$(\omega^2 - k^2 V_1^2)(\omega^2 + i\omega|\Omega|B_{HV} - k^2 V_2^2) = 0. \tag{4.2.8}$$

The first sound wave (first parenthesis in (4.2.8)) is not influenced by the rotation, but the second sound (temperature wave, second parenthesis) experiences an attenuation. Explicitly, it is found, up to the first order in Ω,

$$u_{\pm}^{(2)} \simeq \pm V_2, \qquad k_s^{(2)} = \frac{B_{HV}|\Omega|}{2V_2}. \tag{4.2.9}$$

Thus the velocity of the second sound is not modified, but the attenuation is proportional to the angular velocity, in agreement with experimental data. In fact, since the vortex length density is $2\Omega/\kappa$, the attenuation factor (4.2.9) may be rewritten as

$$k_s^{(2)} = \frac{B_{HV}}{4V_2}\kappa L. \tag{4.2.10}$$

This expression has practical interest, as it allows measuring L from experimental results for k_s, because B_{HV} and V_2 are known.

4.3 Heat Transport Between Two Rotating Cylinders

Another situation of interest is considering two concentric cylinders of radii R_1 and R_2 ($> R_1$), rotating with angular velocities Ω_1 and Ω_2, respectively, and at temperatures T_1 and T_2 [19]. Assume, for instance, that the internal cylinder 1 is at $T_1 > T_2$, so that heat is flowing from 1 to 2 and that the heat flux is small enough so that no turbulence is produced by it, so that L is exclusively due to the relative rotation. In this case, L may be obtained by equating the total circulation between both cylinders, given by

$\Omega_2 R_2\ 2\pi R_2 - \Omega_1 R_1\ 2\pi R_1 = 2\pi\ \left(\Omega_2 R_2^2 - \Omega_1 R_1^2\right)$, to the sum of the circulation of the several vortices, given by $L\kappa\pi\left(R_2^2 - R_1^2\right)$, so that

$$L = \frac{2}{\kappa}\frac{\Omega_2 R_2^2 - \Omega_1 R_1^2}{R_2^2 - R_1^2}. \tag{4.3.11}$$

Note that L may be positive, denoting anti-clockwise rotation of the vortices, or negative, corresponding to clockwise rotation of vortices, with Ω_i taken positive for anti-clockwise.

In this case, Eq. (3.1.19) for the heat flux, using the production (4.2.6) and neglecting the nonlocal term corresponding to the flux $\mathbf{P_q}$, is

$$\tau_1 \frac{\partial \mathbf{q}}{\partial t} + \lambda_1 \nabla T = -\mathbf{q} - \tau_1 B_{HV} \frac{\Omega}{|\Omega|} \times \Omega \times \mathbf{q} + \tau_1 B'_{HV} \Omega \times \mathbf{q}, \tag{4.3.12}$$

where Ω is the local value of the angular velocity. Since $\mathbf{q}$ is orthogonal to Ω, by the symmetry of the situation being considered, this may be rewritten as

$$\tau_1 \frac{\partial \mathbf{q}}{\partial t} + \lambda_1 \nabla T = -\mathbf{q} + \tau_1 B_{HV} |\Omega| \mathbf{q} + \tau_1 B'_{HV} \Omega \times \mathbf{q}. \tag{4.3.13}$$

The second term on the right-hand side of (4.3.13) leads to the second sound attenuation, as seen in Sect. 4.2.

The third term on the right-hand side implies a lateral deviation of $\mathbf{q}$ with respect to the radial direction of ∇T, in such a way that $\mathbf{q}$ will have radial and tangential components q_r and q_θ, given by

$$\tau_1 \frac{\partial q_r}{\partial t} + \lambda_1 \frac{\partial T}{\partial r} = -\left[1 + \tau_1 B_{HV} |\Omega|\right] q_r + \tau_1 B'_{HV} |\Omega| q_\theta, \tag{4.3.14}$$

$$\tau_1 \frac{\partial q_\theta}{\partial t} = -\tau_1 B'_{HV} |\Omega| q_r - \left[1 + \tau_1 B_{HV} |\Omega|\right] q_\theta. \tag{4.3.15}$$

In steady state, when this matrix is inverted, one gets

$$\begin{pmatrix} q_r \\ q_\theta \end{pmatrix} = -\frac{\lambda_1}{\Delta} \begin{pmatrix} 1 + \tau_1 B_{HV} |\Omega| & -\tau_1 B'_{HV} |\Omega| \\ \tau_1 B'_{HV} |\Omega| & 1 + \tau_1 B_{HV} |\Omega| \end{pmatrix} \begin{pmatrix} \dfrac{\partial T}{\partial r} \\ \dfrac{\partial T}{\partial \theta} \end{pmatrix} \tag{4.3.16}$$

with $\Delta = [1 + \tau_1 B_{HV} \Omega]^2 + \tau_1^2 B'^2_{HV} \Omega^2$ the determinant of the matrix in (4.3.16). We have written $\partial T/\partial\theta$ in (4.3.16) instead of taking directly 0 for the sake of generality. In Fig. 4.2, the deviation of the heat flux due to the polarized vortices between two cylinders is sketched [19].

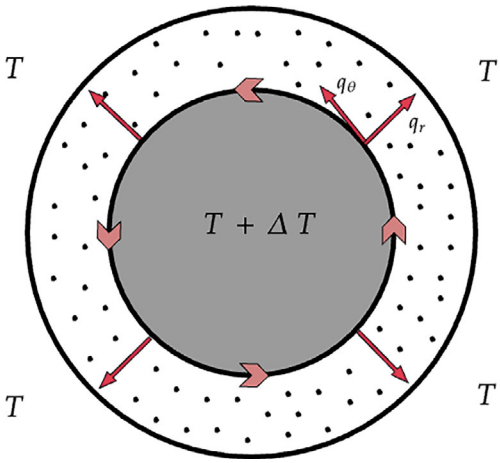


Fig. 4.2 Deviation q_θ of the heat flux with respect to the radial component q_r (arrows) between two concentric cylinders, due to the upward polarized vortices (black dots). The internal cylinder at temperature $T + \Delta T$, rotates anti-clockwise and the external one, at temperature T, is at rest [19]

The antisymmetric matrix relating $\mathbf{q}$ and $-\nabla T$ may be considered as a tensorial antisymmetric effective thermal conductivity. Though Onsager reciprocity relations usually require the thermal conductivity tensor to be symmetrical, the antisymmetry of this tensor is not in contradiction, because the non-diagonal terms are of purely dynamical origin, and vanish for nonrotating systems (i.e. for $\Omega = 0$). This situation is analogous to a Coriolis force acting on the heat carriers along their trajectory between successive collisions in rotating gases.

Since pure rotation produces vortex lines parallel to the rotation axis, the ensemble of vortices may be considered as a completely polarized tangle. Whenever a collection of vortex lines is partially polarized, this deviation effect acting on the heat flux will be present to some degree, the bigger for higher the polarization.

4.4 Rotating Spheres

Neutron stars provide another system where a rotating superfluid and the quantized vortices in it may be of importance. Neutron stars are extremely compact self-gravitating systems which are one of the possible end products of stellar evolution [20]. Though their internal structure is unknown, many physical and astronomical considerations have led to a wide acceptance that there are several concentric distinct layers, rotating at different angular velocities. The surface layer is composed of Fe nuclei and forms a closed-packed solid. The next layer (outer crust) contains heavy nuclei and electrons. The next one, the inner crust, is composed of neutron-rich nuclei, free degenerate neutrons, and degenerate electron gas. The core of the stars is assumed to be a superfluid neutron liquid of huge density, in which rotons plus protons have given electrons plus antineutrinos by means of weak neutral force. The antineutrinos flight away and the whole matter takes the form of neutrons.

The idea that neutrons in the core of such stars may be in superfluid state after pairing of neutrons (fermionic particles) to yield Bosonic component was proposed by Migdal in 1952. He calculated the temperature for the transition to superfluid state, which is of the order of 1 MeV, namely, of $10^{11} - 10^{12}$ K. It is interesting knowing this result because it is usually thought that superfluidity is always related to very low temperatures. Since the temperature of neutron stars at the moment of their formation is less than 10^9 K, which is considerably smaller than the critical temperature, it is possible that neutrons—after suitable pairing—may be in a superfluid state. In 1964, Ginzburg and Kirzhnits suggested, just after the discovery of quantized vortices in rotating Helium II, that in rapidly rotating neutron stars quantized vortex lines could be formed with a quantum of circulation $h/2m_n$, with m_n the neutron mass.

In view to a better understanding of these topics, in 1980, Tsakadze and Tsakadze [21] performed experiments with small glass spheres between 1.5 and 3 mm of radius, full of superfluid helium and accelerated electromagnetically. In these experiments, they studied with particular detail the relaxational process of angular velocity after the rotational acceleration of the sphere.

They observed, among other results, that superfluid may carry out metastable rotations with a non-equilibrium number of vortices. If the number of vortices decreases, their angular momentum may be communicated to the sphere, which may accelerate for a short time.

After fast acceleration of the sphere they observed a long process to establish equilibrium during which there is a redistribution of angular momentum between the vessel and the superfluid. An exponential decrease of the angular velocity of the vessel is observed.

The experiments paid special attention to the dependence of the decay time τ of Ω on temperature, radius R, initial angular velocity Ω_0, and initial jump of the angular velocity $\Delta\Omega_0$. Concerning the variation of τ on temperature it was found that at the lambda temperature 2.17 K there is a jump in τ_0. Such an increase corresponds to a decrease of the interaction between the vessel and the liquid, corresponding to a decrease in the normal component.

Then, it was investigated the dependence of the decay time τ_0 in terms of the radius of the vessel R, the temperature (through the ratio ρ_n/ρ), the initial angular velocity Ω_0, and the initial jump in the angular velocity $\Delta\Omega_0$. The result may be fit as

$$\tau_0 = A\left(\frac{R^2}{\kappa}\right)^{\beta} \Omega_0^{\beta-1} \left(\frac{\rho_n}{\rho}\right)^{-\alpha} \ln\left[1 + C'\Delta\Omega_0\right]. \tag{4.4.17}$$

The method of least squares led to $A = 1.0 \pm 0.1$, $\beta = 0.40 \pm 0.05$, $\alpha = 0.25 \pm 0.01$, and $C' = 5.0 \pm 0.2$.

In a classical theory, one would have expected τ_0 to be proportional to the Eckmann time $(R^2/\nu\Omega_0)^{1/2}$. This may be obtained from expression (4.4.17), replacing quantum of circulation κ, by the kinematic viscosity ν, and taking $1/2$ for the exponent β and 0 for the exponent α. Applying this expression for τ_0 to the case of pulsars leads to a characteristic value of order of 10^7 s, compatible with astrophysical observations of glitches in neutron stars.

In Sect. 5.5, we will specifically deal with the case of pulsars, after having discussed the dynamics of vortex lines in rotating vessels in the presence of a heat flux. In that case the vortex tangle is disordered, but partially polarized. The interaction between disordered turbulent tangle of vortices and the ordered array of vortex lines in rotating systems cannot be avoided in a realistic analysis of glitches in neutron stars.

4.5 Comparison with Rotating Bose–Einstein Condensates

The study of vortices with quantized circulation in Helium II has been extended to the study of quantized vortices in Bose–Einstein condensates of alkali-atomic gases [2, 22, 23]. To compare the behaviour of vortices in both systems, it is important to recall some of the relevant physical differences: BEC are much more dilute, compressible, and cold than liquid helium; the number of particles is much smaller; and instead of physical containers they are kept in optical or magnetic traps. Two other differences are of interest when considering quantized vortices: the long coherence length in BEC is of the order of the size of the trap, so that the effects of the "walls" are felt everywhere in the system, and the atomic mass of the atoms is much higher than that of Helium II atoms (for instance, the rubidium atoms ^{87}Rb have atomic mass 87, 20 times higher than that of ^{4}He). The latter thing means that the quantum of circulation h/m will be some 20 times smaller in ^{87}Rb BEC than in Helium II; also, the core of the quantized vortices, of the order of the atomic size, will be bigger in Rb BEC than in Helium II. Notwithstanding these differences, some analogies are also present, which underline some features common to both kinds of systems.

The BEC are kept in optical or magnetic traps, mathematically represented by a force potential $V(x, y, z)$ of ellipsoidal form, namely, $V(x, y, z) = m(w_x^2 x^2 + w_y^2 y^2 + w_z^2 z^2)/2$, where w_x, w_y, and w_z are the corresponding oscillatory frequencies in the three spatial directions. Rotation may be incorporated making that the X- and Y-axes rotate with a frequency Ω, fixed by the frequency of variation of the external magnetic field, in magnetically rotating traps (an alternative way is stirring the BEC with a laser beam). Though such a trap has not a well-definite value for the diameter of the container, one may take as an effective diameter d the value given by $1/d^2 = w_\perp \kappa$, with $w_\perp^2 = (w_x^2 + w_y^2)/2$, namely, the average transversal frequency of the trap. Usually the frequencies depend on the direction, so that one needs defining the eccentricity of the trap, ϵ, as $\epsilon = (w_y^2 - w_x^2)/(w_y^2 + w_x^2)$. In contrast, the experiments in Helium II usually take circular containers with zero eccentricity. This suggests enlarging the experiments in Helium II by using containers with elliptical transversal section instead of circular transversal section.

To refer to a specific example, consider the situation studied in [24, 25] for a condensate of $2 \cdot 10^4$ atoms of ^{87}Rb for an oblate geometry of the trap ($w_\perp < w_z$) within a magnetically rotating trap. In contrast to rotating Helium II, for which the transversal density L of vortices per unit area increases with Ω after a given threshold

has been crossed, it is observed in BEC that vortices appear for Ω higher than some minimal threshold Ω_1 but that, instead of increasing with Ω, the number of vortices N per unit transversal area A, N/A, reaches a maximum and decreases again, so that beyond some higher threshold value Ω_2 the vortices disappear. The threshold values Ω_1 and Ω_2 are seen to depend on the eccentricity ϵ, in such a way that increasing ϵ decreases Ω_1 and increases Ω_2, thus enlarging the range of values of Ω for which vortices are present. The decrease of the number of vortices after some maximum is not yet clearly explained, but it could be due to a combination of an increase of repulsion between vortices due to higher values of vortex density, and to the higher effect of centrifugal forces for increasing Ω. The repulsion between vortices in BEC (as well as collective effects of vortices) may be higher than in Helium II because of the longer coherence length, and the centrifugal forces may be higher because of the higher values of Ω in BEC experiments. Furthermore, the walls of containers of Helium II are rigid but this is not so in the traps for BEC.

Indeed, typical values of rotation frequency in experiments with Helium II are less than 10 rad/s, whereas in BEC they may be of the order of 300 rad/s. Indeed, the critical threshold value below which vortices are not found is of the order of κ/d^2. For ^{87}Rb BEC, κ is 22 times smaller than for Helium II, and d may be of the order of 1 cm to 10 cm for helium experiments whereas it is of the order of 10^{-3} to 10^3 cm or smaller for BEC traps. Thus, the centrifugal forces acting on the vortices will be much higher for BEC.

This simple qualitative comparison is useful to put a deeper attention to the values of the physical variables intervening in the problem, and to their different effects. Deeper quantitative comparisons will be of much interest.

4.6 Comparison with Magnetic Flux: Quantization in Superconductors

The quantization of circulation of vortices in Helium II is analogous to magnetic flux quantization in superconductivity [3]. Quantized vortices were applied by Abrikosov in 1947 to describe the magnetic phase diagram of type II superconductors. Whereas in superfluids the quantum vortices are linked to orbital angular momentum related to superfluid rotation, in superconductors they are related to quantized magnetic flux. The core of such vortices is made of normal state with size of the order of the London penetration length, surrounded by superconducting state, with a quantum vortex supercurrent circulating around it.

Instead of the circulation of velocity $\int \mathbf{v} \cdot d\mathbf{l}$ in closed lines, in superconductors the quantized quantity is the circulation of the vector potential $\mathbf{A}$ along closed lines, $\oint \mathbf{A} \cdot d\mathbf{l}$. By using Stokes theorem one has

$$\oint \mathbf{A} \cdot d\mathbf{l} = \int \nabla \times \mathbf{A} \cdot d\mathbf{S} = \int \mathbf{B} \cdot d\mathbf{S} = \Phi, \tag{4.6.18}$$

with B the magnetic field and Φ the magnetic flux. But from the London equation relating the electric current $\mathbf{j}$ to the vector potential $\mathbf{A}$ and to the gradient of the phase ϕ_q of the wave collective function, one may write

$$\oint \mathbf{j} \cdot d\mathbf{l} = \oint \left(-\frac{2e^2}{mc}\mathbf{A} + \frac{e\hbar}{m}\nabla\phi \right) \cdot d\mathbf{l} = 0. \tag{4.6.19}$$

Since $\oint \nabla\phi \cdot d\mathbf{l} = \Delta\phi = 2\pi n$, it follows from (4.6.19) that

$$|\Phi| = n\frac{hc}{2e} \tag{4.6.20}$$

with $\dfrac{hc}{2e}$ the quantum of magnetic flux, a combination of fundamental universal constants h, c, e. Its value is, therefore, the same for any superconductors, in contrast to the quantum of circulation in superfluids, which depends on the mass of the particles. The quantization of magnetic flux was predicted by London in 1948 and observed for the first time in 1961.

Relatively small magnetic fields are expelled from superconductors (Meissner effects) but if the magnetic field is sufficiently strong it may destroy the superconductive state by inducing a phase transition to the normal state [3]. However, in some materials (the so-called type II superconductors), it is energetic favourable to form a lattice of quantum vortices, carrying a quantized magnetic flux (4.6.20). Such vortices may be moved by the flow of electric current, leading to energy dissipation and causing the system to have a non-vanishing electrical resistance despite of being in the superconducting state.

Thus, the quantum description of superfluid, as well as the quantum description of superconductors by means of a collective wave function, leads to a quantization of circulation of velocity around vortex lines, and quantization of magnetic flux across surfaces limited by closed lines around vortex lines in superconductors. Analogies of vortices in superfluids and superconductors are an interesting physical topic, but we focus our attention only on superfluids. Vortex states may also be found in some ferromagnetic or antiferromagnetic materials and are relevant for information technology, because they are used to generate bits for information storage and recognition. The description of such vortices is physically and mathematically more complicated than that of the vortices in superfluids and superconductors.

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Chapter 5
Turbulence in Superfluids: Macroscopic Description and Applications to Heat Transport



In Chap. 4, a description of some salient features of quantized vortices in rotating Helium II has been presented. We have restricted our consideration to pure rotation situations, with ordered rectilinear vortices and vortex line density L independent of time. In the present chapter, we consider turbulent states, in which a heat flux higher than some critical value acts as a source of quantized vortices which distribute in the space in the form of a disordered tangle of lines [1–3]. Counterflow experiments (heat propagation without net mass propagation) are a typical way to produce superfluid turbulence for heat fluxes higher than a critical value [4–7]. In this chapter, we consider that the tangle is spatially homogeneous, in such a way that a single scalar function suffices to describe it, namely, the vortex line density L (namely, the total length of vortex lines per unit volume) and we will pay a special attention to the evolution of L in several physical circumstances. A deeper analysis including inhomogeneous situations will be presented in Chap. 6.

There are essentially two methods to measure L: observations of thermal resistance, and of attenuation of the second sound [8–11]. Indeed, appearance of quantized vortices ($L \neq 0$) produces a steep increase of the thermal resistance and of the attenuation coefficient, as well as several other properties, from which L may be derived as seen in Sect. 4.2. This additional resistance and attenuation is due to a friction force between the flow of excitations and a chaotic tangle of quantized vortex filaments (see Fig. 5.1c).

Here, we consider L as an additional independent variable, with its own evolution equation, and we write dynamical equations for L focusing on transition from laminar flow to turbulent flows, combination of rotation and counterflow, and the influence of the walls. To deal with these situations, we adopt a macroscopic phenomenological perspective. The starting point is an analysis of the rates of vortex formation and of vortex destruction, and their relative balance, by a combination of dimensional analysis, experimental observation, and microscopic arguments. In the simplest situation, such an analysis leads to the well-known Vinen equation, which here is generalized to incorporate the influence of walls and of rotation.

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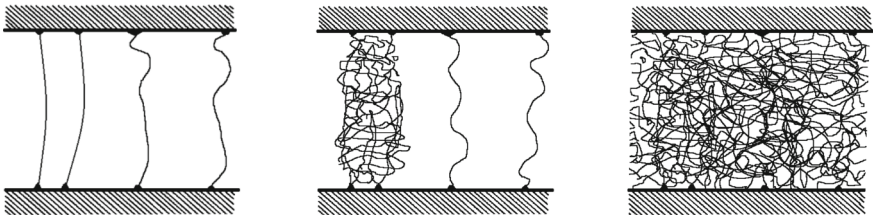


Fig. 5.1 A sketch of the vortices behaviour, following the discussion in Sect. 5.2.2, showing the increasing abundance and complexity of vortices, represented by thin lines. **a** laminar regime ($V_{ns} < V'_{c1}$); **b** turbulent TI regime ($V_{c1} < V_{ns} < V_{c2}$); **c** turbulent TII regime ($V_{ns} > V_{c2}$). Figure from Ref. [28]

5.1 Rates of Vortex Production and Destruction, and Evolution Equation of Vortex Line Density

The evolution of L is described by specifying production and destruction of vortices per unit time. Here we will refer to counterflow situations characterized by heat flux $\mathbf{q}$ or, in alternative notation, by a relative velocity $\mathbf{u}_{ns}$ of normal component with respect to superfluid component, given by

$$\mathbf{u}_{ns} = \frac{\mathbf{q}}{\rho_s T s}, \tag{5.1.1}$$

according to (3.4.85).

For simplicity of notation, and to make a more direct contact with the usual form of the equations in this field, we will use $\mathbf{u}_{ns}$ rather than $\mathbf{q}$, and we also introduce a quantum Reynolds number

$$Rey = \frac{V_{ns} d}{\kappa} = \frac{\rho}{\rho_s} \frac{q}{Ts} \frac{d}{\kappa}, \tag{5.1.2}$$

with V_{ns} the modulus of the spatial average of $\mathbf{u}_{ns}$. Recall that the classical Reynolds number of hydrodynamics is defined as $Rey = \rho V d/\nu$, with ν the kinematic viscosity. The two changes made in the definition (5.1.2) with respect to the classical Reynolds number are: (a) replacing the kinematic viscosity ν by the quantum of circulation κ, which have the same dimensions and (b) replacing the modulus of the barycentric velocity V by the modulus of the thermal velocity V_{ns} (5.1.1). The observed critical values of this Reynolds number depend on the temperature and are 127 at 1.5 K, 112 at 1.6 K, and 96 at 1.7 K (see Table 5.1).

The net time derivative dL/dt is composed of two terms:

$$\frac{dL}{dt} = \left[\frac{dL}{dt}\right]_f - \left[\frac{dL}{dt}\right]_d, \tag{5.1.3}$$

Table 5.1 Critical value of the quantum Reynolds numbers Re_1 and Re_2 for the appearance of TI and TII turbulent regime, respectively, for three different temperatures and wide channels and narrow channels [16, 22]

Wide channels $d \geq 1000\,\mu$ m				Narrow channels $d \leq 100\,\mu$ m			
T	1.5 K	1.6 K	1.7 K	T	1.5 K	1.6 K	1.7 K
Re_1	127	112	96	Re_1	52	46	43
Re_2	226	212	187	Re_2	149	134	125

the first describes the rate of growth of L through vortex production; the second, its rate of decay, because of vortex destruction. In inhomogeneous situations, a transport of vortices should also be considered. One may assume that $[dL/dt]_f$ depends on the quantum of circulation κ, the instantaneous value of L, and the counterflow velocity V_{ns}. Dimensional analysis leads to [6, 12]:

$$\left[\frac{dL}{dt}\right]_f = \kappa L^2\, \Phi_f\left(\frac{V_{ns}}{\kappa L^{1/2}}\right), \tag{5.1.4}$$

where Φ_f is some dimensionless function of its dimensionless argument. The determination of this function is a crucial problem of the phenomenological theory [12, 13]. In the simplest version (5.1.4) is taken as a power of its variable, namely:

$$\left[\frac{dL}{dt}\right]_f = \alpha_n \kappa L^2 \left[\frac{V_{ns}}{\kappa L^{1/2}}\right]^n, \tag{5.1.5}$$

with α_n a dimensionless constant and n a constant exponent.

Vinen, by analogy with the growth of a vortex ring, originally assumed $n = 1$. Another form ($n = 2$) has also been considered [13]. A still more recent proposal is $n = 3$ [14], which together with a diffusive term fits better the results of some numerical simulations for L strongly inhomogeneous in a planar channel geometry.

The form of $[dL/dt]_d$ term, describing the vortex decay, was suggested by Vinen in analogy with classical turbulence as

$$\left[\frac{dL}{dt}\right]_d = -\beta_v \kappa L^2, \tag{5.1.6}$$

with β_v a dimensionless coefficient.

From (5.1.5) with $n = 1$ and (5.1.6), Vinen's equation

$$\frac{dL}{dt} = \alpha_v V_{ns} L^{3/2} - \beta_v \kappa L^2, \tag{5.1.7}$$

follows immediately. From (5.1.5) and (5.1.6), one obtains in general

$$\frac{dL}{dt} = \alpha_n \kappa L^2 \left[\frac{V_{ns}}{\kappa L^{1/2}} \right]^n - \beta_v \kappa L^2, \tag{5.1.8}$$

with n being 1, 2, or 3, or a sum of these different possibilities (fractional exponents have also been considered in [15]). In the steady state, combination of any production term of (5.1.8) with the destruction term (5.1.6) leads to

$$L = \left(\frac{\alpha_i}{\beta_v} \right)^2 \frac{V_{ns}^2}{\kappa^2}. \tag{5.1.9}$$

Therefore, the difference between the proposals for several n is mainly dynamical, and it is through dynamics—or stability, or in inhomogeneous situations—that the relative importance of the three production terms may be decided.

Vinen's Equation (5.1.7)—and the more general form (5.1.8)—is successful in the description of completely developed turbulence, but it cannot describe the full complexity of the different regimes observed experimentally [16, 17]. This limitation may be overcome when the influence of the walls is taken into account as shown below.

5.2 Influence of the Walls on Superfluid Turbulence

The influence of the walls on the vortex dynamics has direct interest when the average separation between neighbour vortices, $\delta \simeq L^{-1/2}$, becomes comparable to the diameter d of the container. These effects are important when vortex line density is small, as in the last phases of vortex decay, or in the transition from laminar to turbulent regimes.

Terms in δ/d are needed when the system is inhomogeneous or when d becomes comparable to δ, in which case the influence of the walls extends deeply into the volume of the system [18–20]. By now, we focus our attention on homogeneous situations for narrow channels or on the late stages of the decay of turbulence [11, 21].

5.2.1 The Several Regimes of Quantum Turbulence

In the 1980s, Tough and coworkers performed many experiments on quantum turbulence of superfluid helium inside channels with different geometries and materials [16, 22–25]. In some experiments where the turbulence is generated by thermal counterflow, a vortex tangle is observed to develop from a low-density state (TI) to a higher density state (TII) that can be associated with the homogeneous

well-developed state. It was found that in both regimes the vortex line density L depends on V_{ns} and d as

$$L_i^{1/2} = \gamma_i \frac{V_{ns}}{\kappa} - \frac{1}{d} b_i, \tag{5.2.10}$$

(with $i = 1$ for I regime and 2 for II regime), with the dimensionless coefficients γ_i and b_i depending on the temperature. The presence of d in (5.2.10) shows that the effects of the size cannot be neglected. For high values of d, the last term in (5.2.10) may be neglected and (5.1.9) is recovered. In some occasions, the transition from laminar to turbulent regimes exhibits metastable laminar regions [16, 17], which also depend on the size of the system.

In this section, we consider a phenomenological extension of (5.1.7) able to describe with a single evolution equation the three stationary regimes observed in counterflow in a cylindrical channel: a laminar regime ($L = 0$) at low V_{ns}, the transition at a critical counterflow velocity V_{c1}^H from the laminar to the turbulent TI regime, a metastability region $V_{c1}^H < V_{ns} < V_{c1}'$, and the turbulent well-developed regime TII at increasing values of V_{ns}.

The transition from $L = 0$ (laminar regime) to $L \neq 0$ (turbulent regime, or with more precision to TI turbulent regime) in cylindrical channels is experimentally found to happen at [16]

$$Rey = \frac{V_{ns} d}{\kappa} \approx 127 \quad (1.5\,\mathrm{K}); \quad Rey = \frac{V_{ns} d}{\kappa} \approx 112 \quad (1.6\,\mathrm{K}); \quad Rey = \frac{V_{ns} d}{\kappa} \approx 96 \quad (1.7\,\mathrm{K}). \tag{5.2.11}$$

Tough and coworkers found that at this point $L^{1/2}$ has a discontinuity of the order $L^{1/2} d \approx 2.5$ for the TI regime, and $L^{1/2} d \approx 10$ for the TII regime [16]. In fact, $L = 0$ may be metastable rather than unstable for some range of V_{ns}. Some values of the critical Reynolds numbers are reported in Table 5.1 for three temperatures, both in wide and in narrow channels and for state TI and state TII. Note that the critical value of the quantum Reynolds number Re_q is smaller for narrow channels.

In [23], the authors extended their studies on the thermal resistance to small square glass tubes for comparison with circular tubes. For these geometries they found the same behaviour (the two turbulent regimes) found in circular tubes, but scaling by the hydraulic diameter of the channels.

5.2.2 *Generalized Evolution Equation for the Vortex Line Density. Production Term*

Vinen's equation (5.1.7) predicts that the critical quantum Reynolds number for the laminar–turbulent transition is zero, i.e. $L \neq 0$ for any value of V_{ns} higher than 0. To obtain a finite threshold for the transition there are several strategies. The first of them is modifying the production term in such a way that

$$\frac{dL}{dt} = \alpha_0 V_{ns} L^{3/2} \left[1 - \omega \frac{L^{-1/2}}{d} \right] - \beta_v \kappa L^2, \tag{5.2.12}$$

with ω a phenomenological positive parameter.

The interpretation of (5.2.12) is that the vortex generation mechanism is inactive within a characteristic distance $L^{-1/2}$ from the wall, but the decay term is left unchanged (see [13, 25–27]).

To better describe in a qualitative way the transition from laminar regime to the turbulent TI regime, one may use, instead of (5.2.12), the slightly more complicated Eq. [28]

$$\frac{dL}{dt} = \alpha_I V_{ns} L^{3/2} \left[1 - \omega \frac{L^{-1/2}}{d} \right] + \alpha_1 \frac{V_{ns}^2}{\kappa} L - \beta \kappa L^2, \tag{5.2.13}$$

being α_I the value of $\alpha_0(V_{ns}, d)$ appearing in (5.2.12) in the laminar and TI regime. This equation is similar to (5.2.12), but with the additional term in V_{ns}^2, corresponding to $n = 2$ in (5.1.5), and whose role will be commented below.

(A) Transition from Laminar to the TI Regime

The steady-state solutions of (5.2.13) are $L_1 = 0$ and

$$L_{2,3} = \frac{1}{2} \frac{\alpha_I V_{ns}}{\beta \kappa} \left[1 \pm \sqrt{1 + \frac{4\beta\kappa}{\alpha_I V_{ns}} \left(\frac{\alpha_1}{\alpha_I} \frac{V_{ns}}{\kappa} - \frac{\omega}{d} \right)} \right]. \tag{5.2.14}$$

The solutions $L_{2,3}$ (corresponding to + and – signs in (5.2.14), respectively) are real only for values of

$$V_{ns} \geq V_{c1} = \frac{4\beta\alpha_I \omega}{\alpha_I^2 + 4\beta\alpha_1} \frac{\kappa}{d}. \tag{5.2.15}$$

In terms of the quantum Reynolds number (5.1.2), this means that turbulence appears for

$$Rey \geq Rey_{c1} = \frac{4\beta\alpha_I \omega}{\alpha_I^2 + 4\beta\alpha_1}. \tag{5.2.16}$$

At the transition, when $Rey = Rey_{c1}$, i.e. for $V_{ns} = V_{c1}$, there is a discontinuity from $L = 0$ to

$$L_{c1}^{1/2} = \frac{2\alpha_I^2 \omega}{\alpha_I^2 + 4\beta\alpha_1} \frac{1}{d}. \tag{5.2.17}$$

The stability of solutions (5.2.14) may be studied from the equation for a perturbation δL of L, given by the differential form of (5.2.13), namely:

$$\frac{d\delta L}{dt} = - \left[2\beta\kappa L - \frac{3}{2} \alpha_I V_{ns} L^{1/2} - \alpha_1 \frac{V_{ns}^2}{\kappa} + \alpha_I \omega \frac{V_{ns}}{d} \right] \delta L, \tag{5.2.18}$$

from which follows that $L_1 = 0$ is stable up to a value Rey' of the quantum Reynolds number given by

$$Rey' = \frac{\alpha_I \omega}{\alpha_1}. \tag{5.2.19}$$

The solution L_2 is stable where it exists, while L_3 is unstable. As it is immediately verified, $Rey^H < Rey'$ (namely, $V_{c1}^H < V_{c1}'$). This means that for $Rey_{c1}^H < Rey < Rey'$ or in other terms, for $V_{c1}^H < V_{ns} < V_{c1}'$, the laminar solution $L_1 = 0$ is metastable. If the term in V_{ns}^2 is omitted in (5.2.13), the laminar regime would be metastable from V_{c1} to infinite. Thus, the observed upper value for metastability (5.2.19) may give the value of the coefficient α_1 if α and ω are known.

(B) Transition from TI to TII Regimes

Equation (5.2.13) describes the transition from laminar to turbulent TI regime, but not the transition from TI regime to the TII regime, taking place at a second critical number Rey_{c2}.

To describe this transition, a dependence of α_0 (the first term in (5.2.12)) on the quantum Reynolds number (5.1.2) may be introduced. Indeed, experimentally it has been seen that α_0, characterizing the slope of the stationary solution for L in terms of V_{ns}, undergoes a step from the value α_I in the TI regime to a value α_{II} in the TII regime. Here we assume

$$\alpha_0(Rey) = \alpha_{c2}\,(1 + c\ \tanh\left[A\,(Rey - Rey_{c2})\right]), \tag{5.2.20}$$

being c and A dimensionless constants such that, for $Rey \ll Rey_{c2}$, i.e. for $V_{ns} \ll V_{c2}$, $\alpha_0 \simeq \alpha_I = \alpha_{c2}(1 - c)$ and for $Rey \gg Rey_{c2}$, i.e. for $V_{ns} \gg V_{c2}$, $\alpha_0 \simeq \alpha_{II} = \alpha_{c2}(1 + c)$, being α_{c2} the value of $\alpha_0(Rey)$ at the second transition. Rey_{c2} denotes the critical quantum Reynolds number for the appearance of TII regime, $2c = \alpha_{II} - \alpha_I$ gives the size of the step of α_0 near V_{c2}, and A characterizes the width of the domain of V_{ns} in which the transition is produced.

A physical motivation for (5.2.20) can be the following: in the laminar regime, when $Rey \ll Rey_{c1}$, in the channel there are some vortex lines (the so-called "remnant vortices" formed when helium was cooled or during previous turbulent flows [29–32]) strongly pinned to the walls. As the counterflow velocity grows, these lines are bent and increase their length, but most of them remain pinned. Helical waves may be propagated along these vortices. With increasing counterflow velocity, the separation between successive coils of these helices reduces. When the counterflow velocity reaches the value $V_{c1} = Rey_{c1}\kappa/d$, small localized arrays of quantized vortices appear. Sometimes, these localized arrays may be generated for Rey numbers beyond Rey_{c1}, and a region of metastability of the laminar regime arises.

In this picture, the TI turbulent regime is inhomogeneous and locally polarized. When the counterflow reaches the critical value Rey_{c2} this state becomes unstable and there is a transition to the homogeneous state TII. This agrees with experimental observations made in [33] and computer simulations [34, 35] which show that in well-developed turbulence the tangle is roughly isotropic.

In the laminar and in the turbulent TI regime, the interaction with the boundary is prevalent, and so the terms dependent on d are dominant, while in the TII regime the terms linked to V_{ns} are dominant. In the transition region, described with the function (5.2.20), $1/d$ contributes to the order, i.e. to the decrease of vortex lines, while V_{ns} contributes to the disorder, i.e. to their increase.

5.2.3 Destruction Term

The corrective contribution in the vortex production term in (5.2.12) is negative because the vortex generation mechanism is inactive within the characteristic distance δ from the wall. However, the walls also play an important role in the vortex decay because of the interaction between a single vortex line and a boundary when their relative distance is less than a characteristic distance Δ_c, proportional to the average intervortex distance δ: $\Delta_c = \lambda_c \delta$.

If Δ is smaller than Δ_c, the wall traps the vortex, but if it is greater than Δ_c, the vortex motion is weakly affected by the wall. Obviously, if the size d of the channel is small, the influence of the walls on the decay processes cannot be neglected. We therefore modify the destruction term (5.1.6), introducing in it a quadratic dependence on $L^{-1/2}/d$, in such a way that Eq. (5.2.13) is generalized as

$$\frac{dL}{dt} = -\beta\kappa L^2 + \left[\alpha_0 V_{ns} - \omega'\beta\frac{\kappa}{d}\right] L^{3/2} + \left[\alpha_1 \frac{V_{ns}^2}{\kappa} - \omega\alpha_0 \frac{V_{ns}}{d} + \omega''\beta\frac{\kappa}{d^2}\right] L, \tag{5.2.21}$$

with $\alpha_0 = \alpha_0(Rey)$ expressed by (5.2.20) (the new terms are those with coefficients ω' and ω''). With this choice of the signs, the destruction term is higher than Vinen's one when the typical separation vortex wall is less than $\Delta_c = \lambda_c d\, \omega'/\omega''$, whereas for higher values it is lower.

It has been shown in Ref. [28] that the stationary solutions of (5.2.21) provide a good description of the experimental results for all the stationary regimes in counterflow superfluid turbulence, with more detail than (5.1.7) and (5.2.13). Putting $y \equiv L^{1/2} d$ the non-zero stationary solution of (5.2.21) is

$$y_I = a_1 Rey - b_1, \tag{5.2.22}$$

in the TI region, with the dimensionless coefficients a_1 and b_1 given by [28]

$$a_1 = \frac{\alpha_I + \sqrt{\alpha_I^2 + 4\beta\alpha_1}}{2\beta}, \qquad b_1 = \frac{\alpha_I\omega + \beta\omega' a_1}{\sqrt{\alpha_I^2 + 4\beta\alpha_1}} \tag{5.2.23}$$

where α_I (and later α_{II}) are the values of $\alpha_0(Rey)$ in (5.2.20) for small and high Rey, respectively.

Table 5.2 Values of the parameters a_1, b_1, a_2, b_2 in (5.2.22) and (5.2.24) and of Rey_{c1} and Rey_{c2} obtained from [16]

T(K)	a_1	b_1	a_2	b_2	Rey_{c1}	Rey_{c2}
1.5	0.076	7.25	0.139	9.99	127.07	219
1.7	0.091	6.92	0.176	8.18	97.74	186

Table 5.3 Values of the parameters ω, ω', and ω'' in (5.2.21) obtained using the data reported in Table 5.2 [28]

T(K)	ω	ω'	ω''
1.5	5.178	3.634	25.1
1.7	3.560	4.259	17.3

For high values of $y = L^{1/2}d$ and $Rey = V_{ns}d/\kappa$, i.e. in the asymptotic developed TII regime, Eq. (5.2.21) may be approximated with its asymptote, obtaining the stable solution

$$y_{II} = a_2 Rey - b_2, \tag{5.2.24}$$

with a_2 and b_2 given by [28]:

$$\alpha_{II} + \sqrt{\alpha_{II}^2 + 4\beta\alpha_1} = 2\beta a_2, \qquad \alpha_{II}\omega + \beta\omega' a_2 = b_2\sqrt{\alpha_{II}^2 + 4\beta\alpha_1} \tag{5.2.25}$$

describing the TII regime.

Relations (5.2.22) and (5.2.24) are usually written as

$$L_I^{1/2} = \gamma_I \frac{V_{ns}}{\kappa} - \frac{1}{d}b_1, \qquad L_{II}^{1/2} = \gamma_{II}\frac{V_{ns}}{\kappa} - \frac{1}{d}b_2, \tag{5.2.26}$$

with $\gamma_I = a_1$ and $\gamma_{II} = a_2$. The coefficients in (5.2.26) have been obtained empirically, so that the values of a_1, a_2, b_1, and b_2 may be derived from experiments. In Table 5.2, their values for $T = 1.5$ K and $T = 1.7$ K are reported, obtained from the experimental data of Martin and Tough [16].

Note that the non-vanishing value of ω' modifies the values of a_i and b_i in (5.2.22) and (5.2.24). The values of ω, ω', and ω'' are reported in Table 5.3.

In Fig. 5.2, the comparison between the prediction of this theoretical model and the experimental data from [16] is reported; with $A = 0.05$ at $T = 1.5$ K and $A = 0.25$ at $T = 1.7$ K in (5.2.20) [28]. Despite the simplifying hypotheses, the agreement between the macroscopic description and experimental observations is good.

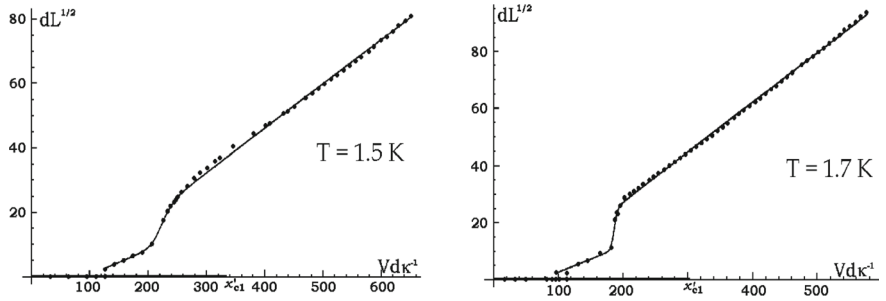


Fig. 5.2 $y = L^{1/2}d$ as function of $Rey = V_{ns}d\kappa^{-1}$ at $T = 1.5$ K and 1.7 K. Data are from Martin and Tough [16]. Lines are determined from Eq. (5.2.20) with $A = 0.05$ at $T = 1.5$ K, and $A = 0.25$ at $T = 1.7$K. Figure from [28]

5.3 Heat Transport in Turbulent Superfluid Helium

Superfluid turbulence implies a very high increase of the thermal resistance of helium, because of the mutual friction between heat flux and vortex lines. Thus, the vortex line density L, as a function of the heat flux and the temperature, plays a relevant role in the description of thermal resistance in turbulent flows.

In Sect. 3.2, we have studied counterflow heat transport in the laminar regime. Since the vortices do strongly contribute to the thermal resistance, the production term σ^q in the evolution Eq. (3.1.19) for $\mathbf{q}$ may be generalized as

$$\sigma^q = -\frac{1}{\tau_1}\mathbf{q} - K_f L\mathbf{q}, \tag{5.3.27}$$

with the friction coefficient between heat flux and vortex lines given by $K_f = (1/3)\kappa B_{HV}$, B_{HV} being the dimensionless Hall–Vinen coefficient [see (4.2.6)]. For $L = 0$ this reduces to (3.1.22). Then (3.2.25) are modified into

$$\nabla p - \eta\nabla^2(\mathbf{v} - \beta T\mathbf{q}) = 0, \tag{5.3.28}$$

$$\nabla T + \beta T^2\nabla p = -\frac{1}{\lambda_1}\mathbf{q} - \frac{K_f L}{\zeta}\mathbf{q}, \tag{5.3.29}$$

Assuming, as in (3.2.27), that λ_1 is very high and q/λ_1 may be neglected, and $\beta = -1/(ST^2)$, the solution of (5.3.28) and (5.3.29) is (3.2.28). The mean value $\bar{q}$ along the pipe of the solution (3.2.28) is then substituted in the equation

$$\Delta T - \frac{1}{S}\Delta p = \frac{K_f l\bar{L}}{\zeta}\bar{q}, \tag{5.3.30}$$

which is Eq. (5.3.29) integrated along the pipe, where l is the length of the pipe. One obtains

$$\Delta T = \frac{8\eta l}{\pi R^4 S^2 T}\dot{Q} + \frac{K_f l}{\pi R^2 \zeta} L \dot{Q}, \tag{5.3.31}$$

where $\dot{Q} = qA$. The first term comes from $\frac{1}{S}\Delta p$ in (5.3.30) and Eqs. (5.3.28) and (5.3.29) and the second term corresponds to friction resistance.

For fully developed turbulence ($\dot{Q}$ high), L is given by $L = \frac{\alpha_v}{\beta_v}\frac{V_{ns}^2}{\kappa^2} \sim \gamma \dot{Q}^2$ (see (5.1.9)), and (5.3.31) becomes

$$\Delta T = \frac{8\eta l}{\pi R^4 S^2 T}\dot{Q} + \gamma \frac{K_f l}{\pi R^2 \zeta}\dot{Q}^3. \tag{5.3.32}$$

Since the second term is much higher than the first one, one gets that $\dot{Q} \sim (\Delta T/l)^{1/3}$, which is the so-called Gorter–Mellink regime.

To describe in detail the transition from Landau regime ($\dot{Q} \sim (\Delta T/l)$) to Gorter–Mellink regime ($\dot{Q} \sim (\Delta T/l)^{1/3}$) one must use (5.2.21) to describe how L changes with $\dot{Q}$ (or with V_{ns}). For $\alpha_1 = \omega = \omega'' = 0$ in (5.2.21) for the sake of higher simplicity, the results are shown in Figs. 5.3 and 5.4 (from [36]). The effective thermal conductivity K_{eff} in Helium II with and without quantum turbulence ($K_{eff-turb}$ and $K_{eff-Landau}$, respectively) is shown in Figs. 5.5 and 5.6 [36], where $\frac{K_{eff-Landau}}{K_{eff-turb}}$ is plotted in terms of $\dot{Q}$ and of the radius of the channel, respectively.

Thus, the generalized Eq. (5.2.21) describing the laminar–TI–TII transitions is useful to describe the full nonlinear complexity of heat transport. This is needed in cryogenic situations because in narrow channels the actual physical situations

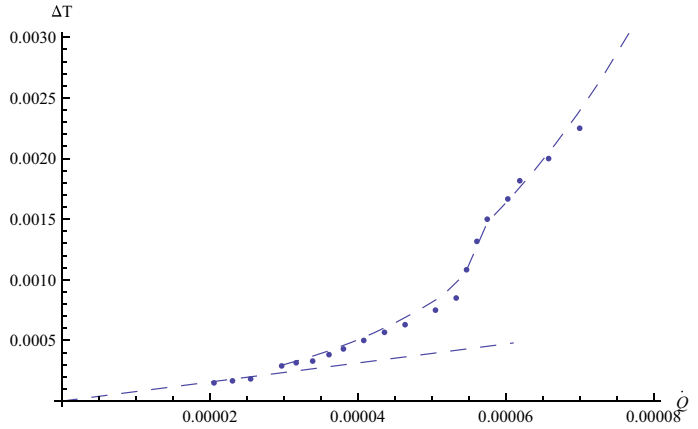


Fig. 5.3 Comparison between the model (5.3.31) relating the temperature difference ΔT and the total heat flow $\dot{Q}$, and the experimental data from Childers and Tough. Data are of Fig. 11 in [37] for $d = 124\,\mu$m at $T = 1.6$ K. The figure is from [36]

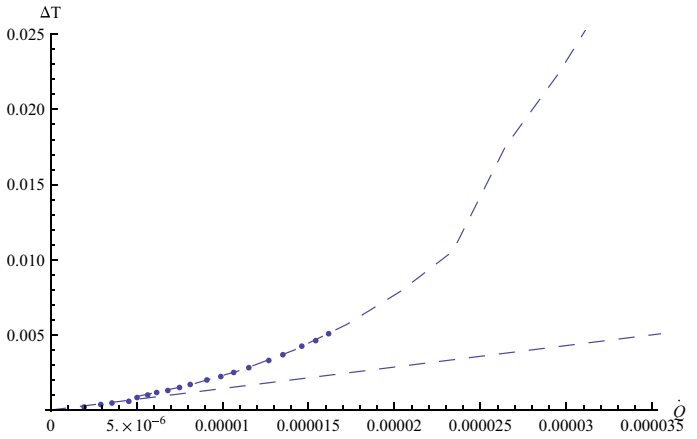


Fig. 5.4 Comparison between the model (5.3.31) relating the temperature difference ΔT and the total heat flow $\dot{Q}$ and the experimental data from Childers and Tough. Data are of Fig. 14 in [37] for $d = 61\,\mu$m at $T = 1.6$ K. The figure is from [36]

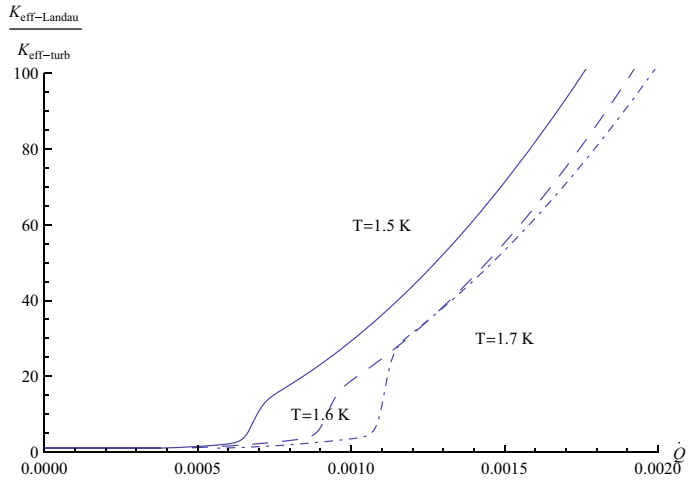


Fig. 5.5 Ratio between the effective thermal conductivity in Helium II without and with vortices versus the heat current $\dot{Q}$ (J/s) using the data from [16] ($d = 1000\,\mu$m at $T = 1.5$ K (solid line), $T = 1.6$ K (dashed line), and $T = 1.7$ K (dot- dashed line)). The transition between TI and TII turbulence corresponds to the elbow of the plot. The figure is from [36]

correspond to the intermediate region, rather than to the laminar regime nor to the fully developed turbulent regime. In fact, in the intermediate regime, hysteresis effects may appear (i.e. the line ΔT in terms of $\dot{Q}$ being different when $\dot{Q}$ is being increased and when it is being decreased). Equations exhibiting some metastable branches (as discussed in (5.2.19)) could in principle describe this behaviour.

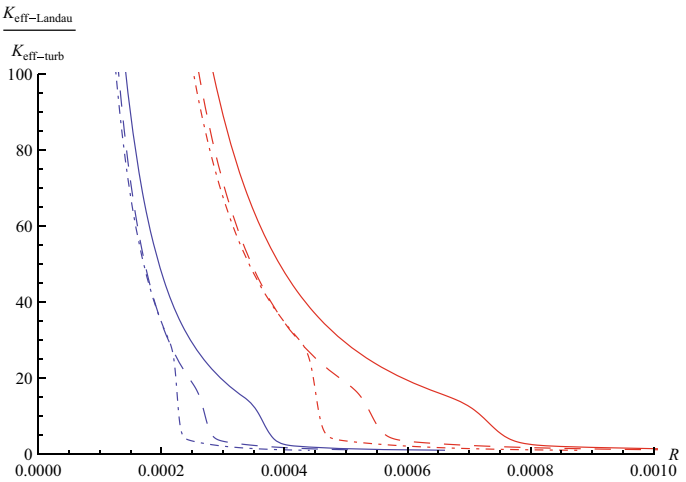


Fig. 5.6 Ratio between the effective thermal conductivity in Helium II without vortices ($K_{eff-Landau}$) and with vortices ($K_{eff-turb}$) versus the radius of the channel (m) using the data from [16] $T = 1.5$ K (solid line), $T = 1.6$ K (dashed line), and $T = 1.7$ K (dot-dashed line). The first three lines (left) refer to $\dot{Q} = 5 \times 10^{-4}$ J/s but the second three lines (right) refer to $\dot{Q} = 10^{-3}$ J/s, respectively. The figure is from [36]

5.4 Influence of Rotation on Superfluid Turbulence

Chapter 4 has been devoted to vortices in rotating superfluids, without heat flux or with heat fluxes low enough to not generate vortices. The combination of rotation with heat flux exhibits a still greater complexity, because both sources of vorticity are not additive (i.e. the total amount of vortex lines is different from that expected if the two sources of vorticity are simply added). A subtle interplay arises between the ordered vortices of rotation, tending to be parallel to the rotation axis, and the disordered ones of counterflow, tending to be distributed isotropically [8, 38–45].

In the presence of counterflow and rotation, the anisotropy of the tangle becomes a dynamical variable, whose description requires an additional evolution equation, which will be examined in Sect. 6.7.

5.4.1 Essential Phenomenology

In [45], it is experimentally found that the ordered array of vortex lines produced by steady rotation and the disordered tangle produced by the counterflow produce a globally anisotropic tangle when rotation and axial counterflow are combined.

The essential experimental observations of [45] may be summarized as follows. The container is rotating at a given angular speed Ω and the heat flow (corresponding

to a counterflow velocity V_{ns}) parallel to the rotation axis is gradually increased. Then one finds a complete absence of the laminar regime, and the presence of two critical velocities $V_{c1}^{H,R}$ and $V_{c2}^{H,R}$, scaling as $\Omega^{1/2}$. In more detail, the observed phenomenology is summarized in the next paragraphs.

For small values of Ω: (a) any rotation seems to eliminate the counterflow critical velocity V_c present in the absence of rotation; (b) for subcritical counterflow velocities, $V_{ns} < V_c$, the difference in L between no rotation and very slow rotation is larger than expected, and grows with velocity; (c) "enough" above the critical V_c, the difference between the line density which L asymptotically approaches and L_H is much less than L_R (L_H and L_R as defined in (5.1.9) and (4.1.3), respectively). This feature increases with V_{ns} and with Ω.

For higher values of Ω: (d) for $V_{ns} \leq V_{c1}^{H,R}$, with $V_{c1}^{H,R}$ a counterflow-rotation critical velocity, L is independent of V_{ns} and proportional to Ω; in particular, $L = 2\Omega/\kappa$, as expected in the absence of counterflow according to (4.1.3); e) for $V_{c1}^{H,R} \leq V_{ns} \leq V_{c2}^{H,R}$, with $V_{c2}^{H,R}$ a second counterflow-rotation critical velocity, a situation similar to (d) is observed, but with L independent of V_{ns} and proportional to Ω, with a slightly different proportionality constant than in the previous situation; f) for $V_{ns} \geq V_{c2}^{H,R}$, L increases with V_{ns} and becomes proportional to V_{ns}^2 at high values of V_{ns}, with a proportionality constant independent of Ω.

The two critical counterflow-rotation velocities scale as $V_{c1}^{H,R} = C_1\sqrt{\Omega}$, $V_{c2}^{H,R} = C_2\sqrt{\Omega}$, with $C_1 = 0.053$ cm sec$^{-1/2}$, $C_2 = 0.118$ cm sec$^{-1/2}$ (at 1.65 K and at saturated vapour pressure). The first transition seems to correspond to the Donnelly–Glaberson instability [21, 46, 47]: excitation of helical waves (Kelvin waves) by the counterflow on the vortex lines induced by rotation. The second one appears to be a transition to a disordered tangle, when the amplitude of the Kelvin waves approaches the average vortex separation so that the different vortex lines reconnect with each other [39, 41]. In the considered range of Ω and V_{ns}, the measured values of L are always less than $L_H + L_R$, and the deviation increases with V_{ns} and Ω. In the limit of large heat flux and slow rotation, the tangle appears to be "polarized". At large V_{ns}, the increase in L is always less than expected, and the deviation increases with V_{ns} and Ω.

5.4.2 *Vortex Line Density Evolution Equation in the Presence of Rotation*

To describe the dynamics of L in terms of V_{ns} and Ω, we consider the experiments in the range of high values of Ω (0.2 Hz $\leq \Omega/2\pi \leq 1.0$ Hz) and for $0 \leq V_{ns}^2 \leq 0.2\,\mathrm{cm}^2/\mathrm{s}^2$. Another situation of interest could be Couette flow between two concentric cylinders rotating at different angular speeds [48], which will be briefly commented in Sect. 7.4.

We model the vortex destruction contribution with (5.1.6), and the production contribution with a term depending on V_{ns}, Ω and L through two dimensionless

combinations:

$$\left[\frac{dL}{dt}\right]_f = \kappa L^2 \phi_f \left[\left(\frac{V_{ns}}{\kappa L^{1/2}}\right), \left(\frac{\Omega}{\kappa L}\right)^{1/2}\right]. \tag{5.4.33}$$

For the function ϕ_f we choose a quadratic dependence on its variables, and write

$$\left[\frac{dL}{dt}\right]_f = \alpha_0 V_{ns} L^{3/2} + \alpha_2 \sqrt{\kappa\Omega} L^{3/2} - \beta_1 \Omega L - \beta_4 V_{ns} \frac{\sqrt{\Omega}}{\sqrt{\kappa}} L, \tag{5.4.34}$$

being α_i and β_i dimensionless constants that may depend on the anisotropy of the tangle and on the temperature T. The term in V_{ns}^2 has been omitted because the values of V_{ns} considered here are not too high. A generalization of this equation including the influence of the walls has been considered in [42]. Such effects are especially relevant for slow rotation and small heat fluxes, but we will not consider them in this book.

Note that, had we restricted (5.4.34) to a linear approximation, the effects of V_{ns} and Ω would have been merely additive, at odds with observations. It is observed that the rotation facilitates the formation of vortices, for small or null counterflow velocities, but it hinders their lengthening for high values of V_{ns} and Ω.

In absence of counterflow ($V_{ns} = 0$), the term $-\beta_1 \Omega L$ reduces vortex formation at high Ω and L. This could be tentatively interpreted as a repulsion between parallel vortex lines. The maximum density of vortex lines would then be $L_{max} = (\beta_1/\alpha_2)^2(\Omega/\kappa)$, corresponding to a minimum separation between vortex lines given by $\delta_{min} = L_{max}^{-1/2} = (\alpha_2/\beta_1)(\kappa/\Omega)^{1/2}$. Thus, the terms in α_2 and β_1 in (5.4.34) would reflect some competition between vortex formation and vortex repulsion.

The negative contribution of the last term in (5.4.34) could be interpreted as a tendency of rotation to orientate and straighten the vortex lines along its direction; thus making increasingly unfavourable the existence of vortex lines orthogonal to V_{ns}.

Substituting expressions (5.4.34) and (5.1.6) in (5.1.3), we obtain [42]

$$\frac{dL}{dt} = -\beta\kappa L^2 + \left[\alpha_0 V_{ns} + \alpha_2 \sqrt{\kappa\Omega}\right] L^{3/2} - \left[\beta_1 \Omega + \beta_4 \frac{V_{ns}\sqrt{\Omega}}{\sqrt{\kappa}}\right] L. \tag{5.4.35}$$

In the absence of rotation, Vinen's equation (5.1.7) is recovered. Equation (5.4.35) provides a good approximation to the experimental observations, in the range of V_{ns} and Ω considered.

For the sake of completeness, in [49] the alternative Vinen equation (Eq. (5.1.8) with $n = 2$) is generalized to combined rotation and counterflow. There, the obtained equation was compared to the generalized Vinen equation (5.4.35) both in the steady and unsteady states. The results were that in the steady state the description based on the alternative Vinen equation is slightly better than that on Eq. (5.4.35), but, in the unsteady state, the temporal scale of the solution of (5.4.35) was closer to the observations (the time required to reach the steady state is less than 10 min [45]) than

the solutions of the alternative Vinen, which run much faster (for more details see [49]).

5.4.3 Pure Rotation, Combined Rotation, and Counterflow

Equation (5.4.35) describes most of the phenomenology mentioned in Sect. 5.4.1. Here we consider two situations: rotation only, and combined rotation and counterflow. Counterflow only need not be considered, as in this case (5.4.35) reduces to (5.1.7).

Rotation Only

For $V_{ns} = 0$ and $\Omega \neq 0$, Eq. (5.4.35) becomes

$$\frac{dL}{dt} = -\beta\kappa L^2 + \alpha_2\sqrt{\kappa\Omega}L^{3/2} - \beta_1\Omega L. \tag{5.4.36}$$

This equation admits only one stable non-zero stationary solution, namely:

$$L_R^{1/2} = \left[\frac{\alpha_2}{2\beta} + \sqrt{\frac{\alpha_2^2 - 4\beta\beta_1}{4\beta^2}}\right]\sqrt{\frac{\Omega}{\kappa}}. \tag{5.4.37}$$

Then, L is proportional to Ω in agreement with experimental results; in view of (4.1.3), the latter imply that

$$\frac{\alpha_2 + \sqrt{\alpha_2^2 - 4\beta\beta_1}}{2\beta} = \sqrt{2}. \tag{5.4.38}$$

Note that rotation only is not enough to determine the values of β_1 and α_2.

Combined Rotation and Counterflow

Looking at experimental data of Fig. 3 of [45] (see Fig. 5.7) for L as a function of V_{ns}^2 it turns that L is almost independent of V_{ns} for $V_{ns} < V_{c2}^{H,R}$, with a steep change around $V_{c1}^{H,R}$ (in other words, for $V_{ns} < V_{c1}^{H,R}$, L in terms of V_{ns}^2 is a horizontal line with a small step in $V_{c1}^{H,R}$). However, there is a variation of the slope near $V_{c2}^{H,R}$. In [42], we commented the microscopic differences between both transitions.

Equation (5.4.35) yields the observed behaviour for $L^{1/2}$ as function of V_{ns}, in the limiting case in which

$$\frac{\beta_1}{\beta} = \frac{\beta_4}{\alpha_0}\left(\frac{\beta_4}{\alpha_0} - \frac{\alpha_2}{\alpha_3}\right). \tag{5.4.39}$$

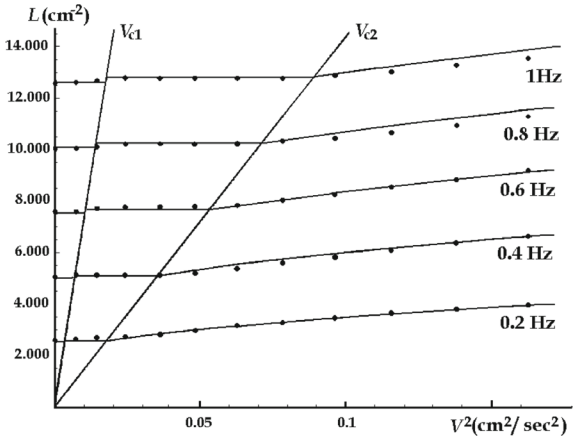


Fig. 5.7 Vortex line density L as function of V_{ns}^2 (Eqs. (5.4.40) and (5.4.41)) for different values of the rotation frequency. The equations of the straight lines are, respectively, $L = L_H(V_{ns}/V_{c1}^{H,R})^2$ and $L = L_H(V_{ns}/V_{c2}^{H,R})^2$. Experimental data are from [45]. The plots are from [42]

Under this hypothesis, the steady-state solutions of (5.4.35) can be written as

$$L^{1/2} = \frac{\beta_4}{\alpha_0}\sqrt{\frac{\Omega}{\kappa}} = mL_R^{1/2}, \tag{5.4.40}$$

$$L^{1/2} = \gamma^{H,R}V_{ns} + \left(\frac{\alpha_2}{\beta} - \frac{\beta_4}{\alpha_0}\right)\sqrt{\frac{\Omega}{\kappa}}, \tag{5.4.41}$$

where $\gamma^{H,R} = \alpha_0/\beta\kappa$ and $m = \beta_4/\sqrt{2}\alpha_0$, with $L_R = 2\Omega/\kappa$. Observe that the value of $\gamma^{H,R}$ differs from the values of $\gamma_H = \left(\frac{\alpha_i}{\beta_v}\right)^2$, introduced in (5.1.9), characterizing the stationary solution in the presence of counterflow only, because α_0 depends on the anisotropy, which is different in the two situations.

Equation (5.4.40) represents a family of straight lines parallel to the V_{ns}-axis, scaling with $\sqrt{\Omega}$. Equation (5.4.41) represents a family of straight lines (for different angular velocities) with the same slope $\gamma^{H,R}$, independent of Ω (see Fig. 5.7). In [42], it was shown that (5.4.40) is stable for $0 < V_{ns} < V_{c2}^{H,R}$, with

$$V_{c2}^{H,R} = \frac{1}{\gamma^{H,R}}\left[2\frac{\beta_4}{\alpha_0} - \frac{\alpha_2}{\beta}\right]\sqrt{\frac{\Omega}{\kappa}}, \tag{5.4.42}$$

while, for values of V_{ns} higher than $V_{c2}^{H,R}$, the solution (5.4.41) is stable. Therefore $V_{c2}^{H,R}$ represents the critical counterflow-rotation velocity, mentioned in point (e) of Sect. 5.4.1, characterizing the transition to a turbulent disordered tangle. This critical velocity scales as $\sqrt{\Omega}$, in agreement with experimental observations. The critical velocity $V_{c1}^{H,R}$ will be commented below.

The experimental data on the steady states of L and Eqs. (5.4.39)–(5.4.41) yield $\beta_4/\alpha_0 = 1.427$, $\alpha_2/\beta = 2.68$, $\alpha_0/\beta = 0.047$, and $\beta_1/\beta = 1.78$ [50] at $T = 1.5$ K. The coefficient β, controlling the rate of evolution of L, cannot be determined from the stationary solutions and we choose for it the value obtained from Vinen equation (5.1.6). In Fig. 5.7, we show the results for the vortex line density L to different values of V_{ns} and Ω, with these four parameters. They furnish a good approximation to more than 60 experimental data, depending on Ω and V_{ns}. However, it does not describe the small step in L near $V_{c1}^{H,R}$, mentioned in point (e) of Sect. 5.4.1. We comment this in [42].

Regarding heat transport along rotating cylinders in liquid helium, the essential equation is again (5.3.30), but taking into account the complex dependence of L on Ω and q, namely:

$$-\nabla T = \frac{8\eta}{S^2 T R^2}\mathbf{q} + \frac{K_f L(\Omega, q)}{\zeta}\mathbf{q}, \tag{5.4.43}$$

with $L(\Omega, q)$ given by the solutions of (5.4.35). In particular, the possibility of controlling Ω independently from q would allow to achieve some control on q at fixed ∇T, or a control of ∇T at fixed q by suitably changing Ω. Another related topic of interest is the combination of rotation and a radial counterflow perpendicular to the rotation axis, instead of counterflow parallel to the rotation axis. Since the polarization and the anisotropy of both cases would be different, comparison of both situations would lead to interesting information on the role of the tangle anisotropy on the effective thermal resistance.

5.5 Heat Transport in Rotating Turbulent Superfluid: Implications for Neutron Stars

Heat transport in rotating superfluids is of interest in neutron superfluid in neutron stars as it has been preliminary commented in Sect. 4.4. In the central regions of these very dense stars, neutrons could form a superfluid, after pairing and becoming a Bosonic system, able to form a Bose–Einstein condensate. This superfluid would experience a combination of axial rotation and radial counterflow. The situation would be considerably more complicated than the cylindrical one analysed in Sect. 5.4, but it is worth of examination because of its astrophysical interest [51, 52].

Since the heat transport depends on the vortex line density, which in turn depends on q and Ω, the analysis is truly challenging. From it, it will depend on the heat transport ability in neutron stars. Transitions in the internal vortex tangle of the star lead to modifications in the effective viscosity in the neutron superfluid, thus implying modifications of the moment of inertia of the star [51, 53].

Neutron stars rotate at a high angular velocity, which decreases in time because of radiative loss of energy. However, from time to time and in a relatively erratic way, there are observed short periods of increase of the angular velocities, known as *glitches*. Since it is believed that the interior of the star is composed of a superfluid of

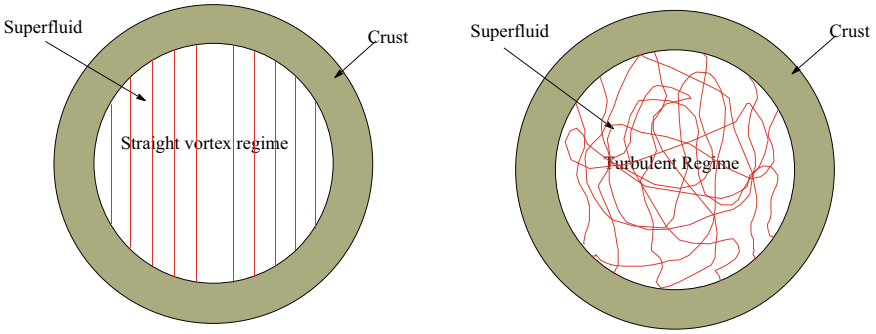


Fig. 5.8 Sketch of vortex lines in rotating neutron stars, whose external solid crust and internal superfluid core are shown. The straight vortex regime (left); Turbulent vortex regime, just after the glitch (right). Figure from [53]

neutrons, the high number of quantized vortices, induced by the rotation and probably by a radial heat flow as well, are arranged parallel to the rotation axis. These vortices may be curved near the spherical boundary of the superfluid, in such a way that they pin normally to such boundary. It is assumed that these vortices may play a role in the dynamics of the glitches. The simplest model assumes that the star may be described as two concentric spherical layers: the internal superfluid sphere and a rigid outer crust, which rotate with different angular velocities and interact with each other (see Fig. 5.8). Glitches could be caused by the relative velocity between the crust and the superfluid and the presence of quantized vortices, as explained below. Indeed, the neutron liquid could not sustain a rigid-body rotation unless it is penetrated by a high number of quantized vortices.

Let us assume that initially the bulk and the crust co-rotate with a common value Ω_0 of their angular velocities. The crust slows down because of the energy radiated away of the star and because of the energy supplied to the internal superfluid component. Vortex lines, which are strongly pinned to the crust, deform from their straight configuration because the crust slows down faster than the superfluid (see Fig. 5.8). The slowing down of the crust increases the difference of angular velocities of the crust and the superfluid. When the difference is too high, the vortices catastrophically unpin from the crust, and transfer angular momentum to the outer crust, which accelerates suddenly (glitches) forming a tangle of vortices (see Fig. 5.8).

These simplified ideas suggest the possible relevance of the quantized vortices. The acceleration period may be mathematically modellized by assuming that the dynamics of the star includes the rotation of the crust, the rotation of the superfluids, and the dynamics of the vortices.

In [53], we have proposed a mathematical model describing the glitches in pulsars combining the quantized vortex lines which fill the superfluid neutron liquid, with the slowing down of the crust. It is

$$I_C \frac{d\Omega_C}{dt} = -k_r^C - \frac{1}{\tau}\frac{I_S I_C}{I_S + I_C}\left(\Omega_C - \frac{\kappa L}{2}\right), \tag{5.5.44}$$

$$I_S \frac{d\Omega_S}{dt} = -k_r^S + \frac{1}{\tau}\frac{I_S I_C}{I_S + I_C}\left(\Omega_C - \frac{\kappa L}{2}\right), \tag{5.5.45}$$

$$\frac{1}{\aleph\,\kappa}\frac{dL}{dt} = -\left(L - \frac{2\Omega_S}{\kappa}\right)\left[L - \left(\frac{\alpha_1}{\aleph}\frac{\Omega_{SC}}{\kappa} - \frac{\beta_1}{2\,\aleph}\frac{\Omega_S}{\kappa}\right)\right], \tag{5.5.46}$$

where Ω_C and Ω_S are the angular velocities of the crust and of the inner superfluid and L the vortex line density in the inner superfluid. I_S and I_C are the moment of inertia of the superfluid core and of the external crust, $\aleph$ is a coefficient describing the mutual friction between the core and the crust, and τ is the characteristic time scale of the coupling between the crust and the superfluid. The two terms k_r^C and k_r^S take into account of the loss of rotational energy due to the radiation.

The appearance of the glitches is mainly related to the outer crust, which allows a rate of growth of the relative angular velocity between crust and superfluid. Two main regimes occur, which we may briefly refer as "a straight vortex regime" and a "turbulent regime", related to the first and the second zeros of the right-hand side of Eq. (5.5.46), namely:

$$L = \frac{2\Omega_S}{\kappa}, \qquad L = \left(\frac{\alpha_1}{\aleph}\frac{\Omega_{SC}}{\kappa} - \frac{\beta_1}{2\,\aleph}\frac{\Omega_S}{\kappa}\right). \tag{5.5.47}$$

In the first regime (first solution in (5.5.47)), crust and superfluid co-rotate in such a way that vortex lines arrange like straight lines parallel to the rotation axis and pinned to the outer crust. The dynamics of the two angular velocities Ω_C and Ω_S is described by the two Eqs. (5.5.44) and (5.5.45) with L given by the first solution of (5.5.47). Because of the slowing down of the crust, the relative velocity increases and acts on the vortices both by means of the mutual friction force (inside the star) and by stretching the vortices in the pinning regions (outer crust). When the amplitude of the Kelvin waves reaches the inter-vortex distance between neighbouring vortices or the tension in the pinning regions becomes strong enough, the topology of the vortices (inside the star) changes, and a disordered vortex tangle originates. The sudden unpinning of the vortices and the decrease of the vortex tension supply momentum to the crust at the cost of the momentum of the superfluid, and produce a sudden spin-up of the angular velocity of the crust, which is in fact the glitch. In terms of the mathematical model (5.5.44)–(5.5.46), this means that the first solution of (5.5.47) becomes unstable and that the system switches from the first to the second solution of (5.5.47), namely, the so-called turbulent regime. Accordingly, the value of L in Eq. (5.5.44) and (5.5.45) will be the turbulent vortex line density (the second in (5.5.47)) and the dynamics of the two angular velocities will be completely different with respect to the first regime.

In [53], four hypotheses have been considered: a) the velocity of the whole star Ω_{av} decreases with the same rule in both regimes; b) there is continuity of the solution Ω_S

and Ω_C in the transition between two consecutive regimes; c) the difference between the two velocities, Ω_S and Ω_C, increases in the straight vortex regime, but decreases in the turbulent regime; d) the value of L increases at $t = \tau$.

The switch from the straight vortex regime to the turbulent vortex regime is characterized by the dimensionless parameter γ

$$\gamma = \frac{1}{2\,\aleph\,\tau\Omega_0} \tag{5.5.48}$$

which naturally arises in the left-hand side of Eq. (5.5.46) when the system of equations is written in dimensionless form, introducing the variables $y_S = \Omega_S/\Omega_0$, $y_C = \Omega_C/\Omega_0$, $z = \kappa L/(2\Omega_0)$, and $q = t/\tau$, Ω_0 being the characteristic angular velocity (note that it is the angular velocity of the crust and of the core, before the spin-down begins). This parameter can be written as

$$\gamma = \frac{1}{\aleph\,\tau}\frac{L_0^{-1/2}}{\kappa L_0^{1/2}} = \frac{\tau'}{\tau} = \frac{1}{\tau}\frac{\delta}{v_l}$$

with $\tau' = \delta/v_l$ the ratio between the initial intervortex space $\delta = L^{-1/2}$ and the typical turbulent superfluid velocity around a vortex $v_l = \aleph\,\kappa L^{1/2}$ [54]. The switch between the two solutions of L is allowed if $\gamma \ll 1$, that is, the characteristic time τ' of the dynamics of vortices is sufficiently smaller than the characteristic time τ of coupling between crust and superfluid.

This is what happens in a pulsar with good approximation: a rough evaluation of τ' can be achieved by assuming that the constant $\aleph$ is of the order of the unit as in superfluid helium. Thus, in the straight vortex regime $\tau' \simeq (\kappa L)^{-1} \equiv (2\Omega_S)^{-1}$. Since Ω_S is of the same order of magnitude of Ω_0 then $\tau' \simeq 0.007$ s for $\Omega_0 = 68\ \mathrm{s}^{-1}$. This implies that the dynamics of L is much faster than the dynamics of the angular velocities of the crust and of the inner part of the star since the characteristic time τ' is smaller than τ.

In [53], the mathematical model has been applied to Vela pulsar, even though our results can be easily applied to any other pulsar. The main data of the Vela pulsar are reported in Table 5.4 and are taken from [55–57]. The exact regions of the star

Table 5.4 The main data of the Vela pulsar are taken from Dodson, Lewis & McCulloch [55], Yu et al. [56] and Padmanabhan [57]

M_{Tot}	$1,2 \cdot (1.98855 \cdot 10^{30})$ Kg
R_{Tot}	10,5 Km
τ	950 days
τ_1	30 s
Ω_0	$70\ \mathrm{s}^{-1}$
$k_r = k_r^C + k_r^S$	$7,37 \cdot 10^{25}$ J
I_{Tot}	10^{38} Kg $\cdot$ m^2

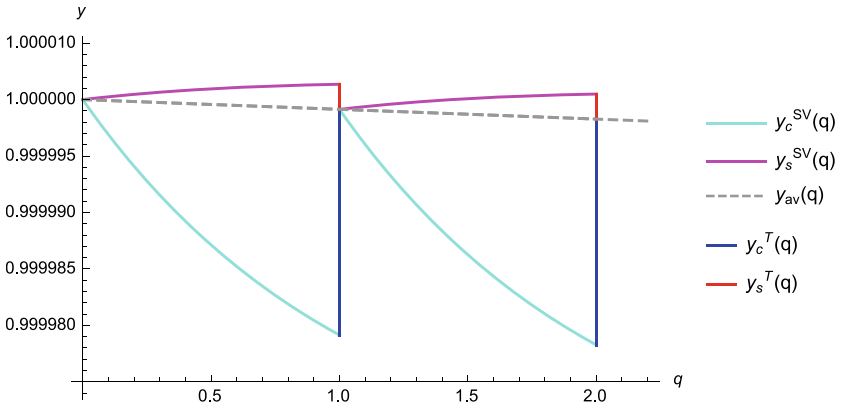


Fig. 5.9 Plots of the dimensionless angular velocities $y_c(q)$ (crust) and $y_s(q)$ (superfluid) as a function of the dimensionless time q both in the straight vortex regime (SV) and in the turbulent regime (T). The jumps in the angular velocities, especially the fast positive jump of the rotation velocity of the crust, reflect the glitches in the rotation of the neutron star. Figure from [53]

which belong to the crust and to superfluid are of course unknown, thus we have some difficulties to find the moments of inertia I_C and I_S. The model works quite well also for different relative choices of I_C and I_S with the condition that $I_S > I_C$. In [53], the inertia momentum $I_C = 10^{-1} \cdot I_{Tot}$ and hence $I_S = I_{Tot} - I_C = 9 \cdot 10^{-1} \cdot I_{Tot}$ have been used.

In Fig. 5.9, the two dimensionless velocities of the crust, $y_C = \Omega_C / \Omega_0$, and of the inner part of the neutron star, $y_S = \Omega_S / \Omega_0$, are plotted in terms of the dimensionless time $q = t/\tau$ both in the straight vortex regime (y_C^{SV} and y_S^{SV}) and in the turbulent regime (y_C^T and y_S^T), and among three consecutive glitches. In Fig. 5.9, it is assumed that the duration and frequency of the glitch are constant, but glitches are in general neither periodic nor with the same intensity. It is quite hard to appreciate the duration of the glitches in these plots because it just lasts $q_1 = 3.65497 \times 10^{-7}$ ($\tau_1 = 30$ s) compared to $q = 1$ ($\tau = 950$ days). For this reason, a more detailed analysis is shown in Fig. 5.10.

According to the scenario explained above, at $q = 1$ the straight vortex regime ceases, because the topology of the vortex lines changes, and vortex tangle takes part in the superfluid. The vortex line density undergoes a sudden increase at $q = 1$ and consequently the angular superfluid velocity decreases and the velocity of the crust increases, as shown in Fig. 5.10, until both velocities coincide again at $q = q_2$ and the straight vortex regime is restored.

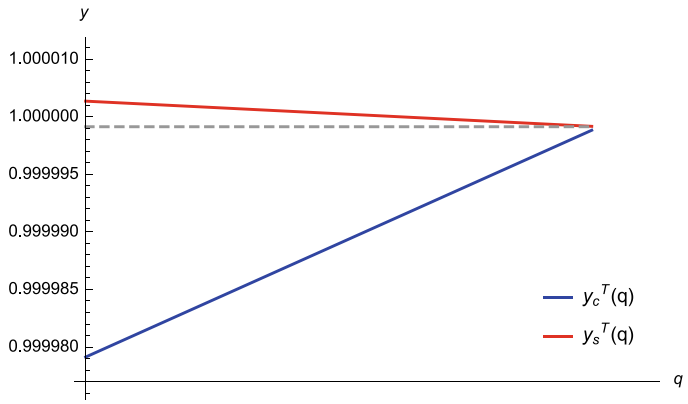


Fig. 5.10 A horizontal blowup of the straight lines in the plots shown in Fig. 5.9, near the dimensionless time $q = 1$, to better show the decrease of angular velocity of the superfluid and the fast increase of the angular velocity of the crust of the neutron star during the glitch. Figure from [53]

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Chapter 6
Coupled Heat and Vortex Flows: Inhomogeneity and Polarization



In this chapter, we consider the interaction between heat flow and quantized vortices, especially in the case that vortices are polarized. Thus, we are considering that the dynamics of vortex lines is not only production and destruction, as considered in Chap. 5, but also diffusive motion as a consequence of inhomogeneities, and drift motion, related to the interaction between heat flow and vortices, either as a drag force in nonpolarized vortices, or as an additional lateral force in polarized vortices. This topic is being the subject of experiments and numerical simulations, and has the additional complexity of the presence of small suspended particles used for visualization of the flow.

It is often assumed that the vortex tangle is practically homogeneous and isotropic, so that the vortex line density L is sufficient to describe it. But when the heat flux is not homogeneous, as it happens, for instance, in situations with axial or spherical symmetry or in channels with inhomogeneous cross section, the related vortex tangle will not be homogeneous neither, and a diffusive contribution will arise in the evolution equation of L, describing the flow of vortices from regions of higher L to those of lower L. Besides this relatively simple extension, other more complicated situations may arise, as in rotating counterflow or Couette and Poiseuille flows, or behind a grid towed across a quiescent superfluid, where compromise between the orienting effect of a local rotation and the randomizing effect of the heat flux makes the vortex tangle partially polarized and anisotropic. In fact, also in pure counterflow situations in cylindrical channels, the vortex tangle is not completely isotropic, as shown in some experiments [1] and simulations [2–8] because the vortex lines tend to orient perpendicularly to the heat flux. Thus, it is necessary for a more detailed geometrical description of the tangle, beyond the scalar vortex line density.

In situations with polarized vortices, as in rotating systems or in plane flows with vortices and antivortices, the local polarization $\mathbf{p}$ exhibits a mutual interaction with the local heat flux $\mathbf{q}$, described by a coupling of their respective evolution equations leading to a lateral force on the vortices and a deviation of the heat flow.

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6.1 Inhomogeneous Superfluid Turbulence

When heat flux is inhomogeneous, the subsequent vortex tangle will also be inhomogeneous, since in the steady state there is roughly a proportionality between L and q^2 [see (5.1.9)]. If L changes from point to point, vortex diffusion should be incorporated to the dynamics of L. Furthermore, gradients of L may contribute to heat transport and $\mathbf{q}$ may exert drag forces or lateral forces on the vortices, thus leading to a much richer dynamics than that considered in Chap. 5. For the sake of generality, we rewrite the equations for $\mathbf{q}$ and L as [9, 10]

$$\dot{\mathbf{q}} + \nabla \cdot \mathbf{J}^{\mathbf{q}} = \sigma^{\mathbf{q}}, \tag{6.1.1}$$

$$\dot{L} + \nabla \cdot \mathbf{J}^{L} = \sigma^{L}, \tag{6.1.2}$$

where the tensor $\mathbf{J}^{\mathbf{q}}$ and the vector $\mathbf{J}^{L}$ are the fluxes of the heat flux and of vortex lines, and $\sigma^{\mathbf{q}}$ and σ^{L} the respective net production terms, for which we take $\boldsymbol{\sigma}^{q} = -K_f L\mathbf{q}$ as in (5.3.27) (neglecting $\mathbf{q}/\tau_1$) and

$$\sigma^{L} = \tilde{A} q L^{3/2} - \tilde{B} L^2, \tag{6.1.3}$$

corresponding to Vinen's equation (5.1.7).

The influence of inhomogeneities is described through $\mathbf{J}^{\mathbf{q}}$ and $\mathbf{J}^{\mathbf{L}}$. To the first order in $\mathbf{q}$, they can be expressed in the form:

$$\mathbf{J}^{\mathbf{q}} = \beta_0(T, L)\mathbf{U}, \qquad \mathbf{J}^{L} = \nu_0(T, L)\mathbf{q}, \tag{6.1.4}$$

with β_0 and ν_0 coefficients whose physical meaning will be clarified below. Note that one could add a term $\nabla\mathbf{q}$ to $\mathbf{J}^{\mathbf{q}}$ and a term ∇L to $\mathbf{J}^{L}$ in (6.1.4), in a higher-order description.

Substituting (6.1.4) in (6.1.1) and (6.1.2), the following system is obtained:

$$\dot{\mathbf{q}} + \zeta\nabla T + \chi\nabla L = -K_f L\mathbf{q}, \tag{6.1.5}$$

$$\dot{L} + \nabla \cdot (\nu_0\mathbf{q}) = \tilde{A} q L^{3/2} - \tilde{B} L^2, \tag{6.1.6}$$

with $\zeta \equiv \partial\beta_0/\partial T$ and $\chi \equiv \partial\beta_0/\partial L$. Coefficients χ and ν_0 describe the nonlocal coupling between the heat flux $\mathbf{q}$ and the vortex line density L. For more details, see Ref. [9].

Equations (6.1.5) and (6.1.6) are enough for the discussion of the diffusion of vortices and of the coupling of second sound and the distortions of the vortex tangle (represented by local inhomogeneities in L), which are especially interesting in some physical situations.

Equation (6.1.6) may be rewritten as

$$\dot{L} + \nabla \cdot (L\mathbf{v}_L) = \tilde{A} q L^{3/2} - \tilde{B} L^2, \tag{6.1.7}$$

with $\mathbf{v}_L$ the drift velocity of the tangle with respect to the steady superfluid background, identified as

$$\mathbf{v}_L = \frac{\nu_0}{L}\mathbf{q}. \tag{6.1.8}$$

Measurements [11] in counterflow superfluid turbulence in cylindrical containers with axis parallel to the heat flux show that the vortex tangle drifts slowly as a whole towards the heater, i.e. opposite to $\mathbf{q}$, so that $\nu_0 \leq 0$.

6.2 Vortex Diffusion and Vortex Density Waves

When $\mathbf{q}$ varies slowly enough that $\dot{\mathbf{q}}$ may be neglected, one obtains from (6.1.5)

$$\mathbf{q} = -\frac{\zeta}{K_f L}\nabla T - \frac{\chi}{K_f L}\nabla L. \tag{6.2.1}$$

The coefficient of the first term is the effective thermal conductivity of the vortex tangle, and the second one describes a coupling between the vortex line density and the heat flux.

6.2.1 Vortex Diffusion

When the temperature gradient vanishes, Eq. (6.1.6) yields after introduction of (6.2.1)

$$\dot{L} = \tilde{A}qL^{3/2} - \tilde{B}L^2 + \frac{\nu_0\chi}{K_f L_0}\nabla^2 L. \tag{6.2.2}$$

This is Vinen's equation (5.1.7) plus a diffusion term, in such a way that the coefficient $D = \dfrac{\nu_0\chi}{K_f L_0}$ may be identified as the vortex diffusion coefficient. This coefficient must be positive, because of the second law. Since $\nu_0 < 0$, as said below (6.1.8), it turns out that χ must be negative too.

Vortex diffusion in inhomogeneous vortex tangles was first considered in [12], where it was numerically found that, at T close to 0 K, $D \approx 10^{-1}\kappa$. Indeed, since the circulation quantum κ has the same units than D, namely, $length^2/time$, it seems natural that it is related to D. In [13], Nemirovskii computed D using a kinetic model and an evolution equation for L in which only the diffusive term appears. He found $D = 2.2\kappa$ at zero temperature, a value higher than the one mentioned before, probably because he considered a pure diffusive situation (results for D at higher temperatures are not yet available, to our knowledge).

In [14], Nemirovskii has also considered the role of vortex diffusion in the propagation of turbulence fronts, and in the decay of inhomogeneous superfluid turbulence [13]. An illustration of vortex diffusion will be explicitly discussed in Sect. 6.3.

6.2.2 *Vortex Density Waves*

If $\dot{\mathbf{q}}$ is not neglected in (6.1.5) and still in the isothermal assumption—for the sake of simplicity—one obtains from (6.1.6) and (6.2.1)

$$\ddot{L} + K_f L_0 \dot{L} = \nu_0 \chi \nabla^2 L + K_f L_0 \sigma^L + \dot{\sigma}^L, \tag{6.2.3}$$

with σ^L standing for (6.1.3). For high frequencies ω such that $\omega \gg K_f L_0$ and ω much higher than the reciprocal of the characteristic time of vortex destruction and formation as described by σ^L in (6.1.3), namely, $\omega \gg 2\tilde{B} L_0 - \frac{3}{2}\tilde{A} q L_0^{1/2}$, (6.2.3) becomes

$$\ddot{L} \approx \nu_0 \chi \nabla^2 L, \tag{6.2.4}$$

that is, we get vortex density waves propagating with speed

$$v_\infty^2 = \nu_0 \chi. \tag{6.2.5}$$

This outlines the physical features of ν_0 and χ, as their product is seen to be related both to the diffusion coefficient (in (6.2.2)) and to the speed of vortex density waves (in (6.2.5)). This would allow to determine this product ($\nu_0 \chi$) from the knowledge of v_∞ or D.

In the study of Sect. 4.1, the interaction between second sound and vortex lines providing the basis to obtain the values of L from the attenuation of low-amplitude second sound it was implicitly assumed that the vortex tangle remains as a static structure. In a more detailed analysis, the tangle is not static and its dynamical back reaction on the applied second sound in the form of vortex density waves (6.2.4) should be considered [15].

A detailed analysis of wave propagation in a channel filled with turbulent Helium II at rest and submitted to a longitudinal stationary heat flux $\mathbf{q}_0$ (counterflow situation) may be found in [15] both for high frequency and low frequency, in order to explore the vortex density perturbations due to the second sound and their back reaction on the speed and attenuation of the second sound. It is found that the speed of the second sound is not much influenced by the presence of vortex waves; in contrast, the speed (6.2.5) of vortex waves is much increased in the presence of high-frequency second sound. The attenuation coefficient of second sound at high frequency is much lower than that of low-frequency waves. Thus, from the practical point of view, it seems that, at high frequency, second sound will not provide much information on the vortex tangle. Instead, at low frequency the back reaction of the vortex tangle on

the heat waves implies remarkable changes in the velocity and the attenuation of the second sound, depending on the relative direction between $\mathbf{q_0}$ and the propagation direction $\mathbf{n}$ [15].

6.3 Radial Turbulent Counterflow

An example of inhomogeneous vortex tangle arises in radial heat flow from a warm wire and the corresponding superfluid turbulence around it. In this case, $\mathbf{q}$ and L depend on the radial distance r to the central axis. As a consequence, vortex diffusion plays a role [12, 14] and, in turn, influences the temperature profile. Because of the geometry of the flow, the heat flux is maximum near the centre and decreases as $q \sim 1/r$. The source of vortices given by the first term on the right-hand side of Eq. (5.1.7) becomes less intense as r increases.

We consider for L Eq. (6.2.2)

$$\dot{L} = \tilde{A} q L^{3/2} - \tilde{B} L^2 + D \nabla^2 L, \tag{6.3.1}$$

and apply it to a steady-state counterflow between two concentric cylindrical walls at different temperatures. This situation is highly inhomogeneous and it is of interest in cryogenic situations, where a cylindrical system is introduced into a cooler superfluid environment [16].

The inner cylinder (the hotter one) is at fixed temperature T_0, and the temperature profile $T(r)$ between both cylinders will depend on the heat flux, the thermal conductivity, and the vortex line density profile $L(r)$. The steady-state condition $\nabla \cdot \mathbf{q} = 0$ implies for radial heat flux q_r (called here as q), the form $q = \dot{Q}/r$, with $\dot{Q}$ the heat supplied per unit time and unit length by the hot cylinder (Fig. 6.1).

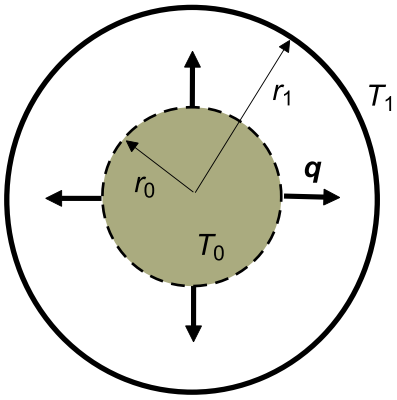


Fig. 6.1 Heat flows radially outwards from the hot inner cylinder of radius r_0 at temperature T_0. The radial temperature profile $T(r)$ depends on the heat flow q and on the vortex density radial distribution. Figure from [17]

Introduction of $q = \dot{Q}/r$ in (6.3.1) and assuming steady-state situation ($\dot{L} = 0$) allows to obtain $L(r)$ from:

$$\tilde{A}\frac{\dot{Q}}{r}L^{\frac{3}{2}} - \tilde{B}L^2 + D\nabla^2 L = 0, \tag{6.3.2}$$

where, for the axial symmetry of the problem, it is $\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}(r\frac{\partial}{\partial r})$. If the temperature gradients are small, $\tilde{A}$, $\tilde{B}$, and D (which in principle depend on temperature) may be taken as constant. In this case, (6.3.2) admits the solution:

$$L(r) = \gamma'^2\frac{\dot{Q}^2}{r^2}, \tag{6.3.3}$$

with

$$\gamma' = \frac{\tilde{A}}{2\tilde{B}}\left(1 + \sqrt{1 + \frac{16\tilde{B}D}{\tilde{A}^2\dot{Q}^2}}\right). \tag{6.3.4}$$

If $D = 0$ one recovers $\gamma' = \frac{\tilde{A}}{\tilde{B}}$, corresponding to Vinen's equation (5.1.7), i.e.

$$L(r) = \frac{\tilde{A}^2}{\tilde{B}^2}\frac{\dot{Q}^2}{r^2}. \tag{6.3.5}$$

The diffusion increases the local values of L as compared with the situation with $D = 0$, especially at relatively low values of the heat flux. This increase may be understood as the result of a diffusive flow of vortices from the inner cylinder (where q is higher and therefore the vortex production is higher) to the outer cylinder, where the vortices disappear when colliding against it. Thus, the term in $\tilde{A}\dot{Q}$ accounts for the "native" vortices, produced at the same point being considered, in contrast with the term in D, which accounts for the "migrating" vortices in that point. For small heat flux or high diffusion, the "migrating" vortex population is higher than the native vortex population.

When the vortex diffusion effects become dominating, from (6.3.4) the vortex length density profile takes the form:

$$L(r) = \frac{4D}{\tilde{B}r^2}. \tag{6.3.6}$$

In this regime, L is independent of the heat flux, but it keeps the same dependence $1/r^2$ on the radius than in (6.3.3) for $D = 0$. In the limit of low $\dot{Q}$, (6.3.2) with $D = 0$ would lead to $L = 0$, but for $D \neq 0$ it leads to (6.3.6).

The observation of different values of L (6.3.3) in the presence or absence of diffusion suggests the possibility of hysteresis when the heat flux is increased from

zero to some maximum value, at a rate sufficiently higher than the diffusion rate, and is subsequently lowered to zero, at a rate sufficiently lower than the diffusion rate. In the first case, diffusion will not have time to act during the process, so that L will be given by (6.3.3)–(6.3.4) with $D = 0$ (the "native" population will predominate), i.e. (6.3.5). In contrast, in the slow process, diffusion will always act and L will be described by (6.3.3)–(6.3.4) with $D \neq 0$.

The situation is shown in Fig. 6.2, where we have considered the dimensionless quantities Lr^2, $\gamma_0 \dot{Q}$ and $\tilde{D}$, defined by $\gamma_0 = \dfrac{\tilde{A}}{2\tilde{B}}$, $\tilde{D} = \dfrac{4D}{\tilde{B}}$ and

$$\gamma' = \gamma_0 \left[1 + \sqrt{1 + \tilde{D} \frac{1}{(\gamma_0 \dot{Q})^2}} \right]. \tag{6.3.7}$$

In this notation, for $D = 0$, one has, from (6.3.5) $Lr^2 = 4(\gamma_0 \dot{Q})^2$.

The initial situation corresponds to $L = 0$ and $\dot{Q}_0 = 0$ (state A). A quick increase of $\bar{\dot{Q}}$ produces the corresponding increase of L (from A to B). From B to C, the heat flux is no longer changed and diffusion plays a central role. When $\dot{Q}$ slowly decreases (from C to D), the final value of L will be given by (6.3.6), i.e. $Lr^2 = \tilde{D}$. This value will not last very much, because in the absence of the driving heat flux, L

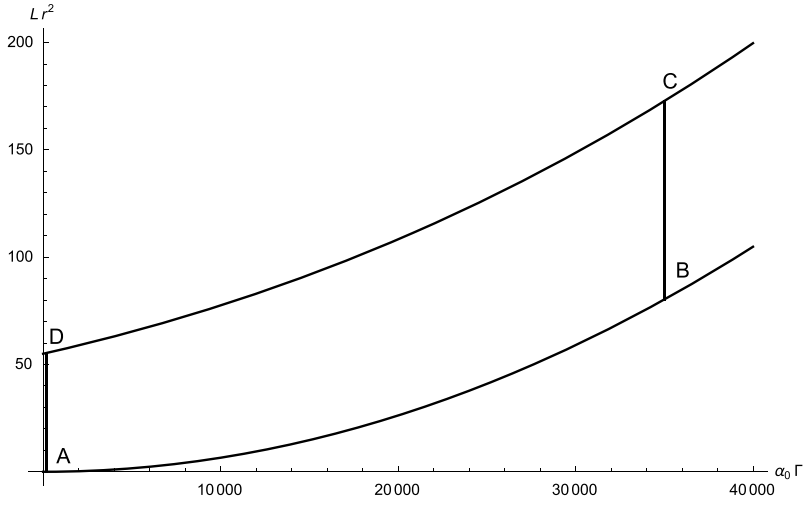


Fig. 6.2 The plot of Lr^2 versus the dimensionless expression for the heat flux $\gamma_0 \dot{Q}$ shows a hysteresis cycle. A to B: The increase of L for fast increasing heat flux corresponds to Eq. (6.3.5); C to D: the behaviour for slow decrease of heat flux corresponds to (6.3.3)–(6.3.4) with $D \neq 0$. The values of $\gamma_0 = 1.28 * 10^{-4}\,\mathrm{s}^3/(\mathrm{cm}^*\mathrm{g})$ for $T = 1.5$ K and $D = 2.2\kappa$ cm^2/s have been taken from [13] for D, from [18] for $\tilde{B}$, and from [19] for $\tilde{A}/\tilde{B}$. Figure from Ref. [17]

will decay from D to A. The paths followed by the vortex line density during increase and decrease of the heat flux will not be the same, which is the defining feature of hysteresis.

When the temperature profile is derived for a given value of the heat flow Q, one finds in some conditions the phenomenon of heat rectification, namely, that when given values of temperature T_H and T_C, for instance, are applied, respectively, to the internal and the external walls, one obtains a value of the outward heat flow Q different than that obtained for the inward flow Q' when the mentioned temperatures are mutually exchanged and, respectively, applied to the external and the internal walls [20].

6.4 Heat Flow and Vortex Flow in Two-Dimensional Channels

We now introduce the vortex polarization which implies an additional coupling between heat flow and vortices, besides the vortex production effect considered in Sect. 6.3. A relatively simple but very rich situation combining heat flow, vortex polarization, and vortex diffusion leading to inhomogeneities is counterflow along two-dimensional channels. In Sect. 4.3, it has been seen how polarized vortices between two rotating concentric cylinders at slightly different temperature produce a lateral deviation of the heat flow, which instead of being purely radial acquires a tangential component related to the rotation. Reciprocally, a heat flow will produce a lateral drift of polarized vortices. This is especially visible in two-dimensional channels, where the vortex lines are replaced by a number of positively and negatively polarized vortex points. If initially these points are homogeneously distributed in the channel, they separate into a positive-rich longitudinal region and a negative-rich longitudinal region after the applied heat flux has been acting on them for some time. This effect is mainly caused by the interactions between the heat flux and vortices. In this section, we study this situation and we make a numerical experiments to visualize after a transient both the distribution of the vortex points and the profiles of the velocities of Helium II involved. In Sect. 6.5, we deal with a more complicated situation including a cylindrical obstacle at the centre of the channel and some suspended particles used for flow visualization (Fig. 6.3).

6.4.1 Heat Flow Interaction with Polarized Vortices

The equation describing the influence of the heat flux on the evolution of polarization $\mathbf{p}$ is [21]

$$\frac{d\mathbf{p}}{dt} = -\nabla \cdot \mathbf{Q}_p + \gamma' \mathbf{p} \times \mathbf{q} + \sigma_q, \tag{6.4.1}$$

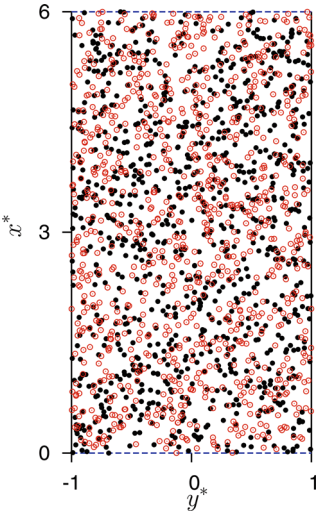


Fig. 6.3 Two-dimensional situations studied in Sects. 6.4 and 6.5. Initially there is a random homogeneous distribution of the vortices in the channel with the same number of positive and negative vortices. The red points mean positive circulation, and the black points negative circulation. Figure from [7]

where γ' is a coefficient. The corresponding term describes a lateral deviation of polarized vortices in the presence of the heat flux $\mathbf{q}$. Such term is, in some sense, the reciprocal of the last term in (4.2.6), where the polarization $\mathbf{p}$, parallel to Ω, produced a deviation of the heat flux proportional to $\mathbf{p} \times \mathbf{q}$. In the second term of (6.4.1), a heat flux produces a lateral deviation of vortices also proportional to $\mathbf{p} \times \mathbf{q}$ (to the right of $\mathbf{q}$ for positive vortices and to the left of $\mathbf{q}$ for negative vortices, in two-dimensional situations). The polarization flux tensor $\mathbf{Q}_p$ is given by

$$\mathbf{Q}_p = \gamma''(\mathbf{qp} + \mathbf{pq}) - D\nabla\mathbf{p} \tag{6.4.2}$$

with D the diffusion coefficient of vortices, and σ_p is the production (or destruction) term of polarized vortices (they may disappear, for instance, when a positive vortex point meets a negative vortex point, or when they arrive to the walls, and they may be produced by the heat flux, which may be numerically simulated by randomly adding vortex points to the channel) [21]. After substituting $\mathbf{Q}_p$ in Eq. (6.4.1), we find

$$\frac{d\mathbf{p}}{dt} = D\nabla^2\mathbf{p} + \gamma''\nabla \cdot (\mathbf{qp} + \mathbf{pq}) + \gamma'\mathbf{p} \times \mathbf{q} + \sigma_q \tag{6.4.3}$$

which also focuses on the diffusive behaviour in the first term and on the lateral deviation in the third term.

6.4.2 Microscopic Model

In this section, we consider two-dimensional numerical experiments which focus the coupling of the heat flux on quantum vortex points (completely polarized vortices with circulation $+\kappa$ or $-\kappa$) and the back reaction of these vortices to the main fields of Helium II. In these numerical experiments, the two-fluid model is used because of the clearer specification of boundary conditions, which are still an open problem in the extended theory [7]. In the two-fluid theory, instead, it is assumed that the normal component behaves as a viscous fluid and the superfluid component behaves as an inviscid fluid.

Consider an infinitely long two-dimensional channel of width D, with x and y the directions along and across the channel, respectively ($-D/2 \leq y \leq D/2$ and periodic boundary conditions imposed at $x = 0$ and $x = L_x$). The numerical experiment is applied to a counterflow channel (heat flow without mass flow) in terms of the two-fluid model described by Eqs. (3.4.89) and (3.4.90) obtained from the one-fluid extended model in Sect. 3.4.2.

The calculation mainly regards the dynamics of the vortex points over the channel (which ideally correspond to the dynamics of vortex lines in three dimensions) and the dynamical equation for the normal component. The velocity field of the superfluid component is instead obtained by means of the superfluid field generated by all vortex points and the counterflow condition $\rho^{(n)}\overline{\mathbf{u}}^{(n)} + \rho^{(s)}\overline{\mathbf{u}}^{(s)} = 0$, where $\overline{\cdots}$ indicates channel-averaged quantities.

According to (3.4.89), the dynamics of the normal component is given by equations $\nabla \cdot \mathbf{u}^{(n)} = 0$ and

$$\frac{\partial \mathbf{u}^{(n)}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{u}^{(n)} = -\frac{1}{\rho}\nabla p - \frac{\rho^{(s)}}{\rho^{(n)}} s^{(n)} \nabla T + \frac{\eta}{\rho^{(n)}} \nabla^2 \mathbf{u}^{(n)} + \frac{1}{\rho^{(n)}} \mathbf{F}_{ns}, \qquad (6.4.4)$$

where $s^{(n)}$ is the entropy of the normal fluid and the barycentric velocity $\mathbf{v}$ is kept in the left-hand side, and the mutual friction term $\mathbf{F}_{ns}$ describes the interaction between the normal component and the quantized vortices [22] (note that the term in $\mathbf{F}_{ns}$ does not appear in (3.4.89) because in Chap. 3 we deal with laminar situations without vortices). The bulk viscosity may be neglected with respect to the other terms.

Quantum vortex lines in a two-dimensional channel are usually described by vortex points of circulation κ (half of them have positive circulation and half have negative circulation) and position $\mathbf{r}_j(t) = \big(x_j(t), y_j(t)\big)$, where $j = 1, \ldots N$, being N the total number of vortices. They move according to Eq. (9.1.6), proposed by Schwarz [3] (see Sect. 9.1), which in two dimensions becomes

$$\frac{d\mathbf{r}_j}{dt} = \mathbf{u}_0^{(s)}(\mathbf{r}_j, t) + \mathbf{u}_i^{(s)}(\mathbf{r}_j, t)$$
$$+\alpha\, \mathbf{s}'_j \times \left(\mathbf{u}^{(n)}(\mathbf{r}_j, t) - \mathbf{u}_0^{(s)}(\mathbf{r}_j, t) - \mathbf{u}_i^{(s)}(\mathbf{r}_j, t)\right)$$
$$+\alpha' \left(\mathbf{u}^{(n)}(\mathbf{r}_j, t) - \mathbf{u}_0^{(s)}(\mathbf{r}_j, t) - \mathbf{u}_i^{(s)}(\mathbf{r}_j, t)\right), \tag{6.4.5}$$

where $\mathbf{s}'_j$ is the unit vector along vortex j (in the positive z-direction if the quantum of circulation is κ and in the negative z-direction if it is $-\kappa$), α and α' are mutual friction coefficients, depending on temperature [23], respectively, related to the Hall–Vinen coefficients B_{HV} and B'_{HV} introduced in (4.2.6) in such a way that $B_{HV} = (\rho/2\rho_n)\alpha$ and $B'_{HV} = (\rho/2\rho_n)\alpha'$ (also given in Table 4.1).

The field $\mathbf{u}^{(n)}(\mathbf{r}_j, t)$ is the normal fluid velocity at position $\mathbf{r}_j$, $\mathbf{u}_i^{(s)}(\mathbf{r}_j, t) = \sum_{k=1\ldots N} \mathbf{u}_{i,k}^{(s)}(\mathbf{r}_j, t)$ is the superfluid velocity field induced by all N vortex points at $\mathbf{r}_j$, and $\mathbf{u}_0^{(s)} = (u_{0x}^{(s)}, 0)$ is the superfluid flow enforcing the counterflow condition $\rho^{(n)}\overline{\mathbf{u}}^{(n)} + \rho^{(s)}(\overline{\mathbf{u}}_0^{(s)} + \overline{\mathbf{v}_{si}}) = 0$ at each channel cross section (for further details see [7]). Thus the superfluid velocity $\mathbf{u}^{(s)}$ at a point $\mathbf{r}$ is simply given by the sum of the contributions of each vortex point $\mathbf{u}_i^{(s)}(\mathbf{r})$ and a constant contribution $\mathbf{u}_0^{(s)}$, satisfying the counterflow condition. More details about the numerical code can be found in [5, 7].

The creation and destruction of vortices is modellized by a "numerical vortex reconnection" procedure described, tested, and used in [5–8, 24]: a couple of vortex points of opposite circulation is removed and then randomly re-inserted in the channel when the distance between the two vortex points becomes smaller than a critical value ϵ_1 or when the distance between a vortex point and a channel wall is less than $\epsilon_2 = 0.5\epsilon_1$ (refer to Ref. [7] and Supplementary Material in Ref. [8] for further insight on this numerical reconnection model).

The dynamical state of the normal fluid is investigated by applying the vorticity-stream function Ψ formulation [25] to (6.4.4) equations in such a way that the velocity $\mathbf{u}^{(n)}$ and the normal vorticity ω_n are defined as $\mathbf{u}^{(n)} = \left(\frac{\partial\Psi}{\partial y}, -\frac{\partial\Psi}{\partial x}\right), \omega_n = \nabla \times \mathbf{u}^{(n)}$.

The Cauchy–Dirichlet problem is

$$\mathbf{u}^{(n)}(x, \pm D/2) = \mathbf{0} \qquad \forall t$$
$$\mathbf{u}^{(n)}(x, y)_{|t=0} = \left(-V_{n0}\left[1 - (2y/D)^2\right], 0\right) \tag{6.4.6}$$

together with the counterflow condition

$$\rho^{(n)}\langle\mathbf{u}^{(n)}\rangle + \rho^{(s)}\langle\mathbf{u}_0^{(s)} + \mathbf{u}_i^{(s)}\rangle = 0. \tag{6.4.7}$$

In a recent paper [26], we have explored the case in which the normal component is not null on the wall in order to investigate the contribution of the heat carriers

reflected by the wall, as it occurs in phonon hydrodynamics when the phonon mean free path is comparable with the size of the system.

For the mutual friction force $\mathbf{F}_{ns}$ one may use the coarse-grained theoretical framework proposed by Hall and Vinen [27], at length scales larger than the average inter-vortex spacing ℓ. For this reason, we distinguish between the fine $(\Delta x, \Delta y)$ grid where $\mathbf{u}^{(n)}$ is numerically determined, and the coarse $(\Delta X, \Delta Y)$ grid on which we define the mutual friction $\widetilde{\mathbf{F}}_{ns}$ [7]

$$\widetilde{\mathbf{F}}_{ns} = \alpha\rho_s\widetilde{\boldsymbol{\omega}}_s \times \left[\widetilde{\boldsymbol{\omega}}_s \times \left(\widetilde{\mathbf{u}}^{(n)} - \widetilde{\mathbf{u}}^{(s)}\right)\right] + \alpha'\rho_s\widetilde{\boldsymbol{\omega}}_s \times \left(\widetilde{\mathbf{u}}^{(n)} - \widetilde{\mathbf{u}}^{(s)}\right) \ , \tag{6.4.8}$$

where $\widetilde{\cdot}$ symbols indicate quantities averaged over coarse grid cells. The second term in (6.4.8) produces a lateral motion of the vortices. After the mutual friction force $\widetilde{\mathbf{F}}_{ns}$ is computed on the coarse grid, it is interpolated on the finer grid via a two-dimensional bi-cubic convolution kernel [28].

The complete list of parameters used in the simulation and the physical relevant quantities are reported in Tables 6.1 and 6.2, expressed in terms of the following units of length, velocity, and time, respectively: $\delta_c = D/2$, $u_c = \kappa/(2\pi\delta_c)$, and $t_c = \delta_c/u_c$ [7]. Hereafter all the quantities are dimensionless, unless otherwise stated. The parameters employed in the simulations, in particular, the channel width D, the average inter-vortex spacing ℓ (determined by the total number of vortex points N as $N\ell^2 = DL_x$), and the normal fluid volume flow rate (imposed by the parameter V_{n0}), have been chosen to make qualitative comparisons with experimental superflow

Table 6.1 Dimensionless physical and numerical parameters used in the simulations in Sect. 6.4.3 in dimensionless units. The units of length, velocity, and time are $\delta_c = D/2 = 4.55 \times 10^{-3}$ cm, $u_c = \kappa/(2\pi\delta_c) = 3.49 \times 10^{-2}$ cm/s, and $t_c = \delta_c/u_c = 0.13$ s

D	2	T	$1.7K$
L_x	6	ρ_s/ρ_n	3.373
N	1876	V_{n0}	553.6
n	156.3	ϵ_1	2.5×10^{-3}
ℓ	0.08	Δt_v	7.5×10^{-6}
η	$1.29 \cdot 10^{-6}$ $Pa\,s$	$s^{(n)}$	0.3972 J/g K
α	0.126	α'	0.0123

Table 6.2 Number of grid points and spacings in dimensionless units of the grids used in the simulations in Sect. 6.4.2

Fine grid		Coarse grid	
n_x	192	N_x	48
n_y	64	N_y	16
Δx	3.125×10^{-2}	ΔX	0.125
Δy	3.125×10^{-2}	ΔY	0.125

studies performed in Prague [29–31]. For further numerical details concerning the numerical model used to perform the simulations (integration methods, timesteps, grids characteristics), refer to Refs. [5–8, 24].

6.4.3 *Results*

The aims of this numerical simulation are to compare the results of the Landau two-fluid model (Chap. 2) and the extended two-fluid model (Sect. 3.4.2), and to investigate low-viscosity effects by means of the two-fluid extended model.

First of all, it is seen that though in the initial situation the distribution of positive and of negative vortices is homogeneous across the channel, in the final steady state the concentration of vortices becomes very inhomogeneous. Indeed, positive vortices are seen to accumulate on the right-hand side of the channel and negative ones on the left-hand side of the channel, because of the action of the second term on the right-hand side of Eq. (6.4.1), expressing the coupling between the heat flux and the polarized vortices, and which separates positive from negative vortices, the first ones going to the right and the second ones to the left of the heat flux.

Comparison of Usual Two-Fluid Model and Extended Two-Fluid Model
In Figs. 6.4 and 6.5, the numerical results with the usual two-fluid model and with the extended two-fluid model are reported, respectively, by assuming the same value for

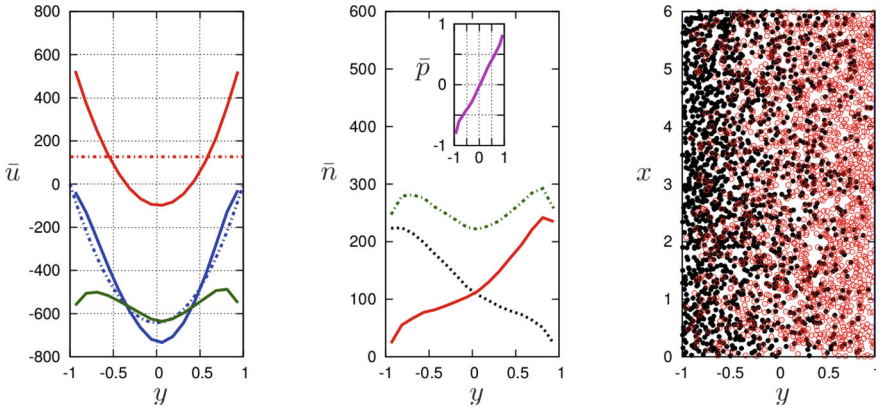


Fig. 6.4 Numerical results for the distribution of vortices submitted to a steady heat flow upwards along the x axis, using for the usual (Landau) two-fluid model: (left) profile of superfluid component $\mathbf{v}_s$ (red line), the normal component $\mathbf{v}_n$ (blue line) compared with the Poiseuille profile (blue dotted line), the flat superfluid component $\mathbf{v}_s$ in the absence of vortices (red dotted line) and the counterflow velocity $\mathbf{v}_{ns} = \mathbf{v}_n - \mathbf{v}_s$ (green line); (middle) the number of vortex points per unit area in the channel (green dotted line), the number of positive vortex points per unit area (red line), the number of negative vortex points per unit area (black dotted line) and the polarization $\mathbf{p}$ (purple line); (right) distribution of positive (red) and negative (black) vortex points inside the channel when a steady state has been achieved. Figure from [7]

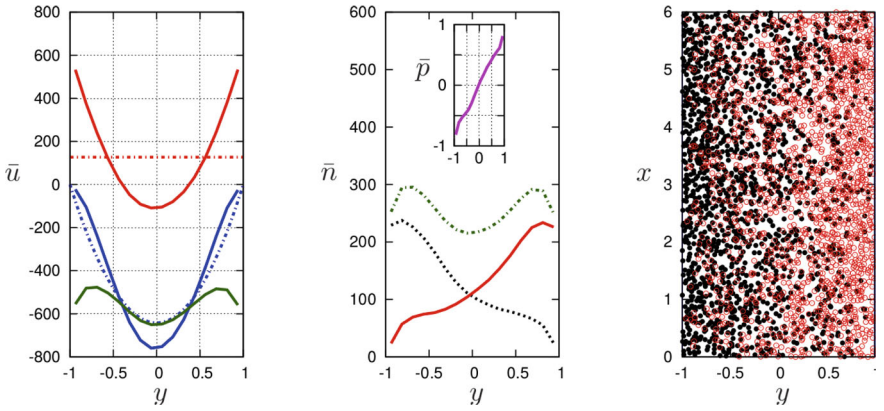


Fig. 6.5 Numerical results for the extended two-fluid model with the same viscosity as in the analysis in Fig. 6.4: (left) profile of superfluid component $\mathbf{u}^{(s)}$ (red line), the normal component $\mathbf{u}^{(n)}$ (blue line) compared with the Poiseuille profile (blue dotted line), the flat superfluid component $\mathbf{u}^{(s)}$ in the absence of vortices (red dotted line) and the counterflow velocity $\mathbf{u}^{(ns)} = \mathbf{u}^{(n)} - \mathbf{u}^{(s)}$ (green line); (middle) the number of vortex points per unit area in the channel (green dotted line), the number of positive vortex points per unit area (red line), the number of negative vortex points per unit area (black dotted line) and the polarization $\mathbf{p}$ (purple line); (right) the distribution of the positive (red) and negative (black) vortex points inside the channel once a statistically steady state has been achieved. Figure from [7]

the viscosity of the respective normal component (as reported in Table 6.1). The figure on the left shows the profile of the velocity of the superfluid (red line) and normal fluid (blue line) components across the channel. The latter is compared with the parabolic profile of the Poiseuille flow (blue dotted line). Even though the simulations are bi-dimensional, both models successfully mimic the experiments [32]. In our simulations, we have not taken into account the superfluid entropy, which, according to Putterman [33], is estimated of about $s_s = 0.02s$ with s the total entropy. If this value of s_s is considered, the values of the two velocities have to be modified as $\overline{\mathbf{u}}^{(s)}_{s_s \neq 0} = 0.98\overline{\mathbf{u}}^{(s)}$ and $\overline{\mathbf{u}}^{(n)}_{s_s \neq 0} = 0.98\overline{\mathbf{u}}^{(n)}$. A non-zero superfluid entropy changes the profile of the two components making the normal component a little bit closer to the results of the experiments.

From a rough comparison between the results shown in Figs. 6.4 and 6.5, we do not notice a strong difference between the results of the two models, apart from that reported above. The explanation for this small difference is found in the strength of the term $\left(\frac{\rho^{(s)}}{\rho^{(n)}}\mathbf{u}^{(s)} \cdot \nabla\right)\mathbf{u}^{(n)}$, which is absent in Eq. (6.4.4) of the usual two-fluid model, because in it $\mathbf{u}^{(n)} \cdot \nabla\mathbf{u}^{(n)}$ appears instead of the term $\mathbf{v} \cdot \nabla\mathbf{u}^{(n)}$ which appears in (6.4.4). For this reason in the next subsection, we consider a numerical experiment with lower normal viscosity, which makes the relative influence of this term more relevant.

To do that we perform a numerical experiment employing the two-fluid extended model with the same parameters as in the previous paragraph, apart from the viscosity,

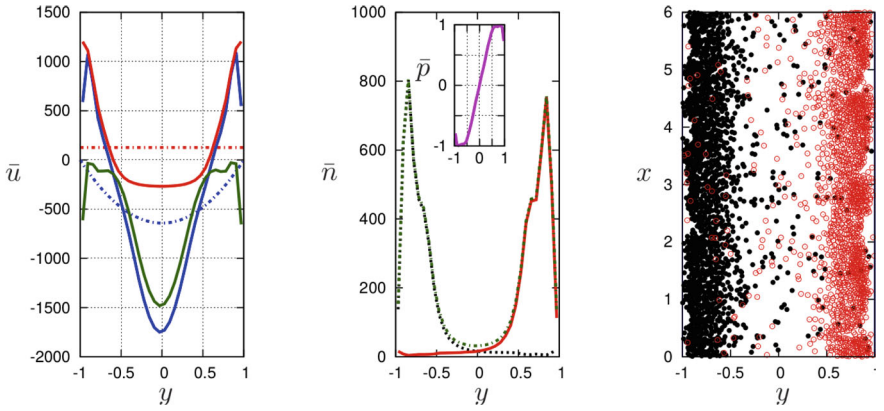


Fig. 6.6 Numerical results for the extended two-fluid model with a viscosity 1/5 of the viscosity assumed in Figs. 6.4 and 6.5: (left) profile of superfluid component $\mathbf{u}^{(s)}$ (red line), the normal component $\mathbf{u}^{(n)}$ (blue line) compared with the Poiseuille profile (blue dotted line) and the counterflow velocity $\mathbf{u}^{(ns)} = \mathbf{u}^{(n)} - \mathbf{u}^{(s)}$ (green line); (middle) the number of vortex points per unit area in the channel (green dotted line), the number of positive vortex points per unit area (red line), the number of negative vortex points per unit area (black dotted line) and the polarization $\mathbf{p}$ (purple line); (right) distribution of positive (red) and negative (black) vortex points inside the channel once a steady state has been achieved. Figure from [7]

which is set equal to 1/5 of its actual value (used in the previous simulations). The numerical results are shown in Fig. 6.6. The smaller value of the viscosity makes the other terms of Eq. (6.4.4) relatively stronger. Note that the net separation of the vortex points in Fig. 6.6 (right) is more drastic than in Figs. 6.4 and 6.5.

The low-viscosity case emphasizes that the smaller viscosity makes the inertial and the mutual friction terms relatively stronger, causing the results shown in Fig. 6.6. In particular, the mutual friction term moves the vortex points farther apart, leading to a vanishing value of the vortex density in the middle of the channel and very large in the proximity of the walls. This produces the observed normal fluid velocity profile, given that the normal fluid volume flow rate is constant, being determined by the constant heat released by the heater.

Furthermore, the inertial term contributes to the steepness of the profile of the normal component because nonlinearity becomes stronger than the viscous term, which fosters the levelling of the velocity profile.

In [26], we have considered the same numerical experiment removing the condition that the normal component is zero on the wall, namely, the normal component can slip along the wall. The results are similar to those shown in this section, but the profiles of the components are flatter than those found for a null value on the wall, especially in the case $\mathbf{u}^{(n)}(x, \pm D/2) = (-250, 0)$.

6.5 Heat Flow and Vortex Flow in Two-Dimensional Channel with a Central Obstacle

In Sect. 6.4, we have analysed heat flux and vortex motion in a two-dimensional channel without any obstacles by numerical experiments. In this section, we include a circular (cylindrical) obstacle at the centre of the channel, in order that the flow lines are curved, in contrast with the straight flow lines in the previous section (see Fig. 6.7). This makes the situation physically richer but more complex than the previous one. In Sect. 6.7, we will also consider the behaviour of small neutrally buoyant suspended particles added in order to visualize the flow. Indeed, the observation of suspended particles in Helium II has become a blossoming technique for the visualization of the flow of the normal component which, because of its viscosity, is able to drag the observation particles [34–37]. However, the presence of vortices may produce some modifications of the motion of particles with respect to the flow of the normal component, which must be examined in detail.

Here we deal with the experiments of Zhang and Van Sciver (ZVS) [38] referring to the flow along a channel having a cylindrical obstacle at the centre. Although their experiments were three dimensional we assume, as starting simplification, a bi-dimensional flow. The situation is sketched in Fig. 6.7 with a heat flux $\mathbf{q}$, high enough to produce vortices, and a finite temperature ($T = 2.03$ K and $T = 1.65$ K), reflecting the relative amount of normal and superfluid components.

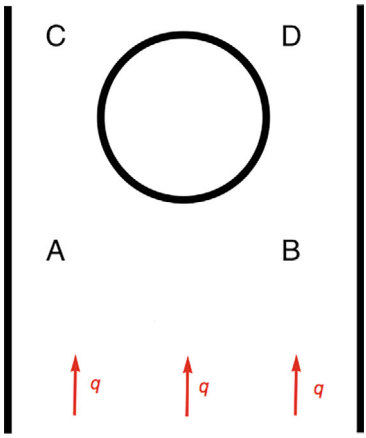


Fig. 6.7 A sketch of the experimental setup performed by Zhang and Van Sciver [38], discussed in Sect. 6.5. The channel is full of superfluid helium, with heat flowing upwards along the vertical axis, and there is a cylindrical obstacle at the center. A, B, C and D are labels for four different regions whose corresponding behaviour is discussed in the text

6.5.1 Experimental Observations

In the experiments by Zhang and Van Sciver [38], depending on the value of the imposed heat flow, one may observe big polarized eddies upflow and downflow of the cylinder, constituted by the local accumulation of polarized vortices (Fig. 6.8).

Zhang and van Sciver (see Fig. 1 in [38]) use a cylinder of 6.35 mm diameter in a channel of width 19.5 mm. They study two situations: (a) $q = 4$ Kw/m^2 at 1.6 K and (b) $q = 11.2$ Kw/m^2 at 2.03 K. In the first situation, they observe two eddies upflow of the cylinder as referred to the direction of the heat flux (regions A and B in Fig. 6.7). The eddy in the right-hand side (region B) is of $-$ type (i.e. clockwise) and that on the left-hand side (region A) is of $+$ type (i.e. counterclockwise). The two mentioned eddies are situated in approximately symmetric positions with respect to the axis of the circle in the direction of the heat flux. Downflow in the direction of the heat flow there are two small eddies just in contact with the rear side of the cylinder (regions C and D in Fig. 6.7). The diameter of the two first eddies (regions C and D) is approximately 1.87 mm. The distance of the centre of the eddies to the centre of the cylinder is 7.47 mm.

In the second situation, analogous results are found, but the eddies are considerably bigger. The two downstream eddies (regions C and D) are much elongated along the direction of the flow; they have 7.47 mm in this direction and 6.35 mm in the transversal direction. The eddies are practically in contact with the rear walls of the cylinder and the mutual separation of the centres of the two eddies is 6.35 mm. The upstream eddies have a characteristic elongation of 11.75 mm and have their centres at some 8.59 mm from the centre of the cylinder, at some 5.75 mm before the axis of the cylinder transversal to the flow.

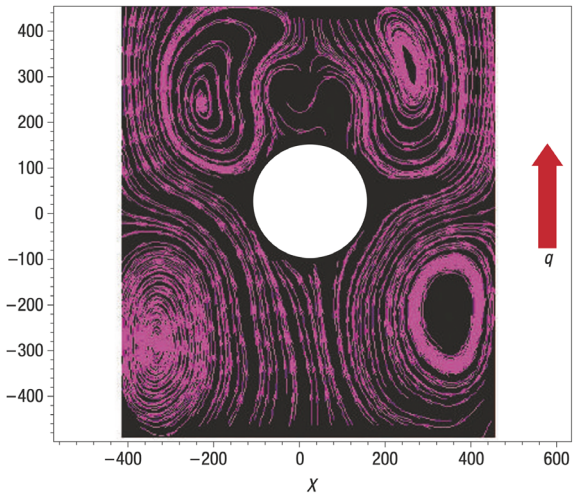


Fig. 6.8 Sketch of the heat flow lines in a two-dimensional channel with a central cylindrical obstacle [38], as the one shown in Fig. 6.7

The eddies [39, 40] could be interpreted as the usual eddies produced in the rear side of a cylinder in a viscous flow with Reynolds number higher than 10 [41, 42]. There are, however, two apparent problems with this interpretation. The first one is that situation (a) is characterized by a Reynolds number of 41000 and situation (b) by a Reynolds number 21000. Thus, one would expect that the rear eddies are larger in the first situation than in the second one. But Zhang et al. used in their definition of the Reynolds number the total mass density $\rho = \rho_n + \rho_s$. If the Reynolds number corresponding to the density of the viscous component is taken, this Reynolds number should be multiplied times ρ_n/ρ, whose value is 0.18 at $T = 1.6$ K and 0.65 at $T = 2.03$ K. The modified Reynolds numbers are then 7400 and 13000, respectively. Thus, it seems logical that the eddies are longer in the second case than in the first one.

The second problem is that, even at these lower Reynolds numbers, the eddies should be larger than actually observed. However, if the velocity of $\mathbf{v}_n$ actually measured from the PIV (particle-image-velocity) is used (namely, $v_n = 26, 3$ mm/s for situation (a) and $v_n = 28, 6$ mm/s for situation (b)) instead of the counterflow velocity, the Reynolds numbers in term of the diameter are of the order of 2000 and 5000 approximatively, i.e. considerably lower than the values given by Zhang et al., which are referred to the counterflow velocity. Then, the size disparity is not so high as that which could be expected from the Reynolds numbers.

The eddies A, B could be the consequence of a separation of + and − vortices because of the heat flow, as seen in Sect. 6.4, but in a more complicated geometry because of the curvature of the flow lines around the obstacles. Indeed, though this kind of separation seems opposite to the one found in Sect. 6.4, where negative vortices accumulated near the left wall and positive ones near the right wall of the channel, further reflection indicates that with respect to the heat flows along each side of the channel, the one on the right of the cylinder (B) accumulates negative vortices at its left, whereas the one on the left of the cylinder (A) accumulates positive vortices at its right.

6.5.2 Vortex Separation Near the Cylinder

In Sect. 6.4, we have seen that a heat flux induces a separation of positive and negative vortices, because of the lateral forces of respectively opposite directions acting on them. But, in the present section, the flow lines are curved, and the separation process of positive and negative vortices is more complicated. In this subsection, we analyse this process. An accumulation of vortex lines of the same sense could induce macroscopic eddies in the motion of the suspended particles. Since the vortex lines are assumed to be equally distributed far away from the cylinder, it is convenient to see how the cylinder contributes to separation and accumulation of + and − vortices.

The central element of our analysis is the equation of motion of the vortices, which the reader can find in Chap. 9 (Eq. (9.1.6)) for more details. Now, we neglect the contribution from the third term in (9.1.6), proportional to α', and we add a further

term $\mathbf{F}_{susp}$ which takes into account the interactions between vortices and suspended particles [43–50] in such a way that (9.1.6) becomes

$$\frac{d\mathbf{s}'}{dt} = \mathbf{v}_s + \mathbf{v}_{si} + \alpha\mathbf{s}' \times (\mathbf{v}_n - \mathbf{v}_s - \mathbf{v}_{si}) + \mathbf{F}_{susp}. \tag{6.5.1}$$

In Eq. (6.5.1) $\mathbf{s}'$ is the tangent vector to the vortex line, $\mathbf{v}_s$ is the superfluid component velocity, $\mathbf{v}_{si}$ is the self-induced velocity, α is the friction coefficient, and $\mathbf{v}_n$ is the velocity of the normal component. Equation (6.5.1) holds in the two-fluid extended model too, where $\mathbf{v}_s$ is replaced by $\mathbf{u}_s$ and $\mathbf{v}_n$ by $\mathbf{u}_n$. The suspended particles in Helium II interact with vortices, through the force $\mathbf{F}_{susp}$.

The self-induced velocity contains the effects of the interactions between vortices, due to their curvature; in the presence of walls, it also contains the effects of the image vortices used to cancel the normal component of the velocity induced by the vortex on the wall. Since we are considering rectilinear vortices perpendicular to the plane in which the flow takes place, the self-induced velocity will be zero, but we must take into account the influence of the image vortices used to describe the effects of the cylinder.

To describe the counterflow on a horizontal plane, we assume that the normal component flows along the y-axis towards increasing values of y with speed $\mathbf{v}_n(x, y)$ and that the superfluid component flows in the opposite direction with speed $\mathbf{v}_s(x, y)$. We assume that far from the cylinder, of radius a, $\mathbf{v}_n$ and $\mathbf{v}_s$ satisfy the counterflow condition of vanishing mass flow, i.e. $\rho_n\mathbf{v}_n + \rho_s\mathbf{v}_s = 0$, ρ_n and ρ_s being the respective mass densities of the normal and superfluid components. Finally, we assume that the superfluid carries along with it + and − vortices, which, far from the cylinder (i.e. for $y >> a$), are mixed in a homogeneous way, with vanishing average polarity (i.e. there are so many + vortices as − vortices everywhere).

To understand the eddy formation near the cylinder we study, first of all, the effect of the cylinder on individual vortices, and we will show that it leads to a separation of + and − vortices downflow of the superfluid motion (or, in other terms, upflow with respect to the normal fluid motion). In Figs. 6.10 and 6.11, the flow lines of $\mathbf{v}_n - \mathbf{v}_s - \mathbf{v}_{spwall}$ are sketched. Because of the latter, namely, the vortex motion induced by the cylinder walls, the + vortices tend to rotate around the cylinder in counterclockwise direction whereas − vortices tend to rotate in the opposite way. We also consider the friction-induced velocity, namely:

$$\frac{d\mathbf{s}'}{dt} = \alpha\mathbf{s}' \times (\mathbf{v}_n - \mathbf{v}_s - \mathbf{v}_{spwall}). \tag{6.5.2}$$

The distributions of $\mathbf{v}_n$, $\mathbf{v}_s$, and $\mathbf{v}_{spwall}$ are qualitatively known, as they correspond to the motion of a perfect fluid and of a viscous fluid around the cylinder, respectively, whereas $\mathbf{v}_{spwall}$ is circular with the direction mentioned above. The vector $\mathbf{s}' \times (\mathbf{v}_n - \mathbf{v}_s - \mathbf{v}_{spwall})$ leads to a separation of + and − vortices because $\mathbf{s}'$ has opposite direction for + and − vortices. The + and − vortices are expected to accumulate in the downflow part of the superfluid flow (regions A and B). Approaching the cylinder,

the vortices begin to deplace laterally in the direction shown, and the effect of the separation will increase with time, i.e. it will be higher downstream than upstream. The separation effect has been clearly seen in Sect. 6.4, namely, the positive vortices move to the right in the heat flux direction and the negative ones to the left. This vortex separation causes a bottleneck in the proximity of the cylinder. The formation of spots of positive and negative vortices leads to the formation of big eddies in front of the cylinder (regions A and B).

Besides the vortex separation effects, the flow of the viscous component will also show some specific features, namely, the formation of two eddies of opposite sign downflow of the direction of the viscous component. The formation of these vortices is expected to begin at Reynolds number $Rey = 2a\rho_n v_n/\eta_n$ of the order 10 [41], and they will increase in size for increasing Reynolds numbers. For $Rey \approx 100$ their size is expected comparable to the diameter a of the cylinder [51].

6.6 Motion of Visualization Particles

Another relevant aspect of superfluid dynamics is the motion of visualizing particles. For a right interpretation of the results from this visualization, it is needed to know what is the relation between such particles and the actual motion of the superfluid [34–37, 52].

In the situation, we are considering some particles of polystyrene (radius 1.7 μm) are immersed into the helium. In a first approximation, we assume that vortices move downwards with velocity $\mathbf{v}_s$ and the particles move upwards with velocity $\mathbf{v}_n/2$, according to detailed discussions on the motion of visualization particles in rectilinear flows. Normal fluid makes the Stokes drag force on the particles

$$\mathbf{F}_{drag} = 6\pi r_p \eta(\mathbf{v}_n - \mathbf{v}_p), \tag{6.6.1}$$

with r the radius of the particle, $\mathbf{v}_p$ the velocity of the particle, and η the viscosity of the normal component [41, 53], and the particle-vortex interaction force is [54]

$$\mathbf{F}_{inter} = \frac{\rho_s \kappa^2}{4\pi} \ln\left(\frac{r_p}{a_0}\right) \hat{\mathbf{u}}, \tag{6.6.2}$$

which refers to a single vortex whose core radius is a_0, $\hat{\mathbf{u}}$ being the unitary vector along $\mathbf{v}_v - \mathbf{v}_p$ with $\mathbf{v}_v$ the vortex velocity (which we assume to be $\mathbf{v}_s$). In rectilinear flows, both forces are collinear but it is not true when streamlines of the flow are not rectilinear. Analysing this point is crucial for the visualization method, in order to see whether the visualization particles directly yield a faithful information on the actual fluid motion, or whether some corrections are needed to infer the actual fluid motion from the observed trajectories of the particles.

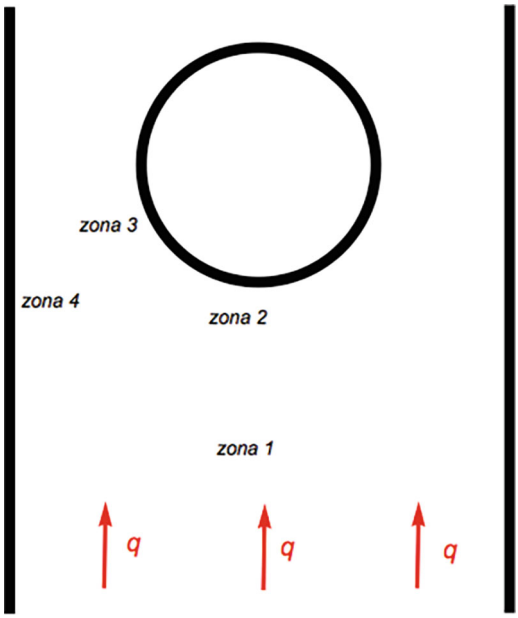


Fig. 6.9 A sketch of the experimental setup performed by Zhang and Van Sciver [38], discussed in Sect. 6.5 for the distribution of the forces on suspended particles in the different zones 1, 2, 3, 4. The channel is full of superfluid helium, with heat flowing upwards along the vertical axis, and there is a cylindrical obstacle at the center

The direction of the several forces, acting on each particle, depends on the zone where the particles are, as shown in Fig. 6.9. The central disc represents the solid cylindrical obstacle at the centre of the channel. In Figs. 6.10 and 6.11, we sketch these forces in the zones referred to in the present figure.

Zhang and Van Sciver performed experiments around the cylinder analysed in Sect. 6.5 in two cases: case (a) heat flux $q = 4\,\mathrm{kW\,m^{-2}}$, $T = 1.6$ K, and $L_0 = 1 \times 10^{10}\,\mathrm{mm^{-3}}$; case (b) heat flux $11.2 = 4\,\mathrm{kW\,m^{-2}}$, $T = 1.2.03$ K, and $L_0 = 2.6 \times 10^{10}\,\mathrm{mm^{-3}}$ [38]. The velocities $\mathbf{v}_n$, $\mathbf{v}_s$, and $\mathbf{v}_p$ in cases (a) and (b) are [46]

$$|\mathbf{v}_p| \simeq \frac{1}{2}|\mathbf{v}_n| \simeq 2.3|\mathbf{v}_s| \quad \text{(case a)}, \qquad |\mathbf{v}_p| \simeq \frac{1}{2}|\mathbf{v}_n| \simeq \frac{1}{4}|\mathbf{v}_s| \quad \text{(case b)}, \tag{6.6.3}$$

where $\rho_n/\rho \simeq 0.18$ (case a) at $T = 1.65$ K) and $\rho_n/\rho \simeq 0.66$ (case b) at $T = 2.03$ K). The ratios of the modulus of the forces (6.6.2) and (6.6.1), respectively, are

$$\frac{F_{inter}}{F_{drag}} \simeq 3.7 \quad \text{(case a)}, \qquad \frac{F_{inter}}{F_{drag}} \simeq 0.65 \quad \text{(case b)}, \tag{6.6.4}$$

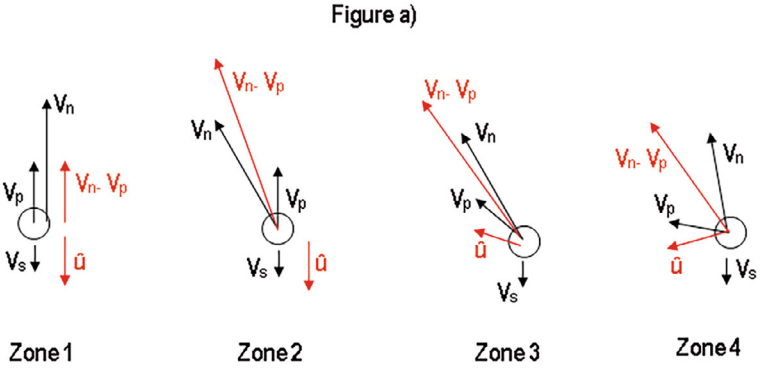


Fig. 6.10 Sketch of the velocities $\mathbf{v}_n$, $\mathbf{v}_s$, and $\mathbf{v}_p$ in the different zones of Fig. 6.7. Red vectors mean the directions of the drag force (collinear with the vector $\mathbf{v}_n - \mathbf{v}_p$), and the vortex-particle interaction force, having direction $\mathbf{v_s} - \mathbf{v_p}$. Figure refers to the case (a) of the Zhang and Van Sciver's experiments: $q = 4$ kW m^{-2}, $T = 1.6$ K, and $L_0 = 1 \times 10^{10}$ mm^{-3} [38]

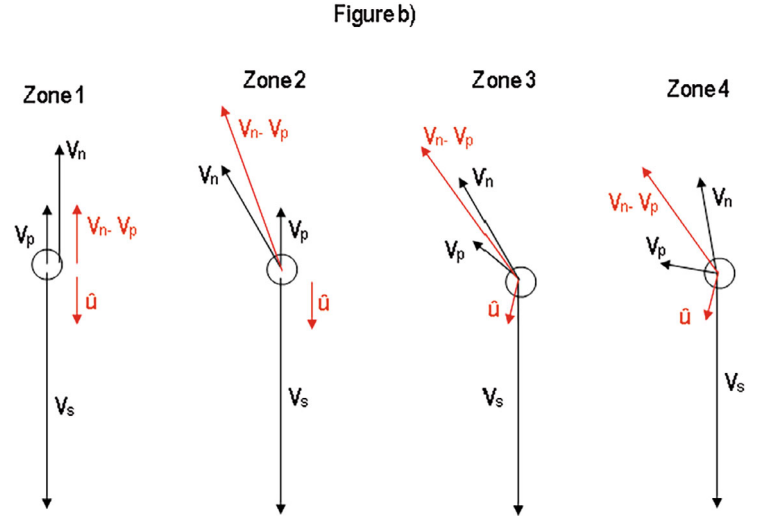


Fig. 6.11 Sketch of the velocities $\mathbf{v}_n$, $\mathbf{v}_s$, and $\mathbf{v}_p$ in different zones of Fig. 6.7. Red vectors mean the directions of the drag force (collinear with the vector $\mathbf{v}_n - \mathbf{v}_p$) and vortex-particle interaction force, having direction $\hat{\mathbf{u}}$. Figure refers to the case (b) of the Zhang and Van Sciver's experiments: $q = 11.2 = 4$ kW m^{-2}, $T = 1.2.03$ K, and $L_0 = 2.6 \times 10^{10}$ mm^{-3} [38]

with $v_n = 26.3$ mm/s (case a) and $v_n = 28.6$ mm/s (case b). It follows that one cannot neglect the influence of $\mathbf{F}_{inter}$ with respect to $\mathbf{F}_{drag}$. This is well known in rectilinear flows, when the force due to the vortices changes the particles velocity with respect to the normal component velocity, but not when these forces are not collinear. The motion of the particle with respect to that of the normal component differs not only in the modulus of the speed, but also in its direction. As pointed before,

$\mathbf{F}_{inter}$ refers to the interaction with a single vortex. The total force will depend on the number of vortices per unit area, which is different in cases a) and b) of Fig. 2 in [38], namely, $L \simeq 1 \times 10^{10}\,\mathrm{m}^{-2}$ (case a) and $L \simeq 2.6 \times 10^{10}\,\mathrm{m}^{-2}$ (case b), because of the different value of the heat flux.

Now, we examine the behaviour of the particles visualized in the experiments of Zhang and Van Sciver [38], following the motion of a particle which moves from the bottom of the plane of figures in the presence of a cylinder in the centre of the channel (Fig. 6.7), and distinguishing the two cases, respectively, in Fig. 6.10 (case a)) and in Fig. 6.11 (case b)). After immersing the particle in the fluid in the zone below the cylinder (Zone 1 of Figs. 6.7, 6.10, and 6.11), it moves upwards under the influence of the heat flux [or, in other term, under the influence of the drag force of the normal component (Stokes force)] (6.6.1) and the interaction force (6.6.2) with the vortices moving in the opposite direction with respect to the particle and normal fluid—both forces are parallel in this zone but of opposite direction. The resulting force acting on the particle will increase or decrease its speed, according to relative value of $\mathbf{F}_{inter}$ and $\mathbf{F}_{drag}$.

When the particle reaches the zone near the cylinder (Zone 2 of Figs. 6.7, 6.10, and 6.11), the normal fluid and the superfluid components change direction following the standard Newtonian viscous behaviour and acquire a horizontal component. So the forces acting on the particle are no longer parallel (drag force has direction $\mathbf{v}_n - \mathbf{v}_p$ whereas $\mathbf{F}_{inter}$ has downward direction parallel to the walls), so the resulting force moves the particle towards the walls. After that, the particle encounters other vortices which act on the particle by the interaction force, whereas the drag force always changes its direction because $\mathbf{v}_n$ and $\mathbf{v}_p$ change directions (see Figs. 6.10 and 6.11). So the resulting force moves some particles downwards, forming big eddies below the cylinder, and other particles follow the normal fluid behind the cylinder.

According to Figs. 6.10 and 6.11, the particle does not follow the motion neither of $\mathbf{v}_n$ nor of $\mathbf{v}_s$, and relating them is not easy. In particular, it may be asked whether the eddies are composed only of particles or whether they are an artefact of the visualization procedure. Even if the normal fluid actually describes some eddies, it could be asked how they are related to the eddies visualized by the motion of the particles. Note that (1) each particle encounters more vortices in case (b) than in case (a), confirming the higher deviation from its trajectory, because the total force due to the eddies will be higher. In fact, in the case a) $L = 1 \times 10^{10}\,\mathrm{m}^{-2}$ which corresponds to the intervortex distance $\delta = 10\,\mu\mathrm{m}$, whereas in the case (b) $L = 2.6 \times 10^{10}\,\mathrm{m}^{-2}$ and intervortex distance $\delta = 6.2\,\mu\mathrm{m}$. Note also that in zones 3 and 4 of Fig. 6.7 the number of vortices per unit area is increased because of the increasing value of V_{ns}. The maximum value of L is in the zone where the distance of the cylinder from the walls is minimum. There we find $L_1 = 1.48\,L$ with $\delta = 8.22\,\mu\mathrm{m}$ for the case (a) and $\delta = 5.01\,\mu\mathrm{m}$ for the case (b). Note also that (2) from relations (6.6.4) one would expect that in case (a) the eddies would be bigger than in case (b) because of the higher value of the vortex-interaction force relative to the drag force. But one has to take also into account that in case (a) the velocity of the particles is much higher than the vortex velocity (corresponding to the superfluid velocity) than in case (b), where the opposite situation is found.

6.7 Geometry of the Vortex Tangle: Polarization and Anisotropy

In Sects. 6.4 and 6.5, we have considered spatial inhomogeneity of the vortex line density and polarization of vortices in a two-dimensional setting. Going to a three-dimensional situations requires some tensorial aspects of the description, which are introduced in this section.

The interest in these geometrical aspects of turbulent tangles, combined with statistical ingredients and with topological and geometrical measures of complexity [55–57] has been enhanced by the increasing measurement abilities and microscopic understanding of the essential features governing the evolution of quantized vortex filaments [2–4].

To deal with these topics it is necessary to define the particular averages (volume averages or vortex length averages) being used in defining or calculating the several quantities. For instance, the counterflow velocity V_{ns} is defined as the modulus of the averaged counterflow velocity vector

$$\mathbf{u}_{ns} = [\mathbf{u}_{ns}]_{av} = \frac{1}{\Lambda} \int \mathbf{u}_{ns} d\Lambda, \tag{6.7.1}$$

over a small mesoscopic volume Λ. A different average is on all vortices contained in the same volume Λ, that is,

$$< \cdot >= \frac{1}{L\Lambda} \int \cdot \, d\xi, \tag{6.7.2}$$

where L is the vortex line density in the volume Λ, ξ is the arc length along the vortex line, and the integral is made on all vortices contained in Λ. The two averages are linked by the relation $[\cdot]_{av} = L < \cdot >$. When dealing with homogeneous situations, the average may be taken in the whole volume of the system, but in inhomogeneous ones it is important to specify which is the small local volume taken as a reference for the average.

The essential role in the geometry and dynamics of a vortex tangle is played by three vectors: $\mathbf{s}'$, the unit vector tangent to the vortex line $\mathbf{s}(\xi)$ describing the vortex line, $\mathbf{s}''$, the local curvature of the vortex line, and $\mathbf{s}' \times \mathbf{s}''$. To describe the information on the statistical distribution of $\mathbf{s}'$, its first and second moments are particularly relevant, namely, the polarity vector $\mathbf{p}$ defined as

$$\mathbf{p} =< \mathbf{s}' >= \frac{1}{\Lambda L} \int \mathbf{s}' \, d\xi, \tag{6.7.3}$$

which describes the prevalent orientation of the lines in the tangle, and which has been used in a two-dimensional setting (where the concept of polarization is rather intuitive) in Sects. 6.4 and 6.5, and the second-order tensor

$$< \mathbf{s}'\mathbf{s}' >= \frac{1}{\Lambda L}\int \mathbf{s}'\mathbf{s}'\, d\xi, \tag{6.7.4}$$

where $\mathbf{s}'\mathbf{s}'$ is the diadic product (in index notation $[\mathbf{s}'\mathbf{s}']_{ij} = s'_i s'_j$) and the integral is made on all vortices contained in a small volume Λ.

The polarity along a given direction is defined as the average projection of $\mathbf{s}'$ along that direction. If one chooses this direction as x-axis, one has

$$p_x =< s'_x >= \frac{1}{\Lambda L}\int \hat{\mathbf{x}}\cdot\mathbf{s}'\, d\xi. \tag{6.7.5}$$

Note that $|\mathbf{p}| \in [0, 1]$ measures the directional anisotropy of the tangent to the vortex lines: in particular, $|\mathbf{p}| = 1$ for a system of parallel vortices and $|\mathbf{p}| = 0$ for isotropic tangles.

Two important measures of the anisotropy of the tangle are the tensors $< \mathbf{U} - \mathbf{s}'\mathbf{s}' >$ and $< \mathbf{W}\cdot\mathbf{s}' >$, where $\mathbf{U}$ is the unit matrix and $\mathbf{W}$ the Ricci tensor, a completely antisymmetric third-order tensor which makes the second matrix of (6.7.6) an antisymmetric matrix. In index and matrix notation, one has

$$[\mathbf{W}\cdot\mathbf{s}']_{ij} = W_{ijk}s'_k, \qquad \mathbf{W}\cdot\mathbf{s}' = \begin{pmatrix} 0 & s'_3 & -s'_2 \\ -s'_3 & 0 & s'_1 \\ s'_2 & -s'_1 & 0 \end{pmatrix}. \tag{6.7.6}$$

Note that $< \mathbf{s}'\mathbf{s}' >$ and $< \mathbf{U} - \mathbf{s}'\mathbf{s}' >$ project any vector on the directions parallel to $\mathbf{s}'$ and orthogonal to $\mathbf{s}'$, respectively. Consider, for instance, their application to the *microscopic* counterflow velocity $\mathbf{u}_{ns}$. One has $\mathbf{s}'\mathbf{s}'\cdot\mathbf{u}_{ns} = \mathbf{s}'(\mathbf{s}'\cdot\mathbf{u}_{ns})$ and, accordingly,

$$(\mathbf{U} - \mathbf{s}'\mathbf{s}')\cdot\mathbf{u}_{ns} = \mathbf{u}_{ns} - \mathbf{s}'(\mathbf{s}'\cdot\mathbf{u}_{ns}) = -\mathbf{s}'\times[\mathbf{s}'\times\mathbf{u}_{ns}]. \tag{6.7.7}$$

Furthermore, $\mathbf{W}\cdot\mathbf{s}'\cdot\mathbf{u}_{ns} = -\mathbf{s}'\times\mathbf{u}_{ns}$.

An alternative way to characterize the anisotropy of the tangle has been used by Schwarz [3, 11], by means of the scalar quantities $I_\parallel$ and $I_\perp$

$$I_\parallel = \frac{1}{\Lambda L}\int [1 - (\mathbf{s}'\cdot\hat{\mathbf{r}}_\parallel)^2]d\xi, \tag{6.7.8}$$

$$I_\perp = \frac{1}{\Lambda L}\int [1 - (\mathbf{s}'\cdot\hat{\mathbf{r}}_\perp)^2]d\xi, \tag{6.7.9}$$

where $\hat{\mathbf{r}}_\parallel$ and $\hat{\mathbf{r}}_\perp$ are the unit vectors for directions parallel and perpendicular to the $\mathbf{u}_{ns}$ direction. In fact, functions $I_\parallel$ and $I_\perp$ are closely related to the tensor $< \mathbf{U} - \mathbf{s}'\mathbf{s}' >$. If one chooses a frame with the first axis in the direction of $\mathbf{u}_{ns}$, and assumes symmetry in the y, z-plane, one obtains

$$< \mathbf{U} - \mathbf{s}'\mathbf{s}' >= \begin{pmatrix} I_{\parallel} & 0 & 0 \\ 0 & I_{\perp} & 0 \\ 0 & 0 & I_{\perp} \end{pmatrix}. \tag{6.7.10}$$

The description of the anisotropy using the tensor $< \mathbf{U} - \mathbf{s}'\mathbf{s}' >$ is equivalent to the Schwarz description, using the two scalar quantities $I_{\parallel}$ and $I_{\perp}$, when the tangle is axially symmetric. But $< \mathbf{U} - \mathbf{s}'\mathbf{s}' >$ is also able to describe the geometry of the tangle in the absence of axial symmetry, as, for example, in channels with high aspect ratio, or in simultaneous presence of counterflow and rotation, when the angular velocity $\mathbf{\Omega}$ is not collinear with the counterflow velocity $\mathbf{u}_{ns}$, as it happens, for instance, in neutron stars. Thus, tensor $< \mathbf{U} - \mathbf{s}'\mathbf{s}' >$ provides a unified description of several aspects of the global geometry of vortex tangle.

Other quantities characterizing the anisotropy of the vortex tangle but based on $\mathbf{s}''$ and on $\mathbf{s}' \times \mathbf{s}''$ instead than on $\mathbf{s}'$ are the vectors

$$\mathbf{I} = \frac{< \mathbf{s}' \times \mathbf{s}'' >}{< |\mathbf{s}''| >}, \qquad \mathbf{J} = \frac{< \mathbf{s}'' >}{< |\mathbf{s}''| >}. \tag{6.7.11}$$

The vector $\mathbf{I}$ describes the information on the distribution of $\mathbf{s}' \times \mathbf{s}''$, linked to the mean direction of the self-induced velocity $\tilde{\mathbf{v}}^{(i)}$, in the LIA approximation. The second vector $\mathbf{J}$ is tied to the vortex tension force [58].

Other statistical specifications refer to the probability of the number of closed vortex loops according to their respective length [55, 59, 60].

6.8 Polarization and Anisotropy in Rotating Counterflow Turbulence

A specially relevant situation of anisotropy is found in superfluid turbulence under the simultaneous influence of counterflow and rotation providing intermediate situations between an isotropic tangle (pure counterflow) and a totally anisotropic array of parallel vortices (pure rotation). We apply the formalism introduced in Sect. 6.7 to the geometrical description of such situation, and to the suitable generalization of the evolution equations of the heat flux and of the vortex length density.

6.8.1 Geometry of the Tangle

In homogeneous counterflow superfluid turbulence, far from the boundaries of the system, the vortex tangle consists mainly of closed loops, with vanishing polarity $\mathbf{p}$. In contrast, in the rectilinear vortex array in a rotating sample, the polarity p is 1 along the rotation axis. However, with counterflow and rotation, one expects some degree of polarization between 0 and 1 [61, 62].

The anisotropy of the tangle may be described using the tensor:

$$\mathbf{\Pi} = \mathbf{\Pi}^s + \mathbf{\Pi}^a, \tag{6.8.1}$$

with

$$\mathbf{\Pi}^s = \frac{3}{2} < \mathbf{U} - \mathbf{s}'\mathbf{s}' >, \qquad \mathbf{\Pi}^a = \frac{3}{2}\frac{\alpha'}{\alpha} < \mathbf{W}\cdot\mathbf{s}' >= \frac{3}{2}\frac{\alpha'}{\alpha}\mathbf{W}\cdot\mathbf{p}, \tag{6.8.2}$$

where α and α' are the coefficients appearing in (6.4.5). In this way, tensor $\mathbf{\Pi}$ contains the main quantities characterizing the anisotropy of the vortex tangle, namely, the first and second moments of $\mathbf{s}'$.

Pure Rotation

In most references about superfluids in rotating cylinders, one assumes that the vortex tangle is characterized by vortex filaments parallel to the rotation axis. But the vortex lines will always exhibit minute deviations with respect to the right line, described by $< (s'_y)^2 >=< (s'_z)^2 >\equiv \alpha''/\alpha \neq 0$, producing some mutual friction force parallel to the rotation axis. If one chooses the rotation axis as the x-axis the tensor Π is

$$\mathbf{\Pi}_R = \frac{3}{2}\left[\left(1-\frac{\alpha''}{\alpha}\right)\begin{pmatrix}0&0&0\\0&1&0\\0&0&1\end{pmatrix} + \frac{2\alpha''}{\alpha}\begin{pmatrix}1&0&0\\0&0&0\\0&0&0\end{pmatrix} + \frac{\alpha'}{\alpha}\begin{pmatrix}0&0&0\\0&0&1\\0&-1&0\end{pmatrix}\right]. \tag{6.8.3}$$

The normalization condition for s' imposes that $< (s'_x)^2 >= 1 - 2\alpha''/\alpha$.

Pure Counterflow

Counterflow superfluid turbulence is usually assumed isotropic; in this case, $\mathbf{\Pi}_H = \mathbf{U}$. But experimental observations [1] and numerical simulations [3] show some degree of anisotropy in the vortex line distribution, with vortices concentrated in planes orthogonal to $\mathbf{u}_{ns}$. Therefore, assuming $\mathbf{u}_{ns}$ in the direction of the x-axis and isotropy in planes orthogonal to it, one can choose for the symmetric part of tensor $\mathbf{\Pi}$ the following expression:

$$\mathbf{\Pi}^s_H = \frac{3}{2}\begin{pmatrix}2a&0&0\\0&1-a&0\\0&0&1-a\end{pmatrix}; \tag{6.8.4}$$

here a is an anisotropy parameter related to the second moments of s' as $< s'^2_x >= 1-2a$, $< s'^2_y >=< s'^2_z >= a$. It is linked to $I_\parallel$ and $I_\perp$, (6.7.8) and (6.7.9), by $I_\parallel = 2a$ and $I_\perp = 1-a$. The coefficient a may be determined through second sound attenuation propagating parallel and orthogonal to $\mathbf{u}_{ns}$.

A possible practical use of (6.8.4) could be in the expression for the thermal resistance of the tangle, by assuming for the production term $\sigma^{\mathbf{q}}$ appearing in (5.3.29) the expression

$$\sigma^{\mathbf{q}} = -\frac{1}{\tau_1}\mathbf{q} - K_f L \mathbf{\Pi}_H^s \cdot \mathbf{q}, \tag{6.8.5}$$

instead of $\sigma^{\mathbf{q}} = -\frac{1}{\tau_1}\mathbf{q} - K_f L\mathbf{q}$. If $\mathbf{q}$ is along the x-axis, $\mathbf{\Pi}_H^s \cdot \mathbf{q} = 2aq_x = (< s_y'^2 > + < s_z'^2 >)q_x$. This would reflect that the mutual resistance between vortex lines and heat flux is maximum when vortex lines are orthogonal to $\mathbf{q}$, and null when they are parallel. Thus, the contribution of the tangle to the thermal resistance would be $La\dot{Q}$ instead of $L\dot{Q}$. If $a \sim \dot{Q}^n$, and since $L \sim \dot{Q}^2$, one would get $(\Delta T)_{\text{tangle}} \sim La\dot{Q} \sim \dot{Q}^{3+n}$, instead of $(\Delta T)_{\text{tangle}} \sim L\dot{Q} \sim \dot{Q}^3$. For $n = 0.4$, one would have the experimental observation $(\Delta T)_{\text{tangle}} \sim \dot{Q}^{3.4}$ by Sato [63] instead of the standard Gorter–Mellink law $(\Delta T)_{\text{tangle}} \sim \dot{Q}^3$. A further explanation of the experimental exponent 3.4 in $(\Delta T)_{\text{tangle}} \sim \dot{Q}^{3.4}$ was proposed in [64] assuming that the coefficient α_n/β_v appearing in the Vinen equation (5.1.8) depends on the quantum Reynolds number as $(V_{ns}d/\kappa)^{0.2}$.

Concerning the antisymmetric part of tensor $\mathbf{\Pi}$ in pure counterflow, one observes that in TI-turbulence some vortex lines may begin and end at the boundaries, thus providing some contribution to the anisotropy. When the turbulence is fully developed, one can neglect the influence of pinned vortices and assume that $\mathbf{\Pi}_H^a$ is null.

Simultaneous Counterflow and Rotation

Under the simultaneous influence of counterflow and rotation, rotation tends to align vortex lines parallel to rotation axis, and counterflow velocity tends to produce a disordered tangle. In these situations, one has partially polarized tangles, and we assume that the total ensemble of vortex lines is a superposition of both contributions [65]

$$\mathbf{\Pi}^s = (1-b)\mathbf{\Pi}_H^s + b\mathbf{\Pi}_R^s, \tag{6.8.6}$$

$$\mathbf{\Pi}^a = (1-c)\mathbf{\Pi}_H^a + c\mathbf{\Pi}_R^a, \tag{6.8.7}$$

with b and c parameters between 0 and 1, depending on $\mathbf{\Omega}$ and $\mathbf{u}_{ns}$, describing the relative weight of the array of vortex lines parallel to $\mathbf{\Omega}$ and the disordered tangle of counterflow.

If $\mathbf{u}_{ns}$ is parallel to $\mathbf{\Omega}$ and there is cylindrical symmetry with respect to the rotation axis, assuming $\alpha'' = 0$ in $\mathbf{\Pi}_H$ and $\mathbf{\Pi}_R$, and choosing $\mathbf{u}_{ns}$ and $\mathbf{\Omega}$ in the direction of the x-axis, one has

$$\mathbf{\Pi}_H^s = \frac{3}{2}\begin{pmatrix} 2a & 0 & 0 \\ 0 & 1-a & 0 \\ 0 & 0 & 1-a \end{pmatrix}, \qquad \mathbf{\Pi}_R^s = \frac{3}{2}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{6.8.8}$$

Therefore, for the symmetric part of $\mathbf{\Pi}$, from (6.8.6) and (6.8.8) we get

$$\mathbf{\Pi}^s = \frac{3}{2} < \mathbf{U} - \mathbf{s}'\mathbf{s}' >= \frac{3}{2} \begin{pmatrix} 2a(1-b) & 0 & 0 \\ 0 & 1-a(1-b) & 0 \\ 0 & 0 & 1-a(1-b) \end{pmatrix}. \quad (6.8.9)$$

This means that $< s_x'^2 >= 1 - 2a(1-b)$, $< s_y'^2 >=< s_z'^2 >= a(1-b)$, and that $I_\parallel = 2a(1-b)$ and $I_\perp = 1 - a(1-b)$. Once the coefficient a is known, from experiments in pure counterflow [1], from (6.8.9) we can obtain b.

Tsubota et al. [66–70] have studied vortices in simultaneous presence of counterflow and rotation. They obtained from numerical simulations the polarization as a function of V_{ns} and Ω. As expected, the polarization decreases with V_{ns} at constant Ω and increases with Ω at constant V_{ns}. However, in proposal (6.8.1) for the description of anisotropy, also the second moments of $\mathbf{s}'$ appear, besides the first moment studied by these authors. Coefficient b appears in the expression for the attenuation of second sound, which is the most usual method to determine the value of L [65].

6.8.2 *Heat Evolution Equation in Anisotropic and Polarized Tangle*

In the presence of a vortex line described by $\mathbf{s}'(\xi)$, one may generalize the production term in the evolution equation for $\mathbf{q}$ as

$$\sigma^{\mathbf{q}} = -\frac{1}{\tau_1}\mathbf{q} + B_{HV} \int \mathbf{s}' \times (\kappa \mathbf{s}' \times \mathbf{q})\, d\xi + B'_{HV} \int \kappa \mathbf{s}' \times \mathbf{q}\, d\xi \quad (6.8.10)$$

integrated over the length of all the vortices. This expression is an extension of (4.2.6) along the length of the vortex lines.

Indeed, $\kappa \mathbf{s}'$ yields the local circulation around the unit length of vortices. The term in B_{HV} yields a dissipation term opposite to $\mathbf{q}$, which we have written in (6.8.5) as $-K_f L \mathbf{\Pi}^s_H \cdot \mathbf{q}$, with $L\mathbf{\Pi}^s_H$ yielding the total length of vortex lines perpendicular to $\mathbf{q}$, which are those contributing to resistance. The third term is null in completely isotropic tangles, but it yields a non-zero contribution in polarized tangles, of the order of $B'_{HV}\ \kappa \mathbf{p} \times \mathbf{q}$. This produces a lateral deviation of the heat flux, which is analogous to the deviation of an electric current in a metal submitted to an external magnetic field (Hall effect). Thus, if $\mathbf{p}$ points to the z-axis and the initial $\mathbf{q}$ points along the x-axis, $\mathbf{p} \times \mathbf{q}$ will indicate a deviation of $\mathbf{q}$ in the y-direction. By using the tensor $\mathbf{\Pi}$, (6.8.10) may be described by

$$\sigma^{\mathbf{q}} = -\frac{1}{\tau_1}\mathbf{q} - K_f L \mathbf{\Pi}^s \cdot \mathbf{q} - K_f L \frac{B'_{HV}}{B_{HV}} \mathbf{p} \times \mathbf{q}. \quad (6.8.11)$$

6.8.3 Vortex Line Density Evolution Equation in Anisotropic and Polarized Tangle

Vinen's equation (5.1.7) describes the evolution of L, in homogeneous well-developed and isotropic counterflow. Vinen's equation [3] has been generalized explicitly incorporating contributions of the tangle anisotropy and polarization [71]

$$\frac{dL}{dt} \simeq \alpha c_1 \mathbf{u}_{ns} \cdot \mathbf{I} L^{3/2} + \alpha' c_1 \mathbf{u}_{ns} \cdot \mathbf{J} L^{3/2} - \alpha \tilde{\beta} c_2 L^2, \tag{6.8.12}$$

with $\mathbf{I}$ and $\mathbf{J}$ defined in (6.7.11), and c_1 and c_2 dimensionless coefficients given by

$$c_1 = \frac{1}{\Lambda L^{3/2}} \int |\mathbf{s}''| d\xi, \qquad c_2 = \frac{1}{\Lambda L^2} \int |\mathbf{s}''|^2 d\xi. \tag{6.8.13}$$

In a non-isotropic tangle, c_1 and c_2 depend on the polarity vector $\mathbf{p}$. Vinen's equation (5.1.7) is obtained immediately putting $\alpha c_1 \cos\theta = \alpha_v$ and $\alpha \tilde{\beta} c_2 = \kappa \beta_v$, where θ is the angle between $\mathbf{u}_{ns}$ and $\mathbf{I} + \dfrac{\alpha'}{\alpha}\mathbf{J}$.

Experiments [1] and numerical simulations [2–4] show that the vortices in the tangle tend to concentrate in planes orthogonal to the external heat flux. This effect is described by the vector $\mathbf{I}$, namely, if $\mathbf{u}_{ns}$ is orthogonal to $\mathbf{I}$, no vortices are produced, in contrast with the original Vinen's equation (5.1.7), which ignores these effects.

The term in $\mathbf{J}$ in (6.8.12), neglected in the standard derivation of Vinen's equation (5.1.8), may play a physical role when L is low, and the influence of the pinned vortices is relatively relevant, as, for instance, in the transition from laminar flow to TI turbulence or in the last phase of the decay of the vortex tangle after that the heat flux (or V_{ns}) has been cancelled.

Starting from (6.8.12)–(6.8.13) it is possible to provide a justification for the coefficient ω in (5.2.13) and of the step in α_1 at the TI-TII transition (5.2.20). Indeed, in the TI turbulent regime the pinned vortices have the mean curvature prevalently in the direction of the counterflow velocity $\mathbf{u}_{ns}$; as a consequence, the prevalent component of $\mathbf{I}$ is orthogonal to $\mathbf{u}_{ns}$. Consequently, in the TI regime the scalar product $\mathbf{I} \cdot \mathbf{u}_{ns}$ is much lower than in the TII regime and the steep change at the transition can be understood observing that at this transition several vortices unpin.

6.8.4 Evolution Equations of Polarization and Anisotropy

In general, evolution equations are needed not only for L, but also for the anisotropy tensor $\mathbf{\Pi}$ (6.8.1). Assume that, under some given values of $\mathbf{u}_{ns}$ and $\mathbf{\Omega}$, tensor $\mathbf{\Pi}$ takes asymptotically a given form $\mathbf{\Pi}_0$ as the ones discussed in 6.8.1. One may guess for $\mathbf{\Pi}^s$ and $\mathbf{\Pi}^a$ relaxational evolution equations (for small enough changes in V_{ns} and Ω) of the form:

$$\tau_s \frac{d\mathbf{\Pi}^s}{dt} + \mathbf{\Pi}^s = \mathbf{\Pi}_0^s(final), \tag{6.8.14}$$

$$\tau_a \frac{d\mathbf{\Pi}^a}{dt} + \mathbf{\Pi}^a = \mathbf{\Pi}_0^a(final). \tag{6.8.15}$$

These equations will describe the evolution of $\mathbf{\Pi}$ towards the final steady state. The relaxation times τ_s and τ_a will depend, in principle, on L; we assume they are proportional to L^{-1} because the relaxation times are expected to be shorter for higher values of L. Dimensional analysis yields $\tau_s = \dfrac{a_s}{\kappa L}$, $\tau_a = \dfrac{a_a}{\kappa L}$, with a_s and a_a numerical constants.

Equations (6.8.14) and (6.8.15) express the dynamics in the absence of an externally imposed heat flux, namely, they are essentially valid for the decay of vortex tangle after turning off the external heat flux. In the presence of a heat flux, Eq. (6.8.15) should contain terms coupling the heat flux and the polarity (related to $\mathbf{\Pi}^a$), as in Eq. (6.4.1).

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Chapter 7
Coupling Between Mass Flow, Heat Flux, and Vortices



In Chaps. 5 and 6, we have focused the attention on situations without net mass flow, i.e. with local barycentric velocity $\mathbf{v} = 0$. Obviously, it is physically important considering situations with net mass flow, which are practically relevant, for instance, in cryogenics situations. Thus, we will give a short introduction to this topic. Furthermore, other forms of turbulence arise for the barycentric velocity higher than some critical value which are analogous to those arising when the heat flux is higher than some critical value.

Though its usual ability to flow along narrow capillaries without resistance, when the barycentric velocity of Helium II exceeds some critical value a resistance appears and motion is dissipated. Indeed, Landau pointed out that superfluidity would be lost beyond a critical velocity related to the spectrum $\epsilon(p)$ of the excitations [energy $\epsilon(p)$ as function of the momentum p] illustrated in Fig. 8.1, by $v_c = minimum\ [\epsilon(p)/p]$. Beyond this critical velocity the production of elementary excitations becomes energetically favourable, leading to the disappearance of the superfluid phase. Another reason may be, in some case, for quantum Reynolds number $\bar{v}d/\kappa$ (analogous to the thermal quantum Reynolds number (5.1.2)) higher than some critical value, with d, the diameter of the channel and κ, the quantum of circulation. This may happen, for instance, in Poiseuille flow along channels, or exciting the superfluid through towed or oscillating grids or with stirring propellers. The appearance of a tangle of quantized vortices acts as a source of momentum dissipation because of the mutual friction force.

To describe situations with $\mathbf{q}$ and $\mathbf{v}$ different from zero beyond the laminar regime it is necessary a hydrodynamic model of quantum turbulence incorporating the interactions between $\mathbf{q}$, $\mathbf{v}$ and the vortex tangle, in two aspects: (a) considering how the mutual friction and vortex tension of the tangle modify the evolution equations for $\mathbf{v}$ and $\mathbf{q}$ and (b) how $\mathbf{q}$ and $\mathbf{v}$ act as sources of quantum turbulence in the evolution equation for L.

The relation between motion, heat flux, and vortices may have other surprising effects. One of them is found in glitches of neutron stars, introduced in Sects. 4.4

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and 5.5. There, the motion consists of the rotation of stars around their axis; the vortices are produced mainly by the rotation, but their transition from an ordered to a disordered pattern is due to the different rotation between the crust and the superfluid inside the star, which produces changes in the inertial moment of the star, and consequently in its rotation rate.

7.1 Forced Convective Heat Transport in Helium II

In many cryogenic applications, there is a mass flow of helium—often an externally forced mass flow—which increases the amount of heat taken from the system. The basic equations for the analysis of heat removal in the presence of forced convection with net velocity $\mathbf{v}$ are

$$\rho\dot{V} = \nabla\cdot\mathbf{v}, \tag{7.1.1}$$

$$\rho\dot{\epsilon} = -\nabla\cdot\mathbf{q} - p\nabla\cdot\mathbf{v} + \sigma^{\epsilon}, \tag{7.1.2}$$

$$\rho\dot{\mathbf{v}} = -\nabla p + \eta\nabla^2\left(\mathbf{v} + \frac{\mathbf{q}}{ST}\right) + +\sigma^{\mathbf{v}}, \tag{7.1.3}$$

$$\tau\dot{\mathbf{q}} = -\mathbf{q} - \lambda_1\nabla T + \frac{\eta\lambda_1}{S}\nabla^2\left(\mathbf{v} + \frac{\mathbf{q}}{ST}\right) + \sigma^{q}, \tag{7.1.4}$$

where the meaning of the symbols is the same as in precedent chapters. The source term σ^{ϵ}, which has not been previously introduced, stands for the supplies of internal energy characterizing the concrete physical problem. They could represent, for instance, heat-dissipating devices whose heat we want to remove from the system. The term $\sigma^{\mathbf{v}}$ denotes external forces driving the motion of the fluid helium, or, in turbulent case, the influence of vortices on the velocity field. Analogously, σ^{q} represents the interaction of vortices with $\mathbf{q}$.

In (7.1.1)–(7.1.4), the upper dot stands for material derivatives, i.e. $\dfrac{\partial}{\partial t} + (\mathbf{v}\cdot\nabla)$. Combining (7.1.1) and (7.1.2) at constant pressure, one obtains [1]

$$\rho\frac{\partial h}{\partial t} + \rho(\mathbf{v}\cdot\nabla)h = -\nabla\cdot\mathbf{q} + \sigma^{\epsilon}, \tag{7.1.5}$$

with $h = u + pV$ being the enthalpy. In Eq. (7.1.5), σ^{ϵ} is given by the particular system $\rho(\mathbf{v}\cdot\nabla)h$, which may also be written as $\rho C_p(\mathbf{v}\cdot\nabla)T$, because $\partial h/\partial T = C_p$, with C_p the specific heat at constant pressure, corresponds to convective transport, and $\mathbf{q}$ is the heat transported inside the system by conduction, or by internal convection.

In the laminar regime, one has for the average heat flux across the transversal section

$$\bar{\mathbf{q}} = -K_{\text{eff}}\nabla T, \tag{7.1.6}$$

where K_{eff} is the effective thermal conductivity, depending of the size and the geometry as, for instance, (3.2.31), (3.2.34), or (3.2.49). In the fully turbulent regime, such average heat flux is described by the Görter–Mellink relation (second term in the right-hand side of expression (5.3.32)):

$$\bar{\mathbf{q}} = -\left[\frac{1}{f(T,\, p)}\nabla T\right]^{\frac{1}{3}}. \tag{7.1.7}$$

The function $f(T,\, p)$ describes the connection between $\mathbf{q}$ and ∇T in the turbulent regime.

7.1.1 Cryogenic Applications of Helium II: An Overview

Heat transport has always been a topic of interest for applications [2–6]. In particular, in superfluid Helium, it is of interest because of four peculiar features: its peculiar ability to flow with very low viscosity and inside very narrow channels, its high thermal conductivity [7–13], the high specific heat, and the fact that it does not solidify at normal pressure (a pressure higher than 20 atm must be applied to solidify it). These are definite advantages over the use of liquid nitrogen, which solidifies at 64 K. Currently, Helium II is often used to cool down and keep at very low temperatures the superconducting magnets used in particle accelerators, the measuring instruments assembled in some satellites, and many other cryogenic operations and in MNR (magnetic nuclear resonance) imaging.

Owing to its features, Helium II will probably find applications in miniaturized devices, too. One of these applications could be the cooling of future quantum computers, i.e. modern devices which aim to use quantum-mechanical phenomena (as, for example, superposition and entanglement) to perform massively parallel operations on data. In fact, quantum computers require low temperatures to keep, as long as possible, a sufficiently high extent of quantum coherence of the global wave function of their constituent qubits. An illustration may be helium flowing between two parallel planes in the presence of an array of cylinders orthogonal to them, that we have considered in Sect. 3.2.2. When the heat flux is higher than a critical value [14–17], quantized vortices appear, which strongly reduce the effective thermal conductivity of Helium II and the ability of removing heat from the system.

In channels with rectangular cross section with high aspect ratio, only one state of turbulence has been experimentally found [18, 19], and which has been source of inspiration for several numerical experiments [20–22]. Here, we consider the limit of validity of the laminar regime, which is the most desirable regime for cooling devices. For this reason, we will refer to the quantum Reynolds number for the critical values in the transition to turbulence [23]. Here we deal with heat removal in a channel filled by hot cylinders without mass flow. In Sect. 7.1.2, these studies are extended to the case with mass flow.

Here we briefly recall the results for heat transfer along cylindrical channel and rectangular channel. The optimum situation would be heat removal in the laminar regime, where the effective thermal conductivity is maximal, but this is not always possible. For this reason, it is convenient to recall the expressions for $\dot{Q}$ in terms of ΔT in more general situations.

(a) Flux Along Channels

In channels with circular cross section, the relation between ΔT and $\dot{Q}$ (the total rate of heat flow along the channel) is [24]

$$\Delta T = \left(\frac{8\eta l}{\pi R^4 S^2 T}\right)\dot{Q} + \left(\frac{K l L}{\pi R^2 \zeta}\right)\dot{Q} \tag{7.1.8}$$

with $\zeta = \lambda_1/\tau_1 = V_2^2 \rho C_V$ as defined below (3.1.19). The first term in (7.1.8) corresponds to the Landau regime (laminar regime) and the second one (if one takes $L \sim \dot{Q}^2$) to the Gorter–Mellink regime (fully developed TII regime).

In Ref. [19], Ladner and Tough considered four long channels with rectangular cross section given by $10b \times b$ with $b = 3.2 \times 10^{-5}$ m, $b = 9.1 \times 10^{-5}$ m, and $b = 9.8 \times 10^{-5}$ m for $1.2\,\mathrm{K} \leq T \leq 1.9\,\mathrm{K}$. They found only one kind of turbulence, that is, the state TIII, which seems to be similar to the state TI in cylindrical channels. In Ref. [25], it was found that for $b/a \to 0$, and $\mathrm{Re}_q < \mathrm{Re}_1$,

$$\dot{Q} = \frac{b^3 a}{12}\left(\frac{S^2 T}{\eta l}\right)\left(1 - 0.63\frac{b}{a}\right)\Delta T \tag{7.1.9}$$

whereas for $\mathrm{Re}_q > \mathrm{Re}_1$ (turbulent regime)

$$\Delta T = \left[\frac{12\eta l}{S^2 T b^2\left(1 - 0.63\frac{b}{a}\right)} + \frac{K l L}{\zeta}\right]\left(\frac{\dot{Q}}{ab}\right). \tag{7.1.10}$$

The critical quantum Reynolds number Re_1 for the appearance of quantum turbulence in this geometry is

$$\mathrm{Re}_1 = \left(\frac{\rho}{\rho_s T S}\right)\left(\frac{\dot{Q}_c}{10b^2}\right)\left(\frac{b}{\kappa}\right), \tag{7.1.11}$$

where $\dot{Q}_c = \dfrac{4\bar{\alpha}}{\gamma_0 D}\dfrac{\rho_s}{\rho} STA$ was experimentally obtained by Ladner and Tough's in Ref. [19]. Thus, expression (7.1.11) becomes

$$\mathrm{Re}_1 = 2.2\left(\frac{\bar{\alpha}}{\gamma_0 \kappa}\right), \tag{7.1.12}$$

Table 7.1 Values of the critical quantum Reynolds numbers Re_1 for the appearance of turbulence in rectangular channels (for $a = 10b$) at several values of temperature

T	1.2 K	1.3 K	1.4 K	1.5 K	1.6 K	1.7 K	1.8 K	1.9 K
Re_1	45	37	31	27	24	23	22	21

where $\bar{\alpha}$ is a constant value of approximately one (see Table 1 in Ref. [19]) and γ_0 is a constant value reported in Table 2 in Ref. [19]. In Table 7.1, the values of Re_1 are obtained by means of the experimental values of $\bar{\alpha}$ and γ_0 [19].

(b) Radial Heat Flux

Another aspect of interest is the appearance of quantum turbulence around a "hot" cylinder of radius R and height b, delivering heat to the surrounding Helium II. To this end, let us assume that the cylinder produces a radial heat flux $q(r) = \dfrac{\dot{Q}}{2\pi r b}$ for $r \geq R$, with r as the distance from the axis of the cylinder. The quantum Reynolds number is a function of the heat flux according to Eq. (7.1.19), i.e.:

$$\mathrm{Re}_q = \left(\frac{\rho}{\rho_s T S}\right)\left(\frac{\dot{Q}}{2\pi r b}\right)\left(\frac{d}{\kappa}\right). \tag{7.1.13}$$

In particular, in the present case, it decreases for increasing r. Because of the strong dependence of Re_q on r in the geometry considered here, quantum turbulence could appear next to the walls of the cylinder for $\mathrm{Re}_q(r) \geq \mathrm{Re}_1$ for $R \leq r \leq \bar{r}_1$. The maximum r_1 for which $\mathrm{Re}_q(r) \geq \mathrm{Re}_1$ is given by

$$r_1 = \mathrm{Re}_1\left(\frac{\rho_s}{\rho}\right)\left(\frac{2\pi b}{d}\right)\left(\frac{T S \kappa}{\dot{Q}}\right) \tag{7.1.14}$$

whereas Helium II is in the laminar regime for $r \geq r_1$. It also could occur that the dissipated heat in the cylinder is so high that $\mathrm{Re}_q(r) > \mathrm{Re}_2$. In this case, we can find two critical radii, r_2 and r_1, such that Helium II exhibits quantum turbulence in the state TII in the region $R \leq r \leq r_2$, in the state TI in the region $r_2 \leq r \leq r_1$ and laminar regime for $r \geq r_1$.

Note that, according to Eq. (7.1.8), the gap ΔT in the turbulent regimes is higher than that in the laminar regime because $\Delta T = \Delta T_{\mathrm{lam}} + \Delta T_{\mathrm{turb}} \geq \Delta T_{\mathrm{lam}}$, and it is higher for higher values of the vortex line density L, and hence in the state TII than in the state TI. In the turbulent regime ($R \leq r \leq r_1$), the temperature gap ΔT is so high because part of the energy is supplied to the quantum vortex tangle. In the laminar regime, instead, all the heat goes away from the cylinder without any dissipation against the quantum vortices. Because the vortex tangle acts as a resistance where energy is dissipated, the turbulent region can be imagined as part of a wider cylinder with an effective radius given by $r_{\mathrm{eff}} \sim r_1$. Indeed, considering that part of the energy is locked in the vortex tangle the effective "hot" cylinder is not the one of

radius R, but that of radius r_{eff}, which includes the cylinder and the turbulent region of Helium II around it. In this case, the radius of the cylinder would become

$$r_{\text{eff}} \equiv \text{Re}_1 \left(\frac{\rho_s}{\rho}\right)\left(\frac{2\pi b}{d}\right)\left(\frac{TS\kappa}{\dot{Q}}\right) \tag{7.1.15}$$

in such a way that the new Reynolds number would be less than the critical value Re_1. A higher radius of the cylinder would mean a higher ϕ in the expression of the effective thermal conductivity in (3.2.38) and in (3.2.39) in Sect. 3.2, as well as a smaller spacing between the cylinders ($c(1-\phi)$), when one is interested to cool the temperature of the cylinder down. From expressions (3.2.38) and (3.2.39), we note that the effective thermal conductivity in the presence of quantum turbulence around the cylinders K^t_{eff} would be smaller for $r_{\text{eff}} > R$, namely, $K^t_{\text{eff}} < K_{\text{eff}}$.

In the context of cooling, the appearance of a vortex tangle is not an efficient strategy. Possible remedies could be: (a) applying an external flow of superfluid helium to get quantum vortices away from the hot cylinder; (b) cover the cylinder by another material in such a way that the heat flux going out from the surface of the "new" cylinder is low enough that quantum Reynolds number is less than Re_1; and (c) engineering the system in such a way the cylinders are not too small.

7.1.2 *Refrigeration with Forced Mass Flow*

In the present subsection, we analyse the refrigeration of an array of heat-producing cylinders between two parallel planes considered in Sect. 3.2.2 but in the presence of forced helium convection with volume flow $\dot{Q}_v \equiv A\mathbf{v}$ (with $A = ab$ is the transversal area of the channel) crossing the system, which was not considered in Sect. 3.2.2. In continuous terms, Eq. (7.1.5) can be written as

$$\rho\frac{\partial}{\partial t}h(T) + \rho v\frac{\partial h(T)}{\partial x} + \frac{\partial}{\partial x}\left[-K_{\text{eff}}\frac{\partial T}{\partial x}\right] = \sigma^{\epsilon}, \tag{7.1.16}$$

where we have assumed that $\mathbf{v} = (v, 0, 0)$, and σ^{ϵ} describing the source terms given by $\sigma^{\epsilon} = \dot{Q}_i{}'$, and $\mathbf{q} = -K_{\text{eff}}\partial T/\partial x$ with K_{eff} given in Sect. 3.2.2.

Since $\dfrac{\partial h}{\partial T} = C_p$, with C_p the specific heat at constant pressure, we obtain the general steady-state solution of (7.1.16)

$$T(x) = c_1 + c_2\, \mathrm{e}^{\frac{\rho C_p v}{K_{\text{eff}}}x} + \frac{\dot{Q}_i{}'}{\rho C_p Q_v}x. \tag{7.1.17}$$

In concrete terms, Eq. (7.1.17) may be expressed as

$$T(x) = \left(T_0 + \frac{\dot{Q}_i'}{\rho C_p Q_v} n \right) \mathrm{e}^{\frac{\rho C_p v}{K_{\mathrm{eff}}}(n+x)} + \frac{\dot{Q}_i'}{\rho C_p Q_v} x, \tag{7.1.18}$$

where we have assumed that Helium II meets the first column of cylinders of the array at $x = -n$ with temperature T_0, and leaves the system at $x = n$.

Solution (7.1.18) is the combination of two different effects: (a) forced convection brings heat in the positive x-direction, whereas (b) heat conduction brings heat towards both positive and negative x-directions [26] (refer to Fig. 3.1). The following two particular cases arise: (a) in the absence of convection ($v = 0$), we recover the parabolic profile for T [26], where heat is carried out towards the lateral exit. (b) The linear solution $T(x) = T_0 + \frac{\dot{Q}'}{\rho C_p v}(x + n)$ is obtained when the convective term balances the dissipated heat $\dot{Q}'$, namely: $\rho v C_p \partial T/\partial x = \dot{Q}'$. For other values of temperature, the exponential form in Eq. (7.1.18) is required.

The maximum possible value of T_N is restricted by the condition that $T_N < T_\lambda$ in order to avoid transition to normal Helium I. Another restriction is that the quantum Reynolds number $\mathrm{Re_q}$, defined by

$$\mathrm{Re}_q = \frac{(\rho/\rho_s)\,(q/ST)\,d}{\kappa}, \tag{7.1.19}$$

has to be smaller than the temperature-dependent critical value $\mathrm{Re_{q,crit}(T)}$ for each cylinder column i, i.e. $\mathrm{Re_q(T_i)} < \mathrm{Re_{q,crit}(T)}$, in order to avoid quantum turbulence.

7.2 The Fourth Sound in Liquid Helium II: Porous Media

One of most striking properties of liquid Helium II is its ability to flow through narrow constrictions without any resistance. By packing a region with fine powder, channels of diameter 100−200 Å or less (known as *superleak*) can be obtained. When Helium II is forced to flow in very thin capillaries or in porous media, a wave in which all thermodynamic fields vibrate is propagated in it; this wave, anticipated by Pellam [27] in 1948 and Atkins [28] in 1959 and detected experimentally in 1962 for the first time [29, 30], is known as fourth sound [31–33].

We consider the flow of Helium II through a container filled with powder. Considering the powder grains as small spheres among which Helium II flows and supposing that the separation between the spheres is sufficiently small, remembering boundary condition we can assert that $u_i^{(2)}$ is zero in every point:

$$u_i^{(2)} = v_i - \beta T q_i = 0. \tag{7.2.20}$$

Owing to this constraint, the fields v_i and q_i are no longer independent. Consequently, their field Eqs. (3.1.17) and (3.1.19) must be modified, adding suitable production terms coming from the interaction of the fluid with the powder. Further, a drag force

must be added $\mathbf{F}^{(d)} = \gamma \frac{\partial \mathbf{v}}{\partial t}$ [34, 35], where the coefficient γ depends mainly on the temperature and the porosity, defined as the ratio of the fluid volume and to the total volume.

The motion of Helium II inside a superleak is ruled by the fields ρ and T, and the vector field v_i, namely, by the evolution equations [32]:

$$\frac{\partial \rho}{\partial t} + \rho \frac{\partial v_j}{\partial x_j} = 0, \tag{7.2.21}$$

$$\frac{\partial T}{\partial t} + \frac{1}{\rho c_V}\left(\frac{1}{\beta T} + T p_T\right)\frac{\partial v_j}{\partial x_j} = 0, \tag{7.2.22}$$

$$\frac{1}{n^2}\frac{\partial v_i}{\partial t} + \frac{1}{\rho}\frac{P-1}{P}\frac{\partial p}{\partial x_i} + \frac{\beta T \zeta}{P}\frac{\partial T}{\partial x_i} + \\ -\frac{\lambda_0}{\rho}\left(1 - \frac{\beta'}{\beta}\right)^2 \frac{P-1}{P}\frac{\partial}{\partial x_i}\frac{\partial v_j}{\partial x_j} + \frac{1}{\tau_1 P} v_i = 0. \tag{7.2.23}$$

Here $P \equiv 1 + \rho\beta^2 T^3 \zeta$ (see (3.3.59)) and $\frac{1}{n^2} = 1 - \frac{(P-1)\gamma}{P}$ where n is a scattering correction [35], which is 1 in ideal superleaks because in them the drag force $\mathbf{F}^{(d)}$ is zero ($\gamma = 0$). For the propagation of the harmonic plane waves, we obtain the following dispersion relation [36]:

$$\frac{\omega^2}{n^2} - K^2\left[\frac{P-1}{P}V_1^2 + \frac{1}{P}V_2^2\left(1 + \beta T^2 p_T\right)^2\right] + iK^2\omega\frac{\lambda_0}{\rho}\left(1 - \frac{\beta'}{\beta}\right)^2\frac{P-1}{P} = 0. \tag{7.2.24}$$

Supposing that dissipation is small ($k_r \gg k_s$), one gets for the phase speed u_4 and the attenuation factor $k_s^{(4)}$ of the fourth sound:

$$u_4^2 = \left(\frac{\omega}{k_r}\right)^2 = n^2\left[\frac{P-1}{P}V_1^2 + \frac{1}{P}V_2^2\left(1 + \beta T^2 p_T\right)^2\right], \tag{7.2.25}$$

$$k_s^{(4)} = n^2\left[\frac{\lambda_0}{2\rho}\frac{\omega^2}{u_4^3}\left(1 - \frac{\beta'}{\beta}\right)^2\frac{P-1}{P} + \frac{1}{2u_4}\frac{1}{\tau_1}\frac{1}{P}\right], \tag{7.2.26}$$

which both depend on n. Taking up to the second order in δ, the phase speed u_4^2 can be expressed in terms of phase speeds of first V_1 and second sound V_2:

$$u_4^2 \cong n^2\left[\frac{P-1}{P}V_1^2 + \frac{1}{P}V_2^2 + 2\frac{P-1}{P c_V T}\sqrt{\frac{V_1^2 - V_2^2}{c_V T}}\delta + \left(\frac{2}{P} - 1\right)\delta^2\right], \tag{7.2.27}$$

which leads to the corresponding propagation mode:

$$\overline{\rho} = \rho\psi, \tag{7.2.28}$$

$$\overline{T} = \frac{1}{\rho c_V}\left(\frac{1}{\beta T} + T p_T\right)\psi, \tag{7.2.29}$$

$$\overline{v}_n = u_4\psi. \tag{7.2.30}$$

$$\overline{q}_n = \frac{1}{\beta T}\left[\frac{R}{R-1}\right]_0 w_4\psi. \tag{7.2.31}$$

Note that the hypotheses $\beta = \beta'$ and $\frac{1}{\tau_1} = 0$ in Eq. (7.2.26) make this wave propagating without attenuation. When $n = 1$ and $p_T = 1$, expression (7.2.27) for the speed of the fourth sound coincides with (2.8.88) because $(P-1)/P = \rho_{/}\rho$ and $1/P = \rho_n/\rho$. Expression (7.2.27) is more general because it includes the effects of $n \neq 1$.

Using the values reported by Maynard [37] for thermostatic data and for velocity of first and second sounds V_1 and V_2, and the values reported in [38, 39] for the attenuations (since the latter are furnished for pressures expressed in atm, the data of Maynard have been brought back in atm, by means of a linear interpolation), the values of u_4 from the one-fluid extended model and the two-fluid modes are given in Table 7.2.

The minimum and maximum values in Table 7.2 refer to the uncertainty of experimental data for the attenuation coefficient of the second sound. In actual experiments, owing to the porosity, the fourth sound velocity results to be lower than the one reported in Table 7.2. Considering data reported in [37], we deduce that the scattering correction $n_{(E)}$, predicted by the one-fluid extended model for $T = 1.2$ K, is $n_{(E)} = \frac{w_4^{(th.)}}{w_4^{(exp.)}} = 1.196$ while using the two-fluid model one obtains $n_{(LK)} = 1.228$ [40]. The theoretical value for n is 1.2 [34].

Using the data reported by Puttermann [38] at standard volume and temperature and for temperatures $T = 1.5$ K and $T = 2.1$ K, the scattering correction assumes the value $n^{(E)} = 1.09$ for $T = 1.5$ K, and $n^{(E)} = 1.2$ for $T = 2.1$ K. Only for $T = 1.5$ K at standard volume and temperature the value $n^{(E)}$ found is not in agreement with experimental data.

Table 7.2 Minimum and maximum values, from this theory, for the velocity of fourth sound in an ideal superleak ($n = 1$) at $T = 1, 2$ K and at various pressures. The values derived by the two-fluid model are from [37]

p (atm)	Extended one fluid model $u_{4\,\min}$ (m s^{-1}) ($n = 1$)	$u_{4\,\max}$ (m s^{-1}) ($n = 1$)	Two-fluid model u_4 (m s^{-1}) ($n = 1$)
5	253.27	261.08	267.7
10	283.75	288.23	294.3
15	307.60	312.18	315.8
20	322.75	329.70	333.5
24	334.43	340.02	345.4

7.3 Hydrodynamical Model of Superfluid Turbulence with Convective Contributions

In turbulent regime both $\mathbf{v}$ and $\mathbf{q}$ may be sources for quantum vortices, and the interactions between the latter and $\mathbf{v}$ and $\mathbf{q}$ have a deep influence on their evolution and in a reduction of the efficiency in heat removing. The equations for $\dot{\mathbf{v}}$ and $\dot{\mathbf{q}}$ in (7.1.3) and (7.1.4) are

$$\begin{cases} \rho\dot{\mathbf{v}} = -\nabla p + \eta\nabla^2\left(\mathbf{v} + \dfrac{\mathbf{q}}{ST}\right) + \sigma^{\mathbf{v}}_{turb} \\ \\ \dot{\mathbf{q}} = -\dfrac{\lambda_1}{\tau}\nabla T + \dfrac{\eta\lambda_1}{S\tau}\nabla^2\left(\mathbf{v} + \dfrac{\mathbf{q}}{ST}\right) + \sigma^{\mathbf{q}}_{turb}. \end{cases} \tag{7.3.32}$$

The source term $\sigma^{\mathbf{v}}$ may have two parts: one corresponding to external momentum supplies, as, for instance, oscillating grids or rotating propellers, which depend on the problem at hand; the other related to the interaction of $\mathbf{v}$ with the tangle, which is the one we will consider here; analogously $\sigma^{\mathbf{q}}$ represents the influence of vortices on $\mathbf{q}$, for which several forms with increasing generality have been provided in (6.8.5), and which is expressed in more generality in (7.3.33).

Then, for $\sigma^{\mathbf{q}}$ and $\sigma^{\mathbf{v}}$ in terms of $\mathbf{q}$ and of L, $\mathbf{\Pi}$, $\mathbf{I}$, and $\mathbf{J}$ (characterizing the tangle according to Sect. 6.1), we consider [41]

$$\sigma^{\mathbf{q}}_{turb} = -\frac{1}{3}\kappa L B_{HV}\mathbf{\Pi}\cdot\mathbf{q} + TS\epsilon_V c_1 L^{3/2}\left[\frac{B_{HV}}{2}\left(\mathbf{I} + \frac{B'_{HV}}{B_{HV}}\mathbf{J}\right) - \mathbf{J}\right], \tag{7.3.33}$$

$$\sigma^{\mathbf{v}}_{turb} = \epsilon_V L < \mathbf{s}'' > = \epsilon_V c_1(\mathbf{p}) L^{3/2}\mathbf{J} = \rho^{(s)}\mathbf{T}, \tag{7.3.34}$$

with B_{HV} and B'_{HV} the Hall–Vinen friction coefficients introduced in (4.2.6) and c_1 the coefficient defined in (6.8.13). The tensor $\mathbf{\Pi}$, as defined in (6.8.1), describes the tangle anisotropy. For $\mathbf{\Pi} = \mathbf{U}$, they reduce to the isotropic tangle. The last term in (7.3.33) discussed in [41] refers to the average curvature of the vortices in the tangle described by the vectors $\mathbf{I}$ and $\mathbf{J}$ of (6.7.11). The expression for $\sigma^{\mathbf{v}}_{turb}$ describes the effects of the so-called vortex tension $\mathbf{T}$, which may be of special interest in situations with curved pinned vortices, whose friction may reduce the velocity of the fluid.

7.3.1 Evolution of $\mathbf{u}^{(n)}$ *and* $\mathbf{u}^{(s)}$

Using (7.3.33) and (7.3.34), Eq. (7.3.32) for $\mathbf{v}$ and $\mathbf{q}$ may be rewritten in terms of equations for the velocities $\mathbf{u}^{(n)}$ and $\mathbf{u}^{(s)}$ introduced in (3.4.75). Assuming $\nabla\cdot\mathbf{v} = 0$ and neglecting $(1/\tau_1)\mathbf{q}$, we get

$$\frac{\partial \mathbf{u}^{(n)}}{\partial t} + (\mathbf{v}\cdot\nabla)\mathbf{u}^{(n)} = -\frac{1}{\rho}\nabla p - \frac{\rho^{(s)}}{\rho^{(n)}} S\nabla T + \frac{\eta}{\rho^{(n)}}\nabla^2\mathbf{u}^{(n)} + \frac{1}{\rho^{(n)}}\mathbf{F}_{ns}, \quad (7.3.35)$$

$$\frac{\partial \mathbf{u}^{(s)}}{\partial t} + (\mathbf{v}\cdot\nabla)\mathbf{u}^{(s)} = -\frac{1}{\rho}\nabla p + S\nabla T - \frac{1}{\rho^{(s)}}\mathbf{F}_{ns} + \mathbf{T}. \quad (7.3.36)$$

After a cumbersome analysis [41], $\mathbf{F}_{ns}$ and $\mathbf{T}$ may be written in terms of L, $\mathbf{\Pi}$, $\mathbf{I}$, and $\mathbf{J}$ as

$$\mathbf{F}_{ns} = \alpha\rho_s\kappa L\left[-\frac{2}{3}\mathbf{\Pi}\cdot\mathbf{u}_{ns} + \tilde{\beta}c_1 L^{1/2}\left(\mathbf{I} + \frac{\alpha'}{\alpha}\mathbf{J}\right)\right], \quad (7.3.37)$$

$$\rho_s\mathbf{T} = \epsilon_V L < \mathbf{s}'' > = \epsilon_V c_1(\mathbf{p})L^{3/2}\mathbf{J}, \quad (7.3.38)$$

where α and α' are the friction coefficients used in (9.1.3) and $\tilde{\beta}$ is defined in (9.1.7). Simple approximate expressions for $\mathbf{F}_{ns}$ have been suggested, as, for instance, $\mathbf{F}_{ns} = \rho_s\kappa\alpha L\mathbf{u}_{ns}$, which is often used in practical problems [7], whereas $\mathbf{T}$ is often omitted.

This analysis goes beyond the gross practical needs described in Sect. 7.1, because it aims to a fundamental framework being able to stand the comparison with delicate and sophisticated experimental analyses and microscopic understanding.

7.3.2 *Comparison with the HVBK Equations of the Two-Fluid Model*

Equations (7.3.33) and (7.3.34) for the source terms $\sigma^{\mathbf{v}}_{turb}$ and $\sigma^{\mathbf{q}}_{turb}$ describe how the anisotropic vortex tangle (described by L, $\mathbf{I}$, $\mathbf{J}$, and $\mathbf{\Pi}$) influences the $\mathbf{v}$ and $\mathbf{q}$ fields. In the two-fluid model, the effects of the vortex tangle on the normal velocity $\mathbf{v}_n$ and the superfluid velocity $\mathbf{v}_s$ are given by the so-called Hall–Vinen–Bekarevich–Khalatnikov (HVBK) equations [17, 42–55]. Here, for sake of simplicity, we consider the incompressible HVBK equations (namely, $\nabla\cdot\mathbf{v}_n = 0$, $\nabla\cdot\mathbf{v}_s = 0$), which are written as

$$\rho_n\frac{\partial \mathbf{v}_n}{\partial t} + \rho_n(\mathbf{v}_n\cdot\nabla)\mathbf{v}_n = -\frac{\rho_n}{\rho}\nabla p_n - \rho_s s\nabla T + \mathbf{F}^{(HVBK)}_{ns} + \eta\nabla^2\mathbf{v}_n, \quad (7.3.39)$$

$$\rho_s\frac{\partial \mathbf{v}_s}{\partial t} + \rho_s(\mathbf{v}_s\cdot\nabla)\mathbf{v}_s = -\frac{\rho_s}{\rho}\nabla p_s + \rho_s s\nabla T - \mathbf{F}^{(HVBK)}_{ns} + \rho_s\mathbf{T}^{(HVBK)}, \quad (7.3.40)$$

where $\mathbf{F}_{ns}$ is the mutual friction force between normal component and the vortex tangle, and $\mathbf{T}$ the vortex tension that in the HVBK equations are, respectively, expressed by [53]

$$\mathbf{F}_{ns}^{(HVBK)} = \rho_s \alpha \hat{\omega} \times [\vec{\omega} \times (\mathbf{V}_{ns} - \tilde{\beta} \nabla \times \hat{\boldsymbol{\omega}})] + \rho_s \alpha' \vec{\omega} \times (\mathbf{V}_{ns} - \tilde{\beta} \nabla \times \hat{\boldsymbol{\omega}}), \tag{7.3.41}$$

$$\mathbf{T}^{(HVBK)} = (\tilde{\beta} \nabla \times \hat{\omega}) \times \vec{\omega} = \tilde{\beta} (\vec{\omega} \cdot \nabla) \hat{\omega}, \tag{7.3.42}$$

where $\vec{\omega}$ is the local averaged superfluid vorticity, $\hat{\omega} = \vec{\omega}/|\omega|$ the unit vector along $\vec{\omega}$, and $\tilde{\beta}$ the vortex tension parameter, defined as [17] $\tilde{\beta} = \dfrac{\epsilon_V}{\kappa \rho_s} = \dfrac{\kappa}{4\pi} \ln\left(\dfrac{c}{a_0 L^{1/2}}\right)$, with ϵ_V the energy per unit length of vortex line. Relations (7.3.37)–(7.3.38) are more general than (7.3.41)–(7.3.42) and better describe the effects on the presence of the vortices of normal and superfluid velocities.

Note that, in contrast to Eqs. (7.3.35) and (7.3.36), where the convective terms were referred to $\mathbf{v}$, in (7.3.39) and (7.3.40) they are, respectively, referred to $\mathbf{v}_n$ and $\mathbf{v}_s$. Furthermore, in (7.3.39) and (7.3.40), p_n and p_s are the effective pressures acting on the normal and the superfluid components, respectively, defined as $\nabla p_n = \nabla p + (\rho_s/2) \nabla V_{ns}^2$ and $\nabla p_s = \nabla p - (\rho_n/2) \nabla V_{ns}^2$, which differ from the ones obtained from the one-fluid model at the end of Sect. 2.7.

7.4 Evolution Equation for the Vortex Line Density in Convective Flows

In Sect. 7.3, we have described how the vortex tangle influences the equations for $\mathbf{v}$ and $\mathbf{q}$. Now, the reverse problem must be considered, i.e. how $\mathbf{v}$ and $\mathbf{q}$ contribute to the dynamics of the tangle. In Chaps. 5 and 6, we have considered how $\mathbf{q}$ (or V_{ns}) influences the equation for dL/dT. Here, we consider the role of the usual vorticity $\nabla \times \mathbf{v}$ as a source of quantum vortices, and write

$$\frac{dL}{dt} = -\beta \kappa L^2 + \left[\alpha_0 V_{ns} + \alpha_2 \sqrt{\kappa} \sqrt{\frac{|\nabla \times \mathbf{v}|}{2}}\right] L^{3/2} - \left[\beta_1 \frac{|\nabla \times \mathbf{v}|}{2} + \beta_4 \frac{V_{ns}}{\sqrt{\kappa}} \sqrt{\frac{|\nabla \times \mathbf{v}|}{2}}\right] L. \tag{7.4.43}$$

This equation comes from the phenomenological equation (5.4.35) replacing Ω by $|\nabla \times \mathbf{v}|/2$. In [56], it was proposed (7.4.43) with $|\nabla \times \mathbf{v}_s|/2$ instead of $|\nabla \times \mathbf{v}|/2$. Here, that equation is further generalized because in the one-fluid extended model quantum vortices are triggered by the fields $\mathbf{v}$ and $\mathbf{q}$.

Thus, (7.4.43) has two different sources of vorticity, namely, $\mathbf{q}$ (related to V_{ns}) and $\nabla \times \mathbf{v}$. Probably a further source of vorticity could be $\nabla \times \mathbf{q}$, which could appear in some kinds of geometries, but here we restrict to $\nabla \times \mathbf{v}$. In Sects. 7.4.1 and 7.4.2, we apply (7.4.43) to plane Couette and Poiseuille flows. In both cases, we assume that the baricentric velocity $\mathbf{v}$ is null on the wall. We are aware that the normal component is usually set zero on the wall instead of $\mathbf{v}$, but this is worth of future investigation [57].

7.4.1 Plane Couette Flow

The Couette flow consists of the flow of a fluid between two concentric cylinders with different angular velocities, or two parallel planes with different linear velocities. The example of two concentric rotating cylinders allows to bring the experience gained in the consideration of rotating cylinders (see Sect. 5.4.3), to the circular and plane Couette flows. Recall that in rotating cylinders a number of rectlinear vortex lines parallel to the rotation axis appear, with $L = 2\Omega/\kappa$, with Ω the angular velocity of the cylinder. The situation is analogous in circular and planar Couette flows, namely, a number of vortex lines parallel to the rotation axis (or to the limiting planes) will appear, when some threshold conditions are reached for the rotational of the velocity.

In the circular Couette flow between two concentric cylinders of inner and outer radii R_1 and R_2, with the inner one rotating at angular velocity Ω and the external one at rest, the rotation speed at a distance r from the centre is $v(r) = Ar + Br^{-1}$, with $A = -R_1\Omega(R_2^2 - R_1^2)^{-1}$ and $B = -R_1^2R_2^2\Omega(R_2^2 - R_1^2)^{-1}$ (the rotating cylinder case is formally obtained putting $R_2 = 0$). The number per unit area of vortex lines parallel to the rotation axis is $L = 2|A|/\kappa$. The minimal threshold Ω^* necessary for the appearance of the first vortex is [58]

$$\Omega^* = \frac{\kappa}{R_1^2}\frac{R_2 + R_1}{\pi(R_2 - R_1)}\ln\left(\frac{2(R_2 - R_1)}{\pi a_0}\right). \tag{7.4.44}$$

This may be physically interpreted by requiring that $\nabla \times \mathbf{v}$, which is of the order of $\pi(R_2^2 - R_1^2)\Omega/2$, be sufficiently higher than κ. The quantity $\Omega R_1^2/\kappa$ may be considered as the barycentric quantum Reynolds number, in such a way that (7.4.44) gives its critical value that depends on the geometry of the problem.

In the planar Couette flow between two plane surfaces at $y = 0$ and $y = d$, such that the second one moves parallel to the first one at the velocity $\mathbf{V}_0$, the velocity has a profile $\mathbf{v} = (v_x(y), 0, 0)$. In this case, $\nabla \times \mathbf{v} = (1/2)(\partial v_x/\partial y)\hat{\mathbf{z}}$. If $v_x(y) = V_0 y/d$, one has $\nabla \times \mathbf{v} = (1/2)V_0/d\ \hat{\mathbf{z}}$. The corresponding vortex line density L will be $L = \dfrac{1}{\kappa}\dfrac{\partial v_x}{\partial y} = \dfrac{V_0}{d\kappa}$.

These two cases are consistent with the generalized Vinen's equation (7.4.43) with $\mathbf{q} = 0$ (namely, $V_{ns} = 0$). Quantized vortices will be parallel to $\hat{\mathbf{z}}$, which plays a role analogous to the rotation axis. These ordered vortex lines could break and form a turbulent tangle in the presence of a heat flux between both plates, in analogy to the combined rotation-counterflow situation considered in Sect. 5.4. The presence of the vortices influences the attenuation coefficient of second sound between both planes, and the thermal resistance between both planes; according to (4.2.10) and (5.3.31), the attenuation coefficient will be proportional to L, and the vortex thermal resistance will be proportional to $L\mathbf{q}$.

The transient situation when the upper plate starts suddenly moving with respect to the lower plate is much more complicated, and requires the combination of the equations for $\dot{L}$, $\dot{\mathbf{q}}$ and $\dot{\mathbf{v}}$.

7.4.2 Plane Poiseuille Flow

Equation (7.4.43) with $|\nabla \times \mathbf{v}| = \frac{1}{2}\left|\frac{\partial v_x}{\partial y}\right|$ may also be applied to plane Poiseuille flow between two quiescent parallel walls at $y = \pm d/2$, driven by a longitudinal pressure gradient. The main difference is that instead of $v_x(y) = V_0 y/d$ the steady velocity profile has the form:

$$v_x(y) = \frac{\triangle p}{2\eta l}\left[\frac{d^2}{4} - y^2\right] = \frac{\triangle p}{\eta l}\frac{d^2}{8}\left[1 - \frac{4y^2}{d^2}\right] = v_{max}\left[1 - \frac{4y^2}{d^2}\right], \qquad (7.4.45)$$

with $\triangle p/l$ the pressure gradient and $v_{max} = (d^2 \triangle p)/(8\eta l)$ the maximum velocity. Also in this case vortex lines appear parallel to the plates in the region where $\nabla \times \mathbf{v}$ is high enough (this is in contrast with Couette flow, where $\nabla \times \mathbf{v}$ is homogeneous everywhere, and vortex lines are uniformly distributed). Since the Poiseuille flow is inhomogeneous, as $\nabla \times \mathbf{v}$ changes with the position, one should incorporate to (7.4.43) the vortex diffusion term, as considered in Sect. 6.2.

Two relevant features of the steady profile of L (if vortex diffusion is not considered) are [59]: the width $2y_c$ of a central zone without vortices, because $\nabla \times \mathbf{v}$ is there too small, and a boundary layer of width y_w also without vortices. To estimate y_c we calculate the circulation of v_x in (7.4.45) in the square between $y = 0$ and $y = y_c$ and equate it to $\kappa \ln(b/a_0)$, where b is a numerical constant of the order of the unity [56]

$$\frac{y_c^3}{d^3} = \frac{\kappa \ln(b/a_0)}{4 d v_{max}}. \qquad (7.4.46)$$

Note that this is only semiquantitative estimation, because the velocity profile could be different from the classical parabolic one, as shown in Fig. 7.1 and discussed in Sect. 6.4.

For the width of the boundary layer y_w, a similar approximate estimation yields [56], up to second order in y_w,

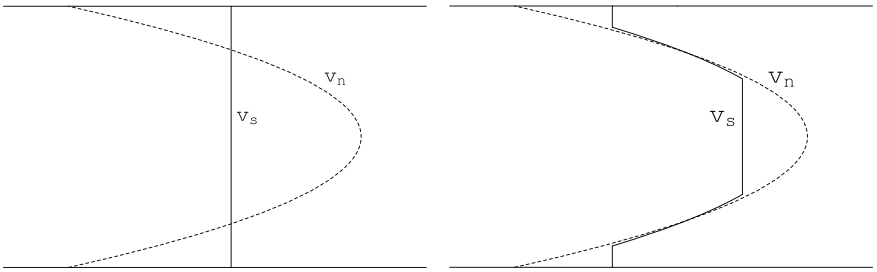


Fig. 7.1 Initial profile **a** and steady profile **b** of the superfluid (continuous line) and normal velocities (dashed line) in plane Poiseuille flow, as discussed in the second paragraph of Subsection 7.4.2. Figure from [56]

$$\frac{y_w^2}{d^2} = \frac{\kappa \ln(b/a_0)}{4 d v_{max}}. \tag{7.4.47}$$

From (7.4.46) and (7.4.47), it is found that y_c and y_w decrease for increasing v_{max} as $y_c \sim v_{max}^{-1/3}$ and $y_w \sim v_{max}^{-1/2}$. Thus for increasing v_{max} (i.e. increasing pressure gradient) the central zone and the outer zone boundary layer free of vortices will become thinner, until the vortices fill the whole space.

Expression (7.4.46) may be used to obtain the conditions for a laminar flow without any vortex. This situation will be found when the width of the central zone without vortices y_c is bigger than $d/2$. This leads to the critical barycentric quantum Reynolds number with the condition $dV_{max}/\kappa \leq 2 \ln(d/(2a_0))$. A more rigorous stability analysis would be necessary to obtain more values of the critical quantum Reynolds number in both situations.

The combination of $\mathbf{q}$ and $\nabla \times \mathbf{v}$ as two sources of vorticity is of interest in cryogenic problems where the convective flow along the channel removes heat from the warmer channel walls. The situation would be analogous in some aspects to the rotating-counterflow analysis in Sect. 5.4, but with the difference that there we considered heat flowing parallel to the rotating axis, whereas here heat flux is perpendicular to the vorticity of the flow.

7.5 Superflow: Normal Fluid Circulation

In this section, we focus on pure superflow, another interesting regime in which the normal fluid is (on the average) at rest with respect to the channel's walls, while the net superfluid flow is non-zero. Pure superflow is easily driven thermally or mechanically by blocking a section of the channel with two superleaks [60–65]. If the applied superflow v_s^{ext} is less than a small critical velocity v_s^c, the system is vortex-free, the normal fluid being quiescent and the superfluid flowing uniformly as schematically shown in Fig. 7.2 (left). The question which we address [66] is what happens when $v_s^{ext} > v_s^c$ and a turbulent tangle of vortices is formed.

Presumably, the vortices via the mutual friction force give rise to large-scale normal fluid circulation as shown in Fig. 7.2 [67].

In this section, we use the same model used in Sect. 6.4. The superfluid velocity $\mathbf{v}_s^0(\mathbf{x}, t) = \left(u_s^0(x, t), 0\right)$ in Eq. (6.4.5) is instead obtained by enforcing at each channel cross section the superfluid flow rate determined by the constant driving superfluid velocity $\mathbf{v}_s^{ext} = \left(v_s^{ext}, 0\right)$, i.e.:

$$u_s^0(x, t) + \langle u_{si} \rangle (x, t) = v_s^{ext}, \tag{7.5.48}$$

where $\mathbf{v}_{si} = (u_{si}, v_{si})$ to ease notation and $\langle \cdot \rangle$ indicates averaging over channel cross sections.

The aim of our numerical simulations is to determine the normal fluid and superfluid velocity profiles across the channel and the spatial distributions of positive and

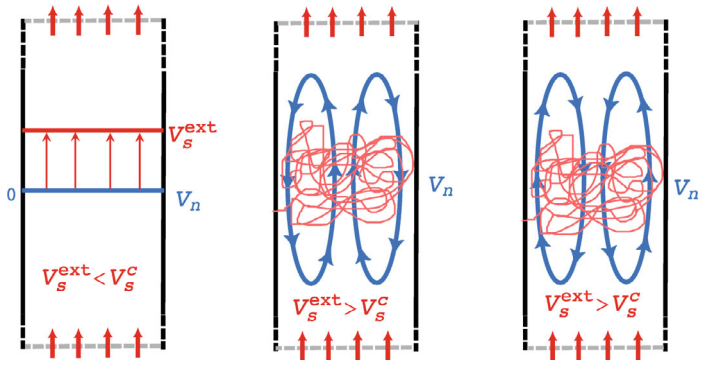


Fig. 7.2 Schematic illustration of the normal fluid (blue) and superfluid (red) velocity profiles in a pure superflow channel in a vortex-free flow (left) submitted to an upwards heat flow. (Middle) and (right): conjectures of normal fluid large flow structures in presence of a vortex tangle. Figure from [67]

negative vortices in the statistically steady-state regime which is achieved after a time interval T_f comparable to the viscous eddy turnover time D^2/ν_n. To stress that these distributions and profiles are meant to be coarse grained over channel stripes of size ΔY, we use the $\overline{\cdot}$ symbols. The key feature emerging from the numerical simulations is the coarse-grained profile of the normal fluid velocity $\overline{u}_n$ in the steady-state regime, reported in Fig. 7.3 (left). The computed profile of $\overline{u}_n$ shows that the back reaction of the motion of the superfluid vortices is effectively capable of driving the motion of the normal fluid whose *local* velocity field may hence be different from zero. We interpret the computed coarse-grained profile of $\overline{u}_n$ as the signature of large-scale normal fluid structures similar to the ones described schematically in Fig. 7.2 (right). It is worth noting that in past experimental studies [66] the existence of these normal fluid large structures has been speculated, even though the normal fluid eddies had opposite vorticity (cfr. Fig. 7.2 (middle)). The discriminant element determining the symmetry of the normal fluid flow pattern, Fig. 7.2 (middle) or Fig. 7.2 (right), is the coarse-grained profile of the streamwise component of the mutual friction force $\overline{F}^x$, illustrated in Fig. 7.3 (right), which is parallel to the driving superfluid velocity $\mathbf{v}_s^{ext}$ and stronger in the near-wall region. As a consequence, the normal fluid in proximity of the walls is accelerated in the direction of $\mathbf{v}_s^{ext}$, while in the central region the normal flow direction is reversed due to the forced re-circulation of the normal fluid arising from the presence of superleaks and the incompressibilty constraint. A similar profile of $\overline{F}^x$ has been recently computed for thermal counterflow [22].

This dynamical equilibrium achieved between the two components of Helium II via the mutual friction coupling is characterized by the polarization of the superfluid vortex distribution, which is the other key feature arising from the numerical simulations. This spatial configuration of the vortex points can be qualitatively observed in an instantaneous snapshot of the vortex configuration in the steady state (Fig. 7.4 (left)) and clearly emerges from the coarse-grained positive and negative

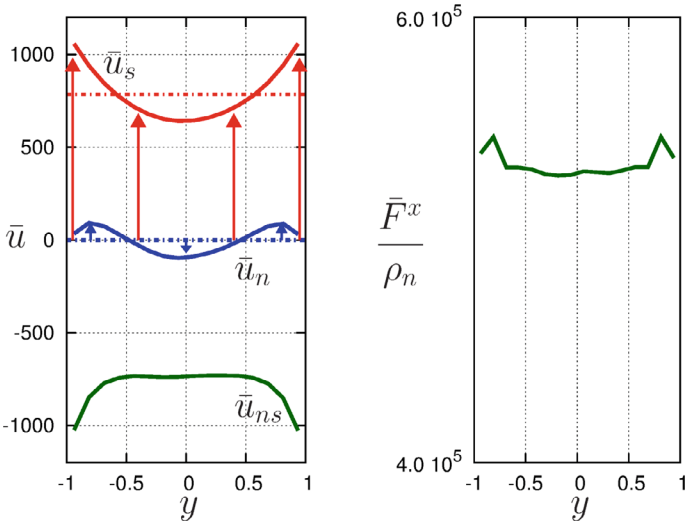


Fig. 7.3 (left): Coarse-grained profiles of superfluid velocity $\overline{u}_s$ (solid red line), normal fluid velocity $\overline{u}_n$ (solid blue line) and velocity difference $\overline{u}_{ns} = \overline{u}_n - \overline{u}_s$ (solid green line) at $t = T_f$. Red and blue dot-dashed lines indicate the initial laminar velocity profiles of the superfluid and the normal fluid, respectively; (right): coarse-grained profile of the streamwise component of the mutual friction force $\overline{F}^x/\rho_n$ at $t = T_f$. Figure from [22]

vortex density profiles illustrated in Fig. 7.4 (right). In the inset of Fig. 7.4 (right) we plot the coarse-grained polarization vector $\overline{\mathbf{p}}(y)$, defined by [56] and introduced in (6.7.3) in order to investigate quantitatively the polarization of the vortex distribution. This polarized pattern directly arises from the vortex-point equations of motion (6.4.5), where the friction term containing α depends on the polarity of the vortex. The idealized three-dimensional dynamics corresponding to this polarity-dependent vortex-point motion is a streamwise flow of expanding vortex rings lying on planes perpendicular to $\mathbf{v}_s^{ext}$ and drifting in the same direction of the latter. This *three-dimensional analogue* is very similar to the one recently illustrated for thermal counterflow [22] and consistent with past three-dimensional analytical [68] and numerical [69–71] investigations.

This polarization of the vortex configuration, which, we stress, is *not* complete, i.e. $|\overline{p}(y)| < 1$, generates a parabolic coarse-grained superfluid velocity profile $\overline{u}_s(y) \sim y^2$ which is reported in Fig. 7.3 (left) and is similar to the coarse-grained profile computed in recent counterflow simulations [22].

It is important to emphasize that our model, although being two-dimensional, recovers an almost constant profile for the total vortex density $\overline{n}(y)$ across the channel (exception made for the near-wall region, cfr. Fig. 7.4 (right)) consistent with experimental measurements performed in Prague [72].

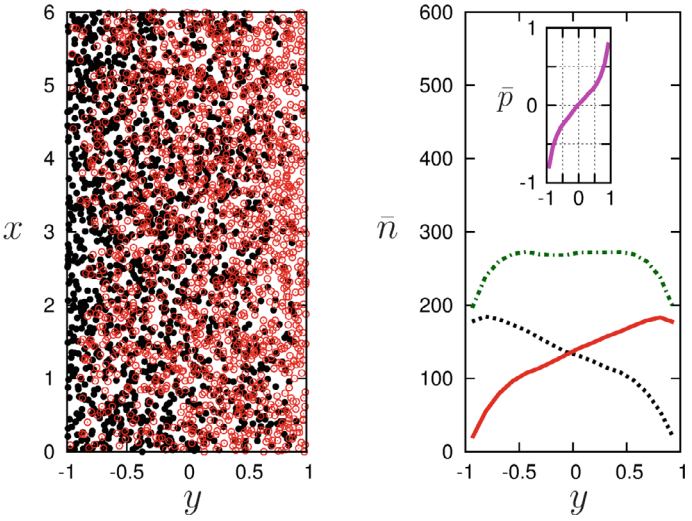


Fig. 7.4 (left): Vortex distribution, red empty (black filled) circles indicate positive (negative) vortices; (right): coarse-grained profiles of positive vortex density $\overline{n}^{+}$ (solid red line), negative vortex density $\overline{n}^{-}$ (dashed black line), and total vortex density $\overline{n}$ (dot-dashed green line) at $t = T_f$. In the inset, we report the corresponding coarse-grained profile of the polarization magnitude $\overline{p}$ (solid magenta line). The fluid is submitted to an upwards heat flow. Figure from [22]

7.6 Neutron Superfluids and Glitches in the Rotation of Neutron Stars

In Sect. 4.4, physical discussion on the structure of neutron stars has been provided, namely, it has been said that a neutron star is made of two components, rotating with different angular velocities and interacting with each other: the superfluid neutron liquid (which corresponds to the superfluid component in the two-fluid model) and the crust (which, in our view, includes also protons and electrons spread inside the star, making the normal component). It has been proposed that the glitches in the rotation of neutron stars are caused by the interaction between the relative velocity between the crust and superfluid and the presence of quantized vortices. Here we provide a simplified quantitative analysis of the problem, beyond the purely qualitative description anticipated in Sect. 4.4.

Following the explanation in Sect. 4.4 [17], the rotation sustains mainly straight vortices, which (in stationary superfluid) must be parallel to the rotation vector and pinned to the crust of the star. However, other arrangements of the vortex array are possible, and a transition between them may cause modifications of the angular momentum of the superfluid.

Pinned and Almost Straight Vortex Regime

In a neutron star, pinned vortices are not strictly straight lines but must be slightly curved near the crust and orthogonal to it. Nevertheless, for the sake of simplicity, in the following this state will be called "straight vortex regime".

Let us suppose that both the crust and the inner of the star co-rotate with an initial common angular velocity Ω_0. The crust (and hence the normal component) slows down due to the loss of energy radiated away from the star or supplied to the superfluid component, through the mutual interaction between the normal component and the vortex lines. In this scenario, vortex lines, which are strongly pinned to the crust, deform their straight configuration because of two main reasons: (a) the crust slows down faster than the superfluid and as a consequence the vortices deform and become longer; (b) the mutual friction force between the two components of the superfluid depends on the distance from the centre of the star and this causes that the mutual friction force is not uniform.

Turbulent Regime

The slowing down of the crust increases the difference between the rotating velocities of crust and superfluid, and hence the distribution of the mutual friction over the quantized vortices. This allows the formation of Kelvin waves over the vortices. When the intervortex distance is of the same order than the amplitude of the waves, vortices unpin from the crust. When the difference between the angular velocities of crust and inner of the star is higher than a critical value, the vortices catastrophically unpin from the crust and change their topology to a disordered turbulent tangle (see Figs. 7.5 and 7.6). This unpinning transfers angular momentum to the outer crust, which accelerates in a sudden way, i.e. there is the glitch (see [74–77]). Thus, according to this model, glitches are caused by two main reasons: (a) the unpinning of the vortices from the crust, which makes the tension forces null (see [41, 78]); (b) the sudden change of the distribution of the mutual friction force caused by the change of topology of quantized vortices.

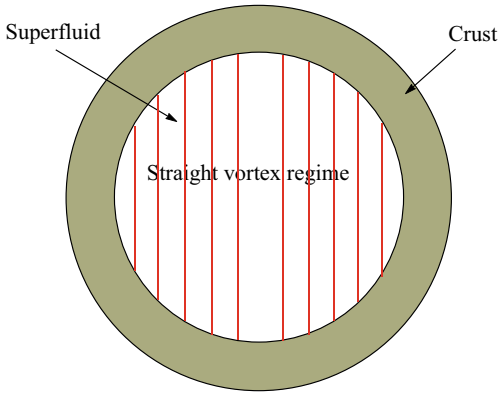


Fig. 7.5 [Color online] Sketch of the neutron star in the straight vortex regime: the quantized vortices are (almost) straight lines, parallel to the rotating axis of the star and pinned perpendicularly to the crust. In the plot only a few number of vortex lines are represented. In fact, owing to the very high angular velocity of the star the density of vortex lines is very high. Figure from [73]

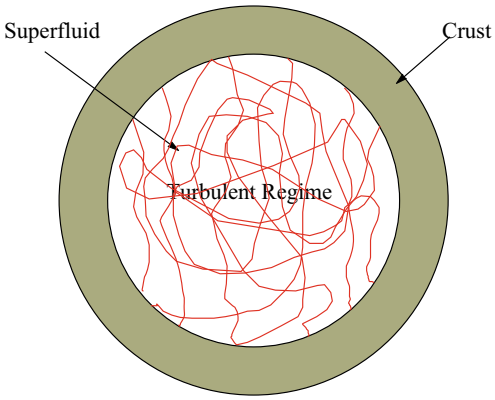


Fig. 7.6 Sketch of the neutron star in the turbulent regime (just after the glitch): most of the quantized vortices are unpinned (the cause of the glitch) in such a way that a vortex tangle forms inside the neutron star. As in Fig. 7.5, in the plot only a few number of vortex lines are represented. Figure from [73]

The growth of the vortex length density per unit volume and the almost absence of tension enforce crust and neutron superfluid to be matched in a very short time τ_1. At the same time, the superfluid neutron quickly slows down until the difference between the two velocities of rotation is less than the critical value. It is a very rapid phenomenon; indeed, due to the fast rotation, the vortex filaments tend to be parallel to the rotating axis, therefore pin to the outer crust, and the outer crust and inner part of the star co-rotate again.

Glitches may vary in size over several orders of magnitude, because one glitch could be caused by the unpinning from some regions of the star, and the next glitch from other regions, where a mild quantum turbulence has been kept.

This is probably the reason why in some neutron stars the size of glitches does not appear to depend on the time since the previous glitches. In fact the straight vortex regime may become turbulent in principle at an arbitrary value of rotation. This trigger mechanism can explain the glitches behaviour of Vela and the other quasi-periodic phenomena in non-young pulsars, but it is in general surpassed by more advanced hypotheses such as "vortex avalanches". We focus our attention on the behaviour of Vela pulsar.

Mathematical Model

The mathematical model proposed consists of three equations for the angular velocity Ω_C, Ω_S of the crust and of the internal superfluid and for the vortex length density in the superfluid, which are [73]

$$\begin{cases} I_C \dfrac{d\Omega_C}{dt} = -k_r^C - \dfrac{1}{\tau}\dfrac{I_S I_C}{I_S + I_C}\left(\Omega_C - \dfrac{\kappa L}{2}\right) \\ I_S \dfrac{d\Omega_S}{dt} = -k_r^S + \dfrac{1}{\tau}\dfrac{I_S I_C}{I_S + I_C}\left(\Omega_C - \dfrac{\kappa L}{2}\right) \\ \dfrac{1}{\aleph\,\kappa}\dfrac{dL}{dt} = -\left(L - \dfrac{2\Omega_S}{\kappa}\right)\left[L - \left(\dfrac{\alpha_1}{\aleph}\dfrac{\Omega_{SC}}{\kappa} - \dfrac{\beta_1}{2\,\aleph}\dfrac{\Omega_S}{\kappa}\right)\right]. \end{cases} \tag{7.6.1}$$

I_C and I_S are the moment of inertia of the crust and of the superfluid, k_r^C and k_r^S are the torque acting on the crust and on superfluid, and $k_r = k_r^C + k_r^S$ is the total torque.

The loss of the energy of the star is due to two different contributions: electromagnetic radiation and the loss of energy at smallest scales in the vortex lines.

At the beginning of the turbulent regime, straight vortices are still pinned and Ω_{SC} reaches its maximum value. Then, the angular velocity of the crust begins to grow, while that of the superfluid is decreasing (with respect to the average velocity of the whole star, because the inner part of the star transfers energy and angular momentum to the crust). Consequently, K_r must have a different sign in the two regimes. Further, at the end of the turbulent regime, the crust and superfluid co-rotate again, and therefore $\Omega_{SC} = 0$. This is reflected by β_1 negative in (7.6.1) [73].

Straight Vortex Regime

In the straight vortex regime, vortex line density L is described by the steady-state value $L = 2\Omega_S/\kappa$, according to (4.1.3), the solutions of Eq. (7.6.1) for Ω_C and Ω_S are

$$\begin{cases} \Omega_C^{(1)}(t) = \Omega_{av}(t) - A(K_C - K_S)(1 - e^{-t/\tau}) \\ \\ \Omega_S^{(1)}(t) = \Omega_{av}(t) + B(K_C - K_S)(1 - e^{-t/\tau}), \end{cases} \tag{7.6.2}$$

where $\Omega_{av}(t)$ is essentially the moment of inertia-weighted average of the angular velocities of the two components, namely:

$$\Omega_{av} = A\Omega_S + B\Omega_C \tag{7.6.3}$$

with

$$K_C = \frac{k_r^C \tau}{I_C \Omega_0}, \quad K_S = \frac{k_r^S \tau}{I_S \Omega_0}, \quad A = \frac{I_S}{I_S + I_C}, B = \frac{I_C}{I_S + I_C}. \tag{7.6.4}$$

At the time $t_0 = 0$, $\Omega_S = \Omega_C = \Omega_0$ and at later times one has

$$\Omega_{av}(t) = \Omega_0 - \frac{k_r^S + k_r^C}{I_S + I_C} t = \Omega_0 - \frac{k_r}{I_S + I_C} t, \tag{7.6.5}$$

which may be written in dimensionless form:

$$y_{av}(q) = 1 - (AK_S + BK_C)\, q, \tag{7.6.6}$$

where $y_{SC} = \Omega_{SC}/\Omega_0$ and $q = t/\tau$.

The second terms in the second and third members of (7.6.5) describe the slow reduction of Ω_{av} as a consequence of the resistive internal torque k_r.

The difference $\Omega_S - \Omega_C$ provides the relative angular velocity of the inner part of the star with respect to the crust

$$\Omega_{SC}^{(1)}(t) = \tau \left(\frac{k_r^C}{I_C} - \frac{k_r^S}{I_S} \right) \left(1 - e^{-\frac{t}{\tau}} \right). \tag{7.6.7}$$

If $K_S = K_C$, then $\Omega_{SC} = 0$ for all t. This would mean that the crust and the superfluid rotate always with the same velocities. This leads us to assume $K_C > K_S$ because the relative velocity Ω_{SC} has to be positive. In fact here the velocity of the crust is lower than the velocity of the inner part of the star, which rotates together with the straight vortices all over distributed.

The relative angular velocity Ω_{SC} in (7.6.7) asymptotically tends to the value $K_C - K_S$. The sudden appearance of glitches, caused by the unpinning of the vortices from the crust, confines the growth of Ω_{SC} to the value (7.6.7) with $t = \tau$. We may compute some parameters of a given star, once we known from astronomic observations the time τ and the angular velocity of the star just after the glitch. Indeed, if $t = \tau$ we get

$$y_{SC}^{(1)}(1) = (K_C - K_S)(1 - e^{-1}). \tag{7.6.8}$$

After the time τ the difference between the two angular velocities reaches its critical value, and the quantized vortices unpin from the crust, transferring some momentum to the crust. This sudden event lets the crust rotate faster because of the small tension of quantized vortices. This scenario depends strongly on the complexity of the interior of the star as well as of the crust.

Turbulent Regime

In this regime, the steady solution of the vortex line density is

$$L = \left(\frac{\alpha_1}{\aleph} \frac{\Omega_{SC}}{\kappa} - \frac{\beta_1}{2\aleph} \frac{\Omega_S}{\kappa} \right). \tag{7.6.9}$$

Following the same procedure for the straight vortex regime, the dynamics of the crust and inner superfluid may be found. In dimensionless form, we assume that $y_S = \frac{\Omega_S}{\Omega_0}$, $y_C = \frac{\Omega_C}{\Omega_0}$ and $z = \frac{\kappa L}{2\Omega_0}$ and $q = \frac{t}{\tau}$, where Ω_0 is the characteristic initial angular velocity.

Then we find

$$\begin{cases} y_C^{(2)} = \frac{A}{D_1} C \exp(-D_1 q) + \frac{b-a}{D_1} y_{av} - A \frac{D_2}{D_1^2} \\ \\ y_S^{(2)} = -\frac{B}{D_1} C \exp(-D_1 q) + \frac{1-a}{D_1} y_{av} + B \frac{D_2}{D_1^2}, \end{cases} \tag{7.6.10}$$

where $\gamma = (2\aleph\tau\Omega_0)^{-1}$ and $a = -\alpha_1/(2\,\aleph)$ and $b = -\beta_1/(4\,\aleph)$, and the parameter C is given by $C = (1-a)c_1 + (a-b)c_2$. Thus, the evolution of $y_{av} = \Omega_{av}/\Omega_0$ is given by

$$y_{av} = Ay_S + By_C = (Ac_2 + Bc_1) - (BK_c + AK_s)q, \tag{7.6.11}$$

which must be equal to (7.6.6). So we equalize and find that $Ac_2 + Bc_1 = 1$.

In (7.6.10), we recognize the contribution of an exponential term and of a linear term. The values c_1 and c_2, inside the parameter C, are constants of integration, which can be found solving Cauchy's problem with $y_C^{(2)}(1) = y_C^{(1)}(1)$ and $y_S^{(2)}(1) = y_S^{(1)}(1)$, namely, imposing that the final values of State 1 coincide with those of State 2.

The physical meaning of this request comes from the request that the microscopic superfluid component is curl-free and from the fact that the straight vortex regime is the most favourable configuration from an energetic point of view, in a rotating container [17]. Thus the straight vortex regime is the final configuration towards which the quantum fluid inside the star points at over the time τ_1. The establishing of this configuration entails the appearance of the vortex tension with the crust and of a disordered topology of the vortex array. Thus, the sudden increase of the angular velocity of crust is due to the appearance of quantum turbulence and it finishes when quantum turbulence is again converted into straight vortex regime, which requires the rigid-body rotation [17].

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Chapter 8
Microscopic Approach to Helium II and λ Phase Transition



Superfluidity and quantized vortices arise in liquid Helium II from macroscopic coherent quantum behaviour of atoms of the system. This chapter provides some essential ideas on the microscopic quantum basis of superfluid helium in order to have some understanding of the basic physics underlying its amazing phenomenology. In this way, we briefly complement the macroscopic approach provided in this book with a glimpse of the microscopic basis. Besides this, we try to clarify the respective bases of the two-fluid and the one-fluid extended models used in this book, from the microscopic perspective.

Nowadays, the quantum microscopic analysis not only of superfluid helium but of the richer field of Bose–Einstein condensates (BEC) has become a very active topic in the analysis of collective quantum systems. We consider the lambda transition from the normal state to the superfluid state and how this transition may be modified in the presence of heat flux and vortices (Sects. 8.3 and 8.4) and the behaviour of second sound near the transition, taking into account its coupling to the order parameter dynamics (Sect. 8.5).

8.1 Semi-microscopic Model of Superfluid Helium II

Following Landau's point of view, superfluid Helium II can be described as a collective ground state (corresponding to the Bose–Einstein condensate) in which quasi-particles, constituted by excitations—phonons and rotons—move. This microscopic point of view is motivated by Bose–Einstein condensation (proposed by Einstein in 1925), namely, in a dilute gas of bosons cooled below a critical temperature a large fraction of bosons occupy the lowest quantum state, and macroscopic coherent quantum phenomena become apparent. In 1938, London proposed to consider superfluidity as a consequence of Bose–Einstein condensation in Helium II.

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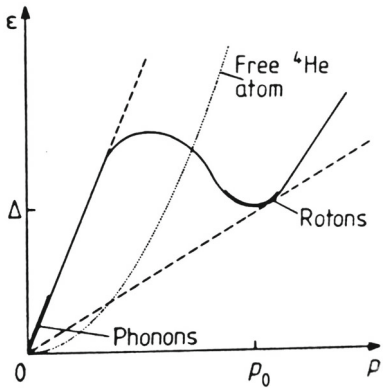


Fig. 8.1 Excitations spectrum in superfluid helium (energy ϵ as a function of momentum p for phonons (linear part) and rotons (bottom of the parabolic part). The parameters Δ and p_0 are the energy and momentum characterizing the local minimum of the spectrum and define the properties of rotons. For the sake of comparison, the energy spectrum of free atoms has been shown as a thin line. Figure from [1]

This view allows to introduce the density and the velocity of the so-called "superfluid component" (a fundamental ingredient of the two-fluid model), and the energy carried by the elementary excitations (which corresponds at a macroscopic level to the heat flux, a fundamental ingredient of the one-fluid model).

The elementary excitations (phonon and rotons) moving in the ground state constitute the normal component of the two-fluid model, and they are bosons. The spectrum of these excitations (energy as a function of wavevector k or momentum p) was derived initially by Landau in 1941 and has the form sketched in Fig. 8.1. At low energies, the spectrum is a straight line, corresponding to phonon dispersion relation. At higher energies the spectrum deviates from the straight line, passes through a maximum and a minimum and it increases again. The excitations near the minimum of the spectrum are the so-called rotons, and have an energy of the form $\varepsilon = \Delta + (p - p_0)^2/2m'$, with Δ an energy gap and m' the effective roton mass related to the curvature of the spectrum at its minimum. The presence of this gap Δ is crucial for the occurrence of superfluidity [1].

A simple theory describing, at a semi-microscopic level, the dynamical behaviour of these quasi-particles is the Gross–Pitaevskii theory that describes a weakly non-ideal Bose gas (with short-range repulsion between the particles) at zero temperature [2, 3]. The Gross–Pitaevskii theory assumes that the collective behaviour of such a gas is described by a single condensate wave function $\psi(\mathbf{x}, t)$ that is a complex function $\psi = |\psi|e^{i\phi}$ [4].

8.1.1 Evolution Equation of the Collective Wave Function

This collective wave function satisfies a nonlinear Schrödinger equation, also called the Gross–Pitaevskii equation. Assuming that ψ varies very slowly at distance of the interatomic scales, the Gross–Pitaevskii equation takes the form [5]:

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi - \mu\psi + V_0|\psi|^2\psi, \tag{8.1.1}$$

where m is the mass of bosons (in our case the atomic mass of Helium II), $\hbar$ is the reduced Planck constant, μ is the chemical potential, and V_0 is the potential of the (repulsive) interatomic forces. For the Bose–Einstein condensate of a weakly interacting gas it is $\mu = nV_0$, where n is the particle density. In this case, Eq. (8.1.1) becomes

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2\psi - V_0\left(n\psi - |\psi|^2\psi\right), \tag{8.1.2}$$

The right-hand side is the energy (kinetic plus potential) of the condensate state. This equation was obtained by Gross [2] and Pitaevskii [3] in 1961 and it is much used in the analysis of Bose–Einstein condensates. The complex conjugate ψ^* of the wave function satisfies the equation:

$$i\hbar\frac{\partial\psi^*}{\partial t} = \frac{\hbar^2}{2m}\nabla^2\psi^* + \mu\psi^* - V_0|\psi^*|^2\psi^*. \tag{8.1.3}$$

The square of the modulus of ψ gives the probability of finding any given state of the system. In particular, it gives the probability of finding particles at each point of the state, and may be interpreted as the number n of particles per unit volume in the condensed state:

$$\psi\psi^* = |\psi|^2 = n. \tag{8.1.4}$$

Therefore $m|\psi|^2$ is the mass density of the condensate that coincides with the total mass density ρ for a weakly interacting atomic gas at low temperature, where almost all atoms are the condensate:

$$m|\psi|^2 = mn = \rho. \tag{8.1.5}$$

According to (8.1.1) and (8.1.3), the rate of change of $\psi\psi^*$ is given by

$$\frac{\partial n}{\partial t} = \frac{\partial}{\partial t}\psi\psi^* = -i\frac{\hbar}{2}\left[\psi^*\nabla\psi - \psi\nabla\psi^*\right]. \tag{8.1.6}$$

By using the quantum-mechanical expression of the momentum per unit volume

$$\tilde{\mathbf{J}} = -i\frac{\hbar}{2}\left[\psi^*\nabla\psi - \psi\nabla\psi^*\right] = \hbar n\nabla\phi, \tag{8.1.7}$$

with ϕ the phase of the wave function, Eq. (8.1.6) becomes

$$\frac{\partial n}{\partial t} = \frac{\partial}{\partial t}\psi\psi^* = -\nabla\cdot\tilde{\mathbf{J}}. \tag{8.1.8}$$

This is a balance equation for the particle number density in the ground state, and appears when inhomogeneities in the phase ϕ are present.

So, we are led to define an "averaged condensate velocity" $\tilde{\mathbf{u}}$ by noting that the mass flux must have the form $\tilde{\mathbf{J}} = \rho\tilde{\mathbf{u}}$. Thus, from (8.1.7) we obtain

$$\tilde{\mathbf{u}} = \frac{\hbar}{m}\nabla\phi, \tag{8.1.9}$$

which describes a potential flow. If the phase ϕ is not singular, from (8.1.9) it follows that the motion of the condensate is irrotational. However, ϕ may have some singular lines—corresponding to vortex lines—described in Sect. 8.2 and Chap. 9.

An evolution equation for the phase ϕ, and then for the "condensate velocity" $\tilde{\mathbf{u}}$, cannot be derived in such a direct way, as Eq. (8.1.8). For the motion of the condensate fraction, in most literature, the following equation (the so-called London equation) is introduced [1, 5]:

$$m\frac{\partial\tilde{\mathbf{u}}}{\partial t} = -\nabla\left(\mu + \frac{1}{2}m|\tilde{\mathbf{u}}|^2\right) \tag{8.1.10}$$

with μ the chemical potential of the superfluid component. This equation (used in Chap. 2) is not directly deduced from (8.1.2), but some additional hypotheses are needed to write it. We present here the derivation made in [1], where the cubic term in (8.1.2) was neglected, for sake of simplicity. Making the change of variable $\psi = \exp[-i\phi]$ in Eq. (8.1.2), and taking the real part of the equation obtained for ϕ, we get

$$\hbar\frac{\partial\phi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\nabla^2 f}{f} - \frac{\hbar^2}{2m}(\nabla\phi)^2 - \mu. \tag{8.1.11}$$

By using Eq. (8.1.5) for $|\psi|^2$ and Eq. (8.1.9) for $\nabla\phi$, we find for the rate of change of the phase:

$$\hbar\frac{\partial\phi}{\partial t} = -\left(\mu + \frac{1}{2}m|\tilde{\mathbf{u}}|^2\right) + \frac{\hbar^2}{2m}\frac{\nabla^2(\sqrt{\rho})}{\sqrt{\rho}}. \tag{8.1.12}$$

Neglecting the last term and taking the gradient of both sides of (8.1.11) we obtain (8.1.10). Other more complete derivations can be found in the literature (see, i.e. [6]).

8.1.2 Heuristic Derivation of the Two-Fluid Model

We briefly recall here the derivation of the standard two-fluid model from this microscopic background. In a non-interacting Bose gas, all the particles condense to the lowest energy level at absolute zero, but in presence of interaction the condensate is partially depleted. Experiments show that in fact only about 9% of helium atoms are in the lowest state. To overcome this difficulty, the two-fluid model assumes that the

superfluid fraction comprises not only the condensate but also an amount of atoms in low (but not lowest) energy levels. The physical picture is that an amount of atoms, although removed from the lowest state by the interactions, still follow the motion of the condensate. This is a conceptual incongruence of the two-fluid model but in general it has not relevant practical consequences. A possible consequence is that though the two-fluid model assumes that the entropy of the superfluid fraction is zero, the fact that it includes some amount of atoms which are not strictly in the lowest energy level makes that the condensate may have a non-vanishing entropy, but the latter is small. Consequently, in the two-fluid model, the mass density of the superfluid component is given by $\psi(\mathbf{x}, t)\psi^*(\mathbf{x}, t) = \frac{\rho_s}{m}$. Analogously, the velocity of the superfluid component is given by $\mathbf{v}_s = <\tilde{\mathbf{u}}> = \frac{\hbar}{m}<\nabla\phi>$, and follows the London equation [1, 5], rewritten as (8.1.10).

The two-fluid model is then build up defining ρ_n and v_n through the relations:

$$\rho = \rho_s + \rho_n, \qquad \rho\mathbf{v} = \rho_n\mathbf{v}_n + \rho_s\mathbf{v}_s \tag{8.1.13}$$

and imposing the conservation of the total mass ρ, of the total momentum $\rho\mathbf{v}$, and of the energy of the "mixture".

To construct a kinetic model of superfluids by using the Gross–Pitaevskii equation, the characteristic length of spatial variations of the wave function (the coherence length, defined as $\hat{\xi} = \hbar/\sqrt{2mV_0n}$) must be $\hat{\xi} \gg d$, being d the interatomic distance [5]. Therefore, the weakly interacting Bose gas is not completely realistic when applied to liquid helium and the interpretation of ρ and $< \tilde{\mathbf{u}} >$ as the true mass density ρ_s and the true velocity $\mathbf{v}_s$ of the "superfluid component" is questionable. Nevertheless, by starting from these considerations, the two-fluid model was formulated and many useful results from it have been obtained.

8.1.3 Heuristic Derivation of the One-Fluid Extended Model

The microscopic background just described is also the starting point to formulate the one-fluid extended model of superfluid helium. Following the point of view of Landau, liquid helium can be seen as a ground state in which the quasi-particles move and carry energy. The microscopic motion of the quasi-particles with respect to the ground state is described by a field that we will denote as $\mathbf{u}_{qp}$; at the macroscopic level this field can be related to the heat flux, as

$$\mathbf{q} =< |\psi|^2 e_{qp}\mathbf{u}_{qp} >, \tag{8.1.14}$$

where e_{qp} is the energy density carried by the quasi-particle and $|\psi|^2$ is the number density of the condensate (indeed the quasi-particles are excitations of the ground state). We can also introduce the counterflow velocity as $\mathbf{U}_{ns} =< \mathbf{u}_{qp} >$; under the hypothesis that $e_{qp}|\psi|^2$ is constant, it is

$$\mathbf{U}_{ns} =< \mathbf{u}_{qp} >= \frac{\mathbf{q}}{|\psi|^2 e_{qp}}. \tag{8.1.15}$$

Equation (8.1.14) is the starting point of the one-fluid model. The hydrodynamical model is then build up adding to the balance equations of mass, momentum, and energy, an evolution equation for the heat flux. In the linear regime, this equation is a relaxational equation for **q**. In the nonlinear regime, the evolution equation for the heat flux can be derived from the kinetic theory of non-ideal gases, where the field **m**, the nonconvective part of the third moment of the phase density, is introduced as an additional variable. This field is linked to the heat flux by the relation:

$$\mathbf{m} = \alpha \mathbf{q} = \alpha(< |\psi|^2 e_{qp} \mathbf{u}_{qp} >), \tag{8.1.16}$$

which furnishes a physical microscopic meaning to the field **m**, α being a temperature-dependent constant.

In general, the two models (two fluid and extended one fluid) cannot be mutually identified with each other because the quantity $\epsilon_{qp}|\psi|^2$ is not constant, but they are closely related to each other and their predictions coincide in a number of physical situations.

8.2 Quantized Vortices

We have shown that extended one-fluid model and two-fluid model have an identical physical background: the Landau microscopic description of Helium II. In both cases, to describe liquid Helium II in the laminar regime, only the modulus of the wave function ψ is considered. If the phase ϕ of the wave function is not singular, from (8.1.9) immediately follows that the motion of the condensate is irrotational, because

$$\nabla \times \tilde{\mathbf{u}} = \frac{\hbar}{m} \nabla \times \nabla \phi = 0. \tag{8.2.17}$$

Note that the complex macroscopic wave function ψ is well defined only when $|\psi| \neq 0$. Indeed, when $|\psi| = 0$ the phase ϕ of the wave function is not defined. The points where $|\psi| = 0$ are defects of the ground state corresponding to vortex lines, and in them ϕ is singular, and the motion of the condensate is no longer irrotational.

In that case, the single-valuedness of the condensate wave function implies that around a closed contour around the vortex line its changes in phase $\Delta\phi$ must be a multiple of 2π, i.e.:

$$\Delta\phi = \oint_\gamma \nabla\phi \cdot d\mathbf{l} = 2k\pi, \qquad k \in \mathbb{Z} \tag{8.2.18}$$

where k is an integer. In that way the circulation Γ around a closed contour is given by

$$\Gamma = \oint_{\gamma} \mathbf{u} \cdot d\mathbf{l} = \frac{kh}{m} \equiv k\,\kappa \qquad k \in \mathbb{Z}. \tag{8.2.19}$$

This equation shows that the circulation is quantized, and that the quantum of circulation is $\kappa = h/m$, with m the atomic mass of helium.

The value $k = 1$ is energetically favourable with respect to higher values of k. Indeed, duplicating Γ means either going from $k = 1$ to $k = 2$ for a given vortex length, or keeping $k = 1$ and doubling the length of the vortex line. Since the energy is proportional to the square of $\mathbf{u}$ (namely, to k^2), the first option requires multiplying the energy times k, whereas the second one requires multiplying it times k^2. Thus $k = 1$ will be found and one concludes that in superfluid Helium II linear defects will have a quantized circulation given by κ.

Putting $\psi = \psi(\mathbf{x}, t) = \mathrm{Re}(\psi)(\mathbf{x}, t) + \mathrm{iIm}(\ (\mathbf{x}, \mathrm{t}))$ (Re and Im stand for the real and imaginary parts of the function ψ), these defects are the solutions of the system $\mathrm{Re}(\psi)(\mathbf{x}, t) = 0$, $\mathrm{Im}(\psi)(\mathbf{x}, t) = 0$ and describe an oriented one-dimensional manifold moving in the fluid that is called quantized vortex. Noting that $\mathrm{Re}(\psi) = (\psi + \psi^*)/2$ and $\mathrm{Im}(\psi) = (\psi - \psi^*)/2$, from Eqs. (8.1.2) and (8.1.3) we deduce that the vortex line equations must be solutions of the following system of partial differential equations:

$$\hbar \frac{\partial}{\partial t} \frac{\psi + \psi^*}{2} = -\frac{\hbar^2}{2m} \nabla^2 \frac{\psi - \psi^*}{2} + \mu \frac{\psi - \psi^*}{2} + +V_0 |\psi^*|^2 - V_0 |\psi^*|^2 \frac{\psi - \psi^*}{2}, \tag{8.2.20}$$

$$\hbar \frac{\partial}{\partial t} \frac{\psi - \psi^*}{2} = +\frac{\hbar^2}{2m} \nabla^2 \frac{\psi + \psi^*}{2} - \mu \frac{\psi + \psi^*}{2} + -V_0 |\psi^*|^2 - V_0 |\psi^*|^2 \frac{\psi + \psi^*}{2}, \tag{8.2.21}$$

with $|\psi|^2 = 0$.

In the literature, one assumes that vortex lines follow the motion laws of a vortex in a fluid (as cosmic strings or other linear oriented defects). Next chapter will be devoted to the dynamics of the vortex lines.

8.3 Transition to Superfluidity in Liquid Helium II

As liquid Helium I is cooled at a constant pressure a temperature appears (usually dubbed as λ temperature) below which superfluid behaviour manifests [1]. Such temperature depends on the pressure and it is $T_\lambda = 2.178\,\mathrm{K}$ at $p = 1$ atm. In principle, the Bose–Einstein condensate in an ideal Bose gas is a third-order condensation, in which neither the internal energy nor the specific heat exhibit a discontinuity. This is in contrast with the observed λ transition, where the specific heat presents a discontinuity, thus being a second-order phase transition. This discrepancy with the Bose–Einstein transition is due to the fact that liquid helium is not a ideal Bose gas, owing to the strong interaction between its particles.

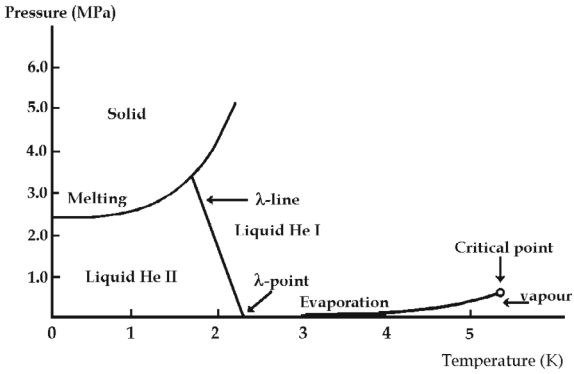


Fig. 8.2 Phase diagram of Helium II. Note the lambda line separating the superfluid region (liquid He II) from the usual viscous region (liquid He I), denoting the change of the lambda critical temperature as a function of the pressure [1]. There are also shown the solid region and the vapour region and the corresponding melting line and evaporation line

A line of λ-points appears at about 2.178 K; indeed, the temperature at which this line occurs depends on the pressure. The phase diagram of ^{4}He is shown schematically in Fig. 8.2. At low temperatures, ^{4}He has four phases. The solid phase cannot exist at vapour pressure; it appears only for pressures above 2.5 MPa and the transition between the liquid and solid phases is first order. The liquid phase exists down to $T = 0$ K. There are two liquid phases: Helium I and Helium II. As the liquid Helium I (a normal heat-conducting fluid) is cooled below a critical temperature, it transforms in superfluid Helium II.

The transition between Helium I and Helium II is a second-order phase transition, corresponding to a continuous symmetry-breaking phase transition. Semimicroscopic models for second-order phase transitions are based on a mathematical physical theory formulated by Landau describing all transformations which involve a broken symmetry and a continuous change in the free energy [7]. The basic ideas are that a phase transition can be characterized by one or more order parameters (quantities which are zero in the disordered phase and non-zero in the ordered phase) and that the free energy can be expanded as a power series in these order parameters.

The Gross–Pitaevskii equation is not able to describe the transition from normal to superfluid state because it is lacking such an order parameter.

8.3.1 Ginzburg–Pitaevskii Equation for Superfluid Transition

The Landau theory of second-order phase transitions [1, 7] was applied to liquid Helium II by Ginzburg and Pitaevskii by taking the complex wave function $\psi(\mathbf{x})$ of the condensate as the order parameter. It is assumed that $\rho_s = m|\psi|^2$, so that the dimensions of ψ are $length^{-3/2}$.

Ginzburg and Pitaevskii consider the fluid at rest in an inertial frame and assume that the free energy density G near the corresponding λ point can be expanded in powers of $|\psi|^2$ and $|\nabla\psi|^2$ as

$$G(\mathbf{x}) = G_I + \frac{1}{2}\alpha|\psi(\mathbf{x})|^2 + \frac{1}{4}\beta|\psi(\mathbf{x})|^4 + \frac{1}{2}\frac{\hbar^2}{m}|\nabla\psi(\mathbf{x})|^2, \tag{8.3.1}$$

where G_I is the free energy density of Helium I, i.e. of usual viscous liquid helium, with α and β suitable coefficients.

To find the stable state at a given temperature and pressure, the total free energy

$$\mathcal{G}(T, \psi) = \int G(\mathbf{x}) d^3\mathbf{x} \tag{8.3.2}$$

must be minimized with respect to ψ.

For minimization with respect to ψ^* the corresponding Euler–Lagrange equation is

$$\frac{\delta G}{\delta \psi^*} = \frac{\partial G}{\partial \psi^*} - \sum_j \frac{\partial}{\partial x_j}\frac{\partial G}{\partial(\nabla_j \psi^*)} = 0, \tag{8.3.3}$$

where $\nabla_j \psi^*$ is the component of the gradient in the j-direction. From (8.3.3) and (8.3.1), one obtains

$$\frac{1}{2}\alpha\psi + \frac{1}{2}\beta|\psi|^2\psi - \frac{\hbar^2}{2m}\nabla^2\psi = 0. \tag{8.3.4}$$

Had we minimized with respect to ψ rather than to ψ^* we would simply have the complex conjugate equation to (8.3.4). In terms of modulus and phase of the wave function, (8.3.4) is written as

$$\alpha|\psi| + \beta|\psi|^3 - \frac{\hbar^2}{m}\left(\nabla^2|\psi| - |\psi|\,|\nabla\phi|^2\right) = 0. \tag{8.3.5}$$

Following Landau theory, one assumes that the coefficients α and β can be expanded in powers of $(T - T_\lambda)$; in particular, we assume $\beta = B > 0$ constant and $\alpha = A(T - T_\lambda)$, with $A > 0$. Equation (8.3.5) is then

$$A(T - T_\lambda)|\psi| + B|\psi|^3 - \frac{\hbar^2}{m}\left(\nabla^2|\psi| - |\psi|\,|\nabla\phi|^2\right) = 0, \tag{8.3.6}$$

and the free energy assumes the form:

$$G(\mathbf{x}) = G_1 + \frac{1}{2}A(T - T_\lambda)|\psi|^2 + \frac{1}{4}B|\psi|^4 + \frac{1}{2}\frac{\hbar^2}{m}\left[(\nabla|\psi|)^2 + |\psi|^2\,|\nabla\phi|^2\right]. \tag{8.3.7}$$

For $T > T_\lambda$, the solution (8.3.7) is $|\Psi| = 0$ and there is not superfluid phase. For $T < T_\lambda$, the solution (8.3.7) for $|\Psi|$ is different from zero and the superfluid phase exists.

8.3.2 *Influence of Vortices and Heat Flux*

The temperature of the lambda transition depends on the applied pressure. Furthermore, the presence of an applied external heat flow or of vortices also involves a change in the transition temperature, analogous to the change produced by a variation of the imposed pressure p. From the thermodynamic point of view, one may obtain the change in the lambda temperature T_λ, induced by an external heat flux $\mathbf{q}$ or by the presence of vortex line L, from the generalized Gibbs equation [8]:

$$Tds = du + pdV - \mu dN - \frac{1}{cV_2T}\mathbf{q}\cdot d\mathbf{q} - \epsilon_V dL, \tag{8.3.8}$$

where V_2 is the second sound speed, c is the specific heat per unit mass at constant pressure, and ϵ_V is the energy of vortices per unit length. By equating μ_I and μ_{II} on the line of lambda points, with I and II standing for Helium I and Helium II, respectively, one obtains (for $L = 0$ and p constant)

$$\frac{\partial \mu_I}{\partial T}dT + \frac{\partial \mu_I}{\partial \mathbf{q}}\cdot d\mathbf{q} = \frac{\partial \mu_{II}}{\partial T}dT + \frac{\partial \mu_{II}}{\partial \mathbf{q}}\cdot d\mathbf{q}. \tag{8.3.9}$$

Since $\partial\mu/\partial T = -s$ and $\partial\mu/\partial q^2 = -(1/2)\partial/\partial\rho\left(\frac{1}{cTV_2}\right)$, one gets

$$\frac{dT_\lambda}{dq^2} = -\frac{1}{2}\frac{T}{\Delta c}\frac{\partial}{\partial\rho}\left(\frac{1}{cTV_2}\right), \tag{8.3.10}$$

where $\Delta c = c_I - c_{II}$. Analogous expression could be obtained for dT_λ/dL. In fact, superfluidity [1, 7, 9] can be destroyed with the increase of the heat flow, or by a high density of quantized vortices. A small amount of vortices may also form at the superfluid [6].

In [10–12], a phase field model for λ-transition in Helium II was proposed in non-stationary and inhomogeneous situations that is able to describe the influence of the pressure and the heat flux on the temperature transition, which produce a shift in the value of the lambda temperature with respect to its value at 1 atm and for vanishing heat flux. In [13], the model was modified to take into account the presence of vortex lines below the λ-transition. These studies are motivated both by theoretical and practical considerations in cryogenic applications [14], because it sets a limit on the maximum heat flux which may be removed through superfluid helium at a given temperature.

8.4 Evolution Equation for the Superfluid Transition

The Ginzburg–Pitaevskii model has also a dynamical counterpart, giving an evolution equation for the order parameter $|\Psi|^2$. Our aim, here, is generalizing Eq. (8.3.6) to non-steady states, and in the presence of an imposed heat flux or of a rotation, and of a small amount of vortex lines. In rotating cylinders, the vortex lines are parallel to the rotation axis, and have an analogy with the lines of magnetic field in a cylinder of a superconductor. The magnetic field modifies the superconductor transition, and the vortex lines may modify the superfluid transition.

The Ginzburg–Pitaevskii model assumes as additional field variable the dimensionless order parameter f, linked to the condensate wave function ψ by the relation:

$$f^2 = m\frac{|\psi|^2}{\rho} = \frac{\rho_s}{\rho}, \tag{8.4.11}$$

with m the atomic mass of helium, ρ the total mass density, and ρ_s the density of the superfluid component.

To describe the superfluid transition under constant pressure p_0 and in the presence of rotation or heat flux, taking into account also inhomogeneities in T, we choose for the free energy density the following expression:

$$\begin{aligned} G = G_1 + G^f + G^L = G_1 + \frac{1}{2}\tilde{a}f^2 + \frac{1}{4}\tilde{b}f^4 + \frac{1}{2}\left|\alpha_1\nabla f + \alpha_2\nabla T + \alpha_3\nabla L\right|^2 \\ + G^L(T, q^2, \Omega). \end{aligned} \tag{8.4.12}$$

In (8.4.12), $G_1 = G_1(p_0, T)$ is the free energy density of Helium I, $G^L = G^L(T, q^2, \Omega)$ the free energy of vortices, formed in presence of heat current or rotation. In (8.4.12), the G^f, coefficients $\tilde{a}$ and $\tilde{b}$, having dimension of energy, are functions of T, Ω, and q:

$$\tilde{a} = \tilde{a}_{p_0}(T, \Omega, q^2), \qquad \tilde{b} = \tilde{b}_{p_0}(T, \Omega, q^2). \tag{8.4.13}$$

Comparing (8.4.12) with (8.3.5) we deduce that in the absence of rotation and heat current, one has

$$a_0(T, 0, 0) = \frac{\rho}{m}\alpha, \qquad \tilde{b}_0(T, 0, 0) = \left(\frac{\rho}{m}\right)^2\beta. \tag{8.4.14}$$

Coefficients α_h are supposed constant quantities.

Following the Ginzburg–Landau theory [7], we minimize the total free energy $\mathcal{G}$ with respect to the order parameter f, obtaining the corresponding Euler–Lagrange equation (analogous to (8.3.3)):

$$\frac{\delta G}{\delta f} = \frac{\partial G}{\partial f} - \sum_j \frac{\partial}{\partial x_j}\frac{\partial G}{\partial(\nabla_j f)} = 0. \tag{8.4.15}$$

When Eq. (8.4.12) is substituted into Eq. (8.4.15), one obtains

$$\frac{\delta G}{\delta f} = \tilde{a}(p_0, T, \Omega, q^2) f + \tilde{b}(p_0, T, \Omega, q^2) f^3 - \nabla \cdot [\tilde{\alpha}_1 \nabla f + \tilde{\alpha}_2 \nabla T + \tilde{\alpha}_3 \nabla L] = 0, \quad (8.4.16)$$

where $\tilde{\alpha}_i = \alpha_1 \alpha_i$.

In non-stationary situations, one assumes that f satisfies an evolution equation of the form:

$$\partial_t f + f^2 \xi \nabla \cdot \mathbf{q} = -\gamma \frac{\delta G}{\delta f}, \quad (8.4.17)$$

where $f^2 \xi \nabla \cdot \mathbf{q}$ denotes the flux of the field f, with ξ the coherence length defined below (8.1.13) while $-\gamma \frac{\delta G}{\delta f}$ is the source term. Coefficient γ—whose dimensions are $J^{-1} s^{-1}$—controls the time evolution of the order parameter f. The steady-state situation corresponds to the minimum of G, where $\partial G / \partial f = 0$.

In our case, using (8.4.16) one has

$$\partial_t f + f^2 \xi \nabla \cdot \mathbf{q} = -\gamma \frac{\delta G}{\delta f} = -\gamma \left[\tilde{a} f + \tilde{b} f^3 - \nabla \cdot (\tilde{\alpha}_1 \nabla f + \tilde{\alpha}_2 \nabla T + \tilde{\alpha}_3 \nabla L) \right]. \quad (8.4.18)$$

Consider Eq. (8.4.18) in the limiting cases $f \to 0$ and $f \to 1$. In the normal phase, when $f = 0$, it must reduce to an identity. Therefore, assuming constant the coefficients α_h, we must put $\tilde{\alpha}_2 = 0$ and $\tilde{\alpha}_3 = 0$, in the considered approximation. Comparing (8.4.18) with (8.3.6), we see that coefficient $\tilde{\alpha}_1$ is linked to the quantum of vorticity κ by $\tilde{\alpha}_1 = \frac{\rho}{4\pi^2} \kappa^2$.

To determine some constrains for coefficients $\tilde{a}$ and $\tilde{b}$, the first observation is that the stationary solutions of Eq. (8.4.18), neglecting spatial inhomogeneities of the field variables, are

$$f = 0, \qquad f^2 = -\frac{\tilde{a}(p_0, T, \Omega, q^2)}{\tilde{b}(p_0, T, \Omega, q^2)}. \quad (8.4.19)$$

The first of them describes the normal phase, and therefore it must be stable above $T_{c(p_0)}$, instead, the second solution describes the superfluid phase, and must be stable below $T_{c(p_0)}$, where $T_{c(p_0)}$ stands for the critical temperature (or λ temperature T_λ) at pressure p_0.

Following Ginzburg–Landau theory [7], we suppose that $\tilde{a}$ depends on the critical temperature $T_{c(p_0)}$, while $\tilde{b}$ is independent of the temperature, then we assume

$$\tilde{a} = \mathcal{A}_{p_0}(q^2, \Omega)(T - T_{c(p_0)}) \quad \text{and} \quad \tilde{b} = \mathcal{B}_{p_0}(q^2, \Omega). \quad (8.4.20)$$

At vapour pressure ($p_0 = 0$) and in absence of heat flux ($q = 0$) or of rotation ($\Omega = 0$) it is $T_{c(0)} = T_\lambda$, $\mathcal{A}_0 = (\rho/m)A$ and $\mathcal{B}_0 = (\rho/m)^2 B$ (A and B are the coefficients introduced in Eq. (8.3.6)).

Experiments to measure the depression of the superfluid transition temperature along the λ-line due to a heat current were performed in [15], for pressure from SVP to $P = 21.6$ bar, in a nonrotating sample ($\Omega = 0$). The critical temperature T_c depends on the imposed pressure p_0, the heat current q (or the rotation Ω), and on the vortex line density L, namely, $T_{c(p_0)} = T_{c(p_0)}(q^2, \Omega, L)$.

In the space of the thermodynamic variables (p, T, q^2) we have a surface of λ-points, of equation $T = T_c(p, q^2)$. At constant pressure p_0, a line of critical points for changing $\mathbf{q}$ is found [15] that we will call $\lambda_{(q)}$-line. Near the λ-transition it can be approximated by

$$T_{c(p_0)} = T_{c(p_0)}(q^2) \simeq T_{p_0}\left[1 - bq^2\right], \tag{8.4.21}$$

with T_{p_0} the temperature transition when $q = 0$. Experimental data in Ref. [15] show that coefficient b is positive.

In rotating samples, one may assume $T_c = T_{c(p_0)}(\Omega) \simeq T_{p_0}\left[1 - d\Omega\right]$, with d a positive coefficient. More complicate expressions are possible, in simultaneous presence of counterflow and rotation.

Recalling the microscopic meaning of the order parameter, $f^2 = \rho_s/\rho$, we can determine the link between $\mathcal{A}_{p_0}$ and $\mathcal{B}_{p_0}$. In fact, with the assumptions (8.4.20) for $\tilde{a}$ and $\tilde{b}$, if we neglect spatial inhomogeneities of the field variables, the non-zero stationary solution of Eq. (8.4.18) is

$$f^2 = \frac{\mathcal{A}_{p_0}(q^2, \Omega)}{\mathcal{B}_{p_0}(q^2, \Omega)}\left(T_{c(p_0)}(q^2, \Omega) - T\right). \tag{8.4.22}$$

Since $\rho_s/\rho \to 1$ when $T \to 0$, then f must be equal to 1 when $T \to 0$. So it is inferred that $\mathcal{B}_{p_0} = \mathcal{A}_{p_0} T_{c(p_0)}$. In the following, for sake of simplicity notation the subscript p_0 in T_c, $\mathcal{A}$ and $\mathcal{B}$ will be dropped out. We obtain the following dynamical GL-type equation for the order parameter f:

$$\partial_t f + f^2 \xi \nabla \cdot \mathbf{q} = -\gamma \frac{\rho}{4\pi^2}\kappa^2 \nabla^2 f - \gamma \mathcal{A} f\left[T - T_c(1 - f^2)\right]. \tag{8.4.23}$$

Note that the source term in this equation depends on the heat flux and vortex line density, through the free energy coefficient $\mathcal{A}$ and the critical temperature T_c.

Furthermore, we have for the free energy

$$G = E - TS = G_1 + \frac{1}{2}\mathcal{A}(T - T_c) f^2 + \frac{1}{4}\mathcal{A} T_c f^4 + \frac{1}{2}\frac{\rho}{4\pi^2}\kappa^2 |\nabla f|^2 + G^L. \tag{8.4.24}$$

Therefore it is

$$\frac{\partial G}{\partial f} = \frac{\partial E}{\partial f} - T\frac{\partial S}{\partial f} = \mathcal{A}(T - T_c) f + \mathcal{A} T_c f^3, \tag{8.4.25}$$

from which one obtains for the internal energy E and the entropy S the expression

$$\frac{\partial E}{\partial f} = -\mathcal{A}T_c f(1-f^2), \qquad \frac{\partial S}{\partial f} = \mathcal{A}f. \tag{8.4.26}$$

Note that in the limit $f \to 1$, Eq. (8.4.23) reduces to an identity for $T = 0$.

According to Eq. (8.4.23) in the absence of the heat flux, the order parameter f decays in a nonlinear way towards the value $f^2 = 1 - (T/T_c)$, which is the equilibrium value of f at temperature T. One could propose more general models but (8.4.23) is enough for our purposes here, which will be the study of the propagation of second sound near the phase transition and its coupling with the dynamics of the order parameter, thus generalizing the results on these topics in Chaps. 3–6, which are valid not too close to the lambda point.

8.5 Second Sound Propagation Near Lambda Transition: Coupling with the Order Parameter

Here, we study second sound propagation near the λ-transition taking into consideration its coupling with the dynamics of the order parameter f, described by (8.4.23) [16]. This is of interest from different perspectives: (a) in contrast to usual steady-state situations, it may incorporate time variations of f; (b) the consideration of f as an independent variable allows us to explore possible static and dynamical couplings between f perturbations and temperature perturbations. Furthermore, one could also consider the possible additional coupling with quantized vortices, which may have an influence on the lambda transition temperature and on the wave propagation, but we will refer the reader to the corresponding references for this more specialized topic.

We consider that the liquid is at rest and at constant pressure p_0. To study second sound propagation below λ-transition, it is sufficient to consider the equations for temperature T, the heat flux $\mathbf{q}$, and the phase field f. The equations are [16]

$$\rho c_p \partial_t T + \nabla \cdot \mathbf{q} = -\rho \epsilon_f \partial_t f, \tag{8.5.27}$$

$$\partial_t \mathbf{q} + \zeta \nabla T = -\frac{1}{\tau_q(f)}\mathbf{q}, \tag{8.5.28}$$

$$\partial_t f + \nu \nabla^2 f = -\frac{1}{\tau_f(q^2)}\left[\frac{T}{T_c(q^2)} - \left(1-f^2\right)\right] f. \tag{8.5.29}$$

The first equation is the energy balance equation, and ϵ_f is the partial derivative of the internal energy with respect to f, and c_p is the constant pressure specific heat. In the absence of heat flux, it can be approximated by $c_p = c_0|T - T_{p_0}|^{-\alpha}$ [12], with $\alpha = 0.0127$ [17]. From Eq. (8.4.26), it is found that $\epsilon_f = -\mathcal{A}T_c f(1-f^2)$.

The second equation is the heat flux evolution equation. An essential feature of such equation is the time derivative of $\mathbf{q}$, with $\tau_q(f)$ the relaxation time. In this equation, the coefficient ζ is related to f as

$$\zeta = \frac{\zeta^*}{f^2}, \tag{8.5.30}$$

in such a way that $\zeta = \zeta^*$ when $f = 1$, i.e. sufficiently below the λ-transition. The quantity $\zeta^* = \lambda_1/\tau_0$ determines the second sound velocity V_2^* sufficiently far from the λ-transition, indeed it is $V_2^* = \sqrt{\zeta^*/\rho c_p}$ as shown in (3.3.56). The parameter $\tau_q(f, L)$ is the relaxation time of the heat flux and we assume that [13]

$$\tau_q(f, L) = \tau_0 \frac{f^2}{1 - f^2(1 - \tau_0 K L)}, \tag{8.5.31}$$

where τ_0 is a constant having the dimension of time. When f goes to zero, expression (8.5.31) reduces to τ_0 and the heat flux equation reduces to classical Fourier law, while, when $f \to 1, \tau_q/f^2$ tends to $1/KL$ with K a coefficient describing the internal friction between quantized vortices and the fluid. This means that when $f = 1$ the only mechanism leading to the decay of $\mathbf{q}$ is the friction with vortices. Here we take $L = 0$.

Equation (8.5.29) describes the dynamics of the order parameter f, and τ_f is the relaxation time of the order parameter f, linked to T_c as

$$\tau_f = \frac{1}{\gamma \mathcal{A} T_c}. \tag{8.5.32}$$

Furthermore, we take

$$\nu = \gamma \frac{\rho}{4\pi^2} \kappa^2. \tag{8.5.33}$$

Note that the dimensions of ν are those of a kinematic viscosity.

A stationary solution of system (8.5.27)–(8.5.29) is $T = T_0$, $\mathbf{q} = (0, 0, 0)$ and $f = f_0$ with $f_0^2 = 1 - \frac{T_0}{T_c}$.

In [13], the propagation of waves below (and near) the superfluid transition is studied, and the influence of f on the second sound. It is seen that the order parameter modifies the speed and the attenuation of the second sound as compared to the results given (3.3.56). In addition to the second sound a new mode is present, corresponding to a perturbation in the order parameter, which is attenuated in a short number of wavelengths.

In the limiting hypotheses $\frac{1}{\tau_q} \to 0$, $\nu_f \to 0$, $\gamma \mathcal{A}_0 = \frac{1}{\tau_f T_c} \to 0$, that is in the absence of dissipative phenomena, these modes propagate decoupled, as shown in Table 8.1 .

Table 8.1 The modes corresponding to null wave and to the second sound wave, respectively

$\omega_1^{(0)} = 0$	$\omega_2^{(0)} = \pm V_2 k_r$
$\tilde{T} = 0$	$\tilde{T} = \chi$
$\tilde{q}_n = 0$	$\tilde{q}_n = \pm V_2 \rho c_p \chi$
$\tilde{f} = \chi$	$\tilde{f} = 0$

In Table 8.1 V_2 denotes the second sound velocity near λ-transition in the absence of dissipative phenomena, given by

$$V_2^2 = \frac{1}{\rho c_p} \frac{\lambda_1}{\tau_0 f_0^2}. \tag{8.5.34}$$

For $f_0 = 1$, this value coincides with that found in Chap. 3, namely, the right term in (3.3.56). The presence of f_0 in the denominator of (8.5.34) seems to create some problem, because at the λ-transition, when f_0 goes to zero the second sound speed goes to zero too. But noting that $c_p = c_0 \left| T_c - T \right|^{-\alpha}$, and $f_0^2 = \frac{T_c - T_0}{T_c}$, we have

$$\lim_{T_0 \to T_c} \frac{\lambda_1 / \tau_0}{(T_c - T_0)^{1-\alpha}} = 0. \tag{8.5.35}$$

From (8.5.35), we deduce that λ_1 / τ_0 must go to 0 too, at the transition temperature, in agreement with experimental results.

In actual situations, second sound and null waves propagate coupled. Following the usual procedure of the perturbation theory, for the order parameter wave, one gets, to second order in f_0:

$$k_r^2 - k_s^2 = \omega_s + 2\mathcal{A}_0 f_0^2 T_c, \tag{8.5.36}$$

$$k_s = \frac{\omega_r}{k_r} \frac{1}{2\nu}. \tag{8.5.37}$$

For ω real (time harmonic dependence), this mode corresponds to perturbations of f which are attenuated within a small number of wavelengths. Indeed, from (8.5.37) we see that, owing to the small value of ν, the attenuation is very strong.

For the second sound wave, the following expressions for speed and attenuation coefficient are found in [13]:

$$u_2 \simeq V_2 \left\{ 1 + \frac{1}{2k_r^2 V_2^2} \frac{\mathcal{A}_0 f_0^2 T_0}{\rho c_p T_c} \frac{1}{[\tau_f]_0} \left(\frac{1}{[\tau_q]_0} - \frac{2}{[\tau_f]_0} f_0^2 - \nu k_r^2 \right) \right\}, \tag{8.5.38}$$

$$k_s \simeq \frac{1}{2V_2} \left[\frac{1}{[\tau_q]_0} + \frac{1}{[\tau_f]_0} \frac{\mathcal{A}_0 f_0^2 T_0}{\rho c_p T_c} \right]. \tag{8.5.39}$$

It is seen how the order parameter f modifies the speed and attenuation of second sound. Note that the correction to the second sound speed is of second order in the dissipative coefficients $1/[\tau_q]_0$, $1/[\tau_f]_0$, and ν, while that of the attenuation coefficient is of first order.

In [16], we have also explored how the waves in f may affect the waves of T, $\mathbf{q}$ and L and vice versa, in the presence of an imposed heat flux. This is of interest near the lambda transition, where the influence of L and $\mathbf{q}$ on the transition temperature may be relevant. However, the topic is too specialized for a general book and the interested reader is referred to [16] for the corresponding results.

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Chapter 9
Microscopic Description of Quantized Vortices



In classical viscous fluids, turbulence consists of many vortices of different size and length chaotically distributed all over the fluid [1–3]. Superfluid turbulence instead consists of a tangle of many vortex filaments, and with quantized circulation, with a circulation quantum given by $\kappa = h/m$ [4–6]. In fact, usual hydrodynamic vortices are also found in turbulent superfluid at scales sufficiently longer than the average size ℓ of quantized vortices, of the order of $\ell \simeq L^{-1/2}$, with L the vortex length per unit volume. At size scales sufficiently bigger than ℓ, turbulence in superfluids shares many features with turbulence in classical viscous fluids but they strongly differ at scales smaller than ℓ, as discussed in Chap. 10. Of course the debate on the analogies and differences between classical and quantum turbulence is far to be well understood.

In Sect. 9.1, we discuss several properties of single vortex lines, in order to contribute to a microscopic understanding of the collective behaviour of vortices which has been described in macroscopic grounds in Chaps. 5, 6, and 7. Both macroscopic and microscopic information are necessary for a basic understanding of the physical bases of our system.

In Sect. 9.2, we consider the classical problem of the propagation of waves along the vortex line. This is a currently open question which finds the first attempt back to the old papers [7, 8].

Other geometrical aspects of the tangle are related to purely microscopic features. Since the vortex lines cannot have free ends, they will be free closed loops moving, breaking, and reconnecting in the volume, or lines pinned to the surface. The statistical distribution of the lengths of vortex loops and the fractal dimension of the vortex tangle are two examples of such microscopic geometrical features (see [9–11]). In Sect. 9.3, two toy models of dynamics allowing to discuss the fractal dimension of vortex tangle are presented.

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9.1 Quantized Vortices in Superfluid Helium II

A quantized vortex line can be thought as a classical vortex line in the superfluid with a thin hollow core of the size of an helium atom, and quantized circulation κ. Its geometrical form is described by a vectorial function $\mathbf{s}(\xi, t)$, with $\mathbf{s}(\xi, t)$ the position of a point on the vortex line (Fig. 9.1). The derivatives of $\mathbf{s}$ with respect to ξ and to t play an essential role in the description of geometry and evolution of the line. The most relevant of them are $\mathbf{s}'$ and $\mathbf{s}''$, respectively, giving the tangent vector and the curvature vector (here and in the following a prime denotes differentiation with respect to ξ). The vectorial product $\mathbf{s}' \times \mathbf{s}''$ is another important vector, usually called the *binormal*. All these three vectors and their relative orientations with respect to the heat flux $\mathbf{q}$ (or to the counterflow velocity $\mathbf{u}_{ns}$) are important in the vortex dynamics.

The behaviour of superfluid helium is different at very low temperature (below 1 K) than at higher temperature (1 K $< T <$ 2.17 K), due to a practical absence of friction nor viscosity, and vortex lines behave as in an ideal fluid. The dynamics of a vortex line in an incompressible non-viscous infinite fluid is governed by the so-called Biot–Savart law (formally analogous to the well-known law of the magnetic field theory) [2]:

$$\dot{\mathbf{s}} = \frac{\kappa}{4\pi} \int \frac{(\mathbf{s} - \mathbf{s}_1) \times d\mathbf{s}}{|\mathbf{s} - \mathbf{s}_1|^3}, \tag{9.1.1}$$

where $\mathbf{s}_1$ is a point in the fluid, and the integral is extended all over the vortex lines. The divergence of this integral when $\mathbf{s}$ goes to $\mathbf{s}_1$ is removed by taking account of the non-vanishing value of the core size, which makes that $\mathbf{s}_1$ cannot coincide with $\mathbf{s}$ (Fig. 9.1).

Equation (9.1.1) can be also applied to the approximate study of the motion of a vortex line in a viscous fluid with very high Reynolds numbers and in a superfluid at temperature lower than 1 K, where normal component is almost absent, and friction effects are practically negligible. At temperature higher than 1 K (but less than the

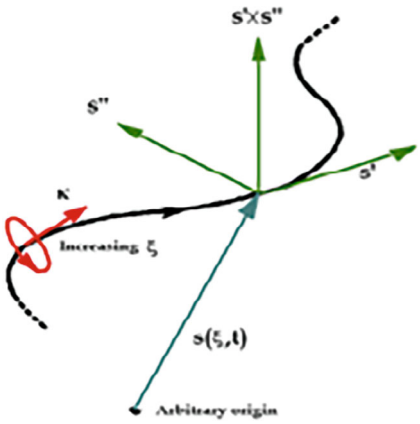


Fig. 9.1 Vortex line, and its tangent $\mathbf{s}'$, curvature vector $\mathbf{s}''$, binormal vector $\mathbf{s}' \times \mathbf{s}''$, and circulation vector $\vec{\kappa}$

critical temperature T_λ), the interaction between vortex lines and quasi-particles (phonons and rotons) modifies the dynamics of the vortex lines.

9.1.1 Microscopic Dynamics of Vortex Lines

Now we briefly recall which forces act on a quantized vortex line and the dynamical equation which rules the motion of each element of the vortex at any temperature.

At such temperatures, each vortex element is subject to two forces, the Magnus force (9.1.2) and the mutual friction force (9.1.3). Magnus force $\mathbf{f}_M$ is generated by the relative flow of superfluid with respect to the vortex [12], and it is expressed by

$$\mathbf{f}_M = \kappa\rho_s \mathbf{s}' \times (\tilde{\mathbf{v}}_L - \tilde{\mathbf{v}}_{sl}), \tag{9.1.2}$$

where $\tilde{\mathbf{v}}_L = \partial\mathbf{s}/\partial t$ is the velocity of the line element at point ξ and $\tilde{\mathbf{v}}_{sl} = \tilde{\mathbf{u}} + \tilde{\mathbf{v}}^{(\mathrm{i})}$ is the "local superfluid velocity", sum of the superfluid velocity at large distance from any vortex line and of the "self-induced velocity", due to the presence of other vortices or other parts of the same vortex (that is, the right-hand side of Eq. (9.1.1)).

The microscopic mutual friction force per unit length can be written as [5]:

$$\mathbf{f}_{MF} = -\alpha\rho_s\kappa\mathbf{s}' \times [\mathbf{s}' \times (\tilde{\mathbf{u}}_{ns} - \tilde{\mathbf{v}}^{(\mathrm{i})})] - \alpha'\rho_s\kappa\mathbf{s}' \times (\tilde{\mathbf{u}}_{ns} - \tilde{\mathbf{v}}^{(\mathrm{i})}), \tag{9.1.3}$$

where α and α' are temperature-dependent coefficients, and they are linked to the Hall–Vinen coefficients B_{HV} and B'_{HV} introduced in (4.2.6) by $\alpha = \dfrac{\rho_n}{2\rho} B_{HV}$ and $\alpha' = \dfrac{\rho_n}{2\rho} B'_{HV}$ [5, 13], and their values are given in Table 4.1.

The force (9.1.3) results from the collision of the quasi-particles (phonons and rotons) with the vortices [4, 5, 14, 15]. The collision cross section is a function of the direction of the quasi-particles drift velocity relative to the vortex line: it is a maximum when the quasi-particles are travelling perpendicular to this line and zero when they move parallel to the line.

Dimensionless coefficients α and α' can be directly measured in experiments. They depend on the temperature and their values are given in Table 4.1.

The force per unit length exerted by the superfluid on a vortex line element is denoted by $\mathbf{f}_{MF}$. The force $-\mathbf{f}_{MF}$ is thus the force on the superfluid exerted by the vortex. If we neglect the inertia of the core, the force balance equation on each line element is

$$\mathbf{f}_M + \mathbf{f}_{MF} = 0. \tag{9.1.4}$$

Substituting (9.1.2) and (9.1.3) in (9.1.4) yields

$$\mathbf{s}' \times (\tilde{\mathbf{v}}_L - \tilde{\mathbf{v}}_{sl}) - \alpha\mathbf{s}' \times [\mathbf{s}' \times (\tilde{\mathbf{u}}_{ns} - \tilde{\mathbf{v}}^{(\mathrm{i})})] - \alpha'\mathbf{s}' \times (\tilde{\mathbf{u}}_{ns} - \tilde{\mathbf{v}}^{(\mathrm{i})}) = 0, \tag{9.1.5}$$

which may be written as

$$\tilde{\mathbf{v}}_L = \frac{\partial \mathbf{s}}{\partial t} = \tilde{\mathbf{v}}^{(\mathrm{i})} + \alpha \mathbf{s}' \times (\tilde{\mathbf{u}}_{ns} - \tilde{\mathbf{v}}^{(\mathrm{i})}) - \alpha' \mathbf{s}' \times [\mathbf{s}' \times (\tilde{\mathbf{u}}_{ns} - \tilde{\mathbf{v}}^{(\mathrm{i})})] + \lambda \mathbf{s}', \qquad (9.1.6)$$

with λ being an undetermined coefficient. In the current literature, it is taken $\lambda = 0$.

Equation (9.1.6) is the equation of motion of a single vortex, considering the influence of the other vortices and the interaction with the quasi-particles, a fundamental result which is usually used both in theoretical and numerical studies (see, for instance, Sects. 6.4 and 6.5). The integration of Eq. (9.1.1) or Eq. (9.1.6) is a hard task to be handled analytically, especially for tracing the trajectory of the vortex. For these reasons, many efforts have been made for simulating the dynamics of many interacting quantized vortices (see Ref. [16] for an exhaustive overview). It is worth to note that the presence of the wall was overcome by Schwarz, and it will be explained better in the next subsection as well as the question of the vortex reconnections, which naturally arises in the Bose–Einstein condensate but an ad hoc procedure is required in superfluid helium, where the Vortex Filament Model (VFM) is used.

In Eq. (9.1.6), the dynamics of the quantized vortices take into account the interactions with the quasi-particles, an arduous problem due to the small sizes of quasi-particles and vortices. In Ref. [17], Galantucci *el.* considered this issue from the microscopical point of view, proposing a method for evaluating this interaction and hence both the dynamics of the vortices and the mutual friction force. They found a slightly different expression for the mutual friction force and for the equation of motion (9.1.6), which, according to their numerical experiments, better describe two physical situations: a superfluid vortex ring moving (a) in an initially stationary normal fluid and (b) in a turbulent normal fluid (more details in [17]).

If one is interested to study analytically the propagation of perturbations along the vortex line (Kelvin waves), an approximation of Eq. (9.1.1) is required. We consider here the two classical approximations, namely, the Local Induction Approximation (LIA) [8, 18] and Fukumoto's approximation [19], which are the first-order and the third-order approximations, respectively, of (9.1.1) in terms of a_0/R (the ratio between the core radius a_0 and the typical curvature radius R).

To the first order in a_0/R, expression (9.1.1) is approximated by the Local Induction Approximation (LIA) [8, 18]:

$$\tilde{\mathbf{v}}^{(\mathrm{i})} = \tilde{\beta} \left[\mathbf{s}' \times \mathbf{s}'' \right]_{s=s_0}, \qquad \text{with} \qquad \tilde{\beta} = \frac{\kappa}{4\pi} \ln \left(\frac{2R}{a_0} \right). \qquad (9.1.7)$$

The intensity of $\tilde{\mathbf{v}}^{(\mathrm{i})}$ is $|\tilde{\mathbf{v}}^{(\mathrm{i})}| = \tilde{\beta}/R$. The self-induced velocity is zero when the vortices are straight lines. The coefficient $\tilde{\beta}$ is linked to the internal energy per unit length of the vortex line ϵ_V (the tension of the vortex line) by $\epsilon_V = \rho_s \kappa \tilde{\beta}$.

Equation (9.1.7) comes out from (9.1.1) when one splits up the integral into an integral over a local neighbourhood around the point $\mathbf{s}_0$, and the rest of the integral, and one neglects the nonlocal part [8]. The choice (9.1.7) for the local part gives the

exact velocity for a planar vortex ring of radius R. In this approximation, vortex line locally moves in the direction of the binormal and its velocity is inversely proportional to the local radius of curvature. The approximation (9.1.7) for the vortex dynamics in a vortex tangle, as proved by Tsubota and collaborators [20], does not work very well for parallel and anti-parallel vortices, but it works quite well for wave propagation (Kelvin waves) along the vortex line.

A more refined approximation is the Fukumoto approximation [19], at third order in a_0/R which is

$$\tilde{\mathbf{v}}_L = \tilde{\mathbf{v}}^{(\mathrm{i})} \simeq \tilde{\beta}\left(k\mathbf{b} + \tilde{\mu}k^3\mathbf{b} + \tilde{\nu}\left((2k_\xi\tau + k\tau_\xi)\mathbf{n} + (k\tau^2 - k_{\xi\xi})\mathbf{b} + k^2\tau\mathbf{t}\right)\right), \quad (9.1.8)$$

where $\tilde{\mu}$ and $\tilde{\nu}$ are constants; $\{\mathbf{t}, \mathbf{n}, \mathbf{b}\}$ is the Frenet reference frame with $\mathbf{t} = \dfrac{\mathbf{s}'}{|\mathbf{s}'|}$, $\mathbf{n} = \dfrac{\mathbf{s}''}{|\mathbf{s}''|}$, and $\mathbf{b} = \dfrac{\mathbf{s}' \times \mathbf{s}''}{|\mathbf{s}' \times \mathbf{s}''|}$; and k and τ are curvature and torsion of the line (see [19] for more details). When $\tilde{\mu} = \tilde{\nu} = 0$, Eq. (9.1.8) simplifies to (9.1.7).

9.1.2 *Evolution of Vortex Tangle: Vortex Reconnections and Vortex Pinning*

Important features of the vortex dynamics are vortex *reconnections*, which was first envisaged by Feynman [21], and then confirmed by the experiments [22–24] and the microscopic models [4, 5, 15, 25–31], and which occur when vortex lines come close enough to each other. In the numerical experiments, reconnections have been achieved by an 'ad hoc" numerical procedure in the so-called Vortex Filament Model (VFM) in order to mimic the real behaviour of the vortex motion and interactions. For this purpose many techniques have been explored and compared finding no significant physical differences [32]. Vortex reconnections instead naturally arise in the Bose–Einstein condensate, where the Gross–Pitaevskii model is used. Very recently, in [24] vortex reconnections between vortices were numerically found and experimentally confirmed.

The same behaviour is seen close the walls, where the vortex lines may pin to the walls (see Fig. 9.2). Vortex reconnections and vortex pinning change the topology of vortex lines: when two vortex filaments approach closely to each other (less than a characteristic distance $L^{-1/2}$) they reconnect, whereas when vortices approach the



Fig. 9.2 Sketch of two basic processes: vortex reconnection and vortex pinning to the walls

walls (less than a characteristic distance $1/2L^{-1/2}$) they pin to the wall. A pinned vortex unpins when the applied flow bends it very strongly; it then undergoes a reconnection and escapes to free space, forming a vortex loop. Schwarz's numerical simulations [33–37] have worked out these effects in detail.

This behaviour may be easily understood if one image that the wall behaves as a mirror for the vortices, and in this sense when the vortex line approach the wall, it behaves as if there were another vortex line to reconnect to. The pinning and unpinning behaviours were used in Sect. 7.6 to explain the glitches in the neutron stars.

These two phenomena are the microscopic basis of the dynamics of the ensemble of vortices constituting the vortex tangles. This is a complex topic that, from a phenomenological perspective, must be captured through simplified but efficient evolution equations for the vortex length density L, the tangle polarity, and the tangle anisotropy parameters. In the simplest case, the gross mesoscopic features of the phenomena are described by the Vinen equation (5.1.7) which may be statistically derived from the microscopic vortex dynamics [37].

9.2 Kelvin Waves

If a vortex is slightly deformed into a helix, the deformation can propagate along the vortex line as a wave called Kelvin wave, as discussed originally by Lord Kelvin in 1880 [7]. The reconnections between vortices cause the appearance of such Kelvin waves, which travel along the vortex line as long-scale perturbation modes of a thin vortex filament [38].

Nonlinearity as that corresponding to the terms in $\tilde{\mu}$ and $\tilde{\nu}$ in (9.1.8), for instance, causes transferring of energy among the different harmonics composing a wave.

Note that inside a vortex tangle many other phenomena occur, as reconnection processes, large-scale deformations caused by the presence of the other vortices and of the walls, escape of loops from the bulk, and so on. In [6, 39, 40], Nemirovskii and Kondaurova have studied numerically and theoretically how various factors contribute to the decay of total vortex length, and indirectly of energy, of interacting quantum vortices inside a bulk.

9.2.1 Frictionless Case

The Local Induction Approximation

Following (9.1.7), a reformulation of (9.1.1) in a suitable extrinsic coordinate space allows us to find an exact nonlinear Kelvin wave solution. Let's assume that the z-axis is the straight configuration of the unperturbed vortex line (see Fig. 9.3) [41, 42]. The position $\mathbf{s}(z, t)$ of a point in the vortex line can be expressed by $\mathbf{s}(z, t) = (\phi(z, t), z)$,

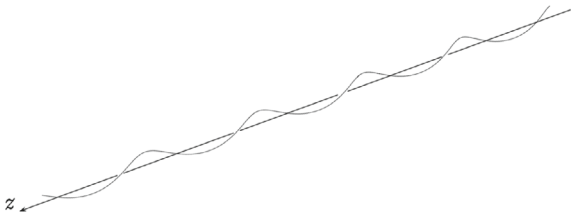


Fig. 9.3 A Kelvin helical wave along a vortex line

where z is the axis of the vortex, and $\phi(z,t) = x(z,t) + \mathrm{i}y(z,t)$ describes the perturbation in the x, y-plane.

In [43], it was shown that, in this frame of reference, (9.1.7) yields

$$\mathrm{i}\phi_t + \tilde{\beta}\phi_{zz}\left(1 + |\phi_z|^2\right)^{-3/2} = 0. \tag{9.2.9}$$

An exact solution of (9.2.9) is [43]

$$\phi(z,t) = C\mathrm{e}^{\mathrm{i}(Kz-\omega t)} \tag{9.2.10}$$

with the nonlinear dispersion relation:

$$\omega = \frac{\tilde{\beta}K^2}{(1 + K^2C|^2)^{3/2}}, \tag{9.2.11}$$

and C a complex constant amplitude, K the wave number, and ω the angular frequency. Relation (9.2.11) has its maximum value at $K = \sqrt{2}\,|C|^{-1}$. A slightly different dispersion relation was obtained in a different way by Sonin [44, 45].

In [43], it was shown that the velocity of this wave (that differs from the phase velocity ω/K) increases for K lower than $0.43|C|^{-1}$, and decreases for K higher than $0.43|C|^{-1}$. It means that for a fixed amplitude the fastest wave corresponds to $K = 0.43|C|^{-1}$, for which the velocity is $v_w = 0.51\tilde{\beta}|C|$.

This result shows that when two reconnections simultaneously occur on a vortex line, the generated Kelvin waves will run with different velocities according to their amplitudes and wave numbers, and this can cause overlapping, interferences, and so on, similar to propagation of pulses in fibre optics [46].

The Fukumoto Approximation

By using the third-order approximation (9.1.8), a generalization of (9.2.9) is obtained [43], leading to the dispersion relation:

$$\omega = \frac{\tilde{\beta}K^2}{(1 + |C|^2K^2)^{3/2}} + K^4\tilde{\beta}\tilde{\nu}, \tag{9.2.12}$$

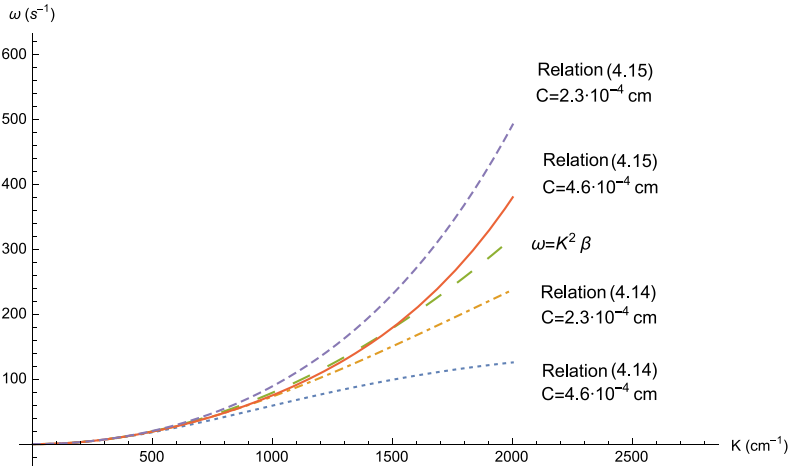


Fig. 9.4 [Color online] Comparison between dispersion relations (9.2.11), (9.2.12) and $\omega = \tilde{\beta} K^2$ for Kelvin waves. Figure from [43]

with ν the coefficient appearing in (9.1.8). This means that the Fukumoto approximation (9.2.12), as compared to the LIA approximation (9.2.11), only contributes to the velocity of the wave, which is higher than for (9.2.11) but does not contribute to the growth or decay of the amplitude of the wave. Note that for $\tilde{\nu} = 0$ (9.2.12) becomes (9.2.11). In [43], a detailed comparison between (9.2.11), (9.2.12) and the usual dispersion relation $\omega = \tilde{\beta} K^2$ was made. It was found that (9.2.11) is enough for studying Kelvin wave propagation for small wave numbers and that for higher wave numbers further terms of the Biot–Savart have to be considered (see Fig. 9.4). These results can contribute to a better understanding of the energy transfer among the harmonics of the Kelvin waves, and of the "Kelvin cascade" [47].

9.2.2 Finite Temperature Case (Friction Case)

At temperature higher than 1 K the friction of the normal component of Helium II with vortices plays a relevant role in the dynamics. The dynamical equation to be considered is then (9.1.6), with the induced velocity $\tilde{\mathbf{v}}^{(\mathrm{i})}$ approximated by (9.1.7) or (9.1.8).

In this case, one must add to (9.2.9) a term $F(\alpha, \alpha', \phi, \ldots)$. In [43], neglecting terms in α and α' which are not linear in $|\phi_z(z, t)|$ the following expression was found:

$$F(\alpha, \alpha', \phi, \ldots) = -\mathrm{i}\alpha\tilde{\beta}\phi_{zz} - u_3(\alpha - \mathrm{i}\alpha')\phi_z + u_n(\alpha - \mathrm{i}\alpha'), \tag{9.2.13}$$

where (u_1, u_2, u_3) are the components of $\tilde{\mathbf{u}}_{ns}$ in the frame of reference $\{x, y, z\}$, and $u_n = u_1 + \mathrm{i}u_2$ and $\overline{u}_n$ is the conjugate of u_n.

The modification of LIA solution (9.2.10) is then [43]:

$$\phi(z,t) = Ce^{-\alpha K t(\tilde{\beta}K - u_3)} e^{\mathrm{i}(Kz - \bar{\omega}(t))} + u_n(\alpha' + \mathrm{i}\alpha)t, \tag{9.2.14}$$

with $\bar{\omega}(t)$ given by $\frac{\mathrm{d}\bar{\omega}(t)}{\mathrm{d}t} = K\left(\frac{\tilde{\beta}K}{(\Delta(t))^{3/2}} + \alpha' u_3\right)$ with $\Delta(t) = |C|^2K^2e^{2\alpha K t(u_3 - \tilde{\beta}K)} + 1$.

The dispersion relation (9.2.11) becomes

$$\omega = \frac{\tilde{\beta}K^2}{\left(|C|^2K^2 + 1\right)^{3/2}} + \alpha'\tilde{\beta}K^4, \tag{9.2.15}$$

which differs from (9.2.11) for the term in α', and which coincides with (9.2.12) for $\nu = \alpha'$.

Note that, according to the first term in (9.2.14) for a fixed counterflow velocity u_3 in the z-direction, Kelvin wave with the wave number $K = u_3/\tilde{\beta}$ is the only one which is free of any friction. This means that the amount of energy dissipated at length scale $K^{-1} = u_3^{-1}\tilde{\beta}$ is equal to the energy supplied to it. For $K < u_3/\tilde{\beta}$ the amplitude, and hence the energy, of the Kelvin wave increases (Donnelly–Glaberson instability), while the opposite behaviour is found for $K > u_3/\tilde{\beta}$. This means that for a given heat flux (related to u_3) all the waves with $K < u_3/\tilde{\beta}$ dissipate less energy than that supplied, and the opposite occurs for $K > u_3/\tilde{\beta}$.

Solution (9.2.14) also confirms that the term proportional to α leads to growth/decay of the amplitude wave (see Figs. 9.5, 9.6, and 9.7), while the one proportional to α' modifies the dispersion relation and hence the velocity of the wave.

In Figs. 9.5, 9.6, and 9.7, we have plotted solution (9.2.14) for three different situations: heat flux parallel, anti-parallel, and orthogonal to the z-axis, at $T = 1.6\,\mathrm{K}$. As concerns the applied counterflow velocity $\tilde{\mathbf{u}}_{ns}$ we refer to Skrbek's experiments [48], which allow us to take $L^{-1/2}/4$ as plausible amplitude for the Kelvin wave.

9.3 Geometry of the Vortex Tangle: Fractal Dimension

In the case of turbulence produced by classical means, i.e. not by counterflow but by towed grid or by rotation, for instance, one will have a transition from classical turbulence (at relatively long vortex length scales, i.e. longer than $\ell = L^{-1/2}$) to quantum turbulence (at relatively small vortex length scales, i.e. shorter than ℓ). L'vov et al. [49] have shown that demanding the same energy flux through the different vortex length scales (quantum and classical) leads to relevant consequences, as the classical and quantum energy spectra at the crossover vortex length scale (at length scales near l) require that the energy at the crossover length scale should exceed

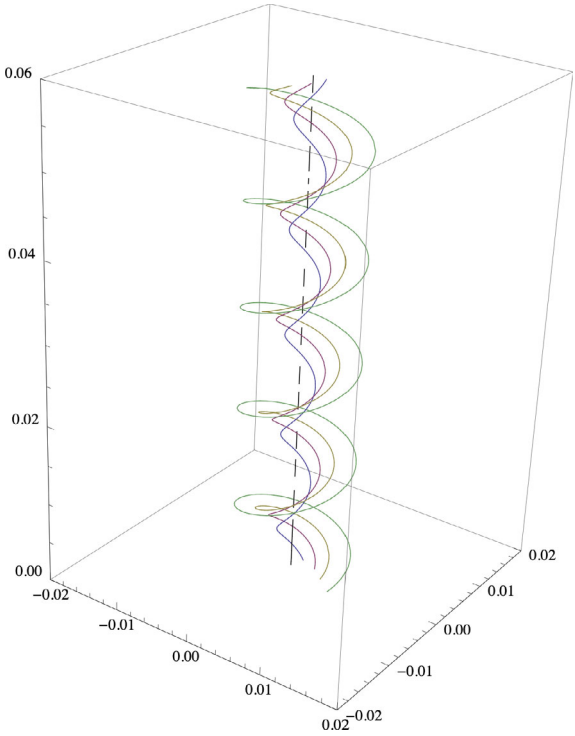


Fig. 9.5 [Color online] Plot of (9.2.14) under the conditions $u_3 > \beta K$, $u_n = 0$ (counterflow velocity parallel to the z-axis), $u_3 = 1.456$ cm/s, and $K = 500$. The amplitude of the Kelvin wave increases as shown plotting the solution at different times: $t_0 = 0$ s (blue line); $t_1 = 0.1$ s (red line); $t_2 = 0.2$ s (yellow line); $t_3 = 0.3$ s (green line). Figure from Ref. [43]

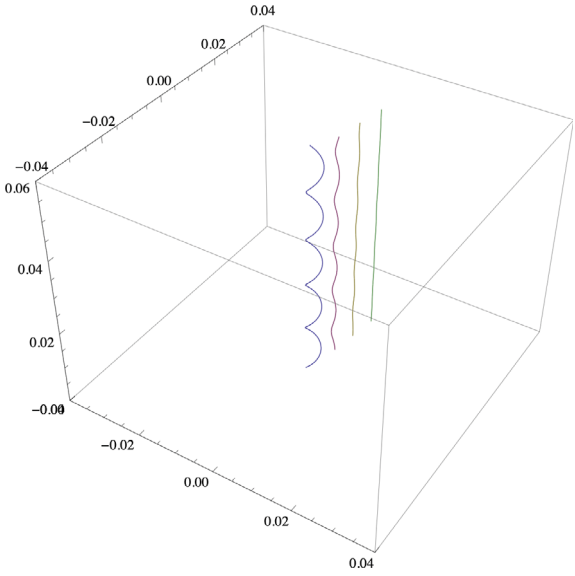


Fig. 9.6 [Color online] Plot of (9.2.14) with counterflow velocity orthogonal to the z-axis ($u_3 = 0$), $u_n = 1.456$ cm/s, and $K = 500$. The solution is shown at different times: $t_0 = 0$ s (blue line); $t_1 = 0.05$ s (red line); $t_2 = 0.1$ s (yellow line); $t_3 = 0.15$ s (green line). Figure from Ref. [43]

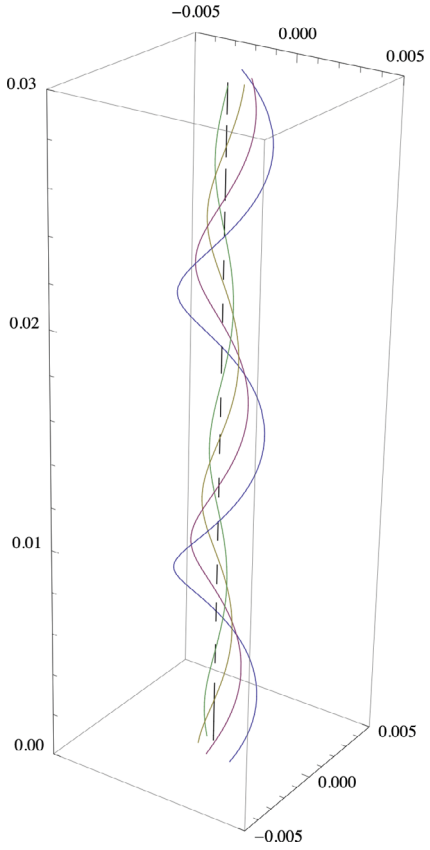


Fig. 9.7 [Color online] Plot of (9.2.14) with $u_3 = -1.456$ cm/s, $u_n = 0$ (counterflow velocity anti-parallel to the z-axis) and $K = 500$. The amplitude of the wave decreases faster than the orthogonal case in Fig. 9.6 and smaller times are chosen: $t_0 = 0$ s (blue line); $t_1 = 0.005$ s (red line); $t_2 = 0.01$ s (yellow line); $t_3 = 0.15$ s (green line). Figure from Ref. [43]

that of the classical Kolmogorov spectrum by a large factor. This interesting topic would lead us too far from the central focus of the present book (for two detailed analyses of the spectrum of Kelvin wave turbulence in superfluids see [50, 51]), but a phenomenological overview of it will be presented in Sect. 10.3.

Further geometrical aspects of the vortex tangle are the statistical properties of the vortex loops. Indeed, refined descriptions of turbulence aim to incorporate information on statistical features of the vortex length distribution, or scaling laws in vortex size, or fractal dimension, and relating them with the power provided to the tangle per unit volume [52]. In particular, Kivotides et al. [53] numerically determined the fractal dimension, and Nemirovskii et al. [54] considered its influence on the energy spectrum of the turbulent velocity field.

The energy transfer from larger to smaller scales is caused by reconnections among vortices, and by Kelvin waves. Vortex reconnections randomize the vortex tangle and initiate the physical mechanisms of the decay of the tangle. The second mechanism is a cascade of Kelvin waves of shorter and shorter wavelengths [47, 53, 55–59] triggered by vortex reconnections. This process can proceed without

significant kinetic energy losses till the ultimate length scale, where energy is radiated away as sound [60] (phonon emission), and dissipated as heat. This length scale is $l_{min} \simeq \left(\kappa^3/\bar{\varepsilon}\right)^{1/4}$, where $\bar{\varepsilon}$ is the power communicated to the system per unit volume, which is proportional to L^2 [60].

If the mentioned processes act in a self-similar way on several orders of spatial lengths, the vortex tangle exhibits fractal features [10, 11, 53, 61], because the net breaking process does not depend on the vortex size along the energy cascade [62].

Simple geometrical models of the fractal dimension of superfluid turbulence representing reconnections and interactions between vortex loops and the subsequent formation at the next generation of new vortex loops have been proposed. These models are partially inspired on the β-model for classical intermittent turbulence [63–66], which include geometrical and energetic aspects, an intuitive view of the actual complicated process, of energy transfer among different length scales.

9.3.1 *Fractal Dimension and Behaviour of the Vortex Energy per Unit Length at Different Scales*

We assume that the tangle can be statistically described as a self-similar hierarchy of loops of different sizes and look for an expression of the fractal dimension D_F of the vortex tangle, and its relation to energy density per unit length. Note that D_F is a property of the ensemble of self-similar loops, and characterizes the self-similarity properties of the tangle and not of individual vortex lines. The generation with $n = 1$ corresponds to the biggest vortices; vortices become more abundant and smaller for increasing n. We call N_n the number of loops at the n-th generation, l_n the size of each loop, and E'_n the energy of each loop.

The fractal dimension D_F of the vortex tangle can be defined by [62, 64]

$$D_F = -\lim_{n\to\infty} \frac{\log(N_n/N_0)}{\log(l_n/l_0)}, \tag{9.3.1}$$

where N_n is the number of steps along a curve (or the number of objects of a given size) and l_n the length of a single step (or the size of a given object).

The geometrical fractal dimension (9.3.1) may be related with the variation of the energy per unit length at different length scales. We assume that $E'_n \propto l_n^{\tilde{\alpha}'}$, where $\tilde{\alpha}'$ is a constant scaling exponent; this means that the energy per unit length is

$$E'_n/l_n \propto l_n^{\tilde{\alpha}'-1}. \tag{9.3.2}$$

Therefore, if $\tilde{\alpha}' > 1$ the contribution to the energy per unit length decreases for lower length scales (shorter l_n) and the opposite is true for $\tilde{\alpha}' < 1$. If $\tilde{\alpha}' = 1$ the energy per unit length is the same at each length scale.

In the low-temperature limit, when the normal component is absent and there is no friction neither sound radiation, the main assumption of the model is that the energy at each loop generation remains unchanged in the breaking and reconnection processes. We call E^{in} this amount of energy, which coincides with the total energy of the tangle. Thus, we impose

$$E_n^{in} = E_{n+1}^{in} \qquad \Rightarrow \qquad N_n E_n' = N_{n+1} E_{n+1}', \tag{9.3.3}$$

with E_n' the energy of each loop at the n-th generation, whereas the total length $L_n = N_n l_n$ may change with n. The underlying physical idea is that very close parts of the vortex lines may interfere with each other, thus leading to a nonlinear relation between the energy and the length.

Under this assumption, the exponent $\tilde{\alpha}'$ is related to the fractal dimension D_F of the tangle. If n is large enough, (9.3.1) implies that $l_n \propto N_n^{-1/D_F}$, thus (9.3.3) leads to

$$N_n^{1-(\tilde{\alpha}'/D_F)} = N_{n+1}^{1-(\tilde{\alpha}'/D_F)}. \tag{9.3.4}$$

In order that this equality holds for any n, one must have $D_F = \tilde{\alpha}'$. This shows the strong connection between the energetic features of the tangle and its geometrical structure, independently of the detailed form of the loops.

In view of the energetic meaning of $\tilde{\alpha}'$ in (9.3.2), the fractal dimension (9.3.1) may be expressed as

$$D_F - 1 = \lim_{n\to\infty} \frac{\log(E_n'/l_n)}{\log(l_n)}. \tag{9.3.5}$$

When the contribution to the energy per unit length at smaller length scales is smaller than that at larger length scales, then $D_F > 1$, whereas in the opposite case $D_F < 1$. The case $D_F < 1$ seems physically unacceptable because it would imply that vortex lines break in objects perhaps similar to Cantor's dust, which would violate the condition that the vorticity is solenoidal (a vortex is a closed loop or terminates on boundaries, but here we have no boundaries). The only acceptable situation seems that the larger length scales contribute to the vortex energy per unit length more than the smaller length scales.

9.3.2 The Random-Walk Model

An alternative simplified fractal model for the vortex tangle at moderately low temperatures ($1\,\mathrm{K} < T < 2.2\,\mathrm{K}$) assumes that the average vortex loop consists of many arches, randomly (but smoothly) connected to each other. These parts depend on the numerous previous reconnections but they are uncorrelated, because Kelvin waves do not have the time to propagate far enough. Thus each loop has a random-walk structure (see Fig. 9.8).

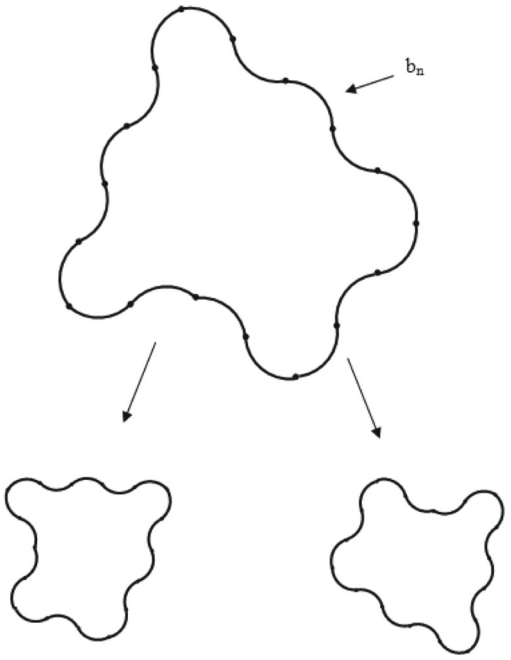


Fig. 9.8 The self-similar vortex loop cascade: each loop of the n-th generation (upper loop) breaks into several loops of the $(n+1)$-th generation (lower loops), which are assumed similar to the respective parent loop. b_n is the length of the elementary arches characterizing the irregular loops on the n-th generation. Figure from [10]

As reference configuration take a collection of N_0 vortex loops of length l_0, with each loop being formed by N_0' small arches of length b_0. The next generation of smaller vortex loops is assumed to be composed of $N_0 r$ vortex loops, of length $l_0/\tilde{\beta}$. Thus, the n-th generation is composed of $N_n = r^n N_0$ vortex loops of length $l_0/\tilde{\beta}^n$. Here, r and $\tilde{\beta}$ are positive numbers higher than 1, whose value depends on the details of the dynamics. At each step, the random-walk structure of each loop is formed by N_n' small arches of length b_n, mimicking the results of successive random reconnections. As scaling rules for N_n' and b_n, the number of small arches and their respective lengths b_n, we assume that each loop, at the n-th generation, has $N_n' = N_0'(r')^n$ arches, and that the length of these arches is scaled as $b_n = b_0/(\tilde{\beta}')^n$, i.e. they might scale in a different proportion than the total length of the loop. We introduce finally two scaling coefficients $\tilde{\alpha}$ and $\tilde{\gamma}$, putting $r' = r^{\tilde{\alpha}}$ and $\tilde{\beta}' = \tilde{\beta}^{\tilde{\gamma}}$.

We assume that in the breaking and reconnection processes the energy E^{in} at each loop generation remains unchanged and that the total length $L_n = N_n N_n' b_n$ may change with n. In the random-walk model [10], we take for the energy of a loop of the n-th generation the expression proposed by Sonin [44] for the energy density distribution in terms of the wavevector K in Kelvin wave cascade

$$E_n' = \Lambda \frac{\rho\kappa^2}{\delta_0^2} \left(\frac{\epsilon L_0^2}{\rho\kappa^3} \right)^{\frac{1}{2p-1}} b_n^{\frac{2p+1}{2p-1}} N_n', \tag{9.3.6}$$

where p is a characteristic dynamical exponent whose most plausible value seems to be $p = 3$.

Both exponents p and $\tilde{\alpha}$ appear in the energy expression (9.3.6): p appears in an explicit way, while $\tilde{\alpha}$ appears implicitly through N_n' and is related to the increment of the number of arches in a loop. If $\tilde{\alpha} > 1$ the number of arches increases faster than the number of loops, namely, smaller loops (belonging to higher values of n) are expected to have a higher number of arches, and the opposite behaviour for $\tilde{\alpha} < 1$.

Since the total number of vortex loops at the n-th generation is N_n, the total energy stored on the vortex loops of the n-th generation will be $E_n^{in} = N_n E_n'$. Imposing $E_{n+1}^{in} = E_n^{in}$ for all values of n, one gets [10]

$$D_F = \frac{2p+1}{2p-1-2\tilde{\alpha}}. \tag{9.3.7}$$

For $p = 3$, it turns out that $D_F = 7/(5 - 2\tilde{\alpha})$ and $\tilde{\alpha} < 2.5$. The range of values of the fractal dimension was [1.35, 1.75] numerically obtained in [61, 67–69] at different temperatures and by different equations. The comparison of our results with the above ones brings to $0 \leq \tilde{\alpha} \leq 0.5$ if $p = 3$. Relating the geometrical properties of the vortex tangle to its dynamical properties, namely, to its evolution equation for L, would be of interest. Indeed, in Eq. (5.1.8) for L there is an exponent n which is taken as $n = 1$ in the usual Vinen expression. It would be of interest to explore whether a geometrical fractal dimension D_F of the tangle could lead to fractional exponent n in the evolution equation for L.

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Chapter 10
Classical Versus Quantum Turbulence: Hierarchies of Fluctuations and Energy Cascades



In this chapter, we deal with a comparison of quantum versus classical turbulence from two different perspectives. The first one is the hierarchy of equations for the turbulent fluctuations of the several physical variables in the system. In classical turbulence, density, velocity, and temperature may fluctuate; in quantum turbulence, also heat flux and vortex line density may independently fluctuate, because they are independent quantities. We will pay attention to the hierarchy of equations for the fast fluctuating parts of these quantities, to its suitable truncation and to the formulation of a $K-\epsilon-L$ model generalizing the classical $K-\epsilon$ model of classical turbulence. This allows a macroscopic treatment and understanding of some features relating both kinds of turbulence.

The second perspective refers to the energy cascade between vortices of different sizes. In classical turbulence, one has the well-known Kolmogorov energy cascade in the inertial region between the stirring scale and the viscous dissipation scale. The Kolmogorov cascade implies a given distribution of the energy density of vortices as a function of the wavevectors, k, related to the reciprocal of the size of the respective vortices. In quantum turbulence instead one has quantized vortex filaments with fixed core having an intervortex distance (or similarly an average size) ℓ of the order of the square root of the vortex length density, namely, $\ell \approx L^{-1/2}$. Thus, one may make the distinction of three different scales: the hydrodynamic one, for sizes higher than ℓ, where the quantization effects are not seen; a quantum region, for sizes smaller than ℓ, where the quantization of vortices plays a fundamental role; and an intermediate region for sizes close to ℓ. The exponents of k in the scaling laws of energy density $E(k)$ versus k in these respective regions are different and they may depend on whether turbulence is mainly excited by the velocity (coflow) or by the heat flux (counterflow).

Our presentation of both aspects will be relatively simple, but this is a field of much current work from microscopic perspectives. Since the present book is mainly devoted to thermodynamic general aspects, more than to detailed microscopic mechanisms, this will be sufficient for an introductory presentation.

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10.1 Dynamics of Turbulent Fluctuations: Hierarchy of Higher-Order Moments

A complete description of turbulence would require a full hierarchy of evolution equations for the average values of the turbulent fluctuating quantities and for their second-, third-, and higher-order moments. To be of practical use, such hierarchy must be truncated at some order of moments. How this hierarchy arises from the basic equations and how it is truncated at low order are the aims of this section.

Let's assume for the sake of simplicity (and to focus the attention on the essential concepts) that the mass density is constant, namely, $\dot{\rho} = 0$, which yields $\nabla \cdot \mathbf{v} = 0$, and assume an isothermal situation, in which the energy balance equation is not necessary. Then, we are left with the following equations for $\mathbf{v}$, $\mathbf{q}$ and L in Helium II:

$$\frac{\partial}{\partial t}\mathbf{v} + \mathbf{v}\cdot\nabla\mathbf{v} + \frac{1}{\rho}\nabla p - \nu\nabla^2\mathbf{v} - \frac{\nu}{\rho s T}\nabla^2\mathbf{q} = 0, \tag{10.1.1}$$

$$\frac{\partial}{\partial t}\mathbf{q} + \mathbf{v}\cdot\nabla\mathbf{q} + \zeta\nabla T - \mu\nabla^2\mathbf{v} - \frac{\mu}{\rho s T}\nabla^2\mathbf{q} = -K_f L\mathbf{q}, \tag{10.1.2}$$

$$\frac{\partial}{\partial t}L + \mathbf{v}\cdot\nabla L - D_L\nabla^2 L = -\beta\kappa L^2 + \alpha q L^{3/2}, \tag{10.1.3}$$

where $\nu = \eta/\rho$ is the classical kinematic viscosity (length^2/time) and $\mu = \dfrac{\nu\zeta}{s}$ is the "heat" kinematic viscosity (length^{4}Kg/time^3). The last terms at the left-hand side of the two first Eqs. (10.1.1) and (10.1.2) suggest to define $\nu_1 \equiv \dfrac{\nu}{\rho s T}$ (time/kg) and $\mu_1 \equiv \dfrac{\mu}{\rho s T}$ (length^2/time), which may be considered as two kinds of thermal viscosity and which simplify the notation. The two first equations come for $\mathbf{v}$ and $\mathbf{q}$ from Eqs. (3.1.17) and (3.1.19) combined with (3.1.20) and (5.3.27) and the third one from Eq. (6.3.1) for L.

To discuss turbulence it is usual writing the several physical quantities of the problem as a time-averaged part and a fast fluctuating part, namely, $\mathbf{v} = \bar{\mathbf{v}} + \mathbf{v}'$, $\mathbf{q} = \bar{\mathbf{q}} + \mathbf{q}'$, $p = \bar{p} + p'$, $T = \bar{T} + T'$, and $L = \bar{L} + L'$. Here, $\bar{\mathbf{v}}$, $\bar{u}$, $\bar{p}$, $\bar{T}$, and $\bar{L}$ are the time average values and $\mathbf{v}'$, p', T', and L' the corresponding fluctuating turbulent contributions. The choice of a suitable time scale for the mentioned average is not an obvious matter and should not be neither too long (in which case the fast contribution scale is subsumed in the average) nor too short (in which case the whole quantity seems to be fluctuating).

For the sake of simplicity, assume that the density does not fluctuate. Because of the linearity of the continuity equation $\nabla \cdot \mathbf{v} = 0$, we find $\nabla \cdot \overline{\mathbf{v}} = 0$ and

$$\nabla \cdot \mathbf{v}' = 0. \tag{10.1.4}$$

Let's first consider the equation of the velocity $\mathbf{v}$, namely, Eq. (10.1.1). By the above substitutions, it becomes

$$\frac{\partial}{\partial t}\overline{\mathbf{v}} + (\overline{\mathbf{v}}\cdot\nabla)\overline{\mathbf{v}} + \overline{(\mathbf{v}'\cdot\nabla)\mathbf{v}'} + \frac{1}{\rho}\nabla\overline{p} - \nu\nabla^2\overline{\mathbf{v}} - \nu_1\nabla^2\overline{\mathbf{q}} = 0, \tag{10.1.5}$$

and by means of the identity $(\mathbf{v}'\cdot\nabla)\mathbf{v}' = \nabla\cdot(\mathbf{v}'\otimes\mathbf{v}') - (\nabla\cdot\mathbf{v}')\mathbf{v}'$ (where $\otimes$ stands for tensorial product) and of (10.1.4) one obtains

$$\frac{\partial}{\partial t}\overline{\mathbf{v}} + (\overline{\mathbf{v}}\cdot\nabla)\overline{\mathbf{v}} + \nabla\cdot\overline{\mathbf{v}'\otimes\mathbf{v}'} + \frac{1}{\rho}\nabla\overline{p} - \nu\nabla^2\overline{\mathbf{v}} - \nu_1\nabla^2\overline{\mathbf{q}} = 0. \tag{10.1.6}$$

We follow the same procedure for Eq. (10.1.2) for $\mathbf{q}$ and obtain

$$\frac{\partial}{\partial t}\overline{\mathbf{q}} + (\overline{\mathbf{v}}\cdot\nabla)\overline{\mathbf{q}} + \nabla\cdot\overline{\mathbf{v}'\otimes\mathbf{q}'} + \zeta\nabla\overline{T} - \mu\nabla^2\overline{\mathbf{v}} - \mu_1\nabla^2\overline{\mathbf{q}} = -K_f\overline{L}\overline{\mathbf{q}} - K_f\overline{L'\mathbf{q}'}, \tag{10.1.7}$$

where we have used the identity $(\mathbf{v}'\cdot\nabla)\mathbf{q}' = \nabla\cdot(\mathbf{v}'\otimes\mathbf{q}') - (\nabla\cdot\mathbf{v}')\mathbf{q}'$ and (10.1.4).

Note that in Eqs. (10.1.6) and (10.1.7) there are three terms dependent on the second moments of $\mathbf{v}'$, $\mathbf{q}'$, and L', namely, $\mathbf{v}'\otimes\mathbf{v}'$, $\mathbf{v}'\otimes\mathbf{q}'$, and $L'\mathbf{q}'$.

Regarding Eq. (10.1.3) for L, since $\ell = L^{-1/2}$ is the inter-vortex space (namely, the average separation between neighbouring vortices) where the interaction between vortices is stronger, we rewrite the right-hand side of (10.1.3) in terms of $\mathcal{L} = L^{1/2} = \ell^{-1}$. Thus, the equation becomes

$$\frac{\partial}{\partial t}L + \mathbf{v}\cdot\nabla L - D_L\nabla^2 L = -\beta\kappa\mathcal{L}^4 + \alpha q\mathcal{L}^3, \tag{10.1.8}$$

where the production term can be seen as $\alpha q L/\ell$ and the destruction term as $\beta\kappa L/\ell^2$.

Inserting the fluctuations in Eq. (10.1.8) for the evolution equation for $\overline{L}$ and considering $L = \overline{L} + L'$, $\mathcal{L} = \overline{\mathcal{L}} + \mathcal{L}'$, one has

$$\frac{\partial}{\partial t}\overline{L} + \overline{\mathbf{v}}\cdot\nabla\overline{L} + \nabla\cdot\overline{\mathbf{v}'L'} - D_L\nabla^2\overline{L} = \alpha q\left(\overline{\mathcal{L}}^3 + 3\overline{\mathcal{L}}\,\overline{\mathcal{L}'^2} + \overline{\mathcal{L}'^3}\right) +$$

$$-\beta\kappa\left(\overline{\mathcal{L}}^4 + 6\overline{\mathcal{L}}^2\,\overline{\mathcal{L}'^2} + 4\overline{\mathcal{L}}\,\overline{\mathcal{L}'^3} + \overline{\mathcal{L}'^4}\right) + \alpha\left(3\overline{\mathcal{L}}^2\,\overline{q'\mathcal{L}'} + 3\overline{\mathcal{L}}\,\overline{q'\mathcal{L}'^2} + \overline{q'\mathcal{L}'^3}\right), \tag{10.1.9}$$

because of the identity $\mathbf{v}'\cdot\nabla L' = \nabla\cdot(\mathbf{v}'L') - (\nabla\cdot\mathbf{v}')L'$ and (10.1.4). In (10.1.9), terms in the products of fluctuations of $\mathcal{L}'$ and $\mathbf{q}'$ appear. As for (10.1.6) and (10.1.7), expressions for these contributions in terms of $\overline{\mathbf{v}}$ and $\overline{\mathbf{q}}$ and $\overline{L}$ or their gradients must be provided.

An evolution equation for the second-order moments $\mathbf{v}'\otimes\mathbf{v}'$, $\mathbf{v}'\otimes\mathbf{q}'$, $\mathbf{q}'\otimes\mathbf{q}'$ and $\mathbf{q}'L'$ could be found by averaging the equations for $\mathbf{v}'\otimes\mathbf{v}'$, $\mathbf{v}'\otimes\mathbf{q}'$, $\mathbf{q}'\otimes\mathbf{q}'$ and $\mathbf{q}'L'$ obtained by multiplying by $\mathbf{v}'$ and $\mathbf{q}'$ the evolution equations for $\mathbf{v}$ and $\mathbf{q}$ and averaging them on time. In such equations would appear third-order moments of

the form $\mathbf{v}' \otimes \mathbf{v}' \otimes \mathbf{v}'$, $\mathbf{v}' \otimes \mathbf{v}' \otimes \mathbf{q}'$, $\mathbf{v}' \otimes \mathbf{q}' \otimes L'$ and so on, as well as second-order moments of the spatial gradients of fluctuations, leading to a hierarchy of higher-order moments, in which moments of order $n+1$ appear in the evolution equation for moments of order n.

The closure of the hierarchy of higher-order moments is crucial for modelling and understanding the main features of turbulence, and there are many different strategies of truncating this hierarchy: use of maximum-entropy methods [1–10] or analogies with well-known models and methods of the kinetic theory of gases [11–14] based on some generalized forms of Boltzmann equation or of lattice-Boltzmann equation [15, 16].

10.2 Closure of the Hierarchy of Equations for Moments

10.2.1 A Short Remainder of the $K-\epsilon$ Model

To truncate the hierarchy of equations for $\overline{\mathbf{v}}$ at lowest order, obtained by averaging the Navier–Stokes equation at successive higher moments of velocity, in classical turbulence it is assumed that the extra turbulent contribution to the evolution equation for $\overline{\mathbf{v}}$, namely, the term in $\overline{\mathbf{v}' \otimes \mathbf{v}'}$ in Eq. (10.1.6), has a form analogous to that of the usual viscous term, but with an effective viscosity of turbulent origin. Thus, it is taken [1, 13, 17, 18]

$$\mathbf{R}_v \equiv \overline{\mathbf{v}' \otimes \mathbf{v}'} = -2\nu_t \nabla^s \overline{\mathbf{v}} \tag{10.2.10}$$

with ν_t the so-called eddy viscosity, which phenomenologically describes the contribution of turbulence to the hydrodynamic resistance to the flow, and $\nabla^s \overline{\mathbf{v}}$ the strain rate tensor defined as $\nabla^s \overline{\mathbf{v}} = (\nabla \overline{\mathbf{v}} + \nabla \overline{\mathbf{v}}^T)/2$. The motivation for such closure is keeping the form of the Navier–Stokes equation, but with a phenomenologically modified viscosity, describing the big increase of the dissipation in turbulent flow as compared to the corresponding laminar flows. Such eddy vorticity is assumed to depend on turbulent features, namely, the kinetic energy per unit mass, $K \equiv \frac{1}{2}\overline{\mathbf{v}' \cdot \mathbf{v}'}$, and the dissipation function $\epsilon \equiv 2\nu \overline{\nabla^s \mathbf{v}' : \nabla^s \mathbf{v}'}$. The dimensions of K are (length$^2 \times$ time^{-2}) and those of ϵ are (length$^2 \times$ time^{-3}); thus, the simplest combination of K and ϵ having the dimensions of viscosity (length$^2 \times$ time^{-1}) is

$$\nu_t = C_\nu \frac{K^2}{\epsilon}, \tag{10.2.11}$$

with C_ν a dimensionless constant of the order of 0.09 [19]. With (10.2.10) and (10.2.11), the evolution Eq. (10.1.6) for $\overline{\mathbf{v}}$ in a classical fluid (i.e. with $\overline{\mathbf{q}} = 0$) becomes closed if evolution equations for K and ϵ are proposed [see, for instance, Eqs. (10.2.23) and (10.2.24)]. For this emphasis on K and ϵ, this model is known as

$K-\epsilon$ model. In superfluid turbulence there are more independent variables than $\mathbf{v}$ (i.e. $\mathbf{q}$ and L) and the situation is conceptually analogous but more cumbersome.

10.2.2 A Generalized $K-\epsilon-L$ Model in Superfluid Helium

Now, we define with specific symbols the terms involving moments of the fluctuations in the evolution equations for $\overline{\mathbf{v}}$ (10.1.6) and $\overline{\mathbf{q}}$ (10.1.7), namely: $\mathbf{R}_v \equiv \overline{\mathbf{v}' \otimes \mathbf{v}'}$ appearing in (10.1.6) and $\mathbf{R}_q \equiv \overline{\mathbf{v}' \otimes \mathbf{q}'}$ and $\mathbf{R}_{qL} \equiv K_f \overline{\mathbf{q}' L'}$ appearing in (10.1.7) (in Sect. 10.2.3 we will consider the evolution equation for $\overline{L}$). In order to close the equations in a way analogous to that used in the $K-\epsilon$ model, these quantities must be given in terms of the average quantities $\overline{\mathbf{v}}$, $\overline{\mathbf{q}}$ and $\overline{L}$ or their gradients.

Closure of Equations for $\overline{\mathbf{v}}$ and $\overline{\mathbf{q}}$

We now take a lowest order closure of Eqs. (10.1.6) and (10.1.7) by expressing the second-order moments appearing in them ($\mathbf{R}_v$, $\mathbf{R}_q$, $\mathbf{R}_{qL}$) in terms of $\overline{\mathbf{v}}$, $\overline{\mathbf{q}}$ and $\overline{\mathbf{L}}$. As in (10.2.10) we impose that the equations for $\overline{\mathbf{v}}$ and $\overline{\mathbf{q}}$ have the same form as those for $\mathbf{v}$ in (10.1.1) and for $\mathbf{q}$ in (10.1.2), but with effective turbulent parameters, by taking

$$\mathbf{R}_v \equiv \overline{\mathbf{v}' \otimes \mathbf{v}'} = -\nu_t \nabla \overline{\mathbf{v}} - \nu_{1t} \nabla \overline{\mathbf{q}} \tag{10.2.12}$$

$$\mathbf{R}_q \equiv \overline{\mathbf{v}' \otimes \mathbf{q}'} = -\mu_t \nabla \overline{\mathbf{v}} - \mu_{1t} \nabla \overline{\mathbf{q}} \tag{10.2.13}$$

$$\mathbf{R}_{qL} \equiv K_f \overline{\mathbf{q}' L'} = K_{ft}\, \overline{q}\, \overline{L}. \tag{10.2.14}$$

In (10.2.12)–(10.2.14), following the usual procedure of the zeroth-order $K-\epsilon$ description of classical turbulence, we have introduced the turbulent parameters ν_t, ν_{1t}, μ_t, μ_{1t}, and K_{ft}, which, in analogy to the $K-\epsilon$ model of classical turbulence, will be phenomenologically expressed in terms of the second moments of $\mathbf{v}'$ and $\mathbf{q}'$

$$K \equiv \frac{1}{2}\overline{\mathbf{v}' \cdot \mathbf{v}'}, \qquad K_q \equiv \frac{1}{2}\overline{\mathbf{q}' \cdot \mathbf{q}'}, \tag{10.2.15}$$

and of the corresponding dissipation functions ϵ and ϵ_q

$$\epsilon \equiv \left[\nu \overline{\nabla \mathbf{v}' : \nabla \mathbf{v}'} + \nu_1 \overline{\nabla \mathbf{v}' : \nabla \mathbf{q}'}\right]. \tag{10.2.16}$$

$$\epsilon_q \equiv \left[\mu \overline{\nabla \mathbf{v}' : \nabla \mathbf{q}'} + \mu_1 \overline{\nabla \mathbf{q}' : \nabla \mathbf{q}'}\right], \tag{10.2.17}$$

related to the second moments of $\nabla \mathbf{v}'$ and $\nabla \mathbf{q}'$ (see (10.2.23) and (10.2.25)). Equations (10.1.6) and (10.1.7) can be rewritten using (10.2.12)–(10.2.14). Thus, the equation for $\overline{\mathbf{v}}$ becomes

$$\frac{\partial}{\partial t}\overline{\mathbf{v}} + (\overline{\mathbf{v}} \cdot \nabla)\overline{\mathbf{v}} + \frac{1}{\rho}\nabla \overline{p} - (\nu + \nu_t)\nabla^2 \overline{\mathbf{v}} - (\nu_1 + \nu_{1t})\nabla^2 \overline{\mathbf{q}} = 0, \tag{10.2.18}$$

and that for $\overline{\mathbf{q}}$ becomes

$$\frac{\partial}{\partial t}\overline{\mathbf{q}} + (\overline{\mathbf{v}} \cdot \nabla)\overline{\mathbf{q}} + \zeta \nabla \overline{T} - (\mu + \mu_t)\nabla^2 \overline{\mathbf{v}} - (\mu_1 + \mu_{1t})\nabla^2 \overline{\mathbf{q}} = -(K_f + K_{ft})\overline{L}\overline{\mathbf{q}}. \tag{10.2.19}$$

The coefficients ν_t, ν_{1t}, μ_t, μ_{1t}, and K_{ft} will be expressed on dimensional grounds in terms of K, ϵ, K_q, and ϵ_q. The dimensions of K are (length^2 time^{-2}), those of K_q are (length^6 kg^2 time^{-6}), those of ϵ are (length^2 time^{-3}), those of ϵ_q are (length^6 kg^2 time^{-7}) and those of μ are (length^4 kg time^{-3}), those of ν_t, μ_{1t} and K_{ft} are (length^2 time^{-1}), those of ν_{1t} are (time kg^{-1}) and those of μ_t are (length^4 kg time^{-3}). A possible proposal for the mentioned coefficients is

$$\nu_t = C_\nu \frac{K^2}{\epsilon}, \qquad \nu_{1t} = \frac{C_{\nu 1}}{\rho s T}\frac{K^2}{\epsilon}, \tag{10.2.20}$$

$$\mu_t = \frac{C_\mu}{\rho s T}\frac{K_q^2}{\epsilon_q}, \qquad \mu_{1t} = \frac{C_{\mu 1}}{(\rho s T)^2}\frac{K_q^2}{\epsilon_q}, \qquad K_{ft} = \frac{C_K}{(\rho s T)^2}\frac{K_q^2}{\epsilon_q}, \tag{10.2.21}$$

with C_ν, $C_{\nu 1}$, C_μ, $C_{\mu 1}$, and C_K dimensionless constants (to be found from the experiments), as it happens with the value of C_ν in the usual $K - \epsilon$ model.

This model is an open proposal rather than a definite one. Expressions (10.2.20), (10.2.21), and (10.2.32) for the turbulent coefficients in terms of K, K_q, and K_L, and ϵ, ϵ_q, and ϵ_L could be further discussed, as well as Eqs. (10.2.24), (10.2.25), and (10.2.35) for the evolution of ϵ, ϵ_q, and ϵ_L. For instance, concerning expressions (10.2.20) and (10.2.21) one could argue that since the terms in ν_t, ν_{1t}, μ_t, μ_{1t} in (10.2.12) and in (10.2.13) couple $\mathbf{v}$ and $\mathbf{q}$, an alternative proposal for ν_{1t} in (10.2.20) and for μ_t in (10.2.21) in terms if K, ϵ, K_q and ϵ_q could be

$$\nu_{1t} = \frac{C'_{\nu 1}}{\sqrt{\rho s T}}\frac{K K_q}{\sqrt{\epsilon \epsilon_q}}, \qquad \mu_t = \frac{C'_\mu}{\sqrt{\rho s T}}\frac{K K_q}{\sqrt{\epsilon \epsilon_q}}. \tag{10.2.22}$$

To close the system of evolution equations, it remains writing the evolution equations for K, ϵ, K_q and ϵ_q, giving the evolution of (10.2.20) and (10.2.21). In the evolution equations for K we take, after a relatively long derivation,

$$\frac{\partial}{\partial t}K + (\overline{\mathbf{v}} \cdot \nabla)K - \nabla \cdot \left[\left(\nu + \nu\alpha_1 + \frac{\nu_t^*}{\sigma_K}\right)\nabla K\right] = -\epsilon + \frac{\nu_t}{\nu}\epsilon_K, \tag{10.2.23}$$

where $\epsilon_K = \nu \nabla \overline{\mathbf{v}} : \nabla \overline{\mathbf{v}} + \nu_1 \nabla \overline{\mathbf{v}} : \nabla \overline{\mathbf{q}}$. Note that there is a relevant conceptual difference between ϵ defined in (10.2.16) and ϵ_K: the former one depends on the gradients of the fluctuating quantities $\mathbf{v}'$ and $\mathbf{q}'$, whereas the latter one depends on the gradients of the average values $\overline{\mathbf{v}}$ and $\overline{\mathbf{q}}$. The physical meaning of the coefficients α_1 and ν_t^* may be seen in [20]. Here, we consider them as experimental fitting parameters.

The ratio K/ϵ has the dimension of time and it is usually considered as a characteristic time for the decay of ϵ. In analogy with what is done in the $K - \epsilon$ model [13], an evolution equation for ϵ can be heuristically obtained by keeping the same structure of the equation of K and by dividing the right-hand side of Eq. (10.2.23) by the characteristic time K/ϵ. This yields

$$\frac{\partial}{\partial t}\epsilon + (\overline{\mathbf{v}} \cdot \nabla)\epsilon - \nabla \cdot \left[\left(\nu + \nu\alpha_1 + \frac{\nu_t^*}{\sigma_\epsilon} \right) \nabla \epsilon \right] = \frac{\epsilon}{K} \left[\frac{\nu_t}{\nu} C_{\epsilon 1} \epsilon_K - C_{\epsilon 2} \epsilon \right], \tag{10.2.24}$$

with $C_{\epsilon 1}$ and $C_{\epsilon 2}$ numerical coefficients. In the $K - \epsilon$ model for viscous fluid ($\alpha_1 = 0$), $C_{\epsilon 1} = 1.44$, $C_{\epsilon 2} = 1.92$ [13]. We tentatively take this equation for our proposal of the model. In fact, the form as that used in (10.2.23) and (10.2.24) has been justified in [13] on grounds of analogy with kinetic theory using thermodynamics arguments.

Following for the second moments of $\mathbf{q}$, the same procedure followed to find the evolution equation for the second moments of $\mathbf{v}'$, the following equations for K_q and ϵ_q are derived

$$\begin{aligned} &\frac{\partial}{\partial t} K_q + (\overline{\mathbf{v}} \cdot \nabla) K_q - \nabla \cdot \left[\left(\mu + \mu\alpha_2 + \frac{\mu_t^*}{\sigma_{Kq}} \right) \nabla K_q \right] \\ &\quad = -\epsilon_q + \frac{\mu_t}{\mu} \epsilon_{Kq} - 2 \left(K_f + K_f^* \right) \overline{L} K_q, \end{aligned} \tag{10.2.25}$$

and

$$\begin{aligned} &\frac{\partial}{\partial t} \epsilon_q + (\overline{\mathbf{v}} \cdot \nabla) \epsilon_q - \nabla \cdot \left[\left(\mu + \mu\alpha_2 + \frac{\mu_t^*}{\sigma_{\epsilon q}} \right) \nabla \epsilon_q \right] \\ &\quad = \frac{\epsilon_q}{K_q} \left[C_{\epsilon q 1} \epsilon_{Kq} + -C_{\epsilon q 2} \epsilon_q - 2 C_{\epsilon q 3} \left(K_f + K_f^* \right) \overline{L} K_q \right], \end{aligned} \tag{10.2.26}$$

with $C_{\epsilon 2q}$, $C_{\epsilon 1q}$, and $C_{\epsilon 3q}$ numerical constants. These equations play a role analogous to those evolution equations for K and ϵ (10.2.23) and (10.2.24). We take them as an explicit illustration of our proposal, but the method could be open to more general evolution equations for ϵ_q.

10.2.3 Evolution Equation and Dissipation Function for $\overline{L}$

After having considered a closure of Eqs. (10.1.6) and (10.1.7) for $\overline{\mathbf{v}}$ and $\overline{\mathbf{q}}$, we consider the fluctuations in the evolution equation for $\overline{L}$ (10.1.9), namely: $\mathbf{R}_{vL} \equiv \overline{\mathbf{v}'L'}$, $\mathbf{R}_{q\mathcal{L}} \equiv \overline{\mathbf{q}'\mathcal{L}'}$, $\mathbf{R}_{\mathcal{L}\mathcal{L}} \equiv \overline{\mathcal{L}'\mathcal{L}'}$, $\mathbf{R}_{\mathcal{L}\mathcal{L}\mathcal{L}} \equiv \overline{\mathcal{L}'\mathcal{L}'\mathcal{L}'}$, $\mathbf{R}_{\mathcal{L}\mathcal{L}\mathcal{L}\mathcal{L}} \equiv \overline{\mathcal{L}'\mathcal{L}'\mathcal{L}'\mathcal{L}'}$, $\mathbf{R}_{q\mathcal{L}\mathcal{L}} \equiv \overline{q'\mathcal{L}'\mathcal{L}'}$ and $\mathbf{R}_{q\mathcal{L}\mathcal{L}\mathcal{L}} \equiv \overline{q'\mathcal{L}'\mathcal{L}'\mathcal{L}'}$. To keep the form of Eq. (10.1.8), we propose the following expressions to close Eq. (10.1.9):

$$\mathbf{R}_{vL} \equiv \overline{\mathbf{v}'L'} = -D_{Lt}\nabla L \tag{10.2.27}$$

$$\mathbf{R}_{q\mathcal{L}} \equiv \overline{q'\mathcal{L}'} = \alpha_t \overline{q}\,\overline{\mathcal{L}} \tag{10.2.28}$$

$$\mathbf{R}_{\mathcal{L}\mathcal{L}} \equiv \overline{\mathcal{L}'\mathcal{L}'} = \beta_t \overline{\mathcal{L}}^2. \tag{10.2.29}$$

Closure in Eq. (10.2.27) follows the same strategy of requiring that the equation for $\overline{L}$ has the same form as that for L, but with an effective diffusion parameter incorporating the influence of $\overline{v'\mathcal{L}'}$.

For the third and fourth moments, we assume that the fluctuations in $\mathcal{L}'$ are Gaussian, namely, $\mathbf{R}_{\mathcal{L}\mathcal{L}\mathcal{L}} \equiv \overline{\mathcal{L}'}^3 = 0, \mathbf{R}_{\mathcal{L}\mathcal{L}\mathcal{L}\mathcal{L}} \equiv \overline{\mathcal{L}'}^4 = 3\mathbf{R}^2_{\mathcal{L}\mathcal{L}}$, and that $\mathbf{R}_{q\mathcal{L}\mathcal{L}} \equiv \overline{q'\mathcal{L}'}^2 = 0$ and $\mathbf{R}_{q\mathcal{L}\mathcal{L}\mathcal{L}} \equiv \overline{q'\mathcal{L}'}^3 = 0$. With the assumptions (10.2.27)–(10.2.29) and introducing them into (10.1.9), it is seen that $\mathbf{R}_{vL}$ modifies the effective diffusion coefficient D_L of the quantized vortices with the turbulent contribution D_{Lt}, while $\mathbf{R}_{qL}$ contributes to the production of quantized vortices and $\mathbf{R}_{LL}$ contributes both to the production and the destruction terms.

In (10.2.27)–(10.2.29), we have introduced the phenomenological turbulent parameters D_{Lt}, α_t, and β_t, which will be expressed in terms of the second moments of L and of $\nabla L'$, namely:

$$K_L \equiv \frac{1}{2}\overline{L' \cdot L'} \tag{10.2.30}$$

and the corresponding dissipation function ϵ_L

$$\epsilon_L \equiv D_L \overline{\nabla L' \cdot \nabla L'}. \tag{10.2.31}$$

This plays an analogous role to ϵ (10.2.16) and ϵ_q (10.2.17) in the equation for K and for K_q, respectively.

In the equation of K_L (see (10.2.34)), the turbulent diffusion coefficient will be

$$D_{Lt} = C_L \kappa^2 \ell^2 \frac{K_L^2}{\epsilon_L}. \tag{10.2.32}$$

The quantities C_L in (10.2.32) and α_t and β_t in (10.2.28) and (10.2.29) are dimensionless constants.

Equation (10.1.9) of the time-averaged vortex line density becomes

$$\frac{\partial}{\partial t}\overline{L} + \overline{\mathbf{v}} \cdot \nabla \overline{L} - (D_L + D_{Lt})\nabla^2 \overline{L} = \tilde{\alpha}\overline{q}\overline{\mathcal{L}}^3 - \tilde{\beta}\kappa\overline{\mathcal{L}}^4 \tag{10.2.33}$$

with $\tilde{\alpha} = \alpha(1 + 3\beta_t + 3\alpha_t)$ and $\tilde{\beta} = \beta(1 + 9\beta_t)$.

After relatively long calculations one obtains for the evolution equation for K_L and ϵ_L

$$\frac{\partial}{\partial t}K_L + \overline{\mathbf{v}} \cdot \nabla K_L - \nabla \cdot \left[\left(D_L + \frac{D_{Lt}}{\sigma_{KL}}\right)\nabla K_L\right] = -\epsilon_L + \frac{D_{Lt}}{D_L}\epsilon_{KL} + (3\alpha\beta_t + \alpha\alpha_t)\overline{q}\,\overline{L}^{5/2} - 4\beta\beta_t\kappa\overline{L}^3 \tag{10.2.34}$$

and

$$\frac{\partial}{\partial t}\epsilon_L + \overline{\mathbf{v}} \cdot \nabla \epsilon_L - \nabla \cdot \left[\left(D_L + \frac{D_{Lt}}{\sigma_{\epsilon L}}\right)\nabla \epsilon_L\right] = = \frac{\epsilon_L}{K_L}\left(-C_{L1}\epsilon_L + C_{L2}\frac{D_{Lt}}{D_L}\epsilon_{KL} + C_{L3}(3\alpha\beta_t + \alpha\alpha_t)\overline{q}\,\overline{L}^{5/2} - 4C_{L4}\beta\beta_t\kappa\overline{L}^3\right) \tag{10.2.35}$$

with C_{Li} dimensionless constants. These equations close the system.

10.2.4 Discussion of the $K - \epsilon - L$ Truncation Model

In Sects. 10.1–10.2.3, we propose a generalized $K - \epsilon$ model applied to superfluid helium. Because of the fields involved in it, we have named this model $K - \epsilon - L$ model. The starting points are the dynamical equations for $\mathbf{v}$, $\mathbf{q}$ and L (10.1.1)–(10.1.3) [21]. Thus, it is natural that second moments of the fluctuations and of the gradient of the fluctuations of $\mathbf{v}'$, $\mathbf{q}'$ and L' appear in this model, namely, K, K_q, K_L, ϵ, ϵ_q, and ϵ_L in an analogous way as K and ϵ appear in the evolution equation for $\mathbf{v}$ in the $K - \epsilon$ model for simple viscous fluids.

The complete $K - \epsilon - L$ model is thus given by the following ingredients:

- The evolution equations for the averaged fields, namely, (10.2.18) for $\overline{\mathbf{v}}$, (10.2.19) for $\overline{\mathbf{q}}$, and (10.2.33) for $\overline{L}$, together with $\nabla \cdot \overline{\mathbf{v}} = 0$.
- The evolution equations for the second moments of the fluctuations and of their gradients, namely, (10.2.23) for K, (10.2.24) for ϵ, (10.2.25) for K_q, (10.2.26) for ϵ_q, (10.2.34) for K_L, and (10.2.35) for ϵ_L.
- The expressions for the phenomenological turbulent coefficients used in the closure, namely, (10.2.20) for ν_t and ν_{1t}, (10.2.21) for μ_t, μ_{1t} and K_{ft} and (10.2.32) for D_{Lt} in terms of K, ϵ, K_q, ϵ_q, K_L, and ϵ_L.

- The numerical constants appearing in the evolution Eq. (10.2.23) and (10.2.24) for K and ϵ, (10.2.25) and (10.2.26) for K_q and ϵ_q, and (10.2.34) and (10.2.35) for K_L and ϵ_L. Those involving K and ϵ are known from the usual viscous fluids [13].
- The numerical constants appearing in (10.2.20), (10.2.21), and (10.2.32) as well as σ_K in (10.2.23), C_ν and $C_{\nu 1}$ in (10.2.20), and σ_{Kq} in (10.2.25), C_ν and $C_{\nu 1}$ in (10.2.21), σ_{KL} in (10.2.34), and $\sigma_{\epsilon L}$ and C_L in (10.2.32).

Of course, these equations strongly depend on the approximations made in the above sections, but they have been suggested by the structure of the equations themselves and keeping in mind the procedure used for the classical turbulence. The second moments of the fluctuating quantities contribute to change the effective viscosity, the diffusion coefficient, and the production/destruction coefficients in their respective equations.

Furthermore, the fluctuations of L are especially complex and they depend on the kind of flow. The studies performed by Nemirovskii in Ref. [22] have shown that different kinds of turbulence (uniformly distributed or bundles of quantized vortices) lead to different expression for the spectrum of L. Thus, a simple proposal as the one made in this paper is not expected to cover all the complexities of the problem, but may provide a useful starting point for a phenomenological modellization. For these reasons, we have chosen the Vinen equation for L with an inertial term, as the simplest model for describing turbulence in a uniformly distributed turbulence. In future analyses, more complex situations should be analysed.

10.3 Energy Cascades in Superfluid Turbulence: From Classical Regime to Quantum Regime

A second perspective in the analysis of turbulence is provided by the energy cascades describing the flow of energy from bigger to smaller vortices. In superfluid helium, the quantized vortices, of characteristic average size $\ell \simeq L^{-1/2}$, set a relevant division of the total cascade into a hydrodynamic region ($k\ell \ll 1$), a quantum region ($k\ell \gg 1$), and an intermediate region ($k\ell \simeq 1$). Furthermore, the cascade may depend on whether the stirring situation corresponds to $\mathbf{q} = 0$, $\mathbf{v} \neq 0$ (coflow) or $\mathbf{q} \neq 0$, $\mathbf{v} = 0$ (counterflow), or both.

In this section, we present a short introduction to some of the main features of the energy spectrum of the superfluid component in Helium II in coflow, counterflow, and superflow.

10.3.1 Coflow

In coflow, the two components of superfluid helium flow essentially in the same direction and, because of the presence of quantized vortices and of the mutual friction

between vortices and normal component, the two components match their velocities in such a way that their relative velocity (counterflow velocity) $\mathbf{v}_{ns} = \mathbf{v}_n - \mathbf{v}_s$ is negligible. Since the heat flux in the two-fluid model is proportional to $\mathbf{v}_{ns}$, namely, $\mathbf{q} = \rho_s ST(\mathbf{v}_n - \mathbf{v}_s)$, with S the entropy per unit volume, then $\mathbf{q} = 0$; thus, in this case, energy is mainly associated to the mean velocity field $\rho\mathbf{v} = \rho_n\mathbf{v}_n + \rho_s\mathbf{v}_s$.

The evolution equation of the velocity $\mathbf{v}$ in the one-fluid theory is Eq. (10.1.1), namely:

$$\frac{\partial}{\partial t}\mathbf{v} + \mathbf{v}\cdot\nabla\mathbf{v} + \frac{1}{\rho}\nabla p - \nu\nabla^2\mathbf{v} - \frac{\nu}{ST}\nabla^2\mathbf{q} = 0, \tag{10.3.36}$$

which is the usual Navier–Stokes equation, ν being the kinematic viscosity plus the contribution of the heat flux in the last term.

Let the energy $E(t) = \dfrac{1}{V}\displaystyle\int\frac{1}{2}\mathbf{v}^2 dV = \int_0^\infty E(k,t)\,dk$, so that $E(k,t)$ gives the fraction of turbulent kinetic energy in modes with wavevectors between k and $d + dk$. In the hydrodynamic regime it is assumed that viscous dissipation does not take place in a wide range of values of k (the so-called inertial range), where the only effect is the breaking of vortices; thus $E(k,t)$ satisfies the continuity equation in the spectral space

$$\frac{\partial}{\partial t}E(k,t) + \frac{\partial}{\partial k}\epsilon(k,t) = 0 \tag{10.3.37}$$

where $\epsilon(k,t)$ is the energy flux between modes with different wavevectors.

In the stationary case, Eq. (10.3.37) leads to $\dfrac{\partial}{\partial k}\epsilon(k) = 0$ and therefore $\epsilon(k) \equiv \epsilon$ will be constant, independent of k and t. Using dimensional arguments, $E(k)$, which has the dimensions $(\text{length})^3(\text{time})^{-2}$ (because it is given by kinetic energy per unit mass per unit wavevector), is a function of the dissipation rate ϵ, which has dimension of $(\text{length})^2(\text{time})^{-3}$, and of the wavevector k (which has dimension of $(\text{length})^{-1}$). Then, from purely dimensional arguments it follows that in this case

$$E(k) = C_{K41}\epsilon^a k^b = C_{K41}\epsilon^{2/3}k^{-5/3} \tag{10.3.38}$$

which is the Kolmogorov cascade and $C_{K41} \simeq 1$ a numerical coefficient. Dimensional arguments are sufficient in this case to lead to the values $a = 2/3$, $b = -5/3$ for the exponents in the second member of (10.3.38).

In the so-called intermittent turbulence, this cascade changes to [4]

$$E(k) \sim \epsilon^a k^b \sim \epsilon^{2/3}k^{-5/3}(kL_0)^\beta, \tag{10.3.39}$$

where L_0 is the characteristic size of the stirring process supplying energy to the system, and the exponent β is related to the fractal dimension of intermittent turbulence which cannot be specified by dimensional analysis. In physical terms, it means that the stirring process has some influence along the whole cascade, instead of being "forgotten" in the intermediate scales of the inertial regime, as in (10.3.38). In the intermittent regime, the average intensity of the turbulent fluctuations of the velocity

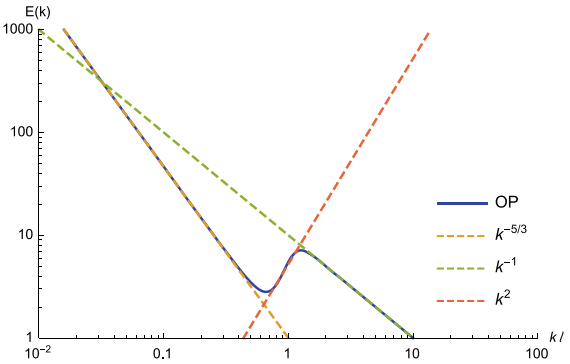


Fig. 10.1 Plot of the energy cascade (10.3.40) for $A_1 = 10$, $A_{K41} = 1$ and $A = 4$. The legend on the right-hand side refers to the power-law behaviour of the straight lines in the Log-Log plot, and OP stands for "Our Proposal" (10.3.40), which reproduces the observed behaviour represented in Fig. 4 of the paper [25]. This figure is Fig. 1 of the paper [24]

is not constant along the time, but it is very strong in some irregular intervals, and very low in the remaining intervals of times.

In superfluid helium, the transfer of energy supplied to the largest scales down to smallest scales is still open to debate. It has been observed that if energy is supplied in the same way as in a classical fluid (using a fork or a grid, for instance), superfluid helium behaves as any classical fluid, namely, the transfer of energy from bigger to smaller vortices follows the classical Kolmogorov spectrum (10.3.38) in the hydrodynamic region $k\ell \ll 1$. A different behaviour is observed for $k\ell \gg 1$, namely, in the quantum regime. It turns out that in such region a k^{-1} behaviour observed instead of $k^{-5/3}$ (in Sect. 10.3.2 we will comment about the k^{-1} behaviour).

In the intermediate region of $k\ell \simeq 1$, it is observed a relatively narrow transition region with equipartition behaviour k^2 [23]. As an interpolation function between the $k^{-5/3}$ and k^{-1} regions, with an intermediate k^2 shoulder, it has been proposed [24]

$$E(k) = \frac{A_1 k^{-1} + A_{K41} k^{-5/3}}{2} \left[1 + \frac{A_1 k^{-1} - A_{K41} k^{-5/3}}{A_1 k^{-1} + A_{K41} k^{-5/3}} \tanh\left[A(k\ell - 1) \right] \right], \tag{10.3.40}$$

where $A_{K41} = C_{K41} \epsilon^{2/3}$, as given in (10.3.38), $A_1 = c_1 (\kappa/\ell)^2$, and c_1 and A dimensionless constants. In Fig. 10.1, there is the plot of (10.3.40). For $A = 4$, this figure qualitatively reproduces the observed behaviour reported in Ref. [25], namely, a transition from $k^{-5/3}$ to k^{-1} behaviour with a k^2 shoulder. In Sect. 10.3.3, we further discuss the physics behind the parameter A.

10.3.2 Counterflow and Superflow

Counterflow and superflow are in some sense analogous situations, because the main important ingredient is the relative velocity between the two components, the counterflow velocity $\mathbf{v}_{ns}$, whereas counterflow is characterized by $\overline{\mathbf{v}} = 0$ and superflow

by $\overline{\mathbf{v}_n} = 0$. In terms of the one-fluid theory, $\overline{\mathbf{v}} = 0$ and $\mathbf{q} \neq 0$ in counterflow and $\overline{\mathbf{v}} = constant$ and $\mathbf{q} \neq 0$ in superflow. Recall that superflow refers to flows in regions between two superleaks (very narrow channels where the superfluid component but not the normal component can flow along them). Thus, instead of being the main field (as in coflow) here $\mathbf{q}$ is the main fluctuating variable.

For the superfluid component, or in heat fluctuations, the relevant physical quantities are κ (the quantum of circulation, having dimensions $(\text{length})^2(\text{time})^{-1}$), the vortex average separation ℓ (related to L as $L^{-1/2}$) and the wavevector k. In these situations, it has been observed a behaviour k^{-3} in the hydrodynamic regime ($k\ell \ll 1$) and k^{-1} in the quantum regime ($k\ell \gg 1$).

From κ, a reciprocal of time can be easily obtained as $(\text{time})^{-1} \sim \kappa/(\text{length})^2$. Then, one may have several different combinations with the dimensions of $E(k, \ell)$, namely, $(\text{length})^3(\text{time})^{-2}$, for instance:

$$E(k) \sim \left(\frac{\kappa}{\ell^2}\right)^2 k^{-3}(\ell k)^{\beta} \tag{10.3.41}$$

or

$$E(k) \sim \frac{\kappa^2}{\ell^2} k^{-1}(\ell k)^{\beta-2} \tag{10.3.42}$$

or, in general terms, one could write

$$E(k) \sim \kappa^a \ell^b k^{-c}. \tag{10.3.43}$$

From dimensional analysis it follows $a = 2, b + c = -1$. Thus, dimensional analysis is not sufficient to derive definite results for the three exponents a, b, and c (remember in any case, that in intermittent turbulence, the exponent β in the cascade (10.3.39) is not given by purely dimensional arguments).

However, the behaviour in k^{-1} in the quantum regions as well as k^{-3} in the counterflow hydrodynamic region may be reasonably interpreted. The exponent k^{-3} may be easily understood for $k \gg \ell$ in counterflow situations. The energy per unit mass of a vortex with quantized circulation κ is $v^2 \sim (\kappa/\ell)^2$, with κ/ℓ being of the order of the speed of the superfluid around the vortex core at a distance ℓ from the core. For $k \gg \ell$, in a surface of area k^{-2} will be k^{-2}/ℓ^2 quantized vortex lines. Thus, the energy per unit mass and unit k will be

$$E(k) \sim \left(\frac{\kappa}{\ell}\right)^2 \frac{k^{-2}}{\ell^2} k^{-1} \sim \kappa^2 \ell^{-4} k^{-3}. \tag{10.3.44}$$

The behaviour in k^{-1} (for $k\ell < 1$) may also be interpreted on the same grounds as k^{-3}, with the difference that the factor in k^{-2}/ℓ^2 giving the number of vortices in the area k^{-2} should be changed to 1, because in the area k^{-2} only a single vortex will be found. Then we simply have $(\kappa/\ell)^2 k^{-1}$. However, in the case of k^{-1} considerably smaller than ℓ one expects to have intermittent behaviour, because in an area k^{-2} one will find sometimes one vortex, and sometimes zero vortices, in a random way.

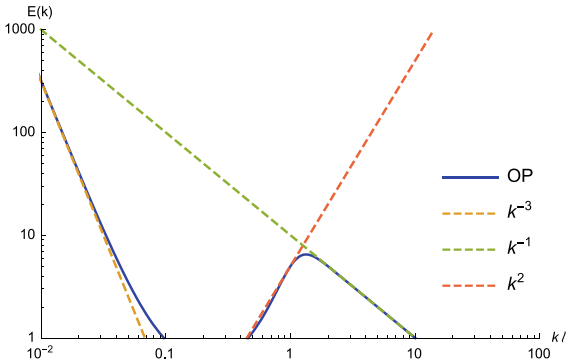


Fig. 10.2 Plot of the energy cascade (10.3.45) for $A_2 = 10$, $A_3 = 10^{-3.5}$ and $A = 2.8$. The legend on the right-hand side explains the power decay of the straight lines in the Log-Log plot, and OP stands for "Our Proposal" (10.3.45), which reproduces the observed behaviour represented in Fig. 4 of the paper [25]. This figure is Fig. 2 of the paper [24]

In (10.3.41) and (10.3.42), we have singled out k scalings in k^{-3} and k^{-1} with the corresponding ℓ^{-4} and ℓ^{-2} scalings, and k^2 in the transition regime, corresponding to scaling in ℓ. Of course, since ℓk is a dimensionless quantity, other exponents for k may be obtained, so that in the quantum case the cascade is not directly given by dimensional arguments. The exponent β in (10.3.41) could be related to some kind of intermittence [26–28]. Alternatively, following the analogy with (10.3.39) intermittence could be associated to a dependence of the kind $(kd)^\beta$, with d being the diameter of the container rather than to $(k\ell)^\beta$ (see the discussion related to Eq. (10.4.54)). Thus, for narrow channels ($\ell \gg d$) one could have $E(k, \ell, d)$.

As an interpolation from the k^{-3} behaviour ($K\ell < 1$) to the k^{-1} behaviour ($K\ell > 1$) (10.3.42) through an intermediate regime in k^2 it has been proposed, in an analogous way to (10.3.40)

$$E(k) \propto \frac{A_1 k^{-1} + A_3 k^{-3}}{2} \left[1 + \frac{A_1 k^{-1} - A_3 k^{-3}}{A_1 k^{-1} + A_3 k^{-3}} \tanh\left[A(k\ell - 1)\right]\right], \qquad (10.3.45)$$

where $A_1 = C_1(\kappa/\ell)^2$ (see (10.3.42) with $\beta = 2$) and $A_3 = C_3\kappa^2/\ell^4$ (see (10.3.42) with $\beta = 0$). Expression (10.3.45) is plotted in Fig. 10.2 for $A_1 = 10$, $A_3 = 10^{-3.5}$, $A = 2.8$. This figure reproduces in a qualitative way the behaviour reported in Fig. 4 in Ref. [25]. For other values of A, instead, the intermediate shoulder regions do not have the equipartition form k^2 (for instance, for $A = 5$, the intermediate region has the form k^3). The limiting expression (10.3.45) for $k\ell \gg 1$ and $k\ell \ll 1$ does not depend on A. In Sect. 10.3.3, we further discuss (10.3.45).

10.3.3 Discussion of the Intermediate Regime

The behaviour of the intermediate regime is not yet completely clear, as in some experiments a k^2 region is not observed. Here, on the tentative basis of the interpolations (10.3.40) and (10.3.45), we briefly discuss alternative possibilities for the intermediate region.

In this section, we discuss the mathematical relation between the value of the parameter A in the interpolation expressions (10.3.40) and (10.3.45) and the exponent m of k^m in the transition zone (the elbow in the plots) as well as some physical comments about the value of A. Let us note that instead of tanh-function used in such interpolations other functions having an analogous qualitative behaviour could be used.

The relation between the coefficient A and the exponent m in interpolation expression (10.3.40) between $k^{-5/3}$ and k^{-1} can be found calculating the slope of the straight line in the log-log plot of the figure. The expression is cumbersome. It follows that for $A = 4$ becomes $m \simeq 2$ and for $A \simeq 0.66$, m reaches its minimum value, $m \simeq -1.4$. The first case corresponds to the transition regime with an equipartition regime; in the second case, the behaviour from $-5/3$ and -1 is always with negative slope, and the slope at $k\ell \simeq 1$ is intermediate between $-5/3$ and -1.

The relation between A and m in interpolation expression (10.3.45) between k^{-3} and k^{-1} can be found following the same procedure used in the interpolation (10.3.40). The result is that for $A \simeq 2.8$ becomes $m \simeq 2$ and for $A \simeq 0.66$, m reaches its minimum value $m \simeq -1.9$. As well as in the previous case, the transition may have an equipartition regime ($m = 2$) or it may be always with a negative slope such that the slope at $k\ell \simeq 1$ is intermediate between -3 and -1.

On physical grounds, we conjecture that the value of A could depend on the ratio between some residence time, t_{res}, of the energy in the transition region, and the characteristic time of exchange of energy, t_{ex}, between the several physical modes of the system.

If $t_{res} \gg t_{ex}$ there is time enough to distribute the energy equally in all physical modes (equipartition). If, instead, $t_{res} \ll t_{ex}$, the energy goes directly from the initial mode to the final mode without time to be redistributed in the several other modes, and a region k^2 will not be achieved.

Thus, we heuristically suggest that A could be related to t_{res}/t_{ex} as

$$A = \frac{4 + 4(t_{res}/t_{ex})^{\beta}}{6.06 + (t_{res}/t_{ex})^{\beta}} \tag{10.3.46}$$

for (10.3.40) and for (10.3.45)

$$A = \frac{2.8 + 2.8(t_{res}/t_{ex})^{\beta}}{4.24 + (t_{res}/t_{ex})^{\beta}} \tag{10.3.47}$$

with β a exponent which is not relevant in the asymptotic regimes.

Indeed, for $t_{res} \gg t_{ex}$ one finds the values corresponding to equipartition (k^2 behaviour) whereas for $t_{res} \ll t_{ex}$ one finds a negative slope intermediate between the negative slopes of the $k\ell \gg 1$ region and the $k\ell \ll 1$ region.

The idea that in the intermediate region there could be a bottleneck in the energy transfer between the $k\ell \gg 1$ and $k\ell \ll 1$ regimes, making $t_{res} \gg t_{ex}$, has been proposed by L'Vov et al. [29]. In their suggestion, the transfer of energy from three-dimensional or two-dimensional excitations ($k^{-5/3}$ and k^{-3} scaling, respectively) at $k\ell \gg 1$ to one-dimensional behaviour (Kelvin wave excitation along one-dimensional vortex lines) for $k\ell \ll 1$ regime would be slow enough to exchange energy between the several modes so having a k^2 intermediate regime.

10.4 From Classical to Quantum Kolmogorov Spectrum

The classical Kolmogorov cascade follows from assuming that $E(\epsilon, k)$ is a function of ϵ, the dissipation rate per unit mass, and k. In classical viscous fluids, $\epsilon = \nu\overline{\nabla \mathbf{v}' : \nabla \mathbf{v}'}$ [13], which may have a more general expression [20]. In the quantum regime with quantized vortices, one may also define a quantum dissipation rate ϵ'. This could follow if instead of the classical dissipation rate ϵ a quantity with the same dimensions ((length^{2}time^{-3})) is used, for instance, [20]

$$\epsilon' \equiv \kappa \left(\frac{\kappa}{\ell^2}\right)^2 = \frac{\kappa^3}{\ell^4}. \tag{10.4.48}$$

This is analogous to the classical dissipation rate $\epsilon = \nu\overline{\nabla \mathbf{v}' : \nabla \mathbf{v}'}$ with ν the kinematic viscosity (with dimension (length)$^2 \times$ (time)$^{-1}$) replaced by κ, and $\nabla \mathbf{v}'$ (with dimension time^{-1}) replaced by κ/ℓ^2, which is the characteristic decay time of L according to Vinen's equation. (Note that the role of κ replacing ν is also used in the definition of quantum Reynolds number related to the appearance of quantum turbulence [30].) Indeed, recall that Vinen's equation for the evolution equation of L has the form [see (5.1.7)]

$$\frac{dL}{dt} = \alpha V_{ns} L^{3/2} - \beta\kappa L^2 = \alpha \frac{V_{ns}}{\ell} L - \beta\kappa \frac{L}{\ell^2}, \tag{10.4.49}$$

with α and β numerical parameters, respectively, related to vortex production and vortex destruction. Thus, in the right-hand side, $\ell^2/(\beta\kappa)$ is the characteristic time of vortex decay. The last member of (10.4.49) follows from the second member of (10.4.49) taking into account that $\ell = L^{-1/2}$. Expression (10.4.48) may be also interpreted as

$$\epsilon' = \left(\frac{\text{energy/mass}}{\text{decaytime}}\right) \propto \frac{\kappa^2 L}{\ell^2/\kappa} = \frac{\kappa^3}{\ell^4}, \tag{10.4.50}$$

where the vortex energy per unit length, ϵ_V is proportional to ρk^2, the energy per unit volume is $\epsilon_V L \simeq \rho\kappa^2 L$, and the energy per unit mass is proportional to $\kappa^2 L$; the decay time is proportional to ℓ^2/κ.

If ϵ' is used instead of ϵ, an approach analogous to (10.3.38) would lead to

$$E(k) \sim \epsilon'^a k^b \sim \epsilon'^{2/3} k^{-5/3}, \tag{10.4.51}$$

which has the same scaling law with k than the classical Kolmogorov law

$$E(k) \sim \kappa^2 \ell^{-8/3} k^{-5/3}. \tag{10.4.52}$$

This may be also obtained from (10.3.42) with $\beta = 4/3$.

Equation (10.4.49) has been generalized [21, 31–39] to take into account the influence of walls. Among the several modifications, the term in $\beta\kappa L/\ell^2$ is replaced by $\beta\kappa L(\ell^{-2} + \beta' d^{-2})$, with d the diameter of the channel. When ℓ becomes longer than d (narrow channels with small d, or weak turbulence with high ℓ), the last term is dominant and the quantum dissipation rate (10.4.50) would be changed to

$$\epsilon' \propto \frac{\kappa^3}{\ell^2 d^2}. \tag{10.4.53}$$

In this case, the energy spectrum (10.4.52) would change to

$$E(k) \sim \kappa^2 \ell^{-4/3} d^{-4/3} k^{-5/3}. \tag{10.4.54}$$

Thus, a dimensional analysis shows that behind a given power law for k, several different dependences on ℓ could be expected, suggesting some aspects of the physical structure of the underlying phenomena depending on the size of the channels. The dependence of $E(k)$ on d for $d \ll \ell$ could be experimentally checked.

In the Kelvin wave regime (at the smallest length scales) and for very low temperature, in such a way that Kelvin waves are not damped down by the normal component, the behaviour k^{-1} for $k\ell \gg 1$, which may be qualitatively understood on dimensional grounds, as seen in Sect. 10.3.2, is not widely accepted. Indeed, in this regime, some intermittence could be expected, as commented below (10.3.44), and detailed dynamical results should be taken into account. In a wide range of scales, Kelvin waves are weakly nonlinear and they can be dealt with the theory of weak-wave turbulence [40, 41]. These studies regard the nonlinear interactions between Kelvin waves starting from the Hamiltonian formulation of the Biot–Savart law six-wave kinetic equation (three waves change into three waves) and lead to the Kelvin wave spectrum [40]:

$$E_{KS} = \frac{C_{KS}\Lambda\kappa^{7/5}\epsilon^{1/5}}{k^{7/5}} \tag{10.4.55}$$

C_{KS} a dimensionless constant, and $\Lambda = \ln(\ell/a_0)$ with a_0 of the order of the atomic size of Helium II. This could follow from (10.3.41) with $\beta = 8/5$ or from (10.3.42) with $\beta = -2/5$.

Instead, a local four-wave theory of Kelvin wave leads to the spectrum [41]:

$$E_{LN} = \frac{C_{LN}\Lambda\kappa\epsilon^{1/3}}{\Psi^{2/3}k^{5/3}}, \tag{10.4.56}$$

where $C_{LN} \approx 0.304$ is a dimensionless constant and $\Psi = 4\pi E_{LN}/\Lambda\kappa^2$. This has the same dependence on k than in Kolmogorov law, namely, $k^{-5/3}$.

Thus, it is important to be aware that up to now the situation of energy cascade is not completely settled, and that dimensional analysis is not able to arrive to definite conclusions, in such a way that microscopic analysis and detailed experiments are very needed in this field. In any case, dimensional analysis may be helpful to check the consistency of the model proposed and it may propose a gross picture of some macroscopically admissible behaviours.

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Chapter 11
Thermodynamics of Vortex Tangles and of Cosmic Strings



In Chaps. 5, 6, 7, we have studied the geometry and the dynamics of turbulent vortex tangles, and their relation to heat flux and to barycentric velocity, either in the sense that both **q** and **v** become a source of quantized vortices beyond some critical values, and in the sense that the vortex tangle in turn contributes to the resistance to heat flow and to mass flow. In this chapter, we deal with another aspect of the tangles, namely, their entropy and temperature. This aspect is not needed for the description of heat transport in the superfluid, but it has an interest on its own.

Indeed, the tangles are geometrical objects and have some energetic contents. This makes that both from the geometrical as the energetic points of view one may ask what is the entropy of the tangle, namely, what is the degree of geometrical "disorder", and of the "disorder" in the distribution of energy among the several degrees of freedom, or among the different kinds of vortex loops.

The entropy and temperature considered here are not related to the entropy and temperature of the underlying superfluid, but may be very different from them, as they have a different physical meaning. Here, we consider the tangle as a dynamical, non-equilibrium physical entity, which continuously exchanges energy with the underlying superfluid as far as the system receives per unit time energy enough to sustain the set of vortices, continuously appearing and disappearing, otherwise, the system of vortices disappears.

Definition for the temperature in terms of the average energy for a system of closed lines suggests an analogy between thermodynamics of quantum vortex lines and thermodynamics of cosmic string loops. Though it does not refer to turbulence, the topic has sufficient physical interest to be briefly presented, in order to show how superfluid dynamics may suggest some aspects of cosmological problems [1]. Indeed, cosmological analogies with some superfluid problems have been pointed out along many years [1]. For instance, some authors have tentatively proposed to describe the cosmic quantum vacuum as a superfluid [2, 3] in which case the vortex loop lines in such superfluid would play a role analogous—in their macroscopic

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consequences—to that of cosmic string loops. In particular, they would modify the evolution of the cosmic expansion rate, as it will be discussed below.

The present analysis is not restricted to cosmic string loops, but we deal with different kinds of physical entities whose basic microscopic energy $u(l)$ may be related to some characteristic length l in a potential form as l^a, with a some exponent. In this way, a single thermodynamic formalism is able to deal with different kinds of systems. In particular, some thermodynamic dualities will arise when one compares entities with $u(l) \sim l^a$ and $u(l) \sim l^{-a}$, as, for instance, cosmic string loops with $u(l) \sim l$ and photons with $u(l) \sim l^{-1}$. Such dualities may tentatively be extended to other situations, as the thermodynamics of macro- and micro-black holes. Several kinds of dualities are of basic relevance in some aspects of theoretical physics, and in M theory establishing connections between the several solutions of superstring theories. The analysis may provide some simple illustrations of some of such dualities.

11.1 Effective Temperature of Turbulent Vortex Tangles

In non-equilibrium systems different degrees of freedom may have different temperatures because they exchange energy very slowly, or because they continuously exchange energy, but are receiving external power in different proportions. There are several ways of defining temperature [4]. At equilibrium all the definitions lead to the same value. In contrast, in non-equilibrium steady states, several definitions lead to different values [4]. This is not a contradiction: every definition is related to some kind of measurement, namely, to some kind of theoretical thermometer, and different kinds of thermometers may lead to different values for the temperature, depending on which kinds of degrees of freedom they are sensitive to.

In this section, we consider three approaches to an effective temperature for turbulent vortex tangles. The aim is to be aware of the diversity of possible proposals, each of them related to some specific physical aspect of the tangle, rather than grasping in a single temperature all the possible aspects. One may try caloric definitions of temperature—related to energy content—entropic definitions—related to energy exchange—or fluctuation-dissipation definitions—related to fluctuations and transport—and still other definitions [5]. Different definitions correspond to different kinds of measurements. Only in equilibrium all the different definitions and measurements lead to the same value for temperature.

From a microscopic perspective, the energy related to the vortices is relatively organized, and it is not counted as internal energy; however, from a macroscopic perspective, since the tangle is a disordered object, this energy is disordered, and it is reasonable to associate to it an effective temperature. Such temperature will be related to the average vortex length density of tangle, which is in turn related to the external heat flux. Thus, for a heat flux zero or lower than the critical value, the vortices would disappear, no tangle would exist, and no temperature related to the tangle would survive.

Definition of effective temperature must give a clear operational definition, and should satisfy that in the energy exchange between vortex tangles with different values of the defined temperature, the energy goes from higher to lower temperatures. In contrast, it is not required that such temperatures coincide with the absolute temperature of the liquid helium.

Here, we basically consider two definitions: one is based on the degree of disorder of the vortices (an entropic definition, Sect. 11.1.1) and another one on the average energy of the vortices (a caloric definition, Sect. 11.1.2). We also briefly mention a definition based on the Einstein relation for vortex diffusion (Sect. 11.1.3), in order to emphasize the diversity of possibilities.

11.1.1 Entropic Temperature of Rotating Tangles

One of the proposals of temperature of quantum vortex tangles originated in the analysis of tangles submitted to rotation with angular velocity Ω and to a heat flux $\mathbf{q}$, as happens, for example, in rotating cylinders or in neutron stars, which has been dealt in Sect. 5.4. Rotation contributes to orientation of vortices along the rotation axis, whereas the heat flux (or counterflow velocity) contributes to the randomization of the direction of the vortices. Thus, considering the case of turbulence in rotating superfluid Helium II, as in the experiment by Swanson et al. [6], Jou and Mongiovì [7] and Tsubota el al. [8] set the analogy of the system of vortices with a system of magnetic dipoles.

Magnetic field contributes to the orientation of the dipoles, whereas temperature randomizes them. Thus, the proportion of dipoles with magnetic dipole μ_d along the magnetic field H in a system at temperature T is given by $\exp(-\mu_d H\cos(\theta)/k_B T)$ with k_B the Boltzmann constant and with θ the angle between the local tangent $\mathbf{s}'$ to the vortex line and the rotation axis. If $T = 0$, all the dipoles are oriented along H. In the rotating superfluid system, the angular velocity Ω plays an orienting role analogous to H. Thus, the proportion of vortices not aligned along Ω may give a measure of an effective temperature related to the heat flux. Indeed, applied to a Langevin system of classical dipoles (all orientations are possible), one has for the average of $\cos\theta$

$$< \cos\theta >= \coth x - \frac{1}{x}, \tag{11.1.1}$$

where $x \equiv \mu_d H/k_B T$. According to dimensional arguments, in the case of vortices a natural analogy with the dimensionless ratio $x = \mu_d H/k_B T$ could be

$$x = a(T)\frac{\Omega\kappa}{v_{ns}^2}, \tag{11.1.2}$$

with v_{ns} the counterflow velocity introduced in two-fluid model (i.e. $v_{ns} = |\mathbf{v}_n - \mathbf{v}_s|$), $a(T)$ a dimensionless coefficient and T the temperature of liquid Helium II. Thus,

Ω may be seen as the analogous of the magnetic field H, κ an analogous of the magnetic dipole, and v_{ns}^2 could be related to T as $T_{tangle} \sim V_{ns}^2$. Since in counterflow $\mathbf{q}$ is related to $\mathbf{v}_{ns}$ as $\mathbf{q} = \rho_s s T \mathbf{v}_{ns}$, (11.1.2) may be rewritten as

$$x = a(T)\rho_s^2 s^2 T^2 \frac{\Omega\kappa}{q^2}, \tag{11.1.3}$$

Tsubota et al. [8] did not mention the idea of non-equilibrium temperature, but they wrote an expression like (11.1.2), with $a(T) = 22/\gamma^2(T)$, a relation obtained from numerical simulations, by looking for the value of $a(T)$ which fits best the numerical results to the relation (11.1.1). Here, γ is a dimensionless coefficient relating the vortex length density $\mathcal{L}$ in steady state to v_{ns} and κ as $\mathcal{L} = \gamma \frac{v_{ns}^{*2}}{\kappa^2}$. An alternative way of writing (11.1.2) is by multiplying numerator and denominator times m then one has

$$x = m \frac{\Omega\kappa}{k_B T_{eff}}. \tag{11.1.4}$$

In (11.1.4), the meaning of m is not sufficiently clear. To clarify it, we discuss several steps leading from $\Omega\kappa/V_{ns}^2$ to $\Omega\kappa/k_B T_{eff}$. To do that, remember that V_{ns}^2 is of the order of $\kappa^2 L$. Taking this and multiplying the numerator and denominator by ρ_s and $L^{-1/2}$, one has

$$\frac{\Omega\kappa}{V_{ns}^2} \sim \frac{\rho_s \Omega\kappa L^{-1/2}}{L} \frac{1}{\rho_s \kappa^2 L^{-1/2}}. \tag{11.1.5}$$

Since the energy per unit length of the vortex line is $\epsilon_V \equiv \frac{\rho_s \kappa^2}{2} \ln\left(\frac{L^{-1/2}}{a_0}\right)$, the denominator $\rho_s \kappa^2 L^{-1/2}$ is of the order of the average energy of the vortex loops, because $L^{-1/2}$ is of the order of the average size of the vortex loops. Thus, defining

$$k_B T = \langle u(\ell) \rangle \tag{11.1.6}$$

relates T to the average energy of the loops $k_B T = \epsilon_V < \ell >= \epsilon_V L^{-1/2}$. This definition will be explored in further detail in Sect. 11.2, and will be summarized in Sect. 11.4.

In the first fraction of the right-hand side, remember that in pure rotation $L = 2\Omega/\kappa$. Thus, one would have $\Omega\kappa/L = \kappa^2/2$, and the fraction would become $(\rho_s/2)\kappa^2 L^{-1/2}$. In this case, the vortices are straight lines parallel to the rotation axis, and $(\rho_s/2)\kappa^2 L^{-1/2}$ would be the energy of a segment of the straight line of length $L^{-1/2}$. Thus, $\Omega\kappa/v_n^2$ gives an indication of the ratio between energy of fragments of vortex lines parallel to Ω and the energy of disordered vortex loops generated by the heat flux.

11.1.2 Temperature of Vortex Bundles with Kelvin Waves

The definition of an effective temperature related to the average energy of vortices has also been applied to the spectrum of perturbations in vortex lines or in bundles of pinned vortices sustaining helical Kelvin waves [9]. In this case, instead of considering the total energy of the vortex line, one focuses the attention on the energy of the increment of length Δl due to waves.

Consider, first, a vortex line with a helical Kelvin wave superimposed on it and be $\langle \Delta l \rangle$ the average increment of length of the vortex line as a consequence of the Kelvin wave. We define as in (11.1.6)

$$k_B T_{eff} = \epsilon_V \langle \Delta l \rangle \tag{11.1.7}$$

with ϵ_V the energy per unit length $\epsilon_V = (\rho_s \kappa^2/2) \ln(L^{-1/2}/a_0)$.

In more explicit terms for Δl, this takes the form:

$$k_B T_{eff} = \epsilon_V \langle (4\pi^2 A^2 + l_0^2)^{1/2} - l_0 \rangle, \tag{11.1.8}$$

with A the amplitude (radius) of the helical wave and l_0 the wavelength of the wave. When $2\pi A \ll l_0$, the right-hand side of (11.1.8) may be approximated up to second order in A/l_0 and (11.1.8) becomes

$$k_B T_{eff} = 2\pi^2 \epsilon_V \left\langle \frac{A^2}{l_0} \right\rangle. \tag{11.1.9}$$

Thus, the situation would be analogous to that of a harmonic oscillator of elastic constant $2\pi^2 \epsilon_V / l_0$ and amplitude of oscillation A, and (11.1.9) could be seen as an "equipartition" of energy for waves of different amplitudes [10].

Now, consider not a single vortex line, but a bundle of vortex lines. In this case, the equipartition situation (hence equal T_{eff} everywhere in the bundle) would lead to a scaling relation of the amplitude A in terms of k (wavevector)

$$A^2(k) \sim k^{-1} \tag{11.1.10}$$

considering that $k = 2\pi/l_0$. This result was obtained microscopically by Kozik and Svistunov [11].

In the opposite extreme situation for which $A \gg l_0$, namely, for solenoidal vortex, the expression (11.1.8) becomes

$$k_B T_{eff} = 2\pi \epsilon_V \langle A \rangle . \tag{11.1.11}$$

In this case, one would have the scaling relation $A(k) \sim constant$, namely, for big A one reaches a plateau of saturation, independent of k.

A condition for the consistent thermodynamic meaning of the temperature of the bundle is that there is sufficient interaction among different excitations in it, leading to an energy exchange among them and, eventually, to a steady-state distribution of energy.

11.1.3 *Kinetic Energy and Einstein Relation for Vortex Diffusion*

One may look deeper into the relation between T_{tangle} and v_{ns}^2, as proposed in (11.1.2)

$$k_B T_{tangle} = \frac{1}{a(T)} m v_{ns}^2, \tag{11.1.12}$$

with $a(T)$ a dimensionless function which may depend on the temperature of the underlying helium fluid, and m the mass of the helium atom. The relation of this temperature with a generalization of the Einstein relation between friction and diffusion coefficient in vortex diffusion was considered [12] (indeed, Einstein relation between temperature, friction coefficient, and diffusion coefficient is often used for the measurement of temperature in shaken granular systems, for instance [4]). In this case, the friction coefficient of a vortex moving inside a dense vortex tangle (namely, a friction dominated by vortex–vortex interaction) is known but not the vortex diffusivity, relating vortex diffusion flux $\mathbf{J}^{\mathcal{L}}$ to the gradient of vortex length density $\mathcal{L}$, namely, $\mathbf{J}^{\mathcal{L}} = -D_{\mathcal{L}} \nabla \mathcal{L}$. It was shown [12] that the effective kinetic temperature (11.1.12) is consistent with

$$k_B T_{tangle} = D_{\mathcal{L}} \zeta_{\mathcal{L}}, \tag{11.1.13}$$

$D_{\mathcal{L}}$ being vortex diffusion coefficient and $\zeta_{\mathcal{L}}$ the friction coefficient of the vortex with the normal component. This does not apply to diluted vortex tangles close to zero Kelvin degree, where the friction is negligible because practically all the "normal component" of the fluid has disappeared.

11.2 Temperature and Entropy of the Vortex Tangle in Terms of the Vortex Line Density

Since vortices cannot have free ends, they can be either pinned vortices (whose ends are on the walls) and vortex loops (closed on themselves and free to move in space). Fully developed turbulence may be considered as a gas of vortex loops in strong mutual interaction, since in this case the number of free loops is much higher than that of pinned vortex lines [13].

The entropy of the vortex tangle could be related to how the energy is distributed in vortex loops of different lengths, and the temperature could be related to the average energy of vortices. If the energy of a peculiar vortex loop is $u(l)$, we take for the temperature T, $k_B T =< u(l) >$, as in (11.1.6).

We assume, furthermore, that $u(l)$ is proportional to l (vortex energy is $u(l) = \epsilon_V l$, with ϵ_V energy per unit length) and that the average length of the loops $< l >$ is proportional to the average separation of the loops, which is of the order of $L^{-1/2}$, namely, $\langle \ell \rangle = \lambda L^{-1/2}$, with λ a dimensionless constant and L the vortex length density. This makes the system invariant under changes of length scales, i.e. if the scale is multiplied by some factor, both the average vortex length and their average separation increase in the same proportion. One gets then the relation between $k_B T$ and L

$$k_B T = \epsilon_V \lambda L^{-1/2}. \tag{11.2.1}$$

The average number of vortices per unit volume will be

$$\frac{N}{V} = \frac{L^{3/2}}{\lambda^3} \tag{11.2.2}$$

and the energy density U/V will be $U/V = (N/V)k_B T = (N/V)k_B T = (\epsilon_V/\lambda^2)L$. From this expression for U in terms of L and from (11.2.2) giving T in terms of L one may obtain the entropy S as a function of L by integrating the fundamental thermodynamic relation:

$$\frac{\partial S}{\partial U} = \frac{1}{T}. \tag{11.2.3}$$

One may write

$$\frac{\partial S}{\partial L} = \frac{\partial S}{\partial U}\frac{\partial U}{\partial L} = \frac{1}{T}\frac{\partial U}{\partial L}. \tag{11.2.4}$$

Since T and U have been obtained in terms of L, the term on the right-hand side of (11.2.4) is known in terms of L and may be integrated to obtain $S(L, V)$

$$S_{turb}(L, V) = \frac{2}{3\lambda} k_B \left(L^{3/2} - L_0^{3/2} \right) V \tag{11.2.5}$$

with L_0 a constant reference length which may be taken as zero.

Since L is known in terms of U, expression (11.2.5) may be written as

$$S(U, V) = \frac{2}{3} k_B \lambda^2 \epsilon_V^{-3/2} U^{3/2} V^{-1/2} \tag{11.2.6}$$

which is the fundamental thermodynamic equation $S(U, V)$ for the tangle. From here, the several thermodynamic quantities of the tangle may be obtained, as, for instance, the thermodynamic pressure, defined by the general thermodynamic relation

$$p/T = (\partial S/\partial V)_U \,. \tag{11.2.7}$$

This leads to $(\partial S/\partial V)_U = -(1/2)(S/V)$ and $(\partial S/\partial U)_V = 1/T = (3/2)(S/U)$ and therefore

$$p = -\frac{1}{3}\frac{U}{V} \tag{11.2.8}$$

which is a negative pressure. Specific heat, free energy, and other thermodynamic quantities could be obtained as well. We will do it and discuss it in depth in the next section, in a more general way not restricted to a linear dependence of $u(l)$ with l. Note, eventually, that the chemical potential of vortex loops, defined as $\mu/T = -(\partial S/\partial N)_U$, is zero because S does not depend on N.

Note that S does not depend on N because N directly depends on L. This is analogous to the behaviour of photons in electromagnetic radiation, where the average number of photons depends on T, but photons themselves are not conserved entities, but are continuously absorbed and emitted.

11.3 Vortex Loops and Cosmic String Loops

Quantized vortices have an energy approximately proportional to their length l, as $u(l) = \epsilon_V l$. Cosmic strings are hypothetical one-dimensional topological defects of the cosmic space time. They may be very long, crossing the whole observable universe or they may be in the form of microscopic closed loops. According to Polchinsky and Rocha [14], some 90% of the strings would be in the form of small loops and 10% in the form of loops near the Hubble scale. Both kinds of strings have an energy proportional to their length [15]. Therefore, some of the features of quantized vortex loops with energy proportional to their length will be applicable to the thermodynamic analysis of string systems.

The energy of cosmic string loops is given by

$$u(l) = \left(\frac{c^4}{a^2 G}\right) l, \tag{11.3.1}$$

with c the speed of light, G the gravitation constant, and a a numerical coefficient. The energy per unit length of the string, namely, $\mu_{cs} = c^4/(a^2 G)$, is called cosmic string tension. In this sense, cosmic string loops are analogous to quantized vortices, whose energy is also proportional to their length. However, we go beyond such an immediate analogy and explore the thermodynamics of physical entities with energy proportional to some power of their characteristic length.

To be more specific, we deal with a generalization of expression (11.3.1) in terms of h, c, and G (namely, Planck constant, speed of light in vacuum, and gravitational constant, three of the most general and significative universal constants of physics). Taking into account the dimensions, we have as possible expressions of the energy

$$u_\alpha(l) = \frac{c^4}{a^2 G}\left(\frac{c^3}{a^2 hG}\right)^{\frac{\alpha-1}{2}} l^\alpha \equiv \mu_{cs} l_0^{1-\alpha} l^\alpha. \tag{11.3.2}$$

with α a constant exponent, μ_{cs} the cosmic string tension (c^4/a^2G), and $l_0 = (a^2 hG/c^3)^{1/2}$ a characteristic length related to the Planck length $l_P = (\hbar G/c^3)^{1/2}$ as $l_0 = a\sqrt{2\pi} l_P$. Since the highest energy experiments performed up to now have explored energies up to 14 TeV (14×10^{13} eV, corresponding to wavelengths of the order of 10^{-19} m), and since Planck length is of the order of 10^{-35} m, the constant a could be of the order up to 10^9 or 10^{10} without contradiction to currently known observations (probably, higher values of a could be indirectly detected in cosmological observations).

For $\alpha = 1$ in the exponent of (11.3.2), one recovers the expression (11.3.1) for the energy of cosmic string loops, and for $\alpha = -1$ (11.3.2) yields $u(l) = hc/l$, which may be interpreted as the energy of photons if l is considered as the wavelength associated to the corresponding wave. Whether some intermediate values of α between 1 and -1 could correspond to some kind of actual physical entity is a question for current speculation.

11.4 Thermodynamics of Cosmic String Loops and Negative Pressure

To go from the energy of the microscopic single component to a thermodynamic formulation of a whole system of components, temperature must be introduced. We will take $\tilde{c} k_B T =< u(l) >$, as in (11.1.6), with $\tilde{c}$ a numerical constant of order 1 and $< \ldots >$ standing for the average over loops of different lengths. Furthermore, we assume as in Sect. 11.2 that the average length of the loops is $< l >= \lambda \mathsf{L}^{-1/2}$, L being the total length of loops per unit volume, which has dimension of length $^{-2}$, λ a dimensionless constant, and $\mathsf{L}^{-1/2}$ the average separation between loops. Note that this relation is scale invariant because if all the sizes of the system are multiplied by a numerical factor, both $< l >$ and $\mathsf{L}^{-1/2}$ will change in the same factor.

Note that the bridge between the microscopic energy $u(l)$ for a single loop and the thermodynamics of a collection of loops has three main ingredients: the use of L as a variable, the definition of T in terms of the average of $u(l)$ (not necessarily linear in l), and the relation between the average of l and the value of L, $< l >= \lambda \mathsf{L}^{-1/2}$.

From these expressions and using the procedure of Sect. 11.2, one obtains for the energy density $u \equiv U/V$ in terms of the temperature T

$$\mathcal{U} \equiv \frac{U}{V} = \frac{A_T}{\lambda} \mathsf{L}^{\frac{2-\alpha}{2}} = \frac{A_T}{\lambda} \left(\frac{\tilde{c} k_B}{A_T} \right)^{-\frac{2-\alpha}{1+\alpha}} T, \tag{11.4.1}$$

with $A_T = \mu_{cs} \beta_{cs} \lambda^{1+\alpha}$, and β_{cs} being a numerical constant related to the detailed form of the statistical distribution of the length of the loops as $< l^{1+\alpha} >= \beta_{cs} < l >^{1+\alpha}$.

Since according to the fundamental Gibbs equation of Thermodynamics one has $\partial s / \partial \mathcal{U} = T^{-1}$, one gets for the entropy density s

$$\frac{\partial s}{\partial \mathsf{L}} = \frac{\partial s}{\partial \mathcal{U}} \frac{\partial \mathcal{U}}{\partial \mathsf{L}} = \frac{1}{T} \frac{A_T}{\lambda} \frac{2-\alpha}{2} \mathsf{L}^{-\alpha/2} = \tilde{c} k_B \frac{2-\alpha}{2\lambda} \mathsf{L}^{1/2}. \tag{11.4.2}$$

Integrating this equation with respect to L one gets for the entropy as a function of L

$$s \equiv \frac{S(\mathsf{L})}{V} = \tilde{c} k_B \frac{2-\alpha}{3\lambda} \left(\mathsf{L}^{3/2} - \mathsf{L}_0^{3/2} \right) + s_0. \tag{11.4.3}$$

Taking $\mathsf{L}_0 = 0$ and $s_0 = 0$ for the sake of simplicity, one may express (11.4.3) in terms of the entropy S of a volume V as

$$S(U, V) = \tilde{c} k_B \frac{2-\alpha}{3\lambda} \mathsf{L}^{3/2} V \sim U^{\frac{3}{2-\alpha}} V^{-\frac{\alpha}{2-\alpha}}, \tag{11.4.4}$$

where relation (11.4.1) between L and $\mathcal{U}$ has been used. For $\alpha = 1$ we recover the expression (11.2.6) for $S(U, V)$.

Giving the entropy S of a system as a function of U, V, and N, namely $S(U, V, N)$, provides the fundamental thermodynamic equation from which all other thermodynamic quantities of the system may be obtained from the methods of equilibrium thermodynamics. Thus, Eq. (11.4.4) has a special relevance. Note that (11.4.4) does not depend on N as it has been commented in Sect. 11.2.

Using the well-known thermodynamic relation $(\partial S_{tot} / \partial V)_U = p/T$ [16], expression (11.4.4) leads to

$$p = -\frac{\alpha}{3} \frac{U}{V}. \tag{11.4.5}$$

When $\alpha = 1$, we get $p = -(1/3)(U/V)$, as in (11.2.8), whereas for $\alpha = -1$ we obtain $p = (1/3)(U/V)$, well known for photon gases. The result (11.4.5) indicates that the negative pressure of systems with $u(l)$ proportional to l has an entropic origin. Indeed, if one keeps the total U fixed but the volume V increases, the total length of cosmic string loops must remain constant, but the average length of cosmic string loops should increase. This means a decrease of the number of cosmic string loops, which leads to a decrease of entropy and therefore to a negative pressure.

A negative thermodynamic pressure may imply relevant consequences in a cosmological context. This is so because, according to Einstein's general relativity, the gravitational source is $(U/V) + 3p$. This is in contrast to Newtonian gravitation,

where the mass (or, equivalently, the energy $u = m\,c^2$) is the only source of gravitation. If $p = -(1/3)(U/V)$, there will not be gravitational interaction at the cosmic scale, whereas if $p < -(1/3)(U/V)$ there will be repulsion instead of attraction. Thus, as discussed in Sect. 11.6, if $p = -(1/3)(U/V)$ the Universe will asymptotically expand at a constant rate [17], whereas if $p < -(1/3)(U/V)$ it will expand in an accelerated way. This possibility is of interest for the physical interpretation of the so-called dark energy, which is thought to produce an acceleration of the cosmic expansion, in contrast with the slowing down expansion expected on classical gravitational grounds, allowing only for attraction. Thus, the idea that the cosmic quantum vacuum has some features analogous to a superfluid with vortices [2] could be useful to modellize cosmic expansion. If α is higher than 1 (in absolute value) cosmic expansion would accelerate, with always increasing rate, instead of tending to a constant expansion rate, as it happens with $\alpha = 1$.

11.5 Thermodynamic Duality of Photons and Cosmic String Loops

The approach presented in Sect. 11.4 allows to establish a duality connection between photon thermodynamics and cosmic string loop thermodynamics. The idea of such duality is appealing, because it connects a microscopic feature—namely, $u(l) \sim l$ (cosmic string loops) and $u(l) \sim l^{-1}$ (photons)—with a corresponding macroscopic feature as $p = -(U/V)/3$ (cosmic string loops) and $p = (U/V)/3$ (photon gas) and other features examined below.

To explore such thermodynamic duality, we rewrite expressions (11.4.1) for $U(T, V)$ and (11.4.4) for $S(T, V)$ for $\alpha = 1$ (string loops) and $\alpha = -1$ (photons) and get

$$S(T, V) = \frac{4}{3}\tilde{a}k_B \left(\frac{k_B T}{hc}\right)^3 V \qquad \text{(photons)}, \tag{11.5.1}$$

$$S(T, V) = \frac{2}{3}\tilde{b}k_B \left(\frac{\mu_{cs}}{k_B T}\right)^3 V \qquad \text{(cosmic string loops)}, \tag{11.5.2}$$

$$U(T, V) = \tilde{a}hc \left(\frac{k_B T}{hc}\right)^4 V \qquad \text{(photons)} \tag{11.5.3}$$

$$U(T, V) = \tilde{b}\mu_{cs} \left(\frac{\mu_{cs}}{k_B T}\right)^2 V \qquad \text{(cosmic string loops)}, \tag{11.5.4}$$

where $\tilde{a}$ and $\tilde{b}$ are numerical constants and μ_{cs} the string tension defined below (11.3.1). The relations $S = T^{-3}V$ (cosmic string loops) and $S = T^3 V$ (photon gas)

imply that in an adiabatic expansion (S constant) a photon gas will cool as $T \sim V^{-1/3}$ whereas the temperature of a gas of cosmic string loops will increase as $T \sim V^{1/3}$.

The thermodynamic duality between the thermodynamics of photons and of cosmic loops comes from the deeper duality relation between photons with $u_l \sim l^{-1}$ and cosmic loops $u_l \sim l$. This is analogous to the T-duality in quantum field theories, or string theories [18–20] in which the behaviour of a theory at a length scale R is analogous to the behaviour of its dual theory at scale $1/R$.

In more concrete terms, if one takes one of the nine spatial dimensions of superstring theory and makes a circle of radius R, a particle travelling around the circle will have a momentum quantized in the form n/R, with n an integer. The number of times a string winds around the circle (the so-called winding number w) describes the energy contribution of the string due to such winding, and it is quantized in the form wR/L_{st}^2, with L_{st}, a constant called string length. Then, a theory with large R and a theory with small R are equivalent by exchanging the role of momentum with that of the winding energy and vice versa, namely, by transforming R into L_{st}^2/R, and exchanging the quantum numbers n and w. In the analysis of the present section, instead, we are working in a three-dimensional space and the dual behaviour we are considering refers to the exchange of hc/l by $(c^4/a^2G)l$ in the energy relation (11.3.1) or, in other terms, to the exchange of l into l_t^2/l, with $l_t^2 = a^2(Gh/c^3) = (al_P)^2$.

Analogously, thermal duality means a symmetry with respect to a characteristic temperature T_c, in such a way that the behaviour of one system at temperature T is analogous to the behaviour of the corresponding dual system at temperature T_c^2/T. In the case examined here, T_c is given by $k_B T_c = (\lambda/\tilde{c}\sqrt{\tilde{a}})^{1/3}(\mu_{cs}hc)^{1/2}$. Since T is related to the average of ℓ, it is logical expecting that a duality between ℓ and $1/\ell$ will imply a corresponding duality between T and $1/T$.

11.6 Cosmological Consequences of Thermodynamics

At the end of Sect. 11.4, it has been commented that the relation between p and U/V has relevant consequences on cosmic expansion. In this section, we explain in detail how this is so. Thermodynamic properties of the contents of the universe are a basic ingredient in cosmology, because the Einstein equation of general relativity relates the properties of the metrics of spacetime as described by the metric tensor $g_{\mu\nu}$ to the contents of the spacetime as

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu}, \tag{11.6.1}$$

where $R_{\mu\nu}$ is the Ricci curvature tensor of $g_{\mu\nu}$, R is the scalar curvature of $g_{\mu\nu}$, $T_{\mu\nu}$ is the stress-energy tensor of the contents of space, Λ is the cosmological constant, and G is the gravitational constant. The tensor Eq. (11.6.1) relates a set of symmetric

4×4 tensors, each of them with 10 independent components (which are reduced to 6 independent components by the 4 Bianchi identities).

For an ideal fluid with four velocity u^μ, the energy-momentum tensor $T^{\mu\nu}$ has the form

$$T^{\mu\nu} = (\rho + p)u_\mu u_\nu - p g_{\mu\nu}, \tag{11.6.2}$$

with ρ energy density and p the thermodynamic pressure.

For a flat homogeneous and isotropic universe, the metric tensor in spherical coordinates has the so-called Robertson–Walker form, such that the invariant four distance s is given by

$$ds^2 = c^2\, dt^2 - a(t)^2[dr^2 + r^2(d\theta^2 + \sin^2\theta\, d\phi^2)], \tag{11.6.3}$$

with $a(t)$ the time-dependent scale factor describing the expansion (or the contraction) of the universe. Introducing (11.6.3) and (11.6.2) into Einstein's equation (11.6.1) one is led after some long calculations to the following evolution equation for the cosmic scale factor a:

$$\frac{\ddot{a}}{a} = -\frac{4\pi}{3} G \Sigma_i (\rho_i + 3p_i), \tag{11.6.4}$$

where the summation is over all constituents of the universe (baryionic matter, radiation, neutrinos, dark matter, and dark energy) whose respective energy density and pressure are denoted by ρ_i and p_i. Note that constituents with $\rho_i + 3p_i > 0$ produce a slowing down of the expansion rate $\dot{a}$ (namely, $\ddot{a} < 0$), whereas those with $\rho_i + 3p_i < 0$ make the expansion rate to increase (namely, $\ddot{a} > 0$), and those with $\rho_i + 3p_i = 0$ do not produce gravitational global effects, nor changes in $\dot{a}$.

For a flat universe, the Hubble ratio $\dot{a}/a$ is related to the total energy density as

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi}{3} G \Sigma_i \rho_i. \tag{11.6.5}$$

Furthermore, the energy conservation law states that for each cosmic component i

$$\dot{\rho}_i = -3\frac{\dot{a}}{a}(\rho_i + p_i). \tag{11.6.6}$$

The first term in the right-hand side is the decrease of mass per unit volume due to expansion, and the second one is the work done by the pressure during the expansion. This equation states that the relation between p and ρ also influences the rate of decrease of the energy density along the cosmic expansion. If $p = -\rho$, the work of expansion would cancel the reduction of energy density, and therefore the energy density would stay constant during the expansion.

Research of systems with the property $p = -w\rho$, with w a numerical coefficient in the interval $1/3 < w < 1$, is an active topic in cosmology as well as in thermodynamics. The value $w = 2/3$ was considered to be close to observational data on cosmic expansion some years ago, but more recent results seem to favour values closer to $w = 1$.

From the cosmological point of view, the most relevant features stemming from the thermodynamic duality between photons and cosmic string loops pointed out in Sect. 11.5 are: (1) cosmic string loops lead to vanishing gravitational effects because $(U/V) + 3p = 0$, thus implying a cosmic expansion with constant rate, whereas photons slow down the cosmic expansion because $(U/V) + 3p = 2(U/V) > 0$; (2) in an adiabatic reversible expansion, i.e. expansion with $S(V, T) = constant$, and assuming $V \sim R^3$ with R the cosmic scale factor, we have $T^3V \sim (TR)^3 \sim constant$ for photons, and $T^3/V \sim (T/R)^3 \sim constant$ for cosmic string loops); this also leads to a generalized Wien's law [21]; and (3) we have $U/V \sim R^{-4}$ for photons and $U/V \sim R^{-2}$ for cosmic string loops. From a cosmological perspective, the thermal duality symmetry between photons and cosmic string loops sets a symmetry between the thermal features of the universe at early period, dominated by radiation, and those at future times, dominated by cosmic strings.

11.7 Duality-Invariant Einstein–Planck Relation and Its Physical Consequences

In Sect. 11.5, we have pointed out microscopic and macroscopic dualities between photons and cosmic string loops, taken as completely different kinds of physical entities. In this section, we go a step beyond and conjecture the possibility of a deeper relation between them, by proposing for $u(l)$ an expression of the form [19]:

$$u(l) = \frac{hc}{al_P} \frac{1}{\left(\frac{l}{al_P} + \frac{al_P}{l}\right)}. \tag{11.7.1}$$

For $l \gg al_P$ and $l \ll al_P$ (11.7.1), respectively, yields the asymptotic forms explored in (11.3.2) for $\alpha = -1$ and $\alpha = 1$, namely:

$$u(l \gg al_P) = \frac{hc}{l}; \qquad u(l \ll al_P) = \left(\frac{c^4}{a^2 G}\right) l. \tag{11.7.2}$$

In physical terms, proposal (11.7.1) may be seen as a duality-invariant relation between behaviour at $l \gg al_P$ and $l \ll al_P$, i.e. under the exchange of l/al_P by al_P/l. This expression may be a purely mathematical interpolation, or it may have a physical meaning currently unknown. Indeed, in the case that the momentum may be related to u as $u(l) = p(l)c$, expression (11.7.1), written as

$$p(l) = \frac{h}{al_P} \frac{1}{\left(\frac{l}{al_P} + \frac{al_P}{l}\right)}, \tag{11.7.3}$$

can be interpreted as a dual-invariant generalization of the de Broglie relation of quantum mechanics. This could suggest a trans-Planckian generalization of quantum physics, namely, a generalization at length scales comparable or smaller than the Planck length l_P or the related quantity al_P. Indeed, one of the deepest open problems in current theoretical physics refers to the compatibility between quantum physics and general relativity. In particular, for l going to zero, the energy of the photons, if given by hc/l, goes to infinity. Furthermore, at very small length scales the quantum fluctuations implied by the Heisenberg principle become very high. However, from an experimental point of view, quantum mechanics has been tested to length scales of the order of 10^{-20} m, still far from the 10^{-35} m of the Planck length. Then, it is logical asking whether at scales smaller than 10^{-20} m some basic relations of quantum mechanics could have some changes. Expression (11.7.3) suggests one of the possible modifications, in which the energy of photons in the limit of very small l would not diverge but become zero, as for $l \ll al_P$, relation (11.7.1) yields $u(l)$ proportional to l.

Relation (11.7.2) (or (11.7.1)) satisfies physical duality in the sense that the behaviour at $1/l$ is symmetrically related to the behaviour at l. This would not be so, for instance, if in the denominator of (11.7.1) one had written $(l/al_P) + (al_P/l)^b$, with b some exponent different from 1. This would be also a possible generalization for the quantum mechanics at trans-Planckian scales, but it would not be symmetrically dual. This does not exclude the possibility, but it is less elegant from the point of view of symmetry. Thus, proposal (11.7.3) is inspired on the duality invariance relations proposed in superstring theories concerning the compact dimensions [22, 23], but it refers to the actual spatial dimensions, instead of proposing higher-order dimensional theories.

This proposal could be in principle checked through particle–antiparticle production or annihilation and in Compton scattering of electrons with very short wavelength photons. According to (11.7.1), the maximum energy of photons would be $E_{max} = hc(2al_P)$, with a less than 10^8; therefore, this maximum should settle some limits on the mentioned processes. At such energies, the energy density associated to a photons, namely, hc/l^4, would become comparable to that of a black hole of this size, and the photon would collapse into a black hole.

Note also that if one interprets relation (11.7.1) as $u(l) = hf(l)$, then the relation for the frequency $f(l)$ becomes

$$f(l) = \frac{c}{al_P} \frac{1}{\frac{l}{al_P} + \frac{al_P}{l}}, \tag{11.7.4}$$

which leads for a phase speed of light to

$$v_f(l) = lf = \frac{c}{1 + \left(\frac{al_P}{l}\right)^2}. \tag{11.7.5}$$

Expression (11.7.5) indicates that the speed of light in vacuum could become smaller than c for very small l.

Interpretation (11.7.3) of p, combined with the use of naive diffraction theory in narrow slits, leads to a modification of Heisenberg's uncertainty expression, as given by [24].

$$\Delta p \Delta x > \frac{h}{1 + \frac{a^2 l_P^2}{l^2}} \sim h\left[1 - \left(\frac{a^2 l_P^2}{h^2}\right) p^2\right], \tag{11.7.6}$$

with Δx and Δp the respective uncertainties in position and momentum. This suggests that for sufficiently short wavelengths the uncertainty could be reduced below the usual Heisenberg's value. Intuitively, this would follow from the idea that according to (11.7.2) the energy of very short wavelength photons would be smaller, and the perturbations on the particle being observed with such photons would be smaller than in the usual theory.

In connection with the spacetime fluctuations arising in quantum versions of gravity, instead of becoming higher and higher at smaller and smaller wavelengths, they would reach a finite maximum for length scales of the order of al_P, and they would decrease for still smaller wavelengths. Thus, maybe some of the incompatibility problems between quantum mechanics and general relativity related to the very strong fluctuations at small-scale lengths could be alleviated in this kind of proposal.

Such a reduction of the uncertainty for very small wavelengths is also found in a completely different theory, the so-called world crystal model [14]. Such a theory, proposed in 1987, considers that space and time are not continuous, but quantized, with a quantum, respectively, related to Planck length and time scales, analogous to an isotropic crystal having a reticle size corresponding to interatomic separation. When the equations for electromagnetism, and other differential equations, are studied in such a discrete spatiotemporal scheme, the corresponding dispersion relation becomes similar to those of waves in crystals, with the Planck length setting the limit of the analogous of the Brillouin zone [25]. Such dispersion relation is linear for relatively long wavelengths, but it curves down and arrives to the limit of Planck length with a zero slope. This would imply that the speed of light in the vacuum at very short wavelengths would become smaller and smaller until arriving to zero at the Planck length scale. In such a theory, Heisenberg's restriction on the product of the uncertainty relations $\Delta x\ \Delta p$ would become less than h for shorter wavelengths, tending to zero at the Planck length.

In contrast, other versions of modified uncertainty relation have been proposed in the context of quantum gravity or of string theories. Up to the second order in the Planck length, they state, for instance,

$$\Delta x > \frac{\hbar}{\Delta p} + al_P^2\left(\frac{\Delta p}{\hbar}\right) \tag{11.7.7}$$

with a a numerical constant of the order of 1. If $l_P = 0$, (11.7.7) reduces to Heisenberg's uncertainty principle. In this expression, the uncertainty product $\Delta x \Delta p$ for l_P different than zero would be higher than in the classical one, instead of being smaller as in (11.7.6). This conclusion is consistent with the idea that the uncertainty of spacetime fluctuations would add to the usual quantum uncertainty in a fixed spacetime. In contrast, in the duality-invariant proposal (11.7.3), the fluctuations of the spacetime would be much less than those expected in string theory or quantum gravity.

11.8 Thermodynamic Duality Between Macro- and Micro-Black Holes

The duality-invariant dispersion relation (11.7.1) also has consequences on the thermodynamics of black holes, which is based on defining an entropy S of the black hole in terms of the area A of its event horizon. Such a theory was initiated by Bekenstein and Hawking in the early 1970s and it was the first consistent theory combining quantum physics and gravitation. Since in the proposal (11.7.1), the Planck length, $l_P = (\hbar G/c^3)^{1/2}$ combining quantum aspects with gravitational aspects, plays a central role, it is natural exploring whether it could contribute in some way to the thermodynamics of black holes.

The energy U of black holes is proportional to their mass $U = Mc^2$, where M can be also expressed in terms of the Schwarschild radius $R_S = 2GM/c^2$. Thus, $U = (c^4/2G)R_S$. This expression has a formal analogy with (11.3.1), as it linearly relates an energy U with a length R_S.

Instead, the black-hole entropy is assumed to be proportional to its area A (given by $4\pi R_s^2$) as

$$S = \frac{k_B A}{4l_P^2} = \frac{k_B \pi R_S^2}{l_P^2}. \tag{11.8.1}$$

This relation was proposed by Bekenstein starting from a theorem by Hawking stating that the total area of a set of black holes can only increase or stay constant.

Since $S = k_B \pi R_S^2/l_P^2 = 4k_B \pi G^2 M^2/l_P^2$, the temperature of a black hole, given by $1/T = \partial S/\partial U$, is

$$\frac{1}{T} = \frac{\partial S}{\partial U} = \frac{1}{c^2}\frac{\partial S}{\partial M} = \frac{4k_B \pi G R_S}{l_P^2}, \tag{11.8.2}$$

and T may be expressed in the well-known Hawking–Bekenstein form:

$$T = \frac{\hbar c^3}{8\pi G k_B M}. \tag{11.8.3}$$

This means that for small black holes T will be very high, in such a way that they will evaporate very fast.

In a form analogous to that presented in (11.7.1), namely, the requirement of duality invariance with respect to the change of l/al_P by al_P/l, we write T in a form invariant with respect to the change of R_S/l_P by l_P/R_S. Then we propose generalizing (11.8.2) as

$$T = \frac{l_P}{4k_B\pi G}\frac{1}{\left[\dfrac{R_S}{l_P}+\dfrac{l_P}{R_S}\right]} \tag{11.8.4}$$

which is analogous to the invariance satisfied by expression (11.7.1). Since R_S is proportional to the mass, this leads to a generalized equation of state relating temperature and mass of the form [24, 26]:

$$2\pi m + \frac{a^2}{16\pi m} = \frac{1}{2\theta}, \tag{11.8.5}$$

where $m = M/M_P$ and $\theta = T/T_P$, with M_P the Planck mass $M_P = \frac{1}{2}\,(\hbar c/G)^{1/2}$ and T_P the Planck temperature $T_P = 2M_Pc^2/k_B$. We have written a^2 in the numerator of the second term of the right-hand side, in order to formally have the Hawking expression for $a = 0$, and the generalized expression (11.8.4) for $a = \pi/2$.

By integrating $\partial S/\partial U = 1/T$, Eq. (11.8.5) leads to a black-hole entropy of the form:

$$S_{bh}(M) = k_B\left[\frac{A}{4l_P^2} + \frac{a^2}{32\pi}\ln\left(\frac{A}{4\pi l_P^2}\right)\right] \tag{11.8.6}$$

with A the surface area of the black hole. When $a = 0$, (11.8.6) reduces to Hawking–Bekenstein expression (11.8.3) for the black-hole entropy.

Note that for high values of m (11.8.5) leads to $m \sim 1/\theta$, as in Hawking expression, but that for m small it leads to $m \sim \theta$. This would have deep consequences for the evaporation of small black soles: instead of being very fast, proportional to $\theta^4 \sim 1/m^4$, it would be much lower since θ would be small and $\theta^4 \sim m^4$. This could imply that, in contrast to Hawking theory, many microscopic black holes could exist, which could be candidates for constituents of dark matter.

Equation (11.8.5) points a thermodynamical dual relation between macro-black holes, with M proportional to T^{-1} for high values of M (the usual Bekenstein–Hawking result), and micro-black holes with M proportional to T for small values of M, which is the duality symmetric of the former one. This relation between macro- and micro-black holes is a special case of thermal duality as that considered in Sect. 11.5. In Fig. 11.1, we illustrate expression (11.8.5). The discontinuous lines plotted in Fig. 11.1 correspond to the branches $m \sim \theta^{-1}$ and $m \sim \theta$. The vertical line intersects the curve of Eq. (11.8.5) at 2 (or 1, or 0) points, corresponding to, respectively, dual states. As seen in Fig. 11.1, there is a maximum value for T for

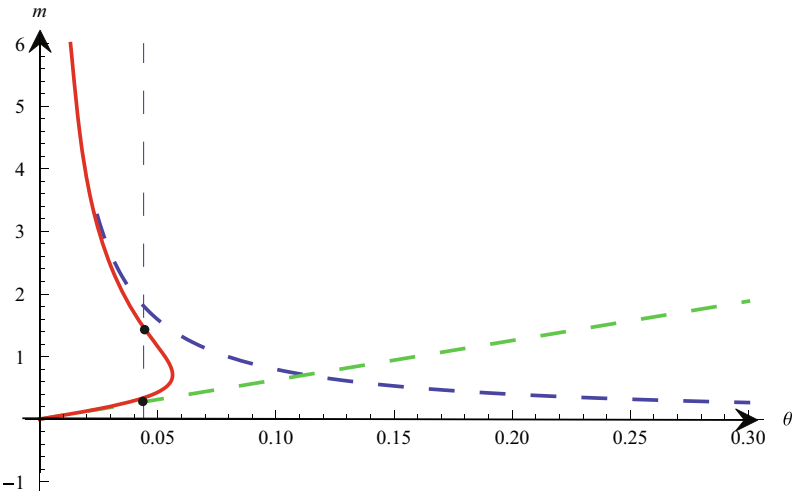


Fig. 11.1 Relation (11.8.5) between the dimensionless mass $m \equiv (M/M_P)$ and dimensionless temperature $\theta \equiv (T/T_P)$ for $a = 0$ (Hawking expression) (discontinuous parabolic line) in the upper curve, for $a = \pi/2$ (solid line) and for the asymptotic expression $m = (a^2\pi/8)\theta$ with $a = \pi/2$ (tilted right discontinuous line). The vertical line corresponds to a given value of T. The intersecting points of it with the continuous line define the respective thermally dual black holes. Figure from [26]

black holes, given by $T_{bh\ max} = c^4 l_P/8G(\pi b)^{1/2}$. At such value of T, M takes the value $MT_{bh\ max} = (b/4\pi)^{1/2} c^2 l_P/G$. In this case, the corresponding thermodynamic state is dual to itself.

In Sect. 11.6, we have related through the thermal duality symmetry some early states of the universe (dominated by photons) to late states of the universe (dominated by cosmic string loops). The thermal duality symmetry examined in this section sets a connection between early state of a macro-black hole to its late microscopic state. In the standard theory of black holes, $R_S \sim M \sim T^{-1}$, in such a way that evaporation, leading to a decrease of R_S, leads to an increase of T and therefore of the rate of evaporation. Eventually, for $R_S \to 0$ this leads to $T \to \infty$, leading to an explosive eventual stage of evaporation in a finite time. In contrast, in the duality set here, the behaviour $R_S \to T^{-1}$ for macro-black holes tends to $R_S \to T$ for micro-black holes. This would imply that the late stages of evaporation would correspond to a low T and to a slow evaporation.

Note that there exists still a further duality symmetry relating thermal equilibrium of macro-black holes and photon gas in a hypothetical finite box, to thermal equilibrium of micro-black holes and cosmic string loop gas, also in a hypothetical finite box [24, 26]. Thus, in Sects. 11.5 and 11.8, we have suggested three thermal duality symmetries: (1) between thermodynamics of a photon gas and of a cosmic string

gas (Sect. 11.5); (2) between thermodynamics of macro-black holes and micro-black holes (Sect. 11.8); and (3) between thermal equilibrium between photons and macro-black holes and thermal equilibrium between cosmic string loops and micro-black holes [24, 26].

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Chapter 12
Perspective and Suggestions. Non-Fourier Heat Transport in Solids



In this chapter, we present a few suggestions and comments for future work. First, we discuss some differences between the usual two-fluid model and the one-fluid extended model (or, in other words, on the macroscopic two-fluid model). Our aim is to provide the researcher interested in such comparison some data about the values of the coefficients to be used in the latter model. Second, it should be desirable to bring some of the results here, especially those referring to the dynamics of quantized vortices and their interaction with the heat flux, to the very active setting of Bose–Einstein condensates. In the introduction we have already commented some of the main differences between these condensates and superfluid helium. And, eventually, since one of the main aspects of the present book is the analysis of non-Fourier heat transport.

The chapter is also devoted to two basic, even still quite open, questions regarding the numerical values of the parameters appearing in this book and regarding the two-fluid model and the one-fluid extended model, as well as the corresponding boundary conditions and in some cases the initial conditions of the fields involved in the latter model. This is a convenient discussion, because practically all interpretations of the superfluid phenomena have been carried out in the two-fluid model, so that the parameters appearing in the literature are those corresponding to that model.

We introduce the state of the art about the knowledge of such parameters, for researchers interested in using the one-fluid extended model which in some cases can be directly found from the Donnelly and Barenghi's paper Ref. [1]. In Chaps. 2–5, we listed the values of some of the relevant quantities. We list them for the use of the reader, and we discuss how the experiments may contribute to find them.

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12.1 Determination of the Physical Parameters

Most of the parameters that Donnelly and Barenghi reported in the paper [1] come up from experiments performed over the last 80 years, just after the formulation of the two-fluid model by Tisza and Landau. This means that all the interpretations of the results, as well as the direction of studies, questions, and experiments, have been influenced by the idea of two fluids dealt with as a classical viscous fluid (normal component) and an Eulerian inviscid fluid (superfluid components), mutually interacting with each other, but without a direct separate visualization of each one. This has been a limitation from one side, but from the other side has contributed to interesting results, and have enlightened the behaviour of superfluid helium by means of hydrodynamical and microscopical arguments.

In Chaps. 5–7, we have seen that the mathematical models may be considered both from the macroscopic and mesoscopic point of view depending on whether the analysis is focused on samples in the absence of vortices—with few vortices or with many vortices inside. Indeed, in the latter situations one usually considers a small volume Λ (containing many vortices) where all the files of the mathematical models are defined. Thus, the parameters of the model depend on the considered viewpoint. Clearly, in the absence of vortices the two viewpoints merge, such as to the so-called Hall–Vinen–Bekarevich–Khalatnikov equations, which in the absence of vortices become the equations proposed by Landau.

12.1.1 Parameters of the Mathematical Model: The Mesoscopic View

First of all, in the one-fluid extended model we do not need to consider two separate densities and two velocity fields, but we can use the total mass density ρ, which is reported in Table 12.1.

In Sect. 3.1.1, and in particular in Eqs. (3.1.5), (3.1.6), and (3.1.7) we have seen the relaxation times τ_0, τ_2, and τ_1 of p_V, $\mathbf{P_v}$, and $\mathbf{q}$, respectively; and bulk viscosity; shear viscosity; and heat conductivity λ_0, η, and λ_1 and the coefficients β and β'. We have seen in Eq. (3.1.7) that when τ_0, τ_1, τ_2, β, and β' are zero, these equations,

Table 12.1 Values of the densities ρ, the entropy s per unit mass, the specific heat c_V per unit mass, and the shear viscosity η. Data from [1, 2]

	2.0 K	1.7 K	1.5 K	1.2 K	1.0 K	0.7 K	0.5 K
ρ(g/cm^3)	0.1456	0.1453	0.1452	0.1451	0.1451	0.1451	0.1451
s(J/g K)	0.9621	0.3950	0.1965	0.0052	0.0016	0.0002	0.0000
c_V (J/g K)	5.190	2.187	1.133	0.335	0.105	0.009	0.002
η(Pa · s x 10^6)	1.468	1.290	1.346	1.736	3.873		

Table 12.2 Values of the velocities V_1, V_2, V_4 from [1, 2]

	2.0 K	1.7 K	1.5 K	1.2 K	1.0 K	0.7 K	0.5 K
V_1(cm/s)	226.2	233.1	235.5	237.3	237.8	238.2	238.2
V_2(cm/s)	16.78	20.33	20.00	18.51	18.93	47.07	130.93
V_4(cm/s)	152.3	205.1	222.2	234.5	236.9	238.1	238.2

respectively, reduce to the Stokes equations for p_V and $\langle \mathbf{P_v} \rangle$, and the Fourier equation for $\mathbf{q}$.

In (3.3.55) and (3.3.56), we have found the expression for the first and second sounds and of their attenuation coefficients. In Ref. [1], some values of the first, second, and fourth sounds have been found using the two-fluid model (some of these values have been reported in Table 12.2).

Thus, in this model, we can use the values of ρ, λ_0, η, and c_V reported in Tables 12.1 and 12.2, even though a further revision of the experiments is suggested in view of the new interpretation. It would be of experimental interest to find the values of τ_1, β, β' and furthermore checking the results of the two models.

The parameters β and β' appear in the general expression (3.1.9) of the entropy flux in extended thermodynamics. It has been seen that in some simple situations β and β' may be chosen to be $\beta = \beta' = -1/(ST^2)$, with S the entropy density per unit volume. In this case, we have also seen that a small amount of entropy is carried out by the superfluid component, as also pointed out in [3]. Clearly, in terms of the one-fluid model it does not matter because there is only one fluid, but in the view of a direct comparison between the two models (and their respective results) it is interesting to understand how to evaluate this amount. From (3.4.73) one finds that

$$\frac{s^{(s)}}{s} = 1 - \frac{\beta^*}{\beta}, \tag{12.1.1}$$

where the asterisk denotes the quantities referring to the case when the superfluid component does not carry entropy. The corresponding second sound velocity can be also written from (3.4.82), namely:

$$V_2 = V_2^* \left(s - s^{(s)}\right). \tag{12.1.2}$$

From (12.1.1) and (12.1.2), $s - s^{(s)}$ and $s^{(s)}$ are found if the velocity of second sound and the entropy s can be obtained. More precisely, the direct measurements of the first, second, and fourth sound speeds and of s, p_T, and c_V help to find the entropy carried by the superfluid component.

In Sect. 7.2, the fourth sound was studied. It was found the speed (7.2.25) and the attenuation (7.2.26). According to these results, it would be easy to perform experiments which would establish the accuracy of the choice $\beta = \beta'$. Indeed, if the latter were true then the wave would be propagated without attenuation (namely, with

$k_s^{(4)} = 0$). For more comments and details, see comments in Eq. (7.2.26). Furthermore, since (2.8.88) and (7.2.26) depend on the small amount of superfluid entropy, then it would be easy to verify the existence of $s^{(s)}$ for the two-fluid model. In [4], the values of β and β' and hence the values of $s^{(s)}$ were determined from experimental data.

12.1.2 The Macroscopic Model

Chapters 2 and 3 deal with superfluid helium in laminar situations, namely, in the absence of quantized vortices, (and hence meso- and macroscopic description) and Chaps. 5–7 are mainly devoted to turbulent situations, and all the fields are defined in a small volume Λ (we may say macroscopic level). The same difference was outlined by Bekarevich and Khalatnikov in their studies, when they considered the presence of the vortices in the two-fluid model [5].

In the one-fluid extended model, the influence of the vortex lines has been introduced in the source term $\sigma^{\mathbf{q}}$ of the heat flux by a tensor which takes into account the different situations at hand (rotating quantum turbulence, counterflow turbulence, and so on), which of course has been averaged in a small volume Λ where many vortices are present [6]. This means that the main fields (velocity, heat flux, density, and so on) have been intrinsically extended to the macroscopic level, namely, averaged in the volume Λ, in the same way as proposed by Bekarevich and Khalatnikov in their studies [5].

The source term $\sigma^{\mathbf{q}}$ in the evolution equation of the heat flux (3.1.19) depends strongly on the experiment at hand and on the presence of turbulence. Indeed, $\sigma^{\mathbf{q}}$ in the laminar regime is

$$\sigma^{\mathbf{q}} = -\frac{1}{\tau_1}\mathbf{q}; \tag{12.1.3}$$

in the presence of rotation is given by expression (4.2.6)

$$\sigma^{\mathbf{q}} = -\frac{1}{\tau_1}\mathbf{q} + B_{HV}\frac{\boldsymbol{\Omega}}{|\Omega|} \times \boldsymbol{\Omega} \times \mathbf{q} + B'_{HV}\, \boldsymbol{\Omega} \times \mathbf{q}, \tag{12.1.4}$$

which is the rewritten version of the macroscopic mutual friction force in the two-fluid model; in the isotropic turbulent regime is (5.3.27)

$$\sigma^{q} = -\frac{1}{\tau_1}\mathbf{q} - K_f L\mathbf{q}, \tag{12.1.5}$$

which comes from the macroscopic mutual friction force in counterflow experiments; and in the anisotropic turbulent regime is (7.3.33)

$$\sigma^{\mathbf{q}}_{turb} = -\frac{1}{3}\kappa L B_{HV}\Pi\cdot\mathbf{q} + TS\epsilon_V c_1 L^{3/2}\left[\frac{B_{HV}}{2}\left(\mathbf{I}+\frac{B'_{HV}}{B_{HV}}\mathbf{J}\right)-\mathbf{J}\right], \quad (12.1.6)$$

incorporating the couplings of $\mathbf{q}$ with quantized vortices in an anisotropic polarized tangle of vortices.

The values of B_{HV} and B'_{HV} have been given in Table 4.1, where we have reported the coefficient α and α' related to B_{HV} and B'_{HV} as $B_{HV} = (\rho/2\rho_n)\alpha$ and $B'_{HV} = (\rho/2\rho_n)\alpha'$. From these values and using the expression $K_f = (1/3)\kappa B_{HV}$, we find the values of K_f.

A further field to be considered is the vortex line density per unit volume L, introduced in Chaps. 5, which appears when quantized vortex lines develop inside Helium II. In this case, we have seen that a further equation has to be included into the model. In (6.1.5) and (6.1.6), we have considered the interplay between heat flux and L, when the variation of the velocity $\mathbf{v}$ may be neglected. In these equations in ζ and χ describe the contribution of the gradients of temperature and of vortex lines to the heat flux, ν_0 takes the back reaction of the heat flux to L, K_f is the mutual friction between vortices and heat flux given by $K_f = \frac{1}{3}\kappa B_{HV}$, and $\tilde{A}$ and $\tilde{B}$ are the so-called production and destruction coefficients of the Vinen equation: $\tilde{A} = \alpha_v/\rho_s T s$ and $\tilde{B} = \beta_v\kappa$ [7–9]. The other three parameters (ζ, χ, and ν_0) are new, and they deserve deeper considerations. In (6.1.8), we have found the drift velocity of the tangle in terms of ν_0, namely, $\mathbf{v}_L = (\nu_0/L)\mathbf{q}$. The new visualization techniques allow to evaluate the tracking velocity of the suspended particles for estimating the velocity of the fluid (free particles) and the vortex lines drift (trapped particles). The last results give information about the velocity $\mathbf{v}_L$, and hence

$$\mathbf{v}_L = \frac{\nu_0}{L}\mathbf{q} \qquad \Rightarrow \qquad \nu_0 = \frac{v_L}{L}q, \quad (12.1.7)$$

where v_L is the experimental velocity of the trapped particles, L is the stationary value, and q is the heat flux along the velocity of the vortex tangle. The velocity of the suspended untrapped particles is usually related to the normal component in the two-fluid model, as $v_{particles} = \frac{1}{2}v_n$ because of the interaction of the particles with the quantized vortices (see Sect. 6.6). From this relation, in the two-fluid model we have in general

$$\mathbf{v}_{particles} = \frac{1}{2}(\mathbf{v} - \beta T\mathbf{q}) \quad (12.1.8)$$

which in our opinion explains better the motion of the suspended particles, as due to the velocity $\mathbf{v}$ (for instance, in the external imposed mass flow) or to the heat flux $\mathbf{q}$ (for instance, in the external imposed heat flow).

In the experiments with external heat flow, it is possible to evaluate the velocity of the untrapped particles, and hence of the heat flux $\mathbf{q}$ from (12.1.8), and of ν_0 from (12.1.7).

The parameter ζ appears in the expression (3.3.56) for the second sound velocity as $V_2^2 = (\zeta/\rho c_p)$, and hence it can be inferred from the evaluation of the speed of

second sound. Its values have been given in Table 5.2. Instead, χ, defined in Eq. (6.1.6), can be found from the vortex diffusion coefficient in Sect. 6.2.1 or from the vortex density waves in Sect. 6.2.2. In Sect. 6.2.1, the vortex diffusion coefficient D is proportional to χ by $D = (\nu_0\chi)/(K_f L_0)$. In literature there are two numerical studies performed for T close to 0 K by Tsubota ($D \approx 10^{-1}\kappa$) in [10] and by Nemirovskii ($D = 2.2\kappa$) in [11]. The values they find are different, but they are proportional to κ. These values allow us to find χ from $\chi = DK_f L_0/\nu_0$, when the other parameters are known.

In Sect. 6.2.2, we have studied the vortex density waves, which have allowed to find at high frequencies the speed (6.2.5): $v_\infty^2 = \nu_0\chi$. This is an interesting result which may contribute to evaluate the value of χ from the experiments, but to our knowledge these kind of waves have not been visualized yet.

12.2 Initial and Boundary Conditions

In this section, we introduce the still open discussion on the boundary conditions and the initial conditions of the main fields of the one-fluid model, which is of interest for performing numerical experiments in Helium II by using this model. This leads to a further difference between these two existing models. Indeed, the hydrodynamical concepts may be directly applied to the two-fluid model because it describes Helium II as composed of two fluids (viscous and ideal), instead the one-fluid extended model considers heat flux as one of the main fields. For this reason, literature and experiments suggest the behaviours of the normal and superfluid component of Helium II on the wall of the system, but less is known about the behaviour of the heat flux on the walls. Some suggestions may be taken from the heat transport in nanosystems.

12.2.1 Initial Conditions

The Cauchy problem associated to Eqs. (3.1.1)–(3.1.3), (6.1.1) and (6.1.2) or to (7.1.1) or to other extensions of the equations proposed in this book requires the knowledge of the initial values for the main fields appearing in the model.

Regarding to (3.1.1)–(3.1.3), we need to set the initial values of the density ρ, the velocity $\mathbf{v}$, the temperature T, the heat flux $\mathbf{q}$, the non-equilibrium pressure p_V, and the deviatoric part of the non-equilibrium stress tensor $\langle \mathbf{P_v} \rangle$. In [12], it was shown that the relaxation times τ_0 and τ_2 of p_V and $\langle \mathbf{P_v} \rangle$ are very small, so we have neglected them. We want to emphasize that there may be experiments where these two relaxation times cannot be neglected and new results may be found, as, for instance, in superfluid helium flowing through nanosystems.

For $\tau_0 = \tau_2 = 0$, we are led to Eqs. (3.1.16)–(3.1.19) for which the Cauchy problem requires the initial density ρ, velocity $\mathbf{v}$, temperature T, and heat flux $\mathbf{q}$. For this

Table 12.3 Initial conditions for the main fields

	Counterflow	Coflow	Superflow
Density $\rho(t=0)$	ρ_0	ρ_0	ρ_0
Temperature $T(t=0)$	T_{bath}	T_{bath}	T_{bath}
Velocity $\mathbf{v}(t=0)$	$\mathbf{v}_0 = 0$	$\mathbf{v}_0 = constant$	$\mathbf{v}_0 =?$
Heat flux $\mathbf{q}(t=0)$	$\mathbf{q}_0 = constant$	$\mathbf{q}_0 = 0$	$\mathbf{q}_0 =?$

reason, we consider the more classical experiments, which may open the discussion for other experiments. We report our proposal in Table 12.3, where it is easy to set the initial conditions for the velocity and heat flux in the case of counterflow (heat flow) and coflow (mass flow), but it is not clear how to choose the initial condition in the case of superflow. If Helium II is thought in terms of the two-fluid model then the classical approach for the viscous fluid can be used, namely, the normal component does not flow inside the superfleak, and we can infer the relation

$$v_0 + \frac{q_0}{\rho_0 S T_0} = 0 \tag{12.2.9}$$

which is velocity of the normal component (Eq. (3.4.75)). The lack of experimental information may lead to use the above relation for numerical experiments, which may be simplified to the conditions $v_0 = q_0 = 0$, for the sake of simplicity. Our aim here is emphasizing that more general situations may be considered without setting (12.2.9) as true a priori.

12.2.2 Boundary Conditions

The arguments of this subsection open new challenges and questions for future experiments and studies, both theoretical and numerical. As already pointed out in the previous subsection, if we abandon the idea that Helium II is made up of two different fluids, then the problem how to set the fields of the one-fluid model quickly arises. Although some suggestions may be borrowed from the two-fluid model, this is not always true, because, for instance, the two-fluid model is not realistic for very low temperature, when the ballistic regime for phonons and rotons takes place.

For these reasons and because of the new advances in the experimental settings—allowing to visualize Helium II close to the walls—we must consider these questions.

In Chaps. 3 and 5, we have used the boundary conditions in our analysis of the effective thermal conductivity of Helium II in the hydrodynamical, ballistic, and turbulent regimes. There we have assumed that heat flux and velocity on the wall are zero in the case of laminar and turbulent regime (namely, $q_{wall} = v_{wall} = 0$) for the sake of simplicity and that only heat flux is not zero in the ballistic regime (namely, $q_{wall} \neq 0$). These assumptions would lead to an inconsistency if we thought

Helium II described by two fluid components. Indeed, in the latter case, it is always assumed that $v_{n\,wall} = 0$ and a slip condition for the superfluid component, which is inconsistent with $q_{wall} = v_{wall} = 0$.

We aim to go beyond the restrictions imposed by hydrodynamical arguments on the two components of Helium II in order to better understand the existing experiments and suggesting new ones. For the field $\mathbf{v}$ we have assumed that it is zero on the wall, but the new forthcoming experiments in Prague may suggest a different outcome where a slight sliding is allowed. Instead, heat flux is related to dynamics of the heat carriers and its value on the walls clearly depends on the collisions of the quasi-particles with the walls. It is plausible that $\mathbf{q}_{wall}$ may depend on the temperature and on the size of the system, because the amount of the quasi-particles decreases with the temperature and with the size of the channel. In phonon hydrodynamics, the collision of phonons against the walls may play an important role if the mean free path of these phonons ℓ_m is comparable to the size of the system (e.g. the channel width D).

In the ballistic regime in Sect. 3.4.2, we have used a relation coming from the hydrodynamics of phonons, namely, expression (3.5.93) which we recall here:

$$\mathbf{q}_{wall} = -C\ell_m \left(\frac{\partial \mathbf{q}}{\partial \xi}\right)_{wall}, \tag{12.2.10}$$

where C is constant depending on the material and the experimental setup, and ξ is the outward-pointing variable. Expression (12.2.10) was used for the first time in Ref. [13] to take into account of the contribution of phonon-boundary scattering (boundary contribution) in the phonon hydrodynamics in nanosystems, and it comes from the microfluidics [14, 15]. This expression is often used in dilute gas of phonons, even though we do not know whether it fulfils the features of Helium II because of the lack of experiments. Further studies and experiments would help to understand better the behaviour of Helium II close to the walls.

It is expected that it is valid when $T < 0.7$ K (ballistic regime in Helium II) [16]. The existence of a slip heat flux at the boundaries of the channel implies that also the flow of the normal component might be non-vanishing at the boundaries. In Ref. [17], we discuss the implications if such slip flow of the normal component at the boundaries also exists when $T > 0.7$ K (the hydrodynamic regime). For these purposes, the same model used in [18] has been consider assuming a slip condition for the normal component $\mathbf{v}_n$.

From a direct comparison between the results found in [17, 18] with the experimental ones by Guo et al. [19], we see that the presence of the slip condition for the normal component, especially the highest sliding, leads to a flatter profile for the normal component, even though it is still parabolic-like.

This means that the velocity of the normal component is not properly zero on the wall and the behaviour of Helium II on the wall requires further studies and experiments.

There is a further interpretation about the results found here, when compared to Guo's experiments, which may be useful for the debate. In our paper, we have

used the usual interpretation of superfluid helium, that it is composed by the two undistinguished components: normal and superfluid. Even though it is well known that it is a toy model proposed by Landau almost a century ago, it is well used for describing the experiments of Helium II.

Another field used in the presence of turbulence is the vortex line density L. According to its definition, the value of this field on the wall is zero, namely, $L_{wall} = 0$. Since the behaviour of the vortex line next to the wall is to move as if another vortex line were on the other side of the wall, we may reflect this behaviour as

$$\frac{\partial L}{\partial \mathbf{n}} = 0 \quad \text{and} \quad \frac{\partial L}{\partial \mathbf{t}} \neq 0,$$

where $\mathbf{t}$ and $\mathbf{n}$ stand for tangent and normal direction with respect to the wall. In fact, the equations should be complemented with a generalization of the Vinen equation applied to zone close to the walls.

12.2.3 The Contribution of Quantized Vortices in the Numerical Experiments

A quantized vortex line can be thought as a classical vortex line in the superfluid component with a thin hollow core of the size of an helium atom, and quantized circulation κ. At temperature higher than 1 K in the presence of normal component, vortex line moves following the equation:

$$\tilde{\mathbf{v}}_L = \frac{\partial \mathbf{s}}{\partial t} = \tilde{\mathbf{v}}^{(\mathrm{i})} + \alpha \mathbf{s}' \times (\tilde{\mathbf{u}}_{ns} - \tilde{\mathbf{v}}^{(\mathrm{i})}) - \alpha' \mathbf{s}' \times [\mathbf{s}' \times (\tilde{\mathbf{u}}_{ns} - \tilde{\mathbf{v}}^{(\mathrm{i})})] + \lambda \mathbf{s}', \quad (12.2.11)$$

where λ is an undetermined coefficient. In the current literature, one assumes $\lambda = 0$. For temperature very close to zero, the above equation reduces to the so-called Biot–Savart law [20]:

$$\dot{\mathbf{s}} = \frac{\kappa}{4\pi} \int \frac{(\mathbf{s} - \mathbf{s}_1) \times d\mathbf{s}}{|\mathbf{s} - \mathbf{s}_1|^3}, \quad (12.2.12)$$

where $\mathbf{s}(\xi, t)$ is the position of a point on the vortex line, $\mathbf{s}_1$ is a point in the fluid, and the integral is extended all over the vortex lines. This integral diverges when $\mathbf{s}$ goes to $\mathbf{s}_1$, for a true line, but divergence is removed by taking proper account of the non-vanishing value of the core size. Equations (12.2.11) and (12.2.12) are used to describe the dynamics of vortex lines, and they may be used both in the two-fluid model and in the one-fluid model by substituting in (12.2.11) $\tilde{\mathbf{u}}_{ns}$ with the local heat flux according to rule (2.1.3). This may be useful for describing the velocity profiles as well as the heat flux profile.

In Sects. 6.4 and 7.5, we have presented the results of numerical simulations for the velocity and vortex density profiles in two-dimensional situations by using the

two-dimensional counterpart of Eq. (12.2.11). As initial conditions on the vortices, it is usual to impose a homogeneous distribution (as it has been done in the two-dimensional case), and also an isotropic distribution, which was not the case in two dimensions. As boundary conditions, it is important to set the interaction of the vortices with the wall. In the two-dimensional case, it was assumed that if the centre of the vortex was closer to the wall that some critical distance, the vortex disappears, and, in order to study steady-state situations, a new vortex was introduced at random in the plane. In three dimensions, the situation is analogous but more complicated, as the vortices form closed loops. Therefore, one could impose that when one point of the loop approaches the wall less than some critical distance, the whole loop disappears and another loop is randomly introduced in the system, with a random orientation. A more exhaustive procedure is the vortex filaments model well explained in literature, as, for instance, in Refs. [21–23].

In these experiments, in Sects. 6.4 and 7.5, we have used the two-fluid model because, as said, it uses a hydrodynamic approach and hence the boundary conditions as well as the initial conditions are prescribed. The same numerical experiments have been realized in a very recent paper [17] by using the two-fluid extended model (coming from the one-fluid model and described in Sect. 3.4.2) but with a slip condition on the wall for the normal component. The profiles in this case are slightly different than those found by the usual two-fluid model, but future experiments may explain better this unclear behaviour of Helium II next to the walls.

12.3 Some Reflections on Non-Fourier Hydrodynamic Heat Transport in Solids

In this book, we have tried to show how rich the phenomenology and the underlying physical basis of non-Fourier heat transport are. We have shown the complexities of the interaction between heat flow and quantum turbulence and their mutual feedbacks, linking in complex ways heat flux and quantum vortex tangles. Though we have devoted all our attention to superfluid helium, it is important to know that several aspects of this analysis are also useful, though in a lesser quantitative degree, in heat transport in solids, especially concerning inertial aspects and nonlocal aspects. In particular, the so-called phonon hydrodynamics [24, 25], where nonlocal contributions analogous to the hydrodynamic viscous contributions appear in the equation for the heat flux, is receiving much attention. Thus, a number of results presented in this book are also applicable to solid systems, especially at the nanoscale. In such systems, the length of the phonon mean free path may be comparable or higher than some characteristic size of the solid system (as the radius of nanotubes or the thickness of thin nanolayers). The effects are not so relevant as in superfluid helium, because the latter is characterized by collective coherent quantum effects, which make that the correlation lengths and relaxation times are very long. Instead, in solid systems, the interaction between electron transport and phonon transport may lead

to very interesting and useful effects which are not found in superfluid helium. Some other aspects of heat transport present in superfluid helium are not known in solids, but may stimulate some questions, as, for instance, the possible role of heat vortices in heat transport, and the possibility of turbulent transport when the perturbations are sufficiently high. Up to now, this situation is very well studied in superfluids, where vortices are quantized, but the effects of heat vortices in solids are just beginning to attract the attention.

Phonon hydrodynamics is considered to have begun in 1963 with a paper by Sussman and Thellung [26], followed by papers by Gurzhi [27], Guyer and Krumhansl [28, 29], and Beck, Meier, and Thellung [30] among others. The physical basis of these papers was the quantum formulation of a Boltzmann equation for phonon by Peierls, on the consideration of two different relaxation times related to resistive phonon collisions and non-resistive phonon collisions, and the observation that even in the case when non-resistive collisions dominate over the resistive ones there is a resistance to heat transport related to a collective motion of the phonons. Papers of that period (from 1963 to 1973, for instance) involved not only to phonon hydrodynamics but also to second sound, and referred to the two-fluid model of Tisza and Landau as one of the sources of their inspiration for two-fluid models of dielectric solids, based, for instance, on polar and nonpolar contributions, or on populations of phonons experiencing resistive or non-resistive collisions.

In a generalized version including convective nonlinearities, a widely used equation for heat transport in solids is of the form [31]:

$$\tau\frac{\partial \mathbf{q}}{\partial t}+\frac{2\tau}{c_v T}\mathbf{q}\cdot\nabla\mathbf{q}=-\mathbf{q}-\lambda\nabla T+\mu\left(\nabla^2\mathbf{q}+2\nabla\nabla\cdot\mathbf{q}\right). \tag{12.3.13}$$

In the absence of the nonlinear terms of the left-hand side, this equation is generically known as Guyer–Krumhansl equation, and its most characteristic feature is the last term in the right-hand side, a nonlocal term in which the coefficient μ plays a role analogous to the viscosity in usual hydrodynamics. The second term on the left-hand side is a convective term, with $\mathbf{q}/(c_v T)$ being the convective average velocity of the phonons, c_v being the specific heat per unit volume.

The phonon-hydrodynamic regime corresponds to situations when the term in $\mathbf{q}$ on the right-hand side may be neglected as compared with those in $\mu\nabla^2\mathbf{q}$; this happens, for instance, in nanotubes whose radius is smaller than the phonon mean free path. In this case, Eq. (12.3.13) becomes analogous to Eq. (3.1.7) for $\mathbf{q}$, when one takes $\mathbf{P_v}$ proportional to $\nabla\mathbf{q}$ and p_V proportional to $\nabla\cdot\mathbf{q}$. The main difference is that the collective quantum effects make that in superfluid helium the nonlocal terms are very relevant. Another difference, as mentioned before, concerns the role of heat vortices. In superfluid helium, such vortices are quantized and their resistive effects on the heat flux may be modellized through a term in $L\mathbf{q}$, with L the vortex length density, as in Eqs. (5.3.27) or (5.3.31) or (10.1.2). Such terms do not appear in usual phonon hydrodynamics in solids. However, non-quantized vortices in phonon hydrodynamics may arise, for instance, as a result of the nonlinear convective terms [32–34]. The exploration of the effects of such vortices on heat transport in solids and their

interaction with the average heat flux would bring phonon hydrodynamics in solids a little bit closer to those in superfluid helium.

Besides the convective nonlinearities, two other kinds of nonlinearities may arise: rheological-like (namely, the term in the Laplacian of the heat flux (analogous to the Newtonian viscous term) could be changed by a term of non-Newtonian form, where nonlinear effects and viscoelastic effects could be taken into account. For instance, one may explore situations in which the flux of the heat flux would not be proportional to $\mu\nabla\mathbf{q}$ but to $\mu(\gamma)\nabla\mathbf{q}$, with γ being related to the second invariant of the $\nabla\mathbf{q}$ tensor; thus, in two-dimensional flows, γ may be taken as the modulus of $d\mathbf{q}/dx$. In usual rheology, this kind of model is known as power-law fluids [33, 35]. Other possibilities inspired on other rheological models could be taken into account [36], thus suggesting a generalization of the presentation we have done in Chap. 3, in the sense that the flux of the heat flux could be explored in further detail rather than taking it as proportional to $\nabla\mathbf{q}$.

Another source of nonlinearities may be the dependence of the several parameters appearing in the equation (the relaxation time, the thermal conductivity, the mean free path) on temperature (very logical and well-accepted possibility) and on the modulus of the heat flux [37]. The latter dependence may follow, for instance, from nonlinear dispersion relations of phonons [38, 39]. Such dependence has been practically dismissed up to know both in solids as in superfluid helium. The effects of this dependence could be dealt with by going to more general forms of thermodynamics, as, for instance, using the extended thermodynamic approach but taken as the additional variable the thermodynamics conjugate of q instead of the heat flux itself. This has been done, for instance, in [40], leading to an evolution equation for the heat flux which displays several analogies with the evolution equation for the electric field in nonlinear optics. Such generalizations could be used not only in the temporal effects (where they lead, in some occasions, to soliton transport [41–43], a topic which would deserve more interest in superfluids too) but also in the nonlocal spatial effects.

Some active lines of research in this fields are: (1) microscopic derivations of the basic equations (from kinetic theory, quantum microscopic formalisms, numerical simulations); (2) realization and analysis of experiments, (in different kinds of materials, of different sizes of the systems, with microstructured systems, with two-dimensional systems); (3) thermodynamic derivation of nonlocal transport laws and their relations with generalized entropy and entropy flux (extended thermodynamics, internal variable theories, gradient theories); (4) exploration of different kinds of boundary conditions (non-slip heat flow along the walls, first-order heat slip flow, second-order slip); (5) consideration of nonlinear generalizations of the equation (temperature-dependent coefficients, nonlinear convective terms, nonlinear relations between the flux of the heat flux and the gradient of the heat flux, dependence of the coefficients on the heat flux itself, as arising from maximum-entropy arguments and from nonlinear dispersion relations for phonons, appearance of heat vortices, soliton heat transport); (6) application of the generalized heat transport theories to nanosystems and microstructured or nanostructured systems and their application

to phononic devices and to thermal metamaterials; (7) transitions between different regimes of heat transport (usual conduction, phonon hydrodynamics, ballistic regimes, second sound); and (8) generalizations of these theories to other kinds of transport, as electronic transport, electronic and phononic transport, thermoelectric devices, and spin waves.

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